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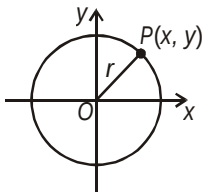
CHAPTER 1

CIRCLE

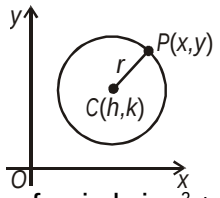
Circle: A circle is a locus of a point whose distance from a fixed point (called centre) is always constant in a Plane .

1. EQUATION OF A CIRCLE IN VARIOUS FORMS

(a) **Central Form:** The circle with centre as origin and radius r has the equation : $x^2 + y^2 = r^2$.



(b) **Standard Form:** The circle with centre (h, k) and radius r has the equation $(x - h)^2 + (y - k)^2 = r^2$.



(c) **General Form:** The general equation of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ with centre as $(-g, -f)$ and radius = $\sqrt{g^2 + f^2 - c}$.

This can be obtained from the eqution $(x - h)^2 + (y - k)^2 = r^2$

$\Rightarrow \quad x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0$

Take $-h = g$, $-k = f$ and $h^2 + k^2 - r^2 = c$.

Condition to define circle -

$g^2 + f^2 - c > 0 \quad \Rightarrow$ Real circle

$g^2 + f^2 - c = 0 \quad \Rightarrow$ Point circle

$g^2 + f^2 - c < 0 \quad \Rightarrow$ Imaginary circle, with real centre, that is $(-g, -f)$

Note :- That every second degree equation in x and y, in which coefficient of x^2 is equal to coefficient of y^2 and the coefficient of xy is zero, always represents a circle.

Illustration 1: For what value(s) of λ, the equation $(10 - \lambda^2)x^2 + (\lambda^2 - 8)y^2 + (3 - \lambda)yx - 10x + 4y + 3 = 0$ represent a real circle.

Solution :

For equation of circle we must satisfy the following 2 conditions

Condition 1 : Coeff. of x^2 = coeff. of y^2

$\Rightarrow \quad 10 - \lambda^2 = \lambda^2 - 8 \;\; \lambda^2 = 9 \qquad \Rightarrow \lambda = \pm 3$

Condition 2 : and coeff. of $xy = 0$

$\Rightarrow \quad 3 - \lambda = 0 \qquad \Rightarrow \; \lambda = 3$

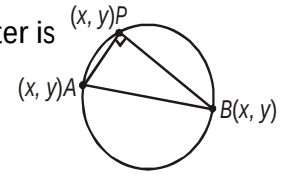
So, the common value is $\lambda = 3$.

(d) The equation of circle with (x_1, y_1) and (x_2, y_2) as extremeties of its diameter is

$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

This is obtained by the fact that angle in a semicircle is a right angle.

$\therefore \quad$ (Slope of PA)(Slope of PB) = –1



$\Rightarrow \quad \frac{y-y_1}{x-x_1}.\frac{y-y_2}{x-x_2}=-1$

$\Rightarrow \quad (x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$

Note that this will be the circle of least radius passing through (x_1, y_1) and (x_2, y_2).

Student Task

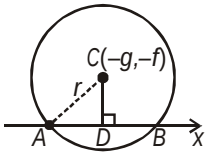
1. If the equation circle is $ax^2+(2a-3)y^2-4x-1=0$ then its centre is
2. If $2x^2+\lambda xy+2y^2+(\lambda-4)x+6y-5=0$ is the equation of a circle then its radius is
3. The equation $x^2+y^2-2x+4y+5=0$ represents

Ans. 1. (2/3, 0) 2. $3\sqrt{2}$ **3. 16**

2. INTERCEPTS MADE BY A CIRCLE ON THE AXES

The intercepts made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on the co-ordinate axes are $2\sqrt{g^2-c}$ (on x-axis) and $2\sqrt{f^2-c}$ (on y-axis) respectively.

If $g^2 > c$ $\Rightarrow$ Circle cuts the x axis at two distinct points.
$g^2 = c$ $\Rightarrow$ Circle touches the x-axis.
$g^2 < c$ $\Rightarrow$ Circle lies completely above or below the x-axis.



$AB = 2AD = 2\sqrt{r^2-CD^2} = 2\sqrt{r^2-f^2} = 2\sqrt{g^2+f^2-c^2-f^2} = 2\sqrt{g^2-c}$

Illustration 2: Find the equation of the circle which touches the axes and whose centre lies on the line $x - 2y = 3$.

Solution : Since the circle touches both the axes, therefore its centre will be $(a, \pm a)$ and radius will be $|a|$, where a is a positive or negative number.
Given line is $x - 2y = 3$ ....(i)

Case I : When centre is $(a, +a)$
Since $(a, +a)$ lies on line a-2a=3 , a=-3 (i)
$\therefore$ centre of the circle is $(–3, –3)$ and radius = $|–3| = 3$.
Hence equation of the circle will be
$(x+3)^2 + (y + 3)^2 = 3^2$ or $x^2 + y^2 + 6x + 6y + 9 = 0$

Case II : When centre is $(a, –a)$
Since, $(a, –a)$ lies on line (i)
$\therefore \quad a + 2a = 3 \qquad \therefore \; a = 1$
$\therefore$ centre of the circle is $(1, –1)$ and radius = $|1| = 1$
Hence equation of the circle will be
$(x – 1)^2 + (y + 1)^2 = 1^1$ or $x^2 + y^2 – 2x + 2y + 1 = 0$

Illustration 3: Find the equation of the circle passing through the origin and cuts off chords of length 4 and 6 on the positive side of x and y-axis respectively.

Solution : If the centre C be (h, k) then the values of h, k will be length of perpendicular CM and CN from C to x- and y-axis respectively.
$\Rightarrow$ M and N are mid points of chords OA and OB
$\Rightarrow$ $OM = 2$, $ON = 2C$ is $(2, 3)$

Also radius $OC = \sqrt{2^2+3^2} = \sqrt{13}$

Hence the equation of the required circle is $(x – 2)^2 + (y – 3)^3 = (\sqrt{13})^2$

Illustration 4: Find the equation of the circle which passes through the points (1, 0), (0, 6) and (3,4).

Solution : Let the equation to the circle be,

$x^2 + y^2 + 2gx + 2fy + c = 0$....(1)

Since, the three points, whose coordinates are given, satisfy this equation, we have

$1 + 2g + c = 0$

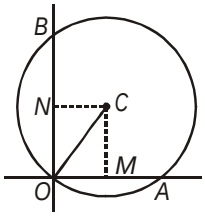
....(2)

$36 + 12f + c = 0$

....(3)

and $25 + 6g + 8f + c = 0$

....(4)



Subtracting (2) from (3) and (3) from (4), we have

$2g + 12f = 35$

and $6g + 20f = 11$

Hence, $f = \frac{47}{8}$ and $g = -\frac{71}{4}$

Equation (2), then gives $c = \frac{69}{2}$

Substituting these values in (1) the required equation is,

$4x^2 + 4y^2 - 142x + 47y + 138 = 0$

Illustration 5: Find the equation of the circle which touches the axis of y at a distance +4 from the origin and cuts off an intercept 6 from the axis of x.

Solution : Any circle is, $x^2 + y^2 + 2gx + 2fy + c = 0$

This meets the axis of y in points given by $y^2 = 2fy + c = 0$

The roots of this equation must be equal and each equal to 4, so that it must be equivalent to

$(y - 4)^2 = 0$

Hence, $2f = -8$ and $c = 16$

The equation to the circle is then

$x^2 + y^2 + 2gx - 8y + 16 = 0$

This meets the axis of x in points given by,

$x^2 + 2gx - 8y + 16 = 0$

i.e., at points distant

$-g + \sqrt{g^2 - 16}$ and $-g - \sqrt{g^2 - 16}$

Hence, $6 = 2\sqrt{g^2 - 16}$

Therefore, $g = \pm 5$, the required equation is, $x^2 + y^2 \pm 10x - 8y + 16 = 0$

Therefore, are therefore, two circle satisfying the given conditions. This is geometrically obvious.

Illustration 6: The abscissa of two points A and B are the roots of the equation $x^2 + 2ax - b^2 = 0$ and the ordinates are the roots of the equation $x^2 + 2px - q^2 = 0$. Find the equation and the radius of the circle with AB as diameter.

Solution : Given equation are

$x^2 + 2ax - b^2 = 0$

....(i)

and $x^2 + 2px - q^2 = 0$

....(ii)

Let the roots of equation (i) be α and β and those of equation (ii) be γ and δ, then

$$\left.\begin{aligned}\alpha+\beta&=-2a\\ \alpha\beta&=-b^2\end{aligned}\right\}\text{and}\quad\begin{aligned}\gamma+\delta&=-2p\\ \gamma\delta&=-q^2\end{aligned}$$

Let $A = (\alpha, \gamma)$ and $B = (\beta, \delta)$

Now equation of the circle whose diameter is AB will be

$(x-\alpha)(x-\beta)+(y-\gamma)(y-\delta)=0$

or $x^2+y^2-(\alpha+\beta)x-(\gamma+\delta)y+\alpha\beta+\gamma\delta=0$

or $x^2+y^2+2ax+2py-(b^2+q^2)=0$ ....(iii)

Centre of circle (iii) is $(-a, -p)$

and radius $\sqrt{(-a)^2+(-p)^2+b^2+q^2}=\sqrt{a^2+b^2+p^2+q^2}$

Student Task

1. A circle touches the y-axis (0, 2) and has an intercept of 4 units on the positiive side of the x-axis. then the equation of the circle is
2. The intercept on the line y = x by the circle $x^2+y^2-2x=0$ is AB. The equation of the circle with AB as a diameter is

Ans. 1. $x^2+y^2-4\left(\sqrt{2x}+y\right)+4=0$ **2.** $x^2+y^2=x+y$

3. PARAMETRIC EQUATIONS OF A CIRCLE

The parametric equations of $(x-h)^2+(y-k)^2=r^2$ are : $x = h + r\cos\theta$; $y = k + r\sin\theta$; $0 \le \theta \le 2\pi$ where (h, k) is the centre, r is the radius and θ is a parameter.

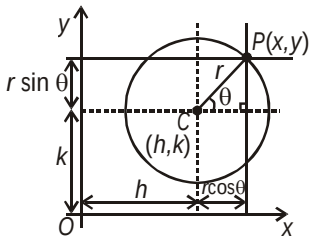


Illustration 7: If the parametric form of a circle is given by

(i) $x = 4 + 5\cos\theta$ and $y = -3 + 5\sin\theta$

(ii) $x = a\cos\alpha + b\sin\alpha$ and $y = a\sin\alpha - b\cos\alpha$

Find its cartesian form.

Solution : (i) The given equation are

$x = 4 + 5\cos\theta$ and $y = -3 + 5\sin\theta$

or $(x+5) = 5\cos\theta$ ....(i)

and $(y+3) = 5\sin\theta$ ....(ii)

Squaring and adding (i) and (ii), then

$(x+4)^2+(y+3)^2=5^2$ or $(y+4)^2+(y+3)^2=25$

(ii) The given equations are

$x = a\cos\alpha + b\sin\alpha$ ....(i)

$y = a\sin\alpha - b\cos\alpha$ ....(ii)

Squaring and additn (i) and (ii), then

$x^2+y^2=(a\cos\alpha+b\sin\alpha)^2+(a\sin\alpha-b\cos\alpha)^2$

$\Rightarrow x^2+y^2=a^2+b^2$

Illustration 8: Show that four points (1, 0), (2, –7), (8, 1) and (9, –6) are concyclic.

Solution : Since the given four points are concyclic, we have to show that they lie on a circle. Let the general equation of circle is

$x^2 + y^2 + 2gx + 2fy + c = 0$....(i)

has three parameters, it is sufficient to obtain the equation of the circle passing through any three of these points. For concyclic, the fourth point should lie on this circle.

Let three points $A(1, 0)$, $B(2, –7)$ and $D(8, 1)$ lie on (i) then

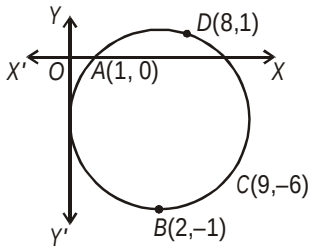
$1 + 0 + 2g + 0 + c = 0$ or $1 + 2g + c = 0$....(ii)

$(2)^2 + (–7)^2 + 2g(2) + 2f(–7) + c = 0$

or $53 + 4g – 14c + c = 0$....(iii)

and $(8)^2 + (1)^2 + 2g(8) + 2f(1) + c = 0$

$\Rightarrow$ $65 + 16g + 2f + c = 0$....(iv)



Now subtracting (2) from (3), we get

$52 + 2g – 14f = 0$ or $26 + g – 7f = 0$....(v)

and subtracting (3) from (4), we get

$12 + 2g + 16f = 0$

$\Rightarrow$ $3 + 3g + 4f = 0$ ....(vi)

Solving (5) and (6), we get

From (2), $1 – 10 + c = 0$

$\therefore$ $c = 9$

Therefore equation of circle passing through these point is

$x^2 + y^2 – 10x + 6y + 9 = 0$

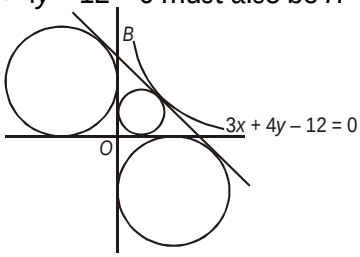
Substituting the fourth point in the equation of this circle, we get

$(9)^2 + (–6)^2 – 10(9) + 6(–6) + 9 = 0$

Hence point $C(9, –6)$ lies on the circle, that is, the four points are concyclic.

Illustration 9: Find the equation for all circles touching x-axis, y-axis and line $3x + 4y – 12 = 0$.

Solution : We can easily observe that there will be four circles touching x-axis, y-axis and the line $3x + 4y – 12 = 0$. The centre of any such circle be the form $(\pm r, \pm r)$ where r is the radius of the circle. If we want to determine the circles whose centres lies in first quadrant and which touch x-axis, y-axis and the line $3x + 4y – 12 = 0$ then centres must be (r, r) $(r > 0)$ and perpendicular distance of (r, r) from the line $3x + 4y – 12 = 0$ must also be r.



$$\Rightarrow \left|\frac{3r + 4r - 12}{5}\right| = r$$

$\Rightarrow$ $7r – 12 = 5r$, $7r – 12 = –5r$

$\Rightarrow$ $r = 6$, $r = 1$

Thus two such circles are

$(x – 6)^2 + (y – 6)^2 = 6^2$, $(x – 1)^2 + (y – 1)^2 = 1^2$

Next in the second quadrant the centre of such a circle must be $(–r, r)$ and

$$\left|\frac{-3r + 4r - 12}{5}\right| = r$$

$\Rightarrow$ $r = –3$, $r = 2$

The value $r = –3$ is inadmissible since $r > 0$. Thus the equation of circle touching axes and the line $3x + 4y – 12 = 0$, in the second quadrant is $(x + 2)^2 + (y – 2)^2 = 2^2$

In the third quadrant $\left|\frac{-3r - 4r - 12}{5}\right| = r$

$\Rightarrow$ $r = 1$, $r = –6$

$\Rightarrow$ Both values are inadmissible.

Thus there is no such circle in the third quadrant.

Finally in the fourth quadrant

$$\left|\frac{3r-4r-12}{5}\right| = r$$

$\Rightarrow \quad r = 3$

The circle in the fourth quadrant must be $(x-3)^2 + (y+3)^2 = 3^2$

Illustration 10: Find the area of an equilateral triangle inscribed in the circle $x^2 + y^2 + 2gx + 2fy + c = 0$.

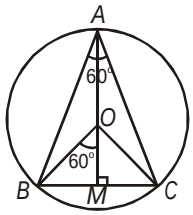
Solution : Given circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ ....(i)

Let O be the centre and ABC be an equilateral triangle inscribed in the circle (i).

$O \equiv (-g, -f)$

and $OA = OB = OC = \sqrt{g^2 + f^2 - c}$ ....(ii)

In ΔOBM, $\sin 60^o = \dfrac{BM}{OB}$



$\Rightarrow \quad BM = OB \sin 60^o = (OB)\dfrac{\sqrt{3}}{2}$

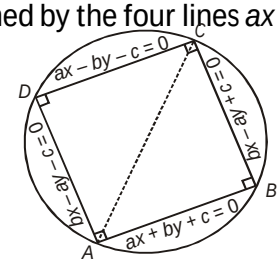
$BC = 2BM = \sqrt{3}\,(OB)$

Area of $\Delta ABC = \dfrac{\sqrt{3}}{4}(BC)^2$ ....(iii)

$= \dfrac{\sqrt{3}}{4} 3(OB)^2$ (from (3))

$= \dfrac{3\sqrt{3}}{4}(g^2 + f^2 - c)^2$ sq. unit

Illustration 11 Find the equation of the circumcircle of the quadrilateral formed by the four lines $ax + by \pm c = 0$ and $bx - ay \pm c = 0$.



Solution : The given lines can be re-written as

$ax + by + c = 0$ ....(i)

$ax + by - c = 0$ ....(ii)

$bx - ay + c = 0$ ....(iii)

$bx - ay - c = 0$ ....(iv)

Equation (i) and (ii) are parallel and equations (iii) and (iv) are also parallel

Slope of (i) or (ii) $= -\dfrac{a}{b} = m_1$ (say)

and Slope of (iii) or (iv) $= \dfrac{a}{b} = m_2$ (say)

Since $m_1 m_2 = -1$

Hence $ABCD$ be a square and AC and BD are the diameters of the circle. After solving, we get

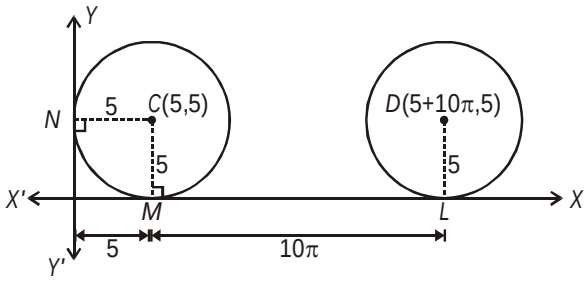
$$A \equiv \left(\frac{bc-ca}{a^2+b^2}, \frac{bc+ca}{a^2+b^2}\right) \text{and} \; C \equiv \left(\frac{ac+bc}{a^2+b^2}, \frac{ac-bc}{a^2+b^2}\right)$$

$\therefore$ Equation of circle is $\left(x-\frac{bc-ca}{a^2+b^2}\right)\left(x-\frac{ac+bc}{a^2+b^2}\right)+\left(y-\frac{bc+ca}{a^2+b^2}\right)\left(y-\frac{ac-bc}{a^2+b^2}\right)=0$

$\Rightarrow \quad x^2+y^2+\left(\frac{2ac}{a^2+b^2}\right)x+\left(\frac{2bc}{a^2+b^2}\right)y=0$

Illustration 12: A circle of radius 5 units touches the co-ordinate axes in first quadrant. If the circle makes one complete roll on *x*-axis along the positive direction of *x*-axis, find its equation in the new position.

Solution : Let *C* be the centre of the circle in its initial position and *D* be its centre in the new position



Since the circle touches the co-ordinate axes in first quadrant and the radius of circle be 5 units.

$\therefore$ Centre of circle is (5, 5)

Moving length of circle = circumference of the circle

$= 2\pi r = 2\pi(5) = 10\pi$

Now centre of circle in new position is $(5 + 10\pi, 5)$ and radius is 5 units, therefore, its equation will be

$(x - 5 - 10\pi)^2 + (y - 5)^2 = 5^2$

or $x^2 + y^2 - 10(1 + 2\pi)x - 10y + 100\pi + 25 = 0$

Student Task

1. Three sides of a triangle have the equations $L_r \equiv y - m, x - c_r = 0; r = 1,2,3.$ Then $\lambda L_2 L_3 + \mu L_3 L_1 = 0,$ where $\lambda \neq 0, \mu \neq 0, v \neq 0,$ is the equation of the circumcircle of the tringle if
2. If the centroid of an equilateral triangle is (1,1) and its one vertex is (-1,2) then the equation of tis circumcircle is
3. The equation of the circle whose one diameter is PQ, where the ordinates of P,Q are the roots of the equation $x^2 + 2x - 3 = 0$ and the abscissae are the roots of the equation $y^2 + 4y - 12 = 0$, is
4. the equation of the incircle of the triangle formed by the axes and the line $4x + 3y = 6$ is

Ans. 1. $\lambda(m_2m_3 - 1) + \mu(m_3m_1 - 1) + v(m_1m_2 - 1) = 0$ **2.** $x^2 + y^2 - 2x - 2y - 3 = 0$

3. $x^2 + y^2 + 4x + 2y - 15 = 0$ **4.** $4(x^2 + y^2 - x - y) + 1 = 0$

4. POSITION OF A POINT WITH RESPECT TO A CIRCLE

The point (x_1, y_1) is inside, outside the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$. according as $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c < 0$ or > 0.

Note :- The greatest and the least distance of a point A from acircle with centre C and radius r is AC + r and AC – r respectively.

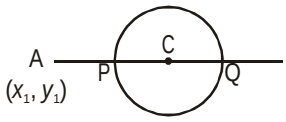
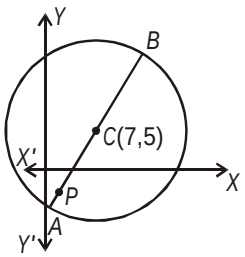


Illustration 13: Find the shortest and largest distance from the point (2, –7) to the circle $x^2 + y^2 – 14x – 10y – 151 = 0$.

Solution :



Let $S \equiv x^2 + y^2 – 14x – 10y – 151 = 0$

$\therefore \quad S_1 = (2)^2 + (–7)^2 – 14(2) – 10(–7) – 151$

$= – 56 < 0$

$\therefore$ $P(2, –7)$ inside the circle

radius of the circle, $r = \sqrt{(7-2)^2 + (–5)^2 + 151} = 15$

$\therefore$ Centre of circle $\equiv (7, 5)$

$\therefore \quad CP = \sqrt{(7-2)^2 + (5+7)^2} = 13$

$\therefore$ **Shortest distance** $= PA = r – CP = 15 – 13 = 2$

and **Largest distance** $= PB = r + CP = 15 + 13 = 28$

LINE AND A CIRCLE

Let $L = 0$ be a line and $S = 0$ be a circle. If r is the radius of the circle and p is the length of the perpendicular from the centre on the line, then :

(i) $p > r$ $\Leftrightarrow$ The line does not meet circle i.e. passes out side the circle.

(ii) $p = r$ $\Leftrightarrow$ The line touches the circle. (It is tangent to the circle).

(iii) $p < r$ $\Leftrightarrow$ The line is a secant of the circle.

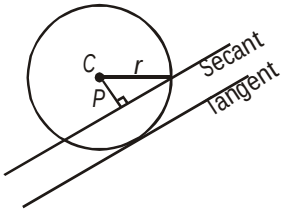
(iv) $p = 0$ $\Leftrightarrow$ The line is a diameter of the circle.

Also, if $y = mx + c$ is line and $x^2 + y^2 = a^2$ is circle then

(i) $c^2 < a^2(1 + m^2)$ $\Leftrightarrow$ The line is a secant of the circle.

(ii) $c^2 = a^2(1 + m^2)$ $\Leftrightarrow$ The line touches the circle. (It is tangent to the circle).

(iii) $c^2 > a^2(1 + m^2)$ $\Leftrightarrow$ The line does not meet the circle i.e. passes outside the circle.



The conditions can also be obtained by solving $y = mx + c$ with $x^2 + y^2 = a^2$ and making the discriminant of the quadratic greater than zero for secant, equal to zero for tangent and less the zero for the last case.

SLOPE FORM OF TANGENT

(i) $y = mx + c$ is always a tangent to the circle $x^2 + y^2 = a^2$ if $c^2 = a^2(1 + m^2)$. Hence, equation of tangent is $y = mx \pm a\sqrt{1+m^2}$ and the point of contact is $\left(-\frac{a^2m}{c}, \frac{a^2}{c}\right)$.

(ii) For $(x-\alpha)^2+(y-\beta)^2=a^2$ equation of tangent will $(y-\beta)=m(x-\alpha)\pm a\sqrt{1+m^2}$

POINT FORM OF TANGENT

(i) The equation of the tangent to the circle $x^2 + y^2 = a^2$ at its point (x_1, y_1) is, $xx_1 + yy_1 = a^2$.

(ii) The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at its point (x_1, y_1) is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$.

Note :- In general the equation of tangent to any second degree curve at point (x_1, y_1) on it can be obtained by replacing x^2 by xx_1 y^2 by yy_1, x by $\frac{x+x_1}{2}$, and y by $\frac{y+y_1}{2}$.

PARAMETRIC FORM OF TANGENT

The equation of a tangent to circle $x^2 + y^2 = a^2$ at $(a\cos\alpha, a\sin\alpha)$ is $x\cos\alpha + y\sin\alpha = a$.

Note :- The point of intersection of the tangents at the points P(a) and Q(b) is $\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$.

Illustration 14 : Find the values of k for which the line $3x - 4y - k = 0$ will touch the circle $x^2 + y^2 - 2x - 4y + 4h = 0$. Find the point of contact also.

Solution : We will solve the problem by two methods. The circle is

$(x - 1)^2 + (y - 2)^2 = 1$

Shifting the origin to (1, 2) the circle becomes $x^2 + y^2 = 1$ and the line becomes

$3x - 4y - (5 + k) = 0$

Now applying $c^2 = a^2 (1 + m^2)$

$$\left(-\frac{5+k}{4}\right)^2 = 1\left(1+\frac{9}{16}\right) \Rightarrow k = 0,\ k = 10$$

If k = 0 then $c = -\frac{5}{4}$

and the point of contact is $\left(-\frac{3/4}{-5/4}, \frac{1}{-5/4}\right)$ or, is $\left(\frac{3}{5}, \frac{-4}{5}\right)$

Now with reference to old origin this point will be $\left(\frac{3}{5}+1, \frac{-4}{5}+2\right)$ i.e. $\left(\frac{8}{5}, \frac{6}{5}\right)$

Alternatively we can proceed as follows. for intersection of line and circle

$$x^2 + \left(\frac{3x-k}{4}\right)^2 - 2x - 4\left(\frac{3x-k}{4}\right) + 4 = 0$$

$\Rightarrow 25x^2 - (6k + 80)x + (k + 8)^2 = 0$

$$\Rightarrow x = \frac{6k+80 \pm \sqrt{(6k+80)^2 - 100(k+8)^2}}{50} = \frac{3k+40 \pm \sqrt{(8k+80)(-2k)}}{25}$$

In case the line happens to be tangent we must have only one value of x.

$\Rightarrow$ Discriminant of the quadratic = 0

$\Rightarrow k = 0,\ k = -10$

Illustration 15: Show that the line $3x - 4y - c = 0$ will meet the circle having centre at (2, 4) and the radius 5 in real and distinct points if $-35 < c < 15$.

Solution : Given line is $3x - 4y - c = 0$ .....(i)

Centre of given circle is (2, 4) and its radius is 5, therefore its equation will be

$(x - 2)^2 + (y - 4)^2 = 52$

or $x^2 + y^2 - 4x - 8y - 5 = 0$ .....(ii)

From (i), $y = \frac{1}{4}(3x - c)$. Putting the value of y in (ii), we get

$$x^2 + \frac{1}{16}(3x - c)^2 - 4x - 8\frac{1}{4}(3x - c) - 5 = 0$$

or $16x^2 + 9x^2 - 6cx + c^2 - 64x - 96x + 32c - 80 = 0$

or $25x^2 - 2(80 + 3c)\,x + c^2 + 32c - 80 = 0$ ..... (iii)

Line (i) will meet the circle (ii) in real and distinct points if discriminant at equation (iii) > 0 i.e., if $4(80 + 3c)^2 - 100\,(c^2 + 32c - 80) > 0$ or $(80 + 3c)^2 - 25(c^2 + 32c - 80) > 0$

or $6400 + 9c^2 + 480c - 25c^2 - 800c + 2000 > 0$

0r $-16c^2 - 320c + 8400 > 0$ or $16c^2 + 320c - 8400 < 0$

or $c^2 + 20c - 525 < 0$

sign scheme for $c^2 + 20c - 525$:

When $c^2 + 20c - 525 = 0$

$$c = \frac{-20 \pm \sqrt{400 + 2100}}{2} = \frac{-20 \pm 50}{2} = 35, 15$$

Therefore, sign scheme for $c^2 + 20c - 525$ is as follows

$\infty \leftarrow$ +Ve -35 +Ve 15 +Ve $\rightarrow \infty$

$\therefore\ c^2 + 20c - 525 < 0 \Rightarrow -35 < c < 15$

Illustration 16: Find the condition that the straight line $lx + my + n = 0$ may touch circle $x^2 + y^2 = a^2$ and also find the co-ordinates of the point of contact.

Solution : given line is $lx + my + n = 0$ ..... (1)

and given circle is $x^2 + y^2 + a^2$ ..... (2)

line (1) will touch the circle (2) if the length of the perpendicular from its centre (0, 0) to line (1) = radius of circle (2)

i.e. if $\frac{|l.0 + m.0 + n|}{\sqrt{l^2 + m^2}} = a$

or $n^2 = a^2(l^2 + m^2)$ This is the required condition.

Second part :

Let line (1) be tangent to circle (2) at (α, β).

Now equation of the tangent to circle (2) at (α, β) is $x\alpha + y\beta - a^2 = 0$ .....(3)

Since (1) and (3) are equations of the same straight line, therefore comparing the coefficients, we get

$$\underset{(i)}{\frac{\alpha}{l}} = \underset{(ii)}{\frac{\beta}{m}} = \underset{(iii)}{\frac{-a^2}{n}}$$

From (i) and (iii), $\alpha = -\frac{la^2}{n}$ and from (ii) and (iii), $\beta = -\frac{ma^2}{n}$

Thus point of contact of line (1) and circle (2) is $\left(-\frac{la^2}{n}, -\frac{ma^2}{n}\right)$

Note : Condition for tangency can be obtained by putting $x = -\frac{la^2}{n}$, $y = -\frac{ma^2}{n}$ in the equation of the circle because point of contact lies on the circle.

Illustration 17: Prove that the tangent to the circle $x^2 + y^2 = 5$ at the point (1, –2) also touches the circle $x^2 + y^2 – 8x + 6y + 20 = 0$ and find its point of contact.

Solution : Given circle are $x^2 + y^2 = 5$ .....(1)

and $x^2 + y^2 – 8x + 6y + 20 = 0$ .....(2)

Given point is (1, – 2).

Now equation of the tangent to circle (1), at (1, –2) will be

$x \cdot 1 + y(– 2) – 5 = 0$ or $x – 2y – 5 = 0$ .....(3)

Center of circle (2) is C (4, –3) and its radius is $\sqrt{5}$

Now length of the perpendicular from C(4, –3) to line (3)

$$\frac{|4 – 2(–3) – 5|}{\sqrt{5}} = \frac{5}{\sqrt{5}} = \sqrt{5} = \text{radius of circle (2)}$$

Hence line (3) also touches circle (2).

Second part :

Let (α, β) be the point of contact of line (3) and circle (2).

Now equation of the tangent to circle (2) at (α, β) is

$$x\alpha + y\beta – 8\left(\frac{x+\alpha}{2}\right) + 6\left(\frac{y+\beta}{2}\right) + 20 = 0$$

or $(\alpha – 4)x + (\beta + 3)y – (4\alpha – 3\beta – 20) = 0$ .....(4)

Comparing equations (3) and (4), we get

$$\frac{\alpha – 4}{\underset{(i)}{1}} = \frac{\beta + 3}{\underset{(ii)}{–2}} = \frac{4\alpha – 3\beta – 20}{\underset{(iii)}{5}}$$

From (i) and (ii), we get $2\alpha + \beta = 5$ ....(5)

and from (i) and (iii), we get $\alpha + 3\beta = 0$ ....(6)

Solving (5) and (6), we get $\alpha = 3$, $\beta = – 1$

Hence point of contact of circle (2) and line (3) is (3, –1).

Illustration 18 Prove that the equation $x^2 + y^2 – 2x – 2\lambda y – 8 = 0$, where λ is a parameter, represents a family of circles passing through two fixed points A and B on the x-axis. Also find the equation of that circle of the family, the tangents to which at A and B meet on the line $x + 2y + 5 = 0$

Solution : Given equation of the family of circles is

$x^2 + y^2 – 2x – 2\lambda y – 8 = 0$ .....(1)

It can be written as $x^2 + y^2 –2x – 2\lambda y-8 = 0$ .....(2)

Clearly (2) is the equation of circles passing through the point of intersection of circle

$x^2 + y^2 – 2x – 8 = 0$

and the line $y = 0$ .....(3)

Putting $y = 0$ in (2), we get $x^2 – 2x – 8 = 0 \therefore x = 4, – 2$

Let A $\equiv$ (4, 0) and B $\equiv$ (–2, 0). Thus (1) represents a family of circles passing through two fixed points A (4, 0) and B (–2, 0) on the x-axis.

Now equation of tangents to circle (1) at (4, 0) and B (–2, 0) are

$4x + y.0 – (x + 4) – \lambda(y + 0) – 8 = 0$ or $3x – \lambda y – 12 = 0$ .....(4)

and $– 2x + y.0 – (x – 2) – \lambda(y + 0) – 8 =$ or $– 3x – \lambda y – 6 = 0$ .....(5)

Solving (4) and (5), we get $x = 1, y = -\frac{9}{\lambda}$

If $\left(1, -\frac{9}{\lambda}\right)$ lies on line $x + 2y + 5 = 0$

then $1 - \frac{18}{\lambda} + 5 = 0$ or $\lambda = 3$

Putting the value of λ in (1), we have the equation of the required circle as $x^2 + y^2 - 2x - 6y - 8 = 0$

Illustration 19: Find the equations of the tangents to the circle $x^2 + y^2 = 9$, which

(i) are parallel to the line $3x + 4y - 5 = 0$

(ii) are perpendicular to the line $2x + 3y + 7 = 0$

(iii) make an angle of 60° with the x-axis.

Solution : (i) Slope of $3x + 4y - 5 = 0$ is $-\frac{3}{4}$

Let $m = -\frac{3}{4}$

and equation of circle is $x^2 + y^2 = 9$

$\therefore$ Equations of tangents $y = -\frac{3}{4}x \pm 3\sqrt{\left(1 + \left(-\frac{3}{4}\right)^2\right)}$

$\Rightarrow 4y = -3x \pm 15$

or $3x + 4y \pm 15 = 0$

(ii) slope of $2x + 3y + 7 = 0$ is $-\frac{2}{3}$

$\therefore$ Slope of perpendicular to $2x + 3y + 7 = 0$ is $\frac{3}{2} = m$ (say)

and given circle is $x^2 + y^2 = 9$

$\therefore$ Equations of tangents perpendicular to $2x + 3y + 7 = 0$ is

$y = \frac{3}{2}x \pm 3\sqrt{1 + \left(\frac{3}{2}\right)^2}$

$\Rightarrow 2y = 3x \pm 3\sqrt{13}$

or $3x - 2y \pm 3\sqrt{13} = 0$

(iii) Since tangent make an angle 60° with the x-axis

$\therefore$ $m = \tan 60^o = \sqrt{13}$

and given circle $x^2 + y^2 = 9$

$\therefore$ Equation of tangents $y = \sqrt{3x} \pm 3\sqrt{1 + (\sqrt{3})^2}$

or $\sqrt{3}x - y \pm 6 = 0$

Alternative Method

(i) Let tangent parallel to $3x + 4y - 5 = 0$ is

$3x + 4y + \lambda = 0$ .....(1)

and circle $x^2 + y^2 = 9$

then perpendicular distance from (0, 0) on (2) = radius

$$\frac{|\lambda|}{\sqrt{(3^2+4^2)}}=3 \text{ or } |\lambda| = 15$$

$\therefore\ \lambda = \pm 15$

From (1), equation of tangents are

$3x + 4y \pm 15 = 0$

(ii) Let tangent perpendicular to $2x + 3y + 7 = 0$ is

$3x - 2y + \lambda = 0$ .....(2)

and circle $x^2 + y^2 = 9$

then perpendicular distance from (0, 0) on (2) = radius

$$\frac{|\lambda|}{\sqrt{3^2+(-2)^2}}=3 \text{ or } |\lambda| = 3\sqrt{3} \text{ or } |\lambda| = \pm 3\sqrt{3}$$

From (2), equations of tangents are $3x - 2y \pm 3\sqrt{3} = 0$

(iii) Let equation of tangent which makes an angle of 60° with the x-axis is

$y = \sqrt{3}x + c$ .....(3) or $\sqrt{3}x - y + c = 0$ and circle $x^2 + y^2 = 9$

then perpendicular distance from (0, 0) to (3) = radius

$$\frac{|c|}{\sqrt{(3)^2+(-1)^2}}=3$$

or $|c| = 6$

or $c = \pm 6$

From (3), equations of tangent are $\sqrt{3}x - y \pm 6 = 0$

Student Task

1. A region in the x-y plane is bounded by the curve $y=\sqrt{25-x^2}$ and the line y = 0. If the point (a, a + 1) lies in the interior of the region then
2. If (2,4) is a point interior tothe circle $x^2+y^2-6x-10y+\lambda=0$ and the circle does not cut the axes at any then λ belongs to the interval
3. The range of values of $\theta\in[0,2\pi]$ for which $(1+\cos\theta,\sin\theta)$ is an interior point of the circle $x^2+y^2=1$ is
4. The line $y=mx+\sqrt{4+4m^2}, m\in R,$ is a tangent to the circle
5. The range of values of m for which the lien $y=mx+2$ cuts the cirlce $x^2+y^2=1$ at distinct or coincident points is
6. The equation of any tangent to the circle $x^2+y^2-2x+4y-4=0$ is

Ans. 1. $a\in(-4,3)$ **2.** (25, 32) **3.** $(2\pi/3, 4\pi/3)$ **4.** $x^2+y^2=4$ **5.** $\left(-\infty,-\sqrt{3}\right]\cup\left[\sqrt{3},+\infty\right)$

6. $y=m(x-1)+3\sqrt{1+m^2}-2$

5. NORMAL

If a line is normal/orthogonal to a circle then it must pass through the centre of the circle. Using this fact normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is $y - y_1 = \frac{y_1 + f}{x_1 + g}(x - x_1)$.

Illustration 20: Find the equation of the normal to the circle $x^2 + y^2 = 2x$, which is parallel to the line $x + 2y = 3$.

Solution : Given circle is $x^2 + y^2 - 2x = 0$

Centre of given circle is (1, 0)

Since normal is parallel to $x + 2y = 3$

let the equation of normal is $x + 2y = \lambda$

Since normal passes through the centre of the circle i.e., (1, 0)then $1 + 0 = \lambda$

$\therefore \lambda = 1$

then equation of normal is $x + 2y = 1$

or $x + 2y - 1 = 0$

Alternative Method :

Equation of normal at (x_1, y_1) of $x^2 + y^2 - 2x = 0$ is

$\frac{x - x_1}{x_1 - 1} = \frac{y - y_1}{y_1 - 0}$ or slope = $\frac{y_1}{x_1 - 1} = m_1$ (say)

since normal is parallel to $x + 2y = 3$

$\therefore$ slope = $-\frac{1}{2} = m_2$ (say)

but given $m_1 = m_2$

$\frac{y_1}{x_1 - 1} = \frac{1}{2}$ or $x_1 + 2y_1 - 1 = 0$

$\therefore$ locus of (x_1, y_1) is $x + 2y - = 0$

PAIR OF TANGENTS FROM A POINT

The length of a tangent from an external point (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $SS_1 = T^2$.

Where

$S \equiv x^2 + y^2 + 2gx + 2fy + c$; $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$

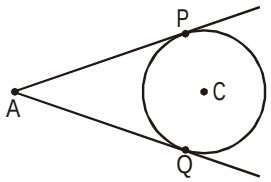


Illustration 21: Let $2x^2 + y^2 - 3xy = 0$ be the equation of a pair of tangents drawn from origin O to a circle of radius 3 with centre in the first quadrant. If A is one of the points of contact find the length of OA.

Solution : The given equation represents lines $y = x$ and $y = 2x$

if 2φ is the angle between them

$\tan 2\varphi = \frac{2-1}{1+2} = \frac{1}{3} = \frac{2\tan\varphi}{1 - \tan^2\varphi} \Rightarrow \tan\varphi = -3 \pm \sqrt{10}$

But, as centre lies in the first quadrant

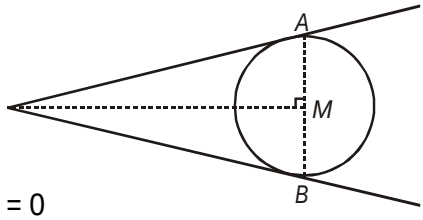
and $0 < \varphi < \frac{\pi}{4}$ $\tan\varphi = -3 + \sqrt{10}$ only. Now OA = 3 cot $\varphi = \frac{3}{-3 + \sqrt{10}}$

Illustration 22: Tangents PA and PB are drawn from P (–1, 2) to the circle $x^2 + y^2 – 2x – 4y + 2 = 0$

(i) Find the lengths PA (or PB)

(ii) Find the area of ΔPAB

(iii) Find the equation of PA and PB



Solution : (i) PA^2 or PB^2

= value of circle equation for $x = – 1$ and $y = 2$

$= 1 + 4 + 2 – 8 = 1 \Rightarrow PA = PB = 1$

(ii) Equation of chord of contact AS must be

$x (– 1) + y (2) – 1 (x – 1) – 2 (y + 2) + 2 = 0$ or $2x + 1 = 0$

Now PM = $\perp$ distance of P(–1, 2) from AB = $\frac{2(-1)+1}{\sqrt{2^2+0^2}} = \frac{1}{2}$

$\Rightarrow$ AM = $\sqrt{PA^2 - PM^2} = \sqrt{1-\frac{1}{4}} = \frac{\sqrt{3}}{2}$

$\Rightarrow$ Area of $\Delta PAB = \frac{1}{2}$ base. height = $\frac{\sqrt{3}}{2} \times \frac{1}{2} = \frac{\sqrt{3}}{4}$ square units.

(iii) Equation of any line through P(–1, 2) is $y – 2 = m (x + 1)$ or $y – mx – 2 – m = 0$ If this happens to be tangent to the given circle the length from the centre (1, 2) on the line must be equal to the radius $\sqrt{3}$.

$\Rightarrow \frac{|2-m-2-m|}{\sqrt{1+m^2}} = \sqrt{3} \Rightarrow 4m^2 = 3m^2 + 3 \Rightarrow m = \pm \sqrt{3}$

Thus the equations of tangents PA and PB are $y – 2 = \sqrt{3} (x + 1)$ and $y – 2 = – \sqrt{3} (x + 1)$

Note : Part (iii) can also be done by using $S S_1 = T^2$

We already know the values of S_1 and T in this problem

$S_1 = 1$ (since $(\text{length})^2$ of PA or PB)

$T = 2x + 1$ (equation of chord of contact)

Therefore combined equation of PA nd PB is

$(x^2 + y^2 – 2x – 4y + 2) . 1 = (2x + 1)^2$ or $– 3x^2 + y^2 – 6x – 4x + 1 = 0$

Let $y^2 – 3x^2 – 6x – 4y + 1 (y = \sqrt{3} x – c_1) (y \pm \sqrt{3} x + c_2)$

On comparing coeffs. of x, y and constant terms, we get

$– 6 = – c_2 \sqrt{3} – c_1 \sqrt{3}$, $– 4 = c_2 – c_1$, $1 = c_1 c_2$

On solving first two we get $c_1 = 2 – \sqrt{3}$, $c_2 = 2 + \sqrt{3}$

These values satisfy the third equation $c_1 c_2 = 1 \Rightarrow$ The assumption (1) is justified

$\Rightarrow$ Lines are $y = \sqrt{3} x + 2 + \sqrt{3}$ and $y = – \sqrt{3} x + 2 – \sqrt{3}$

LENGTH OF A TANGENT AND POWER OF A POINT

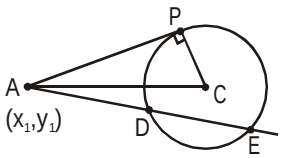
The length of a tangent from an external point (x_1, y_1) to the circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ is given by $L = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c} = \sqrt{S_1}$

Square of length of the tangent from the point P is also called the power of point w.r.t. a circle.

Power of a point w.r.t. a circle remains constant.

Power of a point P is positive, negative or zero according as the point 'P' is outside, inside or on the circle respectively.



AP = length of tangent
$AP^2 = AD \cdot AE$

Illustration 23: If the length of tangent from (f, g) to the circle $x^2 + y^2 = 6$ be twice the length of the tangent from (f, g) to circle $x^2 + y^2 + 3x + 3y = 0$ then will $f^2 + g^2 + 4f + 4g + 2 = 0$?

Solution : According to the question

$$\sqrt{(g^2 + f^2 - 6)} = 2\sqrt{(f^2 + g^2 + 3f + 3g)}$$

On squaring $g^2 + f^2 - 6 = 4f^2 + 4g^2 + 12f + 12g$
or $3f^2 + 3g^2 + 12f + 12g + 6 = 0$
or $f^2 + g^2 + 4f + 4g + 2 = 0$
which is true $\therefore$ yes.

Student Task

1. A foot of the normal form the point (4,3) to a cirlce is (2,1), and a diameter of the circle has the equation 2x-y =2. Then the equation of the circle.
2. The line $\lambda x + \mu y = 1$ is a normal to the circle $2x^2 + 2y^2 - 5x + 6y - 1 = 0$ if
3. The number of feet of normals from the point (7, -4) to the circle $x^2 + y^2 = 5$ is

Ans. 1. $x^2 + y^2 - 2x - 1 = 0$ **2.** $4 + 6\mu = 5\lambda$ **3.** 2

6. DIRECTOR CIRCLE, CHORD OF CONTACT

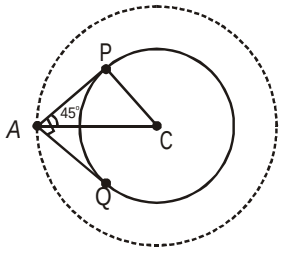
DIRECTOR CIRCLE

The locus of the point of intersection of two perpendicular tangents is called the director circle of the given circle. The director circle of a circle is the concentric circle having radius equal to $\sqrt{2}$ times the original circle.

Proof :

Equation of director of circle $x^2 + y^2 = a^2$ is

$x^2 + y^2 = 2r^2$



$AC = r \operatorname{cosec} 45^\circ = r\sqrt{2}$

CHORD OF CONTACT

If two tangents PT_1 and PT_2 are drawn from the point $P(x_1, y_1)$ to the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, then the equation of the chord of contact $T_1 T_2$ is $T = 0$

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

Note : *Here R = radius; L = Length of tangent.*

(a) Chord of contact exists only if the point 'P' is not inside.

(b) Length of chord of contact $T_1T_2 = \frac{2LR}{\sqrt{R^2+L^2}}$.

(c) Area of the triangle formed by the pair of the tangents and its chord of contact = $\frac{RL^2}{R^2+L^2}$.

(d) Tangent of the angle between the pair of tangents from $(x_1, y_1) = \left(\frac{RL}{R^2-L^2}\right)$

(e) Equation of the circle circumscribing the triangle PT_1T_2 is $(x - x_1)(x + g) + (y - y_1)(y + f) = 0$.

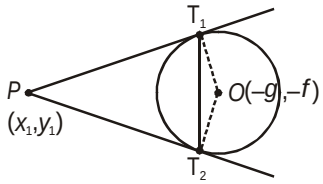


Illustration 24: Find the length of the common chord of the two circles
$x^2 + y^2 - 2x - 4y = 0$
$x^2 + y^2 = 9$

Solution : The equation of the common chord AB is $2x + 4y - 9 = 0$
Since centre of C_2 is (0, 0). The perpendicular length from C_2 to the common chord AB is

$$= \left|\frac{9}{\sqrt{2^2+4^2}}\right| = \frac{9}{\sqrt{20}}$$

$$\Rightarrow AM = \sqrt{C_2A^2 - C_2M^2} = \sqrt{9 - \frac{81}{20}} = \frac{\sqrt{99}}{2\sqrt{5}}$$

$$\Rightarrow \text{Length of common chord} = 2AM = \sqrt{\frac{99}{5}}$$

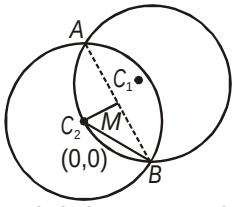


Illustration 25: Tangents are drawn at those points of the circle $x^2 + y^2 - 9 = 0$ where it is intersected by the circle $x^2 + y^2 - 2x - 4y = 0$. Find the intersection point of these tangents.

Solution : Let the circles intersect at A and B and tangents to C_1 are drawn at A and B which intersect at P (α, β) then the line AB has two status. At first it is a common chord of two circles and secondly it is the chord of contact when tangents are drawn from P (α, β) to $x^2 + y^2 - 9 = 0$
$\Rightarrow$ The lines $2x + 4y - 9 = 0$ and $x\alpha + y\beta - 9 = 0$ are coincident.

$$\Rightarrow \frac{\alpha}{2} = \frac{\beta}{4} = \frac{-9}{-9} \Rightarrow \alpha = 2, \beta = 4$$

Thus tangents intersect at (2, 4).

Illustration 26: Find the equation of the chord of $x^2 + y^2 - 6x + 10y - 9 = 0$ which is bisected at (–2, 4).

Solution : The equation of the required chord is
$-2x + 4y - 3(x - 2) + 5(y + 4) - 9 = 4 + 16 + 12 + 40 - 9$
$-5x + 9y - 46 = 0$ or $5x - 9y + 46 = 0$

Illustration 27: Find the middle point of the chord intercepted on line $lx + my + n = 0$ by the circle $x^2 + y^2 = a^2$.

Solution : Let (x_1, y_1) be the middle point of the chord intercepted by the circle $x^2 + y^2 = a^2$, on the line $lx + my + n = 0$. Then equation of the chord of the circle $x^2 + y^2 = a^2$, whose middle

points is (x_1, y_1), is
$xx_1 + yy_1 - a^2 = x_1^2 + y_1^2 - a^2$
or $xx_1 + yy_1 = x_1^2 + y_1^2$ .....(1)
Clearly $lx + my + n = 0$ and (1) represented the same line,

$$\frac{x_1}{l} = \frac{y_1}{m} = \frac{x_1^2 + y_1^2}{-n} = \lambda \text{ (say)}$$

$\therefore\ x_1 = l\lambda$ .....(2)
$y_1 = m\lambda$
and $x_1^2 + y_1^2 = -n\lambda$
or $l^2\lambda^2 + m^2\lambda^2 = -n\lambda$ (from (2))

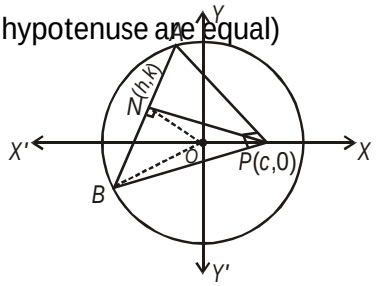
$$\therefore\ \lambda = \frac{n}{l^2 + m^2}$$

so from (2), $x_1 = \frac{nl}{l^2 + m^2}, y_1 = -\frac{mn}{l^2 + m^2}$

Hence, the required points is $\left(-\frac{nl}{l^2 + m^2}, -\frac{mn}{l^2 + m^2}\right)$

Illustration 28: Find the locus of middle points of chords of the circle $x^2 + y^2 = a^2$, which subtend right angle at the point $(c, 0)$.

Solution : Let $N(h, k)$ be the middle point of any chord AB, which subtend a right angle at $P(c, 0)$
Since $\angle APB = 90^o$ $\therefore\ NA = NB = NP$
(since distance of the vertices from middle point of the hypotenuse are equal)
or $(NA)^2 = (NB)^2 = (h - c)^2 + (k - 0)^2$ .....(1)
But also $\angle BNO = 90^o$
$\therefore\ (OB)^2 = (ON)^2 + (NB)^2$
$\Rightarrow -(NB)^2 = (ON)^2 - (OB)^2$
$\Rightarrow -[(h - c)^2 + (k - 0)^2] = (h^2 + k^2) - a^2$
or $2(h^2 + k^2) - 2ch + c^2 - a^2 = 0$
$\therefore$ Locus of $N(h, k)$ is $2(x^2 + y^2) - 2cx + c^2 - a^2 = 0$



POLE AND POLAR

(i) If through a point P is the plane of the circle there be drawn any straight line to meet the circle in Q and R, the locus of the point of intersection of the tangents at Q and R is called the Polar of the point P; also P is called the Pole of the Polar.

(ii) The equation to the polar of a point $P(x_1, y_1)$ w.r.t. the circle $x^2 + y^2 = a^2$ is given by $xx_1 + yy_1 = a^2$, and if the circle is general then the equation of the polar becomes $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$ i.e. $T = 0$. Note that if the point (x_1, y_1) be on the circle then the tangent and polar will be represented by the same equation. Similarly if the point (x_1, y_1) be outside the circle then the chord of contact and polar will be represented by the same equation.

(iii) Pole of a given line $Ax + By + C = 0$ w.r.t. circle $x^2 + y^2 = a^2$ is $\left(-\frac{Aa^2}{C}, -\frac{Ba^2}{C}\right)$.

(iv) If the polar of a point P pass through a point Q then the polar of Q passes through P.

(v) Two lines L_1 and L_2 are conjugate of each other if Pole of L_1 lies on L_2 and vice versa. Similarly two points P and Q are said to be conjugate of each other if the polar of P passes through Q and vice-versa.

Illustration 29: Find the equation of the polar of the point (2, –1) with respect to the circle $x^2 + y^2 – 3x + 4y – 8 = 0$.

Solution : Given circle is $x^2 + y^2 – 3x + 4y – 8 = 0$ .....(1)

Given point is (2, –1) let P ≡ (2, –1). Now equation of the polar of point P with respect to circle (1) is

$$x.2 + y(-1) - 3\left(\frac{x+2}{2}\right) + 4\left(\frac{y-1}{2}\right) - 8 = 0$$

or $4x – 2y – 3x – 6 + 4y – 4 – 16 = 0$ or $x + 2y – 26 = 0$

Illustration 30: Find the pole of the line $3x + 5y + 17 = 0$ with respect to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$

Solution : Given circle is $x^2 + y^2 + 4x + 6y + 9 = 0$ .....(1)

and given line is $3x + 5y + 17 = 0$ .....(2)

Let P (α, β) be the pole of line (2) with respect to circle (1).

Now equation of polar of point P(α, β) with respect to circle (1) is

$x\alpha + y\beta + 2(x + \alpha) + 3(y + \beta) + 9 = 0$

or $(\alpha + 2)x + (\beta + 3)y + 2\alpha + 3\beta + 9 = 0$ .....(3)

Now lines (2) and (3) are same, therefore,

$$\frac{\alpha+2}{\underset{(i)}{3}} = \frac{\beta+3}{\underset{(ii)}{5}} = \frac{2\alpha+3\beta+9}{\underset{(iii)}{17}}$$

From (i) and (ii), we get $5\alpha + 10 = 3\beta + 9$ or $5\alpha – 3\beta = – 1$ .....(4)

From (i) and (iii), we get $17\alpha + 34 = 6\alpha + 9\beta + 27$ or $11\alpha – 9\beta = – 7$ .....(5)

Solving (4) and (5), we get $\alpha = 1$, $\beta = 2$

Hence required pole is (1, 2).

Illustration 31: Show that the polars of the points (1, – 2) with respect to the circles $x^2 + y^2 + 6x + 5 = 0$ and $x^2 + y^2 + 2x + 8y + 5 = 0$ coincide. Prove also that there is another point, the polars of which with respect to these circles are the same and find its co-ordinates.

Solution : Given circles are

$x^2 + y^2 + 6y + 5 = 0$ .....(1)

and $x^2 + y^2 + 2x + 8y + 5 = 0$ .....(2)

Let P ≡ (1, 2)

Now polar of points P (1, –2) with respect to circle (1) is

$x.1 + y(– 2) + 3(y – 2) + 5 = 0$ or $x + y – 1 = 0$ .....(3)

Again polar of point P (1, – 2) with respect to circle (2) is

$x.1 + y(– 2) + x + 1 + 4(y – 2) + 5 = 0$

or $2x + 2y – 2 = 0$ or $x + y – 1 = 0$ .....(4)

From (3) and (4) it follows that the polars of point (1, – 2) with respect to circles (1) and (2) are same.

Second part :

Let Q(α, β) be a point the polars of which with respect to circles (1) and (2) are same.

Now equation of polars of points P(α, β) with respect to circles (1) and (2) are respectively.

$x\alpha + y\beta + 3(y + \beta) + 5 = 0$ or $\alpha x + (\beta + 3)y + 3\beta + 5 = 0$ .....(5)

and $x\alpha + y\beta + (x + \alpha) + 4(y + \beta) + 5 = 0$

or $(\alpha + 1)x + (\beta + 4)y + \alpha + 4\beta + 5 = 0$ .....(6)

Now equations (5) and (6) are identical

$$\therefore \quad \frac{\alpha+1}{\underset{(i)}{\alpha}} = \frac{\beta+4}{\underset{(ii)}{\beta+3}} = \frac{\alpha+4\beta+5}{\underset{(iii)}{3\beta+5}}$$

From (i) and (ii), we have
$\alpha\beta + \beta + 3\alpha + 3 = \alpha\beta + 4\alpha$ or $\alpha - \beta = 3$ .....(7)
From (i) and (iii), we have
$3\alpha\beta + 3\beta + 5\alpha + 5 = \alpha^2 + 4\alpha\beta + 5\alpha$
or $\alpha^2 + \alpha\beta - 3\beta - 5 = 0$
or $(\beta + 3)^2 + (\beta + 3)\beta - 3\beta - 5 = 0$ [(from (7)]
or $2\beta^2 + 6\beta + 4 = 0$ or $\beta^2 + 3\beta + 2 = 0 \therefore \beta = -1, -2$.
From (7), when $\beta = -1$, $\alpha = 2$ and when $\beta = -2$, $\alpha = 1$.
Thus points are (2, – 1), therefore, required point Q will be (2, – 1).

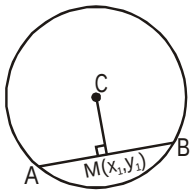
Illustration 32: Let C be the centre of a circle. The lines L_1 and L_2 are the polars of points A and respectively with respect to the circle. Perpendiculars AM and BN are dropped from A to the line L_2 and from B to line L_1. Prove that CA : CB = AM : BN.

Solution : Let the circle be $x^2 + y^2 - a^2 = 0$ .....(1)
$A \equiv (x_1, y_1)$ and $B \equiv (x_2, y_2)$
Since C is the centre of circle (1)
$\therefore$ C $\equiv$ (0, 0)
Now Polars of A and B w.r.t circle (1) are respectively
$xx_1 + yy_1 - a^2 = 0$ ... (2) and $xx_2 + yy_2 - a^2 = 0$ ..... (3)

$$\therefore \frac{AM}{BN} = \frac{|x_1x_2 + y_1y_2 - a^2|}{\sqrt{x_2^2 + y_2^2}} + \frac{|x_2x_1 + y_2y_1 - a^2|}{\sqrt{x_1^2 + y_1^2}} = \frac{\sqrt{x_1^2 + y_1^2}}{\sqrt{x_2^2 + y_2^2}} = \frac{CA}{CB}$$

EQUATION OF THE CHORD WITH A GIVEN MIDDLE POINT

This equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ in terms of its mid point $M(x_1, y_1)$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ which is designated by $T = S_1$.



Note :-
(i) The shortest chord of a circle passing through a point M inside the circle is the one whose middle point is M.
(ii) The chord passing through a point M inside the circle and which is at a maximum distance from the centre is a chord with middle point M.

Student Task

1. The equation of the circle with the chord $y = 2x$ of the circle $x^2y^2 - 10x = 0$ as its diameter is
2. The locus of the middle points of chords of length 4 of the circle $x^2 + y^2 = 16$

Ans. 1. $x^2 + y^2 = 2x + 4y$ **2.** a circle of radius $2\sqrt{3}$

7. EQUATION OF THE CHORD JOINING TWO POINTS OF CIRCLE

The equation of chord *PQ* to the circle $x^2 + y^2 = a^2$ joining two points *P*(*a*) and *Q*(*b*) on it is given by the equation of a straight line joining two point *a* and *b* on the circle $x^2 + y^2 = a^2$ is

$$x \cos\frac{\alpha + \beta}{2} + y \cos\frac{\alpha + \beta}{2} = a \cos\frac{\alpha - \beta}{2}$$

Illustration 33: Find the equation of circle passing through (2, 0) and (3, – 1) and cutting a chord length 4 units on *y*-axis.

Solution : From the equation $x^2 + y^2 - 5x + y + 6 + \lambda(x + y - 2) = 0$ we conclude that length of the

chord intercepted on y-axis = $2\sqrt{\left(\frac{1+\lambda}{2}\right)^2-(6-2\lambda)}$ (applying $2\sqrt{f^2-c}$)

$\Rightarrow 2\sqrt{\left(\frac{1+\lambda}{2}\right)^2-(6-2\lambda)}=4$

$\Rightarrow \lambda = 3, \lambda = -13$

Thus one such circle is $x^2 + y^2 - 5x + y + 6 + 3(x + y - 2) = 0$

$\Rightarrow x^2 + y^2 - 2x + 4y = 0$

Illustration 34: Show that any chord that arises as an intersection of a circle through $A(2, 0)$ and $B(3, -1)$ and the circle $x^2 + y^2 - 2x + 6y - 11 = 0$ passes through a fixed point. Find the co-ordinates of that point.

Solution : Any circle through A (2, 0) and B (3, –1) is $x^2 + y^2 - 5x + y + 6 + \lambda(x + y - 2) = 0$

This will cut the circle $x^2 + y^2 - 2x + 6y - 11 = 0$ in a chord whose equation is

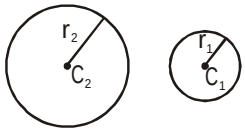
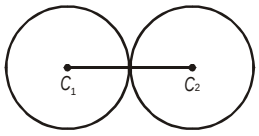
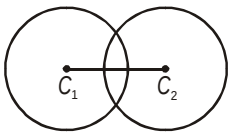
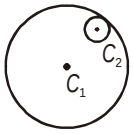
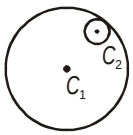
$-3x - 5y + 17 + \lambda (x + y - 2) = 0$

It is clear that these chords are concurrent at a point whose co-ordinates are given by

$-3x - 5y + 17 = 0, x + y - 2 = 0$

We easily get the fixed point as $\left(-\frac{7}{2}, \frac{11}{2}\right)$

COMMON TANGENTS TO TWO CIRCLES

Case	Number of Tangents	Condition
(i) 	4 common tangents (2 direct and 2 transverse)	$r_1 + r_2 < C_1C_2$
(ii) 	3 common tangents	$r_1 + r_2 = C_1C_2$
(iii) 	2 common tangents	$\lvert r_1 - r_2\rvert < C_1C_2 < r_1 + r_2$
(iv) 	1 common tangent	$\lvert r_1 - r_2\rvert = C_1C_2$
(v) 	No common tangent	$C_1C_2 < \lvert r_1 - r_2\rvert$

(Here C_1C_2 is distance between centres of two circles).

Illustration 35: Examine if the two circles $x^2 + y^2 - 2x - 4y = 0$ and $x^2 + y^2 - 8y - 4 = 0$ touch each other externally or internally.

Solution : Given circles are $x^2 + y^2 - 2x - 4y = 0$ ....(i)

and $x^2 + y^2 - 8y - 4 = 0$ ....(ii)

Let A and B be the centres and r_1 and r_2 the radii of circles (i) and (ii) respectively, then

$A \equiv (1,2), B \equiv (0,4), r_1 = \sqrt{5}, r_2 = 2\sqrt{5}$

Now $AB = \sqrt{(1-0)^2 + (2-4)^2} = \sqrt{5}$

and $r_1 + r_2 = 3\sqrt{5}. |r_1 - r_2| = |\sqrt{5} - 2\sqrt{5}| = \sqrt{5}$

Thus $AB = |r_1 - r_2|$, hence the two circles touch each other internally.

Illustration 36: Prove that the circle $x^2 + y^2 + 2ax + c^2 = 0$ and $x^2 + y^2 + 2by + c^2 = 0$ touch each other if $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$.

Solution : Given circles are $x^2 + y^2 + 2ax + c^2 = 0$ ....(i)

and $x^2 + y^2 + 2by + c^2 = 0$ ....(ii)

Let A and B be the centres of circles (i) and (ii) respectively and r_1 and r_2 be their radii, then

$A \equiv (-a, 0), B \equiv (0, b), r_1 = \sqrt{a^2 - c^2}, r_2 = \sqrt{b^2 - c^2}$

The two circles (i) and (ii) will touch each other externally or internally according

$AB = |r_1 + r_2|$ or $AB = |r_1 - r_2|$

i.e. $AB^2 = (r_1 + r_2)^2$ or $AB^2 = (r_1 - r_2)^2$

Thus the two circles will touch each other if

$AB^2 = (r_1 \pm r_2)^2$ or $a^2 + b^2 = r_1^2 + r_2^2 \pm 2r_1r_2$

or $a^2 + b^2 = a^2 - c^2 + b^2 - c^2 \pm 2\sqrt{a^2 - c^2}\sqrt{b^2 - c^2}$

or $2c^2 = \pm 2\sqrt{a^2 - c^2}\sqrt{b^2 - c^2}$

$C^2 = \pm\sqrt{a^2 - c^2}\sqrt{b^2 - c^2}$

or $c^4 = (a^2 - c^2)(b^2 - c^2)$ or $c^4 = a^2b^2 - c^2b^2 - a^2c^2 + c^4$

or $c^2b^2 + a^2c^2 = a^2b^2$

or $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$ [dividing by $a^2b^2c^2$]

Illustration 37: Prove that $x^2 + y^2 = a^2$ and $(x - 2a)^2 + y^2 = a^2$ are two equal circles touching each other. Find the equation of circle (or circles) of equal radius touching both the circles.

Solution : Given circles are

$x^2 + y^2 = a^2$ ....(i)

and $(x - 2a)^2 + y^2 = a^2$ ....(ii)

Let A and B be the centres and r_1 and r_2 the radii of the two circle (i) and (ii) respectively, then

$A \equiv (0, 0), B \equiv (2a, 0), r_1 = a, r_2 = a$

Now $AB = \sqrt{(0 - 2a)^2 + 0^2} = 2a = r_1 + r_2$

Hence the two circles touch each other externally.

Let the equation of equal circle touching circles (i) and (ii) be

$(x - \alpha)^2 + (y - \beta)^2 = a^2$ ....(iii)

Its centre C is (α, β) and radius $r_3 = a$

Since (iii) touches (i)

$\therefore AC = r_1 + r_2 = 2a$. [Here $AC \neq |r_1 - r_3|$ as $r_1 - r_3 = a - a = 0$]

or $AC^2 = 4a^2$ or $\alpha^2 + \beta^2 = 4a^2$ ....(iv)

Again since circle (iii) touches (ii)

$\therefore$ $BC = r_2 + r_3$ or $BC^2 = (r_2 + r_3)^2$

$\therefore$ $(2a - \alpha)^2 + \beta^2 = (a + a)^2$ or $a^2 + b^2 - 4a\alpha = 0$

or $4a^2 - 4a\alpha = 0$ [from (iv)]

$\therefore$ $\alpha = a$ and from (iv), $\beta = \pm\sqrt{3}\,a$

Thus required circles are $(x - a)^2 = (y \mp \sqrt{3}a)^2 = a^2$

or $x^2 + y^2 - 2ax \mp 2\sqrt{3}ay + 3a^2 = 0$

Note :-* (i)** ***The direct common tangents meet at a point which divides the line joining centre of circles externally in the ratio of their radii.
Transverse common tangents meet at a point which divides the line joining centre of circles internally in the ratio of their radii.

(ii) ***Length of an external (or direct) common tangent and internal (or transverse) common tangent to the two circles are given by :*** $L_{ext} = \sqrt{d^2 - (r_1 - r_2)^2}$ ***and*** $L_{int} = \sqrt{d^2 - (r_1 + r_2)^2}$, ***where d = distance between the centres of the two circles and*** r_1, r_2 ***are the radii of the two circles. Note that length of internal common tangent is always less than the length of the external or direct common tangent.***

Illustration 38: Determine the number of common tangents to the two circles $C_1 : x^2 + y^2 = 25$, $C_2 = x^2 + y^2 - 4x + 6y + 4 = 0$ and find their lengths.

Solution : The centres are (0, 0), (2, 3) and radii are 5 and 3.

$$d = \sqrt{13}, r_1 = 5, r_2 = 3$$

We can note that $r_1 - r_2 < d < r_1 + r_2$

$\Rightarrow$ The circles intersect at two distinct real points.

$\Rightarrow$ There are two direct common tangents.

Their lengths = $\sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{13 - (5-3)^2} = 3$ units

Illustration 39: Find all the common tangents to the circles $C_1 : x^2 + y^2 + 22x - 4y - 100 = 0$ and $C_2 : x^2 + y^2 - 22x + 4y + 100 = 0$. Find their lengths also.

Solution : (C_1 and C_2 will also denote the centre of the two circles)

Note that centre of C_1 is (–11, 2) and its radius is 15.

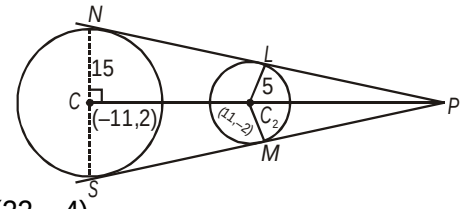
The centre of C_2 is (11, –2) and its radius is 5.

Let the common tangent meet circle C_1 at N and C_2 at L.

Produce NL so that it intersects the line joining C_1 and C_2 at P.

Since the triangle PNC_1 and PLC_2 are similar.

$$\frac{C_1P}{C_2P} = \frac{r_1}{r_2} = \frac{15}{5} = 3$$



Since P lies externally. The co-ordinates of P are

$$\left(\frac{15\times 11 - 5(-11)}{15-5}, \frac{15\times(-2) - 5\times 2}{15-5}\right)$$ i.e. P is (22, –4)

Any line through (22, –4) may be taken as $y + 4 = m(x - 22)$ or $y + mx + 4 + 22m = 0$

Since it is a tangent to C_2, we must have $\perp$ from centre of C_2 = radius C_2

$$\Rightarrow \quad \frac{-2 - 11m + 4 + 22m}{\sqrt{1 - m^2}} = \pm 5 \qquad \Rightarrow m = \frac{7}{24}, -\frac{3}{4}$$

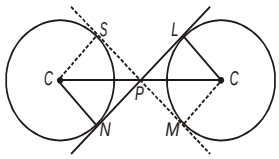
$\Rightarrow$ The direct common tangents are $y + 4 = \frac{7}{24}(x - 22)$ and $y + 4 = -\frac{3}{4}(x - 22)$

or $7x - 24y = 250$ and $3x + 4y + 4y = 50$....(i)

Again for transverse common tangents

We note that common tangent NL in intersected by C_1C_2 at P and $\frac{C_1P}{PC_2} = \frac{r_1}{r_2} = \frac{15}{5}$

$\Rightarrow$ P is $\left(\frac{15\times11+5(-11)}{15}, \frac{15\times(-2)+5\times2}{15+5}\right)$ i.e. P is $\left(\frac{11}{2}, -1\right)$



As before we get $m = -\frac{24}{7}$ or $\frac{4}{3}$

Transverse common tangents are $y + 1 = -\frac{24}{7}\left(x - \frac{11}{2}\right)$ and $y + 1 = \frac{4}{3}\left(x - \frac{11}{2}\right)$.

or $24x + 7y = 125$ or $4x - 3y = 25$ .....(ii)

Note : If we take one of the common tangents (i) and solve it with C_1 and C_2. The distance between the points of contact will be the length of the direct common tangent. But the length can be found without finding these ponts and without finding the equation of common tangents. Indeed lengths of the direct common tangents = $\sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{(-11-11)^2 + (2+2)^2 - (15-5)^2} = \sqrt{400} = 20$

The length of the transverse common tangent = $\sqrt{d^2 - (r_1 - r_2)^2}$ = 10

Illustration 40: Find the coordinates of the point at which the circles $x^2 + y^2 - 4x - 2 + 4 = 0$ and $x^2 + y^2 - 12x - 8y + 36 = 0$ touch each other. Also find the equation of common tangents touching the circles in distinct points.

Solution :

I Circle	**II Circle**
Centre $A(2, 1)$	$B(6, 4)$

Radius $r_1 = \sqrt{4+1-4} = 1$ $r_2 = \sqrt{36+16-36} = 4$

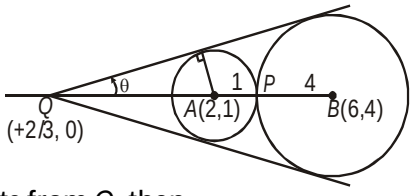
$AB = \sqrt{16+9} = \sqrt{25} = 5 = 1 + 4 = r_1 + r_2$

$\therefore$ the two circles touch externally P, the point of contact, divides AB internally in the ratio 1 : 4 hence P is $\left(\frac{14}{5}, \frac{8}{5}\right)$ and Q which divides AB externally in the same ratio is $\left(\frac{2}{3}, 0\right)$.

The tangent from Q to the circle I will also be tangent to the circle II and hence will be a common tangent touching the two circles in two distinct points.

If θ be the angle that this tangent (actually there are two) makes with AB, then

$$\sin\theta = \frac{1}{AQ} = \frac{1}{\sqrt{1+\frac{16}{9}}} = \frac{3}{5}$$



$\therefore$ $\tan\theta = 3/4$

Now if m be the slope of one of the common tangents from Q, then

$$\frac{3}{4} = \tan\theta = \frac{m - \frac{3}{4}}{1 - m\frac{3}{4}} \left(\text{slope of } AB = \frac{3}{4} \right)$$

$16m - 12 = 12 + 9m$

$7m = 24 \Rightarrow m = \frac{24}{7}$

If m be the slope of the other common tangent $\frac{m' - \frac{3}{4}}{1 + m'\frac{3}{4}} = \frac{-3}{4} \Rightarrow m' = 0$

The two common tangents are $y - 0 = 0 \Rightarrow y = 0$
i.e., the x-axis

and $y - 0 = \frac{24}{7}\left(x - \frac{2}{3} \right)$

$7y = 24x - 16 \Rightarrow 24x - 7y - 16 = 0$

Student Task

1. The equation of the locus of the middle point of a chord of the circle $x^2 + y^2 = 2(x + y)$ such that the pair of lines joining the origin to the point of intersection of the chord and the circle are equally inclined to the x-axis is
2. The locus of the middle points of chords of length 4 of the circle $x^2 + y^2 = 16$ is

Ans. 1. $x + y = 2$ **2.** a circle of radius $2\sqrt{3}$

8. ORTHOGONALITY OF TWO CIRCLES

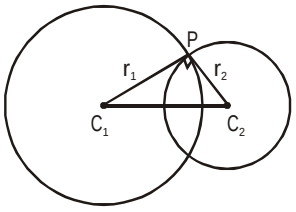
Two circles $S_1 = 0$ and $S_2 = 0$ are said to be orthogonal or said to intersect orthogonally if the tangents at their point of intersection include a right angle. The condition for two circles to the orthogonal is :

$2g_1g_2 + 2f_1f_2 = c_1 + c_2$

Proof : $(C_1C_2)^2 = (C_1P)^2 + (C_2P)^2$

$\Rightarrow (g_1 - g_2)^2 + (f_1 - f_2)^2 = g_1^2 + f_1^2 - c_1 + g_2^2 + f_2^2 - c_2$

$\Rightarrow 2g_1g_2 + 2f_1f_2 = c_1 + c_2$



Note :- (i) The centre of a variable circle orthogonal to two fixed circles lies on the radical axis of two circles.

(ii) If two circles are orthogonal, then the polar of a point P on first circle w.r.t. the second circle passes through the point Q which is the other end of the diameter through P. Hence locus of a point which moves such that its polar w.r.t. the circles $S_1 = 0$, $S_2 = 0$ and $S_3 = 0$ are concurrent in a circle which is orthogonal to all the three circles.

Illustration 41: Show that the circles $x^2 + y^2 - 2x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 6 = 0$ cut each other orthogonally.

Solution : Given circles are $x^2 + y^2 - 2x - 6y - 12 = 0$ .....(1)
and $x^2 + y^2 + 6x + 4y - 6 = 0$ .....(2)

We know that the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and
$x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$
will cut each other orthogonally if $2gg_1 + 2ff_1 = c + c_1$
Here $g = -1$, $f = -3$, $c = -12$
and $g_1 = 3$, $f_1 = 2$, $c_1 = -6$
Now $2gg_1 + 2ff_1 = 2(-1)(3) + 2(-3).2 = -18 = c + c_1$
Hence circles (1) and (2) cut each other orthogonally

Illustration 42: If $S = 0$ and $S_1 = 0$ are the two circles with radii a and a_1 respectively. Show that the circles $\frac{S}{a} \pm \frac{S_1}{a_1}$ intersect at right angles.

Solution : Let A and B be the centre of the two circles. We take AB as x-axis and its middle point as the origin. Let AB = $2k$, then A $\equiv (-k, 0)$, B $\equiv (k, 0)$.

Now the equation of the two circles are



$S \equiv (x-k)^2 + y^2 - a^2 = 0$ and $S_1 \equiv (x+k)^2 + y^2 - a_1^2 = 0$

Also $\frac{S}{a} \pm \frac{S_1}{a_1} = 0 \Rightarrow a_1S \pm aS_1 = 0$

$$\Rightarrow \begin{cases} a_1[(x-k)^2 + y^2 - a^2] + a[(x+k)^2 + y^2 - a_1^2] = 0 \\ a_1[(x-k)^2 + y^2 - a^2] - a[(x+k)^2 + y^2 - a_1^2] = 0 \end{cases}$$

$$\Rightarrow \begin{cases} (a+a_1)(x^2+y^2) - 2kx + (a_1 - a) + (k^2 - aa_1)(a_1 + a) = 0 \\ (a_1 - a)(x^2+y^2) - 2kx(a_1 + a) + (k^2 + aa_1)(a_1 - a) = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x^2 + y^2 - 2k\left(\frac{a_1 - a}{a_1 + a}\right)x + k^2 - aa_1 = 0 \quad .....(1) \\ x^2 + y^2 - 2k\left(\frac{a_1 + a}{a_1 - a}\right)x + k^2 + aa_1 = 0 \quad .....(2) \end{cases}$$

Here $g = -\left(\frac{a_1 - a}{a_1 + a}\right)k$, $f_1 = 0$, $c = k_2 - aa_1$

and $g_1 = -\left(\frac{a_1 + a}{a_1 - a}\right)k$, $f_2 = 0$, $c_1 = k^2 + aa_1$

Now $2gg_1 + 2ff_1 = 2k^2 + 0 = 2k^2 = c + c_1$

Hence circles $\frac{S}{a} + \frac{S_1}{a_1} = 0$ and $\frac{S}{a} - \frac{S_1}{a_1} = 0$ intersect at right angles.

Illustration 43: Prove that the two circles which pass through the points $(0, a)$ and $(0, -a)$ and touch the line $y = mx + c$ will cut orthogonally if $c^2 = a^2(2 + m^2)$.

Solution :

Let a circle be $x^2 + y^2 + 2gx + 2fy + \lambda = 0$ .....(1)
Since circle (1) passes through $(0, a)$ and $(0, -a)$.
$\therefore$ $a^2 + 2fa + \lambda = 0$ and $a^2 - 2fa + \lambda = 0$
Solving these equations, we get $f = 0$, $\lambda = -a^2$
Since line $mx - y + c = 0$ touches circle (1)
$\therefore$ radius of the circle = length of the perpendicular from the centre $(-g, -f)$ to the line.

$\therefore \sqrt{g^2+f^2-\lambda} = \dfrac{|-mg+f+c|}{\sqrt{1+m^2}}$

$\therefore \sqrt{g^2+a^2} = \dfrac{|-mg+c|}{\sqrt{1+m^2}}$ [$\therefore f = 0$ and $\lambda = -a^2$]

or $(1 + m^2)(g^2 + a^2) = (-mg + c)^2$
or $g^2 + 2mcg + (1 + m^2)a^2 - c^2 = 0$ ....(2)
Let g_1 and g_2 be the roots of eqn. (2) $\therefore g_1 + g_2 = -2mc$, $g_1g_2 = (1 + m^2)a^2 - c^2$
Corresponding to the two values of g the two circles are
$x^2 + y^2 + 2g_1x - a^2 = 0$ and $x^2 + y^2 + 2g_2x - a^2 = 0$
They will cut orthogonally if $2g_1g_2 + 2f_1f_2 = c_1 + c_2$
i.e. if, $2\{(1 + m^2)\, a^2 - c^2\} + 2.0.\ 0 = -a^2 - a^2$
or $(1 + m^2)a^2 - c^2 + a^2 = 0$; or $c^2 = (2 + m^2)a^2$.

Illustration 44: Find the angle between the circles
$S : x^2 + y^2 - 4x + 6y + 11 = 0$ and $S' : x^2 + y^2 - 2x + 8y + 13 = 0$

Solution : Centres and radii of circles S and S' are

$C_1(2, -3)$, $r_1 = \sqrt{2}$, $C_2(1, -4)$, $r_2 = 2$
Distance between centres, $d = |C_1C_2|$

$= \sqrt{(2-1)^2+(-3+4)^2} = \sqrt{2}$

If angle between the circles is θ, then

$\cos(180^o - \theta) = \left(\dfrac{r_1^2 + r_2^2 - d^2}{2r_1r_2}\right)$

$\cos(180^o - \theta) = \left(\dfrac{2+4-2}{2.\sqrt{2}.2}\right) = \dfrac{1}{\sqrt{2}}$

$180^o - \theta = 45^o$
$\theta = 135^o$

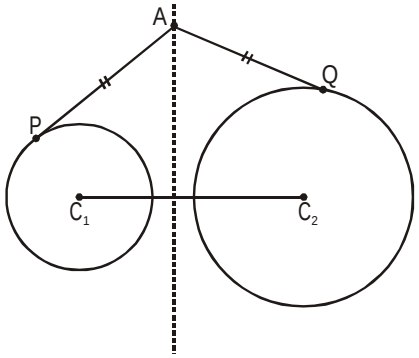
Student Task

1. The locus of the centres of circles passing through the origin and intersecting the fixed circle $x^2 + y^2 - 5x + 3y - 1 = 0$ orthogonally is
2. Find the equation of the circle which cuts the circles $x^2 + y^2 - 4x - 6y + 11 = 0$ and $x^2 + y^2 - 10x - 4y + 21 = 0$ orthogonally and has $2x + 3y = 7$ as diameter.
3. A circle passes through the origin and has its centre on y =x, If it cuts $x^2 + y^2 - 4x - 6y + 10 = 0$ orthogonally, then find the equation of the of the circle.

Ans. 1. a straight line of the slope $\dfrac{3}{5}$ **2.** $x^2 + y^2 - 4x - 2y + 3 = 0$ **3.** $x^2 + y^2 - 2x - 2y = 0$

9. RADICAL AXIS AND RADICAL CENTRE

The radical axis of two circles is the locus of points whose powers w.r.t. the two circles are equal.
The equation of radical axis of the two circle $S_1 = 0$ & $S_2 = 0$ is given by
$S_1 - S_2 = 0$ i.e. $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$



The common point of intersection of the radical axes of three circles takes two at a time is called radical centre of three circles. Note that the length of tangents from radical centre to the three circles are equal.

Note :-

(a) If two circles intersect, then the radical axis is the common chord of the two circles.

(b) If two circles touch each other then a radical axis is the common tangent of the two circles at the comon point of contact.

(c) Radical axis is always perpendicular to the line joining the centres of the two circles.

(d) Radical axis will pass through the mid point of the line joining the centres of the two circles only if the two circles have equal radii.

(e) Radical axis bisects a common tangent between the two circles.

(f) A system of circles, every two which have the same radical axis, is called a coaxal system.

(g) Pairs of circles which do not have radical axis are concentric.

Illustration 45: Find the radical center of three circles described on the three sides $4x - 7y + 10 = 0$, $x + y - 5 = 0$ and $7x + 4y - 15 = 0$ of a triangle as diameters.

Solution : Since the radical centre of three circles described on the sides of a triangle as diameters is the orthocentre of the triangle.

$\therefore$ Radical centre = orthocentre

Given sides are $4x - 7y + 10 = 0$ ....(1)

$x + y - 5 = 0$ ....(2)

$7x + 4y - 15 = 0$ ....(3)

Since lines (1) and (3) are perpendiculars the point of intersection of (1) and (3) is (1, 2), the orthocentre of the triangle. Hence radical centre is (1, 2).

Illustration 46: Find the equation of the system of circles co-axial with the circles

$x^2 + y^2 + 4x + 2y + 1 = 0$

and $x^2 + y^2 - 2x + 6y - 6 = 0$

Also, find the equation of that particular whose centre lies on the radical axis.

Solution : Given circles are

$S_1 \equiv x^2 + y^2 + 4x + 2y + 1 = 0$

$S_2 \equiv x^2 + y^2 - 2x + 6y - 6 = 0$

$\therefore$ Radical axis is $S_1 - S_2 = 0$

i.e., $6x - 4y + 7 = 0$

Now system of co-axial is

$S_1 + \lambda (s_1 - s_2) = 0$

$\Rightarrow (x^2 + y^2 + 4x + 2y + 1) + \lambda (6x - 4y + 7) = 0$

$\Rightarrow x^2 + y^2 + 2x (2 + 3\lambda) + 2y (1 - 2\lambda) + 1 + 7\lambda = 0$

Its central $[-(2 + 3\lambda), -(1 - 2\lambda)]$ lies on (1)

$\therefore$ $6 \times -(2 + 3\lambda) - 4 \times -(1 - 2\lambda) + 7 = 0$

or $-12 - 18\lambda + 4 - 8\lambda + 7 = 0$
or $-26\lambda - 1 = 0$

$\lambda = -\frac{1}{26}$

Substituting the value of λ in (2), the equation of circle is

$$x^2 + y^2 + 2x\left(2 - \frac{3}{26}\right) + 2y\left(1 + \frac{2}{26}\right) + 1 - \frac{7}{26} = 0$$

$\Rightarrow 26(x^2 + y^2) + 98x + 56y + 9 = 0$

Illustration 47 : If A, B,C be the centres of three co-axial circles and t_1, t_2, t_3 be the lengths of the tangents from any point, prove that $\overline{BC}.t_1^2 + \overline{CA}.t_2^2 + \overline{AB}.t_3^2 = 0$

Solution : Let the equations of three circles are $x^2 + y^2 + 2g_ix + c = 0$, $i = 1, 2, 3$,
According to the question $A \equiv (-g_1, 0)$, $B \equiv (-g_2, 0)$, $C \equiv (-g_3, 0)$
Let any point be $P(h, k)$

$\therefore\ t_1 = \sqrt{h^2 + k^2 + 2g_1h + c}$

$t_2 = \sqrt{h^2 + k^2 + 2g_2h + c}$

$t_3 = \sqrt{h^2 + k^2 + 2g_3h + c}$

and $\overline{AB} = (g_1 - g_2)$

$\overline{BC} = (g_2 - g_3)$

and $\overline{CA} = (g_3 - g_1)$

Now $\overline{BC}.t_1^2 + \overline{CA}.t_2^2 + \overline{AB}.t_3^2$

$= \sum (g_2 - g_3)\ (h^2 + k^2 + 2g_1h + c)$
$= (h^2 + k^2 + c) \sum (g_2 - g_3) + 2h \sum g_1\ (g_2 - g_3)$
$= (h^2 + k^2 + c)\ (g_2 - g_3 + g_3 - g_1 + g_1 - g_2) + 2h\ \{g_1\ (g_2 - g_3) + g_2\ (g_3 - g_1) + g_3\ (g_1 - g_2)$
$= (h^2 + k^2 + c)\ (0) + 2h(0) = 0$
which proves the result.

Student Task

1. Find the radical centre of the three circles $x^2 + y^2 + 4x + 7 = 0;\ 2x^2 + 2y^2 + 3x + 5y + 9 = 0$; $x^2 + y^2 + y = 0$ Find the length of the tangent from the radical centre to the second circle.
2. The angle between the radical axis and the line joining the centres of the circles $x^2 + y^2 + 2gx + 2fy + c = 0$ and $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ is

Ans. 1. 2 **2.** $\frac{\pi}{2}$

10. FAMILY OF CIRCLES

This article is aimed at obtaining the equation of a group of circles having a specific characteristic. For example, the equation $x^2 + y^2 + 4x + 2y + \lambda = 0$ where λ is arbitrary, represents a family of circles with fixed centre (–2, –1) but variable radius. We have the following results for some other families of circles.

(a) The equation of the family of circles passing through the points of intresection of two circles $S_1 = 0$ and $S_2 = 0$ is : $S_1 + K S_2 = 0$
($K \neq -1$, provided the co-efficient of x^2 and y^2 in S_1 and S_2 are same)

(b) The equation of the family of circles passing through the point of intersection of a circle S = 0 and a line $L = 0$ is given by $S + KL = 0$.

(c) The equation of a family of circles passing through two given points (x_1, y_1) and (x_2, y_2) can be written in the form :

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + K\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$ where K is a parameter.

(d) The equation of a family of circles touching a fixed line $y - y_1 = m(x - x_1)$ at the fixed point (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + (y - y_1)^2 + K[y - y_1 - m(x - x_1)] = 0$, where K is a parameter.

(e) Family of circles circumscribing a triangle whose sides are given by $L_1 = 0$; $L_2 = 0$ and $L_3 = 0$ is given by : $L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$ provided co-efficient of $xy = 0$ and co-efficient of x^2 = co-efficient of y^2.

(f) Equation of circle circumscribing a quadrilateral whose side in order are represented by the line $L_1 = 0$, $L_2 = 0$, $L_3 = 0$ and $L_4 = 0$ are $\mu L_1L_3 + \lambda L_2L_4 = 0$ where values of μ and λ can be found out by using condition that co-efficient of x^2 = co-efficient of y^2 and co-efficient of $xy = 0$.

Illustration 48: Find the equation of a circle which passes through the intersection of the circle $x^2 + y^2 - 9 = 0$, $x^2 + y^2 + 2x + 4y + 3 = 0$ and also passes through (0, 0).

Solution : The equation of the required circle must be of the form
$x^2 + y^2 - 9 + \lambda (x^2 + y^2 - 2x - 4y + 3) = 0$
If this passes through (0, 0) we must have
$-9 + 3\lambda = 0 \Rightarrow \lambda = 3$
The required circle, is $x^2 + y^2 - 9 + 3(x^2 + y^2 - 2x - 4y + 3) = 0$
or $2x^2 + 2y^2 - 3x - 6y = 0$

Illustration 49: Find the equation of a circle which passes through the intersection of the circles $x^2 + y^2 - 2x + 4y - 3 = 0$, $x^2 + y^2 - 6x - 8y + 5 = 0$ and
(i) whose centre lies on y-axis
(ii) whose centre lies on the line $2x + y = 7$
(iii) whose diameter is the common chord of given circles.

Solution : The required circles must be of the form
$x^2 + y^2 - 2x + 4y - 3 + \lambda(x^2 + y^2 - 6x - 8y + 5) = 0$ .....(1)
which, on arranging becomes
$(1 + \lambda)x^2 + (1 + \lambda)2 - 2(1 + 3\lambda)x + 2(2 - 4\lambda)y + 5\lambda - 3 = 0$

or $x^2 + y^2 - 2.\dfrac{1+3\lambda}{1+\lambda}x + 2.\dfrac{2-4\lambda}{1+\lambda}y + \dfrac{5\lambda-3}{1+\lambda} = 0$

The centre = $\left(\dfrac{1+3\lambda}{1+\lambda}, \dfrac{4\lambda-2}{1+\lambda}\right)$

For part (i), we must have $\dfrac{1+3\lambda}{1+\lambda} = 0$

$\Rightarrow \lambda = -\dfrac{1}{3}$. On putting $\Rightarrow \lambda = -\dfrac{1}{3}$ at (1)

we get the answer to part (i)
For part (ii) we must have

$$2\left(\frac{1+3\lambda}{1+\lambda}\right)+\left(\frac{4\lambda-2}{1+\lambda}\right)=7 \Rightarrow \lambda=\frac{7}{3}$$

(iii) The common chord of the two given circle is $4x + 12y - 8 = 0$ or $x + 3y - 2 = 0$ Now the centre $\left(\frac{1+3\lambda}{1+\lambda}, \frac{4\lambda-2}{1+\lambda}\right)$ must lie on this common chord since common chord is the diameter

$$\Rightarrow \frac{1+3\lambda}{1+\lambda}+3\left(\frac{4\lambda-2}{1+\lambda}\right)-2=0 \Rightarrow \lambda = \frac{7}{13}$$

Illustration 50: Find the equation of a circle passing through the points (2, 0), (3, – 1) and (2, 5).

Solution : The equation of circles passing through first two points (2, 0) and (3, – 1) may be taken as

$$(x-2)(x-3)+(y-0)(y+1)+\lambda\begin{vmatrix} x & y & 1 \\ 2 & 0 & 1 \\ 3 & -1 & 1 \end{vmatrix}=0$$

Or $x^2 + y^2 - 5x + y + 6 + \lambda(x + y - 2) = 0$
If this passes through (2, 5) we must have
$4 + 25 - 10 + 5 + 6 + 5\lambda = 0 \Rightarrow \lambda = -6$
Thus the required circle is $x^2 + y^2 - 5x + y + 6 - 6(x + y - 2) = 0$
$\Rightarrow x^2 + y^2 - 11x - 5y + 18 = 0$

Student Task

1. The locus of the centres of the circles passing through the intersection of the circles $x^2+y^2=1$ and $x^2+y^2-2x+y=0$
2. The locus of the centre of a circle touching the lines $x+2y=0$ and $x-2y=0$ is
3. The common chord of the circle $x^2+y^2+6x+8y-7=0$ and a circle passing through the origin, and touching the line $y=x$, always passes through the point

Ans. 1. a line whose equation is $x+2y=0$ **2.** xy=0 **3.** (1/2, 1/2)

MISCELLANEOUS EXAMPLES

Example 1 :

For what value(s) of λ, the equation $(10 - \lambda^2)x^2 + (\lambda^2 - 8)y^2 + (3 - \lambda)yx - 10x + 4y + 3 = 0$ represent a real circle.

Solution :

For equation of circle we must satisfy the following 2 conditions.

Condition 1 : Coefficient of x^2 = coefficient of y^2

$\Rightarrow \quad 10 - \lambda^2 = \lambda^2 - 8 \Rightarrow \lambda^2 = 9 \Rightarrow \lambda = \pm 3$

Condition 2 : and coefficient of $xy = 0$

$\Rightarrow \quad 3 - \lambda = 0 \qquad \Rightarrow \lambda = 3$

So, the common value is $\lambda = 3$.

Example 2 :

Let a circle be given by $2x(x - a) + y(2y - b) = 0$ $(a \neq 0)$. Suppose it is possible to draw two distinct chords from (a, $b/2$) on the circle such that each is bisected by x-axis. Show that the condition for this is $a^2 > 2b^2$.

Solution :

Since the circle touches both the axes, therefore its centre will be $(a, \pm a)$ and radius will be $|a|$, where a is a positive or negative number.

Given lines is $x - 2y = 3$ ....(i)

Case I : When centre is $(a, \pm a)$

Since $(a, \pm a)$ lies on line (i)

$\therefore$ Centre of the circle is $(-3, -3)$ and radius = $|-3| = 3$

Hence equation of the circle will be

$(x + 3)^2 + (y + 3)^2 = 3^2$ or $x^2 + y^2 + 6x + 6y + 9 = 0$

Case II : When centre is $(a, -a)$

Since $(a, -a)$ line on line (i)

$\therefore \quad a + 2a = 3$

$\therefore \quad a = 0$

$\therefore$ Centre of the circle is $(1, -1)$ and radius = $|1| = 1$

Hence equation of the circle will be

$(x - 1)^2 + (y + 1)^2 = 1^2$ or $x^2 + y^2 - 2x + 2y + 1 = 0$

Example 3 :

Find the equation of the circle passing through the origin and cuts off chords of length 4 and 6 on the positive side of x and y-axis respectively.

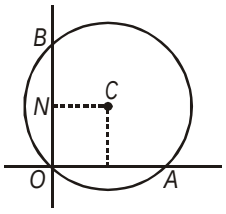
Solution :

If the centre C be (h, k) then the values of h, k will be length of perpendicular CM and CN from C to x- and y-axis respectively.

$\Rightarrow$ M and N are mid points of chords OA and OB

$\Rightarrow$ $OM = 2$, $ON = 3$

$\Rightarrow$ C is $(2, 3)$



Also radius $OC = \sqrt{2^2 + 3^2} = \sqrt{13}$

Hence the equation of the required circle is $(x - 2)^2 + (y - 3)^2 = (\sqrt{13})^2$

Example 4 :

Find the equation to the circle which passes through the points (1, 0), (0, 6) and (3, 4).

Solution :

Let the equation to the circle be,

$x^2 + y^2 + 2gx + 2fy + c = 0$ ....(i)

Since, the three points, whose coordinates are given, satisfy this equation, we have

$1 + 2g + c = 0$ ....(ii)

$36 + 12f + c = 0$ ....(iii)

and $25 + 6g + 8f + c = 0$ ....(iv)

Subtracting (ii) from (iii) and (iii) from (iv), we have

$2g + 12f = 35$

and $6g + 20f = 11$

Hence, $f = \frac{47}{8}$ and $g = -\frac{71}{4}$

Equation (ii), then gives $c = \frac{69}{2}$

Substituting these values in (i) the required equation is,

$4x^2 + 4y^2 - 142x + 47y + 138 = 0$

Example 5:

Prove that the circle $x^2 + y^2 - 6x - 4y + 9 = 0$ bisects the circumference of the circle $x^2 + y^2 - 8x - 6y + 23 = 0$

Solution :

Given circles are

$S_1 \equiv x^2 + y^2 - 6x - 4y + 9 = 0$ ....(1)

and $S_2 \equiv x^2 + y^2 - 8x - 6y + 23 = 0$ ....(2)

Equation of common chord of circles (1) and (2) which is also the radical axis of circle (1) and (2) is

$S_1 - S_2 = 0$

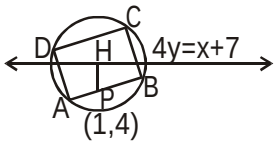
or $2x + 2y - 14 = 0$ or $x + y - 7 = 0$ ....(3)

Centre of circle (2) is (4, 3)

Clearly line (3) passes through the point (4, 3) and hence line (3) is the diameter of circle (2). Hence circle (1) bisects circumference of circle (2).

Example 6:

One of the diameters of the circle circumscribing the rectangle ABCD is $4y = x + 7$. If A and B are the points (–3, 4) and (5, 4) respectively, find the area of the rectangle.



Solution :

Given $A \equiv (-3, 4)$ and $B \equiv (5, 4)$

Let P be the middle point of AB, then $P \equiv (1, 4)$

Equation of AB is $y - 4 = \frac{4-4}{-3-5}(x+3)$ or $y = 4$, clearly

AB is parallel to x-axis, therefore perpendicular bisector of AB will be parallel to y-axis and since it passes through the point P (1, 4) therefore its equation will be $x = 1$. .....(1)

Also equation of one diameter of circle is $4y = x + 7$ .....(2)

Solving (1) and (2), we get $x = 1$, $y = 2$

If H be the centre of the circle circumscribing the rectangle ABCD, then $H \equiv (1, 2)$

$\therefore$ length of perpendicular from H to AB i.e., $HP = \sqrt{(1-1)^2 + (4-2^2)} = 2$

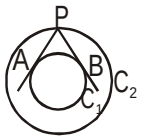
$\therefore$ one side of the rectangle = 2HP = 4

Also other side of rectangle = AB = $\sqrt{(-3-5)^2+(4+4)^2}$

Now area of rectangle ABCD = 4 × 8 = 32 units

Example 7:

C_1 and C_2 be two concentric circles. The radius of C_2 being twice that of C_1. From a point P on C_2 tangents PA and PB are drawn to C_1. Prove that the centroid of the triangle PAB lies on C_1.



Solution :

Let the equations of C_1 and C_2 be

$x^2 + y^2 = a^2$, $x^2 + y^2 = 4a^2$

From any point P $(2a\cos\theta, 2a\sin\theta)$ tangents PA and PB are drawn then AB will be chord of contact whose equation must be

$x.\ 2a\cos\theta + y.\ 2a\sin\theta = a^2$ or $x\cos\theta + y\sin\theta = a/2$ ....(i)

If A be (x_1, y_1) and B be (x_2, y_2) must be roots of

$$x^2 + \left(\frac{\frac{a}{2} - x\cos\theta}{\sin\theta}\right)^2 = a^2$$ (since A, B be points of intersection of AB and C_1)

$$x^2 - a\cos\theta\, x + \left(\frac{a^2}{4} - a^2\sin^2\theta\right) = 0 \Rightarrow x_1 + x_2 = a\cos\theta$$

We similarly get $y_1 = y_2$ = as $\sin\theta$ (on eliminating x term)

Therefore the co-ordinates of centroid G of triangle PAB

$$= \left(\frac{2a\cos\theta + x_1 + x_2}{3}, \frac{2a\sin\theta + y_1 + y_2}{3}\right) = (a\cos\theta, a\sin\theta)$$

Which obviously lies on C_1.

Example 8:

Let $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ be a given circle. Find the locus of foot of the $\perp$ drawn from origin upon any chord of S = 0 which subtend a right angle at the origin.

Solution :

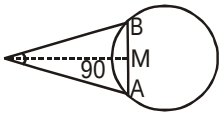
Let AB be any arbitrary chord subtending 90° at origin whose equation is

$y = mx + d$ .....(i)

Then equation of OA and OB can be obtained by homogenizing equation of S with the help of (i)

Thus equation of OA and OB is

$$x^2 + y^2 + 2gx\left(\frac{y-mx}{d}\right) + 2fy\left(\frac{y-mx}{d}\right) + c\left(\frac{y-mx}{d}\right)^2 = 0$$ ....(ii)



Since OA $\perp$ OB (given) in equation (ii), coeff. of x^2 + coeff. of $y^2 = 0$

$$\Rightarrow 1 - \frac{2gm}{d} + \frac{cm^2}{d^2} + 1 + \frac{2f}{d} + \frac{c}{d^2} = 0$$ ....(iii)

Now equation of the line $\perp$ to AB and passing through origin must be

$y = -\frac{1}{m}x$ .....(iv)

Since the point M whose locus is to be determined is point of intersection of OM and AB the required locus must be eliminate of (i), (iv) and (iii). On solving (i) and (iv) for *m* and *d*,

we get $m = -\frac{x}{y}$ and $d = \frac{x^2 + y^2}{y}$

on putting *m* and *d* in (iii), we get

$$1 - 2g\left(\frac{-x}{x^2+y^2}\right) + c\left(-\frac{x}{x^2+y^2}\right)^2 + 1 + 2f\frac{y}{x^2+y^2} + \frac{cy^2}{(x^2+y^2)^2} = 0$$

which simplifies to $x^2 + y^2 + gx + fy + \frac{c}{2} = 0$ (on multiplying by $(x^2 + y^2)^2$).

Example 9:

A circle touches the line $y = x$ at P such that OP = $4\sqrt{2}$. The circle contains (– 10, 2) in its interior and the lengths of its chord on the line $x + y = 0$ is $6\sqrt{2}$. Determine the equation of the circle.

Solution :

Let $y = x$ cuts circle at A and B and M is the mid point of AB. Then if C be the center of the circle CPOM must be a rectangle.

($\angle$ CMO = 90°, CPO = 90°, $(y = x) \perp (y = -x)$

$\Rightarrow$ CM = OP = $4\sqrt{2}$

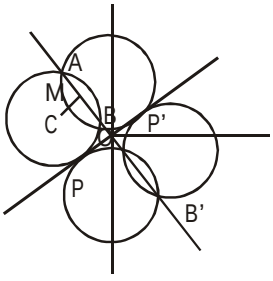
$\Rightarrow$ radius CA = $\sqrt{CM^2 + MA^2} = \sqrt{(4\sqrt{2})^2 + (3\sqrt{2})^2} = 5\sqrt{2}$

If C be (h, k) then $\perp$ from (h, k) on $y = x$ must be equal to radius

$\Rightarrow \frac{k-h}{\sqrt{2}} = \pm 5\sqrt{2} \Rightarrow k - h = \pm 10$ ....(i)

Again $\perp$ from (h, k) $y = -x$ must be equal to $4\sqrt{2}$

$\Rightarrow \frac{k+h}{\sqrt{2}} = 4\sqrt{2} \Rightarrow k + h = \pm 8$ ....(ii)



In all, there are four possibilities in which equations (i) and (ii) can exist

k – h	k + h	Solution	
+ 10	+ 8	$h = -1, k = 9$	The circle is $(x + 1)^2 + (y – 9)^2 = (5\sqrt{2})^2$ this doesn't contain (–10, 2) in its interior
+ 10	– 8	$h = – 9, k = 1$	The circle is $(x + 9)2 + (y + 1)2 = (5\sqrt{2})^2$ and which satisfies all given

condition.

– 10	+ 8	$h = 9, k = -1$	The circle does not contain (– 10, 2)
– 10	– 8	$h = 1, k = -9$	The circle does not contain (–10, 2)

Thus the only circle satisfying the condition of the problem is $(x + 9)^2 + (y + 1)^2 = (5\sqrt{2})^2$

Example 10:

A ray is drawn from origin of cut the given $x^2 + y^2 = 2ax$ ($a > 0$) at B. From B equal segments BM and BN of constant length b are laid of in either direction. As the ray revolves the points M and N describe a curve (limacon of Pascal). Find its equation.

Soluton :

Let $y = mx$ be any ray OB drawn from O then for B

we solve $y = mx$ and $x^2 + y^2 - 2ax = 0$. We easily get $x = \frac{2a}{1+m^2}, y = \frac{2am}{1+m^2}$

Let us now put equation of OB in parametric from with respect to point B (since M and N are situated at a distance b from B).

Indeed equation of OB is $y - \frac{2am}{1+m^2} = m\left(x - \frac{2a}{1+m^2}\right)$

or $\frac{x - \frac{2a}{1+m^2}}{1/\sqrt{1+m^2}} = \frac{y - \frac{2am}{1+m^2}}{m/\sqrt{1+m^2}}$

For the points M and N we can equate to $\pm b$ and the squared result will be satisfied by both M and N. Taking first expression equal to b and on squaring, we get $\left(x - \frac{2a}{1+m^2}\right)^2 = \frac{b^2}{1+m^2}$

But M and N satisfy the simple relation $y = mx$ (since they lie on OB)

Therefore on putting $m = \frac{y}{x}$, we get a pure relation between abscissa and co-ordinates of M (or N) as $\left(x - \frac{2a}{1+\frac{y^2}{x^2}}\right)^2 = \frac{b^2}{1+\frac{y^2}{x^2}}$

which easily simplifies to $(x^2 + y^2 - 2ax)^2 = b^2 (x^2 + y^2)$ and which is the equation of limacon of Pascal.

Example 11 :

Consider two circles $C_1 : x^2 + y^2 = r_1^2$ and $C_2 : x^2 + y^2 = r_2^2$ ($r_2 > r_1$). Let 'A' be a fixed point on the circle C_1 say A (r_1, 0) and 'B' a variable point on the circle C_2. The line BA meets the circle C_2 again at C. Find

(a) show that $OA^2 + OB^2 + BC^2 \in [5r_2^2 - 3r_1^2, 5r_2^2 + r_1^2]$

(b) the locus of mid point of AB, 'O' being the origin

Solution :

A must be (r_1, 0). Let B be the point ($r_2 \cos\theta$, $r_2 \sin\theta$), it is evident that the length of BC will be maximum when BC is the diameter of C_2 and minimum when BC is tangent to C_1 at A.

$\Rightarrow$ max BC = $2r_2$, min BC = $2\sqrt{r_2^2 - r_1^2}$

Now, let D' be the mid-point of AB

$\Rightarrow D \equiv \left(\frac{r_1 + r_2\cos\theta}{2}, \frac{r_2\sin\theta}{2}\right) = (h,k)$ (say)

$\Rightarrow \sin\theta = \frac{2k}{r^2}, \cos\theta = \frac{2h - r_1}{r_2} \Rightarrow 4k^2 + (2h - r_1)^2 = r_2^2$

Locus of 'D' is, $\left(x - \frac{r_1}{2}\right)^2 + y^2 = \frac{r_2^2}{4}$

Example 12 :

Prove that the square of the length of the tangents drawn from a point on one circle to another circle is equal to twice the product of the perpendicular distance of that point from the radical axis of the two circles and the distance between their centres.

Solution :

Without loss of generality, the equation of the two circles can be taken as
$x^2 + y^2 = a^2$ and $(x - c)^2 + y^2 = b^2$
$\therefore$ Radical axis of the two circles is given by $(x - c)^2 - x^2 - b^2 + a^2 = 0$ ... (1)

$\Rightarrow -2cx + c^2 + a^2 - b^2 = 0 \Rightarrow x = \frac{c^2 + a^2 - b^2}{2c}$

Let P ($a\cos\theta$, $a\sin\theta$) be any point on the first circle, then

PA = $\sqrt{(a\cos\theta - c)^2 + a^2\sin^2\theta - b^2} = \sqrt{a^2 - b^2 - c^2 - 2ac\cos\theta}$

Also, PM = $\frac{c^2 + a^2 - b^2}{2c} - a\cos\theta = \frac{c^2 + a^2 - b^2 - 2ac\cos\theta}{2c}$ and $C_1C_2 = c$

$\Rightarrow 2(PM).(C_1C_2) = \frac{c^2 + a^2 - b^2 - 2ac\cos\theta}{c}.c = c^2 + a^2 - b^2 - 2ac\cos\theta = (PA)^2$.

Hence, $(PA)^2 = 2(PM).(C_1C_2)$

Example 13 :

Tangents are drawn from the point (h, k) to the circle $x^2 + y^2 = a^2$. Prove that the area of the triangle formed by them and their chord of contact is $\frac{a(h^2 + k^2 - a^2)^{\frac{3}{2}}}{h^2 + k^2}$

Solution :

Given circle is $x^2 + y^2 = a^2$ .....(1)
The equation of the chord of contact AB of tangents drawn from P(h, k) to the circle (1) is $xh + yh = a^2$ ... (2)

We have to find the area of ΔPAB. From P(h, k) draw PL $\perp$ AB.

Now PL = $\frac{h^2 + k^2 - a^2}{\sqrt{h^2 + k^2}}$ ....(3)

Here $h^2 + k^2 - a^2 > 0$ $\because$ P(h, k) lies outside circle (1)

Also PA = $\sqrt{h^2 + k^2 - a^2}$ ..... (4)

$\therefore$ $AL^2 = AP^2 - PL^2 = (h^2 + k^2 - a^2) - \frac{(h^2 + k^2 - a^2)^2}{h^2 + k^2}$

$= \frac{(h^2 + k^2 - a^2)(h^2 + k^2) - (h^2 + k^2 - a^2)^2}{h^2 + k^2}$

$$\therefore \text{ AL} = \sqrt{\frac{(h^2+k^2-a^2)(h^2+k^2-h^2-k^2+a^2)}{(h^2+k^2)}}$$

$$= \frac{a(h^2+k^2-a^2)^{\frac{1}{2}}}{\left(h^2+k^2\right)^{\frac{1}{2}}}$$

Now the area of ΔAPB = $\frac{1}{2}$.AB.PL = AL.PL

$$= \frac{a(h^2+k^2-a^2)^{\frac{1}{2}}}{\left(h^2+k^2\right)^{\frac{1}{2}}} \cdot \frac{(h^2+k^2-a^2)}{(h^2+k^2)^{\frac{1}{2}}} = \frac{a(h^2+k^2-a^2)^{\frac{3}{2}}}{h^2+k^2}$$

Example 14 :

Find the equation of a circle which is coaxial with the circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ and $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these circles.

Solution :

Equation of the given circles are

$S_1 \equiv x^2 + y^2 - x + 3y - \frac{3}{2} = 0$ ....(1)

$S_2 \equiv x^2 + y^2 + 4x + 2y + 1 = 0$ .....(2)

$\therefore$ The radical axis of circles (1) and (2) is

$S_1 - S_2 = O$ or $-5x + y - \frac{5}{2} = 0$ or $10x - 2y + 5 = 0$ .....(3)

$\therefore$ Required circle will have the equation of the form

$x^2 + y^2 + 4x + 2y + 1 + k(10x - 2y + 5) = 0$

or $x^2 + y^2 + 2(2 + 5k)x + 2(1 - k)y + (1 + 5k) = 0$ .....(4)

Its centre is $(-2 - 5k, k - 1)$. From question it lies on line (3)

$\therefore$ $10(-2 - 5k) - 2(k - 1) + 5 = 0$; or $-52k - 13 = 0$;

$$\therefore k = -\frac{1}{4}$$

Putting the value of k in (4), we get

$$x^2 + y^2 + 2\left(2 - \frac{5}{4}\right)x + 2\left(1 + \frac{1}{4}\right)y + \left(1 - \frac{5}{4}\right) = 0$$

$$\text{or } x^2 + y^2 + \frac{3}{2}x + \frac{5}{2}y - \frac{1}{4} = 0$$

or $4(x^2 + y^2) + 6x + 10y - 1 = 0$ .....(5)

This is the required circle.

Example 15 :

Show that the locus of a point such that the ratio of its distances from two given points is a constant, is a circle. Hence show that this circle can not pass through the given points.

Solution :

Let A and B be the two given points. Let AB be $2a$. We take middle point O of AB as the origin and OB as x-axis, then

$A \equiv (-a, 0)$, $B \equiv (a, 0)$

Let P (α, β) be the point whose locus is to be found.

Given $\frac{PA}{PB}$ = constant = k (say)

$$\therefore \quad \frac{\sqrt{(\alpha+a)^2+\beta^2}}{\sqrt{(\alpha-a)^2+\beta^2}} = k$$

or $(\alpha + a)^2 + \beta^2 = k^2(\alpha - a)^2 + k^2\beta^2$

or $\alpha^2 + 2a\alpha + a^2 + \beta^2 = k^2\alpha^2 + k^2a^2 - 2ak^2\alpha + k^2\beta^2$

or $(k^2 - 1)\,\alpha^2 + (k^2 - 1)\,\beta^2 - 2(k^2a + a)\,\alpha + (k^2 - 1)\,a^2 = 0$

$$\text{or } \alpha^2 + \beta^2 - 2a\left(\frac{k^2+1}{k^2-1}\right)\alpha + a^2 = 0$$

Hence locus of point P (α, β) is $x^2 + y^2 - 2a\frac{k^2+1}{k^2-1}x + a^2 = 0$ ....(1)

Clearly (1) represents a circle

If circle (1) passes through A $(-a, 0)$ then $a^2 + 2\frac{k^2+1}{k^2-1}a^2 + a^2 = 0$

$$\text{or } 2a^2 + 2a^2\frac{k^2+1}{k^2-1} = 0 \Rightarrow 4a^2k^2 = 0$$

$\therefore$ either $a = 0$ or $k = 0$ none of which is possible.

Similarly if B $(a, 0)$ lies on circle (1), then

$$2a^2 + 2a^2\frac{1+k^2}{1-k^2} = 0 \Rightarrow 4a^2 = 0 \;\therefore\; a = 0 \text{ (not possible)}$$

Hence circle (1) can not pass through A and B.

Example 16 :

The circle $x^2 + y^2 - 4x - 4y + 4 = 0$ is inscribed in a triangle which has two of its sides along the co-ordinates axes. The locus of the circumcentre of the triangle is

$x + y - xy + k\sqrt{x^2+y^2} = 0$, find k.

Solution :

Given circle is $x^2 + y^2 - 4x - 4y + 4 = 0$

Its centre is C(2, 2) and radius is $\sqrt{4+4-4} = 2$

Let the hypotenuse of the triangle meet OX and OY at A and B respectively

Let $A \equiv (a, 0)$, $B \equiv (0, b)$

area of $\Delta AOB = \Delta OCA + \Delta OCB + \Delta ACB$

$$\therefore \quad \frac{1}{2}ab = \frac{1}{2}a.2 + \frac{1}{2}b.2 + \frac{1}{2}\sqrt{a^2+b^2}.2$$

or $2a + 2b + 2\sqrt{a^2+b^2} = ab$

Let P (α, β) be the circumcentre of ΔOAB ....(2)

Since ΔOAB is right angled, its circumcentre will be the mid-point of the hypotenuse

$$\therefore \quad \alpha = \frac{a}{2} \text{ and } \beta = \frac{b}{2}$$

i.e, $a = 2\alpha$ and $b = 2\beta$

From (2), $4a + 4b + 2.2\sqrt{\alpha^2+\beta^2} = 4\alpha\beta$

or $\alpha + \beta + \sqrt{\alpha^2+\beta^2} = \alpha\beta$

Hence the equation of the locus of P (α, β) is $x + y - xy + \sqrt{x^2+y^2} = 0$

Example 17.

Lines $5x + 12y - 10 = 0$ and $5x - 12y - 40 = 0$ touch the circle C_1 of diameter 6. If the centre of C_1 lies in the first quadrant, find the equation of circle C_2 which is concentric with C_1 and cuts intercepts of length 8 on these lines.

Solution :

Given lines are

$5x + 12y - 10 = 0$ ....(1)

and $5x - 12y - 40 = 0$ ....(2)

diameter of given is 6. Let C (α, β) be its centre, then its equation will be $(x-\alpha)^2 + (y-\beta)^2 = 3^2$

Since lines (1) and (2) touch circle (3)

$$\therefore \underset{(i)}{\frac{|5\alpha+12\beta-10|}{13}} = \underset{(ii)}{\frac{|5\alpha-12\beta-40|}{13}} = \underset{(iii)}{3}$$

From (i) and (ii), $5\alpha + 12\beta - 10 = \pm(5\alpha - 12\beta - 40)$

$\therefore \alpha = 5$ (taking negative sign)

and $\beta = -\frac{5}{4}$ (taking positive sign)

But centre lies in the first quadrant, therefore only $\alpha = 5$ is acceptable.

From (i) and (iii), we have $\frac{|5\alpha+12\beta-10|}{13} = \pm 3$

or $\frac{5\times5+12\beta-10}{13} = \pm 3$

or $12\beta + 15 = \pm 39 \therefore \beta = 2, -\frac{9}{2}$

But centre of circle lies in the first quadrant $\therefore \beta = 2$

Hence the centre of the circle C_1 is C (5, 2). According to question centre of the required circle C_2 is also C (5, 2).

It is clear that radius of the required circle $CA = \sqrt{CD^2+AD^2} = \sqrt{3^2+4^2} = 5$

where CD = length of the perpendicular from C to line (1) and D is the mid-point of chord AB of length 8

Hence equation of the required circle will be

$(x-5)^2 + (y-2)^2 = 5^2$

or $x^2 + y^2 - 10x - 4y + 4 = 0$

Example 18 :

If the circle C_1, $x^2 + y^2 = 16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of the maximum length and has slope equal to $\frac{3}{4}$, show that the co-ordinates of the centre of the circle C_2 are either $\left(\frac{9}{5}, -\frac{12}{5}\right)$ and $\left(-\frac{9}{5}, \frac{12}{5}\right)$

Solution :

Given, circle C_1 is $x^2 + y^2 = 16$ ....(1)

Its centre is O (0, 0) and radius is 4. According to question C_2 is another circle of radius 5. Let AB be the common chord of circles C_1 and C_2. Since this common chord AB is of maximum length, therefore AB must be the diameter of the smaller circles i.e., of circle C_1. Let H (α, β) be the centre of circle C_2.

According to question slope of AB = $\frac{3}{4}$

$\therefore$ Equation of AB will be $y - 0 = \frac{3}{4}(x - 0)$

or $3x - 4y = 0$ ....(2)

Also slope of OH = $\frac{\beta - 0}{\alpha - 0} = \frac{\beta}{\alpha}$

Since AB $\perp$ OH $\therefore$ $\frac{\beta}{\alpha}.\frac{3}{4} = -1$ or $4\alpha + 3\beta = 0$ ....(3)

Again AH = 5, AO = 4 $\therefore$ HO = 3

or $\frac{|3\alpha - 4\beta|}{5} = 3$ or $3\alpha - 4\beta = \pm 15$ ....(4)

When $3\alpha - 4\beta = 15$ .....(6)

Solving (3) and (6), we get $\alpha = -\frac{9}{5}, \beta = \frac{12}{5}$

Thus the centre of circle C_2 are either $\left(\frac{9}{5}, -\frac{12}{5}\right)$ or $\left(-\frac{9}{5}, \frac{12}{5}\right)$

Second method :

Equation of circle C_1 is $S_1 \equiv x^2 + y^2 - 16 = 0$ ....(1)

Let the equation of circle C_2 be

$(x - \alpha)^2 + (y - \beta)^2 = 25$

or $S_2 \equiv x^2 + y^2 - 2\alpha x - 2\beta y + \alpha^2 + \beta^2 - 25 = 0$ .....(2)

Equation of common chord of circles C_1 and C_2 will be $S_1 - S_2 = 0$

or $2\alpha x + 2\beta y - \alpha^2 - \beta^2 + 9 = 0$ ....(3)

Its slope = $-\frac{\alpha}{\beta} = \frac{3}{4}$ (according to question)

or $4\alpha + 3\beta = 0$ ....(4)

Let the length of the perpendicular from O to line (3) i.e., line AB be p, then $\frac{|-\alpha^2 - \beta^2 + 9|}{\sqrt{4\alpha^2 + 4\beta^2}} = p$

AB will be maximum if $p = 0$ i.e. if $\alpha^2 + \beta^2 = 9$ .....(5)

From (4), $\beta = -\frac{4}{3}\alpha$, putting the value of β in equation (5), we get

$\alpha^2 + \frac{16}{9}\alpha^2 = 9$ or $\alpha = \pm\frac{9}{5}$

$\therefore$ from (4), $\beta = \pm\frac{12}{5}$

Thus centre of circle C_2 are $\left(\frac{9}{5}, -\frac{12}{5}\right)$ or $\left(-\frac{9}{5}, \frac{12}{5}\right)$

Example 19 :

Find the equation of the circle passing through (–4, 3) and touching the lines $x + y = 2$ and $x – y = 2$

Solution :

Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ ....(1)

Since it passes through (–4, 3)

$\therefore$ $25 – 8g + 6f + c = 0$ ....(2)

Since circle (1) touches the given lines

$$\therefore \quad \underset{(i)}{\frac{|-g-f-2|}{\sqrt{2}}} = \underset{(ii)}{\frac{|-g+f-2|}{\sqrt{2}}} = \underset{(iii)}{\sqrt{g^2+f^2-c}}$$

From (i) and (ii), we have $– g – f – 2 = \pm (– g + f – 2)$ $\therefore$ $f = 0$ or $g = – 2$

When $f = 0$, from (ii) and (iii) we have

$(g + 2)^2 = 2(g^2 – c)$ or $g^2 – 4g – 4 – 2c = 0$ ....(3)

On putting $f = 0$ in (2) we get $c = 8g – 25$ .....(4)

Putting in (3), we get $g^2 – 20g + 46 = 0$

$\therefore$ $g = 10 \pm 3\sqrt{6}$ and hence from (4), $c = 55 \pm 24\sqrt{6}$

When $g = – 2$ from (ii) and (iii), we have

$f^2 = 2(4 + f^2 – c)$ or $f^2 – 2c + 8 = 0$ .....(5)

From (2), $c = – 6f – 41$

Putting in (5), we get

$f^2 + 12f + 82 + 8$ or $f^2 + 12f + 90 = 0$

$$\therefore \quad f = \frac{-12 \pm \sqrt{144-360}}{2} \text{ (imaginary)}$$

Hence no circle in this case is possible.

$\therefore$ The required circles are

$x^2 + y^2 + 2(10 \pm 3\sqrt{6})\, x + (55 \pm 24\sqrt{6}) = 0$s

Example 20 :

A circle touches the line $y = x$ at a point P such that $OP = 4\sqrt{2}$, where O is the origin. The circle contains the point (– 10, 2) in its interior and the length of its chord on the line $x + y = 0$ is $6\sqrt{2}$. Determine the equation of the circle.

Solution :

Equation of OP is $y = x$

Let $P \equiv (h, h)$

Given, OP $4\sqrt{2}$ $\therefore$ $h^2 + h^2 = 32$

or $h^2 = 16$ $\therefore h = \pm 4$

Thus $P \equiv (4, 4)$ or $(–4, –4)$

Let C (α, β) be the centre of the circle.

Case I : When $P \equiv (4, 4)$

Slope of CP = $\frac{\beta - 4}{\alpha - 4}$ and slope of OP = 1

Since CP $\perp$ PO $\therefore \frac{\beta-4}{\alpha-4}.1 = -1$ or $\alpha + \beta = 8$ ....(2)

Let *a* be the radius of the circle

Then $CQ^2 + \left(3\sqrt{2}\right)^2 = a^2 \therefore \left(\frac{\alpha+\beta}{\sqrt{2}}\right)^2 + 18 = a^2$

or $a^2 = \left(\frac{8}{\sqrt{2}}\right)^2 + 18 = 50 \quad \therefore \quad a = 5\sqrt{2}$ [$\therefore a > 0$]

Again CP = $a \therefore \frac{|\alpha-\beta|}{\sqrt{2}} = a = 5\sqrt{2}$

or $|\alpha - \beta| = 10 \therefore \alpha - \beta = \pm 10$ ....(3)

Solving (2) and (3), we get

$\alpha = 9, \beta = -1$ or $\alpha = -1, \beta = 9$

$\therefore$ C $\equiv$ ((9, – 1) or C $\equiv$ (– 1, 9)

Let H $\equiv$ (– 10, 2)

When C $\equiv$ (9, –1), $CH^2 = 19^2 + (-3)^2 = 361 + 9 = 370 > a^2$

When C $\equiv$ (–1, 9), $CH^2 = 9^2 + 7^2 = 81 + 49 = 130 > a^2$

Since H lies inside the circle,

$\therefore$ neither (9, – 1) nor (– 1, 9) is the centre of the circle.

Case II : When P $\equiv$ (–4, –4)

Slope of CP = $\frac{\beta+4}{\alpha+4}$

Since CP $\perp$ PO \ $\left(\frac{\beta+4}{\alpha+4}\right).1 = -1$ or $\alpha + \beta = -8$ .....(4)

Again $a^2 = CQ^2 + \left(3\sqrt{2}\right)^2 = \left(\frac{\alpha+\beta}{\sqrt{2}}\right) + 18$

$= \left(-\frac{8}{\sqrt{2}}\right)^2 + 18 = 50 \ \therefore a = 5\sqrt{2}$

Now CP = $a \therefore \frac{|\alpha-\beta|}{\sqrt{2}} = 5\sqrt{2}$ or $\alpha - \beta = \pm 10$ ....(5)

Solving (4) and (5), we get

C $\equiv$ (– 9, 1) or C $\equiv$ (1, – 9)

when C $\equiv$ (– 9, 1) $CH^2 = (1)^2 + (1-2)^2 = 2 < a^2$

when C (1, – 9), $CH^2 = (11)^2 + (-11)^2 = 242 > a^2$

Thus C $\equiv$ (– 9, 1) and a = $5\sqrt{2}$

Hence equation of the required circle is

$(x + 9)^2 + (y - 1)^2 = 50$

or $x^2 + y^2 + 18x - 2y + 32 = 0$

Example 21 :

A circle of radius 5 metres is having its centre at A at the origin. Two circles II and III with centres at B and C of radii 3 and 4 metres respectively touch the circle I and also touch the *x*-axis to the right of A. Find the equations to any two common tangents to the circles II and III.

Solution :

Given A is the origin which is the centre of circle I, AX and AY are the x and y axis respectively. B and C are the centres of circles II and III respectively and their radii are 3 and 4 respectively.
Since circles I and II touch each other externally $\therefore$ AB = 5 + 3 = 8
and since circles I and III touch each other externally $\therefore$ AC =5 + 4 = 9
Let circles II and III touch x-axis at D and E respectively

Then AD = $\sqrt{8^2-3^2}=\sqrt{55}$ and AE = $\sqrt{9^2-4^2}=\sqrt{65}$

Since both circles I and II touch x-axis, therefore, y = 0 (x-axis) is their common tangents. Let BC meet y = 0 at H. Then one more common tangents will pass through H and H will divided BC internally or externally in the ratio 3 : 4 according as circles I and II lie in different quadrants (first and fourth) or in the same quadrant (first or fourth).

Case I : When circles II and III lie in first and fourth quadrants respectively.

In the case B $\equiv \left(\sqrt{55},3\right)$ and C $\equiv \left(\sqrt{65},-4\right)$

Now from similar ΔBDH and ΔCEH

$$\frac{DH}{HE}=\frac{BD}{CE}=\frac{3}{4}$$

Hence H divides DE internally in the ratio 3 : 4

$$\therefore\ H\equiv\left(\frac{3\sqrt{65}+4\sqrt{55}}{7},0\right)$$

Now equation of any line through H will be $y = m\left(x-\frac{3\sqrt{65}+4\sqrt{55}}{7}\right)$

or $7mx - 7y - m\,(3\sqrt{65}+4\sqrt{55}\,) = 0$ ...(1)

If line (1) is tangents to circle II, then $\frac{|7m\sqrt{55}-7.3-m(3\sqrt{65}+4\sqrt{55}|}{7\sqrt{1+m^2}}=3$

or $[3(\sqrt{55}-\sqrt{65}\,)m - 21]^2 = 9 \times 49\,(1 + m^2)$

or $9(\sqrt{55}-\sqrt{65}\,)^2m^2 + 441 - 126(\sqrt{55}-\sqrt{65}\,)m$
$= 441 + 441\,m^2$

or $[9(\sqrt{55}-\sqrt{65}\,)^2 - 441]m^2 - 126(\sqrt{55}-\sqrt{65}\,)m = 0$

$$\therefore\ m=0,\frac{126(\sqrt{55}-\sqrt{65})}{[9(\sqrt{55}-\sqrt{65})^2-441]}=\frac{126(\sqrt{55}-\sqrt{65}}{9(71-10\sqrt{143})}$$

$$=\frac{14(\sqrt{55}-\sqrt{65}}{71-10\sqrt{143}}$$

Thus from (1) equation of two common tangents in this case are

$$y = 0 \text{ and } y=\frac{14(\sqrt{55}-\sqrt{65}}{71-10\sqrt{143}}\left(x-\frac{3\sqrt{65}+4\sqrt{55}}{7}\right)$$

Note : There will be two other common tangents to circles II and III which will meet the line joining their centres B and C at P and P will divide BC externally in the ratio 3 : 4.

$$\therefore\ P \equiv \left(\frac{3\sqrt{65}-4\sqrt{55}}{3-4}, \frac{3(-4)-4.3}{3-4}\right)$$

i.e. $P \equiv (4\sqrt{55}-3\sqrt{65}, 24)$

Case II : When both circle II and III lie in the first quadrant. In this case

$B \equiv (\sqrt{55}, 3)$ and $C \equiv (\sqrt{65}, 4)$

one common tangents is $y = 0$ meeting BC at H, and H will divide BC externally in the ratio 3 : 4

$$\therefore\ H \equiv \left(\frac{3\sqrt{65}-4\sqrt{55}}{3-4}, \frac{3.4-4.3}{3-4}\right)$$

i.e. $H \equiv (4\sqrt{55}-3\sqrt{65}, 0)$

Now equation of any line through H will be

$y = m(x - 4\sqrt{55}-3\sqrt{65})$

or $mx - y - m(4\sqrt{55}-3\sqrt{65}) = 0$ ....(2)

If line (2) is tangent to circle II, then $\frac{|m\sqrt{55}-3-m(4\sqrt{55}-3\sqrt{65})|}{\sqrt{1+m^2}} = 3$

or $[3(\sqrt{65}-\sqrt{55})m - 3]^2 = 9(1 + m^2)$

or $[9(\sqrt{65}-\sqrt{55})^2 - 9]m^2 - 18(\sqrt{65}-\sqrt{55})m = 0$

$$\therefore\ m = 0, \frac{2(\sqrt{65}-\sqrt{55})}{119-10\sqrt{143}}$$

Thus from (2), equation of two common tangents in this case are

$y = 0$ and $y = \frac{2(\sqrt{65}-\sqrt{55}}{119-10\sqrt{143}}(x - 4\sqrt{55}+3\sqrt{65})$

Case III : When circles II and III both lie in the fourth quadrant.

In this case $B \equiv (\sqrt{55}, -3)$ and $C \equiv (\sqrt{65}, -4)$

Let BC meet x-axis at H, then H will divide BC externally in the ratio 3 : 4

$$\therefore\ H \equiv \left(\frac{3\sqrt{65}-4\sqrt{55}}{3-4}, \frac{3(-4)-4(-3)}{3-4}\right)$$

i.e., $H \equiv (4\sqrt{55}-3\sqrt{65}, 0)$

Now equation of any line through H will be

$y = m(x - 4\sqrt{55}+3\sqrt{65})$ .....(2)

or $mx - y - m(4\sqrt{55}-3\sqrt{65}) = 0$

This line will be tangents to circle II if $\frac{|m\sqrt{55}+3-m(4\sqrt{55}-3\sqrt{65})|}{\sqrt{1+m^2}} = 3$

or $[3(\sqrt{65}-\sqrt{55})m + 3]^2 = 9(1 + m^2)$

or $9(120 - 10\sqrt{143})\ m^2 + 9 + 18\ (\sqrt{65}-\sqrt{55})m = 9 + 9m^2$

or $9[119 - 10\sqrt{143})\, m^2 + 18\,(\sqrt{65} - \sqrt{55})m = 0$

$$\therefore\ m = 0, -\frac{2(\sqrt{65} - \sqrt{55})}{119 - 10\sqrt{143}}$$

Thus from (2), equation of two common tangents will be

$$y = 0 \text{ and } y = -\frac{2(\sqrt{65} - \sqrt{55})}{119 - 10\sqrt{143}}(x - 4\sqrt{55} + 3\sqrt{65})$$

Example 22 :

Prove that the limiting points of the system
$x^2 + y^2 + 2gx + c + \lambda(x^2 + y^2 + 2fx + k) = 0$

subtend a right angle at the origin, if $\frac{c}{g^2} + \frac{k}{f^2} = 2$

Solution :

By limiting point we mean that the circle reduces to a point i.e. radius becomes zero.
The given circle is $x^2 + y^2 + 2gxc + c + \lambda(x^2 + 2fy + k) = 0$
or $(1 + \lambda)x^2 + (1 + \lambda)y^2 + 2gx + 2fy\lambda + c + k\lambda = 0$

$$\text{or } x^2 + y^2 + \frac{2g}{(1+\lambda)}x + \frac{2f\lambda}{(1+\lambda)}y + \frac{c + k\lambda}{(1+\lambda)} = 0 \quad \text{.....(1)}$$

$$\text{Its centre is } \left(\frac{-g}{1+\lambda}, \frac{-f\lambda}{1+\lambda}\right) \text{.....(2)}$$

Radius of circle (1) is = 0

$$\Rightarrow \sqrt{\frac{g^2}{(1+\lambda)^2} + \frac{f^2\lambda^2}{(1+\lambda)^2} - \frac{(c + k\lambda)}{(1+\lambda)}} = 0$$

or $\lambda^2(f^2 - k) - \lambda(k + c) + g^2 - c = 0$
which is a quadratic in λ_1 and λ_2.

$$\therefore\quad \lambda_1 + \lambda_2 = \frac{k + c}{f^2 - k} \text{ and } \lambda_1\lambda_2 = \frac{g^2 - c}{f^2 - k}$$

then limiting points are (from 2)

$$A\left(\frac{-g}{1+\lambda_1}, \frac{-f\lambda_1}{1+\lambda_1}\right) \text{ and } B\left(\frac{-g}{1+\lambda_2}, \frac{-f\lambda_2}{1+\lambda_2}\right)$$

But given that AB subtend a right angle at the origin.
$\therefore$ Slope of OA × slope of OB = – 1

$$\Rightarrow \left(\frac{\frac{-f\lambda_1}{1+\lambda_1}}{\frac{-g}{1+\lambda_1}}\right) \times \left(\frac{\frac{-f\lambda_2}{1+\lambda_2}}{\frac{-g}{1+\lambda_2}}\right) = -1$$

$$\text{or } \frac{f\lambda_1}{g} \times \frac{f\lambda_2}{g} = -1$$

or $f^2\lambda_1\lambda_2 + g^2 = 0$

or $f^2\frac{(g^2-c)}{(f^2-k)}+g^2=0$

or $2g^2f^2 - cf^2 - kg^2 = 0$

or $2 = \frac{c}{g^2}+\frac{k}{f^2}$

Example 23 :

Find the equations of tangents drawn from P (1, – 2) to the circle $x^2 + y^2 + 4x + 2y = 0$.

Solution :

Proceeding by formula, $SS_1 = T^2$ we have

$S \equiv x^2 + y^2 + 4x + 2y = 0$

$S_1 = (1)^2 + (-2)^2 + 4(1) + 2(-2) = 5$

$T = x \times 1 + y(-2) + 2(x + 1) + 1(y - 2) = 3x - y$

$\Rightarrow$ Combined equation of tangents is $5(x^2 + y^2 + 4x + 2y) = (3x - y)^2$

$\Rightarrow 2x^2 - 2y^2 - 3xy - 10x - 5y = 0$

We can also proceed directly. Any line through (1, – 2) is $y + 2 = m(x - 1)$

or $y - mx + 2 + m = 0$ ....(1)

If this is tangents to the circle

$x^2 + y^2 + 4x + 2y = 0$ then length of the perpendicular drawn from centre (–2, – 1) to the line (1) must be equal to radius which is $\sqrt{5}$.

$\Rightarrow \left|\frac{-1+2m+2+m}{\sqrt{1+m^2}}\right| = \sqrt{5}$ $\Rightarrow 2m^2 + 3m - 2 = 0$

$\Rightarrow m = -2, m = -\frac{1}{2}$

Thus the two tangents are

$y + 2 = -2(x - 1), y + 2 = \frac{1}{2}(x - 1)$

which, on simplifications are $y + 2x = 0$, $2y - x + 5 = 0$

Note : In the second method care should be taken about the infinite root, since the line $y = mx + c$ (m variable) does not include vertical lines therefore in case a vertical line happens to be a tangents the value of m must be infinity. This will be automatically discovered when the quadratic equation in m degenerates into a linear equation. The following examples will illustrate it well.

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic: 1 Equation of a Circle in Various forms

Q.1 The equation of the circle passing through (4, 5) having the centre at (2, 2) is:
(a) $x^2 + y^2 + 4x + 4y - 5 = 0$
(b) $x^2 + y^2 - 4x - 4y - 5 = 0$
(c) $x^2 + y^2 - 4x - 13 = 0$
(d) $x^2 + y^2 - 4x - 4y + 5 = 0$

Q.2 The radius of the circle $4x^2 + 4y^2 - 10x + 5y + 5 = 0$ is
(a) $\frac{3\sqrt{5}}{8}$ (b) $\frac{3\sqrt{5}}{7}$
(c) $3\sqrt{5}$ (d) 5

Q.3 The centre of the circle passing through the points (0, 0), (a, 0) and (0, b) is
(a) (a, b) (b) $\left(\frac{a}{2}, \frac{b}{2}\right)$
(c) $\left(-\frac{a}{2}, \frac{b}{2}\right)$ (d) ($-a$, $-b$)

Q.4 The equation of circle centred at (1, 2) and passing through (4, 6) is
(a) $x^2 + y^2 + 2x - 4y - 20 = 0$
(b) $x^2 + y^2 + 2x + 4y - 20 = 0$
(c) $x^2 + y^2 - 2x - 4y - 20 = 0$
(d) $x^2 + y^2 + 2x + 4y + 20 = 0$

Q.5 The centre of the circle $(x - a)(x - c) + (y - b)(y - d) = 0$ is
(a) (a, a) (b) (c, d)
(c) $\left(\frac{a+c}{2}, \frac{b+d}{2}\right)$ (d) none of these

Topic: 2 Intercepts Made by a Circle on the Axes

Q.6 The centre of a circle passing through the points (0, 0), (1, 0) and touching the circle $x^2 + y^2 = 9$ is
(a) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (b) $\left(\frac{1}{2}, \frac{3}{2}\right)$
(c) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (d) $\left(\frac{1}{2}, -\sqrt{2}\right)$

Q.7 One of the diameter of the circles $x^2 + y^2 - 12x + 4y + 6 = 0$ is given by
(a) $x + y = 0$ (b) $x + 3y = 0$
(c) $x = y$ (d) $3x + 2y = 0$

Q.8 What will be the equation of a circle whose centre is (3, –1) and which intercept chord of 6 unit length on straight line $2x - 5y + 18 = 0$ is
(a) $x^2 + y^2 - 6x + 2y - 28 = 0$
(b) $x^2 + y^2 + 6x - 2y - 28 = 0$
(c) $x^2 + y^2 + 4x - 2y + 24 = 0$
(d) $x^2 + y^2 + 2x - 2y - 12 = 0$

Q.9 The equation of the circle passing through origin is $x^2 + y^2 - 6x + 2y = 0$. The equation of one of its diameters is
(a) $x + 3y = 0$ (b) $x + y = 0$
(c) $x = y$ (d) $3x + y = 0$

Q.10 The equation of the diameter of circle $(x - 2)^2 + (y + 1)^2 = 16$ which bisects the chord cut off by the circle on the line $x - 2y - 3 = 0$ is
(a) $2x + y - 3 = 0$ (b) $2x + y + 3 = 0$
(c) $2x - y - 3 = 0$ (d) none of these

Q.11 The circle $x^2 + y^2 + 4x - 7y + 12 = 0$ cuts an intercept on y-axis of length
(a)3 (b)4 (c)7 (d)1

Q.12 The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle of area 154 square units. Then the equation of the circle is
(a) $x^2 + y^2 + 2x - 2y = 62$
(b) $x^2 + y^2 + 2x - 2y = 47$
(c) $x^2 + y^2 - 2x + 2y = 47$
(d) $x^2 + y^2 - 2x + 2y = 62$

Q.13 The maximum number of points with rational co-ordinates on a circle whose centre is $\left(\sqrt{3}, 0\right)$ is
(a) One (b) two
(c) four (d) infinite

Q.14 The circle of radius 1, touching the pair of lines $12x^2 - 25xy + 12y^2 = 0$, $x > 0$ has the equation
(a) $x^2 + y^2 + 10x + 10y + 49 = 0$
(b) $x^2 + y^2 - 10x - 10y + 49 = 0$
(c) $x^2 + y^2 + 10x - 10y + 49 = 0$
(d) None of these

Topic: 3 Parametric Equations of a Circle

Q.15 The equation of the circle passing through (2, 0) and (0, 4) and having the minimum radius is
(a) $x^2 + y^2 - 2x - 4y = 0$
(b) $x^2 + y^2 - 2x + 4y = 0$
(c) $x^2 + y^2 + 2x - 4y = 0$
(d) $x^2 + y^2 + 2x + 4y = 0$

Q.16 A square is inscribed in the circle $x^2 + y^2 - 2x + 4y + 3 = 0$ and its sides are parallel to the co-ordinates axis then one vertex of the square is
(a) $(1+\sqrt{2}, -2)$ (b) $(1-\sqrt{2}, -2)$
(c) $(1, -2 + \sqrt{2})$ (d) none of these

Q.17 The circle described on the line joining the points (0, 1), (a, b) as diameter cuts the x-axis at points whose abscissa are roots of the equation
(a) $x^2 + ax + b = 0$ (b) $x^2 - ax + b = 0$
(c) $x^2 + ax - b = 0$ (d) $x^2 - ax - b = 0$

Q.18 The locus of the centre of the circle for which one end of the diameter is (3, 3) while the other end lies on the line $x + y = 4$ is
(a) $x + y = 3$ (b) $x + y = 5$
(c) $x + y = 7$ (d) $x + y = 9$

Q.19 The line $4x + 3y - 4 = 0$ divides the circumference of the circle centred at (5, 3), in the ratio 1 : 2. Then the equation of the circle is
(a) $x^2 + y^2 - 10x - 6y - 66 = 0$
(b) $x^2 + y^2 - 10x - 6y + 100 = 0$
(c) $x^2 + y^2 - 10x - 6y + 66 = 0$
(d) none of these

Q.20 If $f(x + y) = f(x)\, f(y)$ for all x and y, $f(1) = 2$ and $\alpha_n = f(n)$, $n \in N$ then the equation of the circle having (a_1, a_2) and (a_3, a_4) as the ends of its one diameter is:
(a) $(x - 2)(x - 8) + (y - 4)(y - 16) = 0$
(b) $(x - 4)(x - 8) + (y - 2)(y - 16) = 0$
(c) $(x - 2)(x - 16) + (y - 4)(y - 8) = 0$
(d) $(x - 6)(x - 8) + (y - 5)(y - 6) = 0$

Q.21 The circle $x^2 + y^2 = 4$ cuts the circle $x^2 + y^2 + 2x + 3y - 5 = 0$ in A and B, the centre of the circle AB as diameter is:
(a) (0, 0) (b) $\left(\frac{2}{13}, \frac{3}{13}\right)$
(c) $\left(\frac{4}{13}, \frac{6}{13}\right)$ (d) (2, –1)

Q.22 If the lines $2x - 3y - 5 = 0$ and $3x - 4y = 7$ are diameters of a circle of area 154 square units then then the equation of the circle is
(a) $x^2 + y^2 + 2x - 2y - 62 = 0$
(b) $x^2 + y^2 + 2x - 2y - 47 = 0$
(c) $x^2 + y^2 - 2x + 2y - 47 = 0$
(d) $x^2 + y^2 - 2x + 2y - 62 = 0$

Q.23 The area bounded by the circles $x^2 + y^2 = r^2$, $r = 1, 2$ and rays given by $2x^2 - 3xy - 2y^2 = 0$, $y > 0$ is
(a) $\pi/4$ sq. units (b) $\pi/2$ sq. units
(c) $3\pi/4$ sq. units (d) π sq. units

Q.24 A line drawn through a fixed point $P(\alpha, \beta)$ to cut the circle $x^2 + y^2 + r^2$ at A and B. Then $PA.PB$ is equal to
(a) $(\alpha + \beta)^2 - r^2$ (b) $\alpha^2 + \beta^2 - r^2$
(c) $(\alpha + \beta)^2 + r^2$ (d) None of these

Topic: 4 Position of a Point with respect to a Circle

Q.25 The maximum distance of the point (4, 4) from the circle $x^2 + y^2 - 2x - 15 = 0$ is
(a)10 (b)9
(c)5 (d)none of these

Q.26 The shortest distance from the point (2, –7) to the circle $x^2 + y^2 - 14x - 10y - 151 = 0$ is equal to
(a) 2 (b) 5
(c) 3 (d) none of these

Q.27 Four distinct points ($2k$, $3k$), (1, 0), (0, 1) and (0, 0) line on a circle when
(a) All values of k are integral
(b) $0 < k < 1$
(c) $k < 0$
(d) For two values of k

Q.28 If the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ cuts the co-ordinates axes in concyclic points then
(a) $a_1a_2 = b_1b_2$ (b) $a_1b_1 = a_2b_2$
(c) $a_1b_2 = a_2b_1$ (d) none of these

Q.29 The value of k for which two tangents can be drawn from (k, k) to the circle $x^2 + y^2 + 2x + 2y - 16 = 0$ is
(a) $k \in R^+$ (b) $k \in R^-$
(c) $k \in (-\infty, -4) \cup (2, \infty)$
(d) $k \in (0, 1]$

Q.30 The number of integral points (a point whose abscissa as well as ordinate are intrgers) lying inside the circle $x^2 + y^2 - 4x - 6y + 9 = 0$ is
(a) 10 (b) 12
(c) 9 (d) none of these

Q.31 The value(s) of m for which the line $y = mx$ lies wholly outside the circle $x^2 + y^2 - 2x - 4y + 1 = 0$, is (are)

(a) $m \in (-4/3, 0)$ (b) $m \in (-4/3, 0)$
(c) $m \in (0, 4/2)$ (d) none of these

Q.32 The co-ordinates of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$ which is farthest from the origin are
(a) (9, 3) (b) (8, 5)
(c) (12, 4) (d) None of these

Topic: 5 Normal

Q.33 The straight line $y = mx + c$ cuts the circle $x^2 + y^2 = a^2$ at real points if

(a) $\sqrt{a^2(1+m^2)} \le |c|$

(b) $\sqrt{a^2(1-m^2)} \le |c|$

(c) $\sqrt{a^2(1+m^2)} \ge |c|$

(d) $\sqrt{a^2(1-m^2)} \ge |c|$

Q.34 Tangents are drawn to the circle $x^2 + y^2 = a^2$ at the points, where $x\cos\alpha + y\sin\alpha = p$ meets it. find their point of intersection.

(a) $\frac{a^2\cos\alpha}{p}, \frac{a^2\sin\alpha}{p}$

(b) $\frac{a\cos\alpha}{p}, \frac{a^2\sin\alpha}{p}$

(c) $-\frac{a^2\cos\alpha}{p}, -\frac{a^2\sin\alpha}{p}$

(d) $-\frac{a^2\cos\alpha}{p}, -\frac{a\sin\alpha}{p}$

Q.35. The tangent at any point P on the circle $x^2 + y^2 = 4$, meets the coordinate axes in A and B, then find the locus of the midpoint of AB.

(a) $x^2 + y^2 = x^2y^2$ (b) $x^2 - y^2 = x^2y^2$

(c) $y^2 - x^2 = x^2y^2$ (d) None of these

Topic: 6 Director Circle, Chord of Contact

Q.36 The locus of the point, such that tangents drawn from it to the circle $x^2 + y^2 - 6x - 8y = 0$ are perpendicular to each other, is
(a) $x^2 + y^2 - 6x - 8y - 25 = 0$
(b) $x^2 + y^2 + 6x - 8y - 25 = 0$
(c) $x^2 + y^2 - 6x - 8y + 25 = 0$
(d) none of these

Q.37 If $y = 2x$ be the equation of a chord of the circle $x^2 + y^2 = 2ax$, then the equation of the circle, of which this chord is diameter, is
(a) $2(x^2 + y^2) - 5a(x + 2y) = 0$
(b) $x^2 + y^2 - 2a(x + 2y) = 0$
(c) $5(x^2 + y^2) - 2a(x + 2y) = 0$
(d) none of these

Topic: 7 Equation of the Chord Joining Two point of Circle

Q.38 Equation of chord AB of circle $x^2 + y^2 = 2$ passing through $P(2, 2)$ such that $PB/PA = 3$, is given by
(a) $x = 3y$ (b) $x = y$
(c) $y - 2 = \sqrt{3}(x - 2)$ (d) none of these

Q.39 The length of the chord out off by $y = 2x + 1$ from the circle $x^2 + y^2 = 2$ is

(a) $\frac{5}{6}$ (b) $\frac{6}{5}$ (c) $\frac{6}{\sqrt{5}}$ (d) $\frac{\sqrt{5}}{6}$

Q.40 The coordinates of middle point of the chord $2x - 5y + 18 = 0$ cut off by the circle $x^2 + y^2 - 6x + 2y - 54 = 0$ is
(a)(1, 4) (b)(– 4, 2) (c)(4, 1) (d)(6, 6)

Topic: 8 Orthogonality of two Circles

Q.41 If a circle passes through the point (a, b) and cuts the circle $x^2 + y^2 = k^2$ orthogonally then the equation of the locus of its centre is
(a) $2ax + 2by - (a^2 + b^2 + k^2) = 0$
(b) $2ax + 2by - (a^2 - b^2 + k^2) = 0$
(c) $x^2 + y^2 - 3ax - 4by + (a^2 + b^2 - k^2) = 0$
(d) $x^2 + y^2 - 2ax - 3by + (a^2 - b^2 - k^2) = 0$

Q.42 The circles $x^2 + y^2 + xy + y = 0$ and $x^2 + y^2 + x - y = 0$ intersect at an angle of
(a)$\pi/6$ (b)$\pi/4$ (c)$\pi/3$ (d)$\pi/2$

Topic: 9 Radical Axis and Radical Centre

Q.43 Find the radical axis and the limiting points of the system of co-axial circles

$3(x^2 + y^2) - 16x - 14y + 39$

$+\lambda(x^2 + y^2 - 5x - 5y + 13) = 0$

(a) (2,3) (3, 2) (b) (-2,3) (-3,2)
(c) (-2, -3) (-3,-2) (d) (2, -2) (4,1)

Q.44 Find the equation of the circles which pass through the intersection of the circles

$x^2 + y^2 = 4$ and $x^2 + y^2 - 2x - 4y + 4 = 0$

having radius $2\sqrt{2}$.

(a) $5x^2 + 5y^2 + 4x + 8y - 36 = 0$

(b) $x^2 + y^2 + 4x + 8y - 36 = 0$

(c) $x^2 + y^2 + 2x + 3y - 15 = 0$

(d) None of these

Topic: 10 Family of Circles

Q.45 The equation of the through the intersection of the lines $2x + 3y + 4 = 0$ and $6x - 3y + 12 = 0$ and normal to the circle $x^2 + y^2 - 4x - 12 = 0$, is
(a) $x = y$ (b) $x = 0$
(c) $y = 0$ (d) none of these

Q.46 The equation of circle touching the line $2x + 3y + 1 = 0$ at the point $(1, -1)$ and passing through the focus of the parabola $y^2 = 4x$ is
(a) $3x^2 + 3y^2 - 8x + 3y + 5 = 0$
(b) $3x^2 + 3y^2 + 8x - 2y + 5 = 0$
(c) $x^2 + y^2 - 3x + y + 6 = 0$
(d) none of these

Q.47 The points $(4, -2)$ and $(3, b)$ are conjugate with respect to the circle $x^2 + y^2 = 24$, if $b =$
(a)6 (b)–6 (c)12 (d)–4

Q.48 The equation of the circle with centre at the x-axis and touching the line $3x + 4y - 11 = 0$ at the point $(1, 2)$ is
(a) $x^2 + y^2 - x - 4 = 0$
(b) $x^2 + y^2 + 2x - 7 = 0$
(c) $x^2 + y^2 + x - 6 = 0$
(d) none of these

Q.49 The centres of a set of circles, each of radius 3, lie on the circle $x^2 + y^2 = 25$. The locus of any point with such circle is
(a) $4 \le x^2 + y^2 \le 64$ (b) $x^2 + y^2 \le 25$
(c) $x^2 + y^2 \ge 25$ (d) $3 \le x^2 + y^2 \le 9$

Section-B
(Assertion-Reason)

Q.1 **Statement – 1 :** $(n \ge 3)$ for a circles the value of a for which the number radical axis if equal to the number of radical centre is 5.
Statement – 2 : If no two of a circle are concentric, No three of the centre are collinear then number of possible radical centres is nc_3.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1:** The circle $x^2 + y^2 + 2ax + c = 0$, $x^2 + y^2 + 2by + c = 0$ touch if $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$.
Statement-2: Two circles with centre C_1, C_2 and radii r_1, r_2 touch each other if $r_1 + r_2 = C_1C_1$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1:** Number of circles passing through (1, 4), (2, 3), (–1, 6) is 1.
Statements-2: Though 3 non collinear points in a plane only one circle can be drawn.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** The circle described on the segment joining the pionts (–2, –1) (0, –3) as diameter cuts the circle $x^2 + y^2 + 5x + y + 4 = 0$ orthogonally.
Statement–2 : (–2, –1) and (0, –3) are conjugate points with respect to the circle $(x - h)^2 + y^2 + 5x + y + 4 = 0$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement–1 :** The common tangents of the circles $x^2 + y^2 + 2x = 0$, $x^2 + y^2 - 6x = 0$ form an equilateral triangle.
Statement–2 : The given circles touch each other externally.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Year Question)

Q.1 If one of the diameters of the circle $x^2+y^2-2x-6y+6=0$ is a chord to the circle with centre (2,1), then the radius of the circle is

(a) $\sqrt{3}$ (b) $\sqrt{2}$ (c) 3 (d) 2

Q.2 The centre of circle inscribed in square formed by the lines

$x^2-8x+12=0$ and $y^2-14y+45=0$, is

(a) (4,7) (b) (7, 4) (c) (9, 4) (d) (4, 9)

Q.3 Let AB be a chord of the circle $x^2+y^2=r^2$ subtending a right angle at the centre. Then, the locus of the centroid of the ΔPAB as P moves on the circle, is

(a) a parabola (b) a circle
(c) an ellipse
(d) a pair of straight lines

Q.4 the lines $2x-3y=5$ and $3x-4y=7$ are diameters of a circle of area 154 sq units. Then the equation of this circle is

(a) $x^2+y^2+2x-2y=62$

(b) $x^2+y^2+2x-2y=47$

(c) $x^2+y^2-2x+2y=47$

(d) $x^2+y^2-2x+2y=62$

Q.5 AB is a diameter of a circle and C is any point on the circumfernce of the circle. Then,

(a) the area of ΔABC is maximum when it is isosceles

(b) the area of ΔABC is minimum when it is isosceles

(c) the perimeter of ΔABC is minimum when it is isosceles

(d) None of the above

Q.6 The centre of the circle passing through the point (0, 1) and touching the curve $y=x^2$ at (2, 4) is

(a) $\left(-\frac{16}{5},\frac{27}{10}\right)$ (b) $\left(-\frac{16}{7},\frac{53}{10}\right)$

(c) $\left(-\frac{16}{5},\frac{53}{10}\right)$ (d) None of these

Q.7 If one of the diameters of the circle, given by the equation, $x^2+y^2-4x+6y-12=0$, is a chord of a circle S, whose centre is at (-3, 2), then the radius of S is

(a) $5\sqrt{2}$ (b) $5\sqrt{3}$ (c) 5 (d) 10

Q.8 The number of common tangents to the circles $x^2+y^2-4x-6y=0$

and $x^2+y^2+6x+18y+26=0$ is

(a) 1 (b) 2 (c) 3 (d) 4

Q.9 Let C be the circle with centre at (1,1) and radius 1. If T is the circle centred at (0, y) passing through origin and touching the circle C externally, then the radius of T is equal to

(a) $\frac{\sqrt{3}}{\sqrt{2}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

Q.10 If the circle $x^2+y^2+2x+2ky+6=0$ and $x^2+y^2+2ky+k=0$ intersect orthogonally, then k is

(a) 2 or -3/ 2 (b) -2 or -3/ 2
(c) 2 or 3/2 (d) -2 or 3/ 2

Q.11 The ΔPQR is inscribed in the circle $x^2+y^2=25$. If Q and R have coordinates (3,4) and (-4, 3) respectively, then $\angle QPR$ is equal to

(a) $\pi/2$ (b) $\pi/3$ (c) $\pi/4$ (d) $\pi/6$

Q.12 The number of common tangents to the circles $x^2+y^2=4$

and $x^2+y^2-6x-8y=24$ is

(a) 0 (b) 1 (c) 3 (d) 4

Q.13 The angle between a pair of tangents drawn from a point P to the circle $x^2+y^2+4x-6y+9\sin^2\alpha+13\cos^2\alpha=0$ is 2α. the equation of the locus of the point P is

(a) $x^2+y^2+4x-6y+4=0$

(b) $x^2+y^2+4x-6y-9=0$

(c) $x^2+y^2+4x-6y-4=0$

(d) $x^2+y^2+4x-6y+9=0$

Q.14 If the two circles $(x-1)^2+(y-3)^2=r^2$ and $x^2+y^2-8x+2y+8=0$ intersect in two distinct points, then

(a) $2<r<8$ (b) $r<2$
(c) $r=2$ (d) $r>2$

Q.15 If a circle passes through the point (a, b) and cuts the circle $x^2+y^2=k^2$ orthogonally, then

the equation of the locus of its centre is

(a) $2ax+2by-\left(a^2+b^2+k^2\right)=0$

(b) $2ax+2by-\left(a^2-b^2+k^2\right)=0$

(c) $x^2+y^2-3ax-4by+a^2+b^2-k^2=0$

(d) $x^2+y^2-2ax-3by+\left(a^2-b^2-k^2\right)=0$

Q.16 Let ABCD be a quadrilateral with area 18, with side AB parallel to the side CD a d AB= 2CD. Let AD be perpendicualr to AB and CD. If a circle is drawn inside the quadrilateral ABCD touching all the sides, then its radius is

(a) 3 (b) 2 (c) $\frac{3}{2}$ (d) 1

Q.17 If the tangent at the point P on the circle $x^2+y^2+6x+6y=2$ meet P on the circle $x^2+y^2+6x+6y=2$ meets the straight line $5x-2y+6x+6y=2$ meets the straight line $5x-2y+6=0$ at a point Q on the Y-axis, then the length of PQ is

(a) 4 (b) $2\sqrt{5}$ (c) 5 (d) $3\sqrt{5}$

Q.18 Let PQ and RS be tangents at the extremities of the diameter PR of a circle of radius r. If PS and RQ intersect at a point X on the circumference of the circle, then 2r equals

(a) $\sqrt{PQ.RS}$ (b) $\frac{PQ+RS}{2}$

(c) $\frac{2PQ.RS}{PQ+RS}$ (d) $\sqrt{\frac{PQ^2+RS^2}{2}}$

Q.19 The circle passing through (1,-2) and touching the axis of x at (3,0) also passes through the point

(a) (-5, 2) (b) (2,-5)
(c) (5,-2) (d) (-2, 5)

Q.20 The circle passing through the point (-1, 0) and touching the Y-axis at (0,2) also passes through the point

(a) $\left(-\frac{3}{2},0\right)$ (b) $\left(-\frac{5}{2},2\right)$

(c) $\left(-\frac{3}{2},\frac{5}{2}\right)$ (d) (-4, 0)

Q.21 the locus of the centre of circle which touches $(y-1)^2+x^2=1$ externally and also touches X-axis, is

(a) $\left\{x^2=4y,y\geq 0\right\}\cup\left\{(0,y),y<0\right\}$

(b) $x^2=y$

(c) $y=4x^2$

(d) $y^2=4x\cup(0,y),y\in R$

Q.22 If two distinct chords, drawn from the point (p, q) on the circle $x^2+y^2=px+qy$ (where, $pq\neq 0$) are bisected by the X-axis, then

(a) $p^2=q^2$ (b) $p^2=8q^2$
(c) $p^2<8q^2$ (d) $p^2>8q^2$

Q.23 The locus of the centre of circle, which touches externally the circle $x^2+y^2-6x-6y+14=0$ and also touches the Y-axis, is given by the equation

(a) $x^2-6x-10y+14=0$

(b) $x^2-10x-6y+14=0$

(c) $y^2-6x-10y+14=0$

(d) $y^2-10x-6y+14=0$

Q.24 The centre of a circle passing through the points (0, 0), (1, 0) and touching the circle $x^2+y^2=9$ is

(a) (3 / 2, 1 / 2) (b) (1/ 2, 3/ 2)
(c) (1/ 2, 1/ 2) (d) 1/ 2, $-2^{1/2}$)

Q.25 The equation of the circle passing through (1,1) and the points of intersection of $x^2+y^2+13x-3y=0$ and $2x^2+2y^2+4x-7y-25=0$ is

(a) $4x^2+4y^2-30x-10y=25$

(b) $4x^2+4y^2+30x-13y-25=0$

(c) $4x^2+4y^2-17x-10y+25=0$

(d) None of the above

Q.26 Two circles $x^2+y^2=6$ and $x^2+y^2-6x+8=0$ are given. Then the equation of the circle through their points of intersection and the point (1,1) is

(a) $x^2+y^2-6x+4=0$

(b) $x^2+y^2-3x+1=0$

(c) $x^2+y^2-4y+2=0$

(d) None of the above

Q.27 The centres of those circles which touch the circle, $x^2+y^2-8x-8y-4=0$ externally and also touch the X-axis, lie on

(a) a circle
(b) an ellipse which is not a circle a hyperbola
(c) a hyperbola
(d) a parabola

Q.28 The locus of the mid-point of the chord of contact of tangents drawn from points lying on the straight line $4x-5y=20$ to the circle $x^2+y^2=9$ is

(a) $20\left(x^2+y^2\right)-36x+45y=0$

(b) $20\left(x^2+y^2\right)+36x-45y=0$

(c) $36\left(x^2+y^2\right)-20x+45y=0$

(d) $36\left(x^2+y^2\right)+20x-45y=0$

Q.29 Tangents drawn from the point P (1, 8) to the circle $x^2+y^2-6x-4y-11=0$ touch the circle at the Points A and B. The equation of the circumcircle of the Δ PAB is

(a) $x^2+y^2+4x-6y+19=0$

(b) $x^2+y^2-4x-10y+19=0$

(c) $x^2+y^2-2x+6y-29=0$

(d) $x^2+y^2-6x-4y+19=0$

Q.30 The locus of the mid-point of a chord of the circle $x^2+y^2=4$ which subtends a right angle at the origin, is

(a) $x+y=2$ (b) $x^2+y^2=1$

(c) $x^2+y^2=2$ (d) $x^2+y^2=2$

Section-D
(Board pattern)

Q.1 Find equation of circle which touches both the coordinate axes in first quadrant and having radius 2 units is.

Q.2 A circle of radius 5 units, touches both the coordinate axes in first quadrant, is made to roll along the positive direction of *x* axis by one complete roll. Fine centre of circle in new position.

Q.3 The equation of circle of radius 5 units passing through (2, 3) and whose centre lies on positive *x* axis is.

Q.4 Find the equation of circle passing through (2, 2) and (4, 6) and having least radius.

Q.5 Find the equation of circle passing through (5, 4) and whose centre is point of intersection of lines $x + 6 = 2$, $x - y = 2$.

Q.6 Find the centre and radius of the circle $2x^2 + 2y^2 - 12x + 8y - 6 = 0$.

Q.7 Find the equation of the circle with centre (4, 3) and passess through the point (0, 0).

Q.8 Find the equation of circle passing through origin and making equal intercepts of length *a* on both the co-ordinate axes.

Q.9 Find the equation of a circle with centre(1, 2) and cuts a chord of 6 units length which is at 4 units distance from the centre.

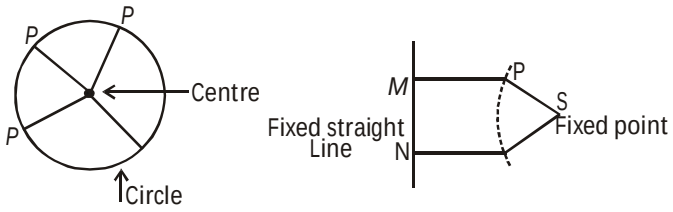
Q.10 Does the point (2, –3) lie, inside outside or on the circle $x^2 + y^2 = 16^2$.

CHAPTER 2

PARABOLA

1. DEFINITION

We have seen in the chapter of Circle that a circle is the locus of a point, which moves in a plane so that its distance from a fixed point remains always constant.



Now let us consider a fixed point and a fixed straight line. A point moves in a plane in such a way that the ratio of its distances from a fixed point to that of form a fixed straight line remains always constant then the locus of the point is called a conic section or a conic. The fixed point does not lie on the fixed straight line

i.e., $\frac{SP}{PM}$ = constant, for all positions of P, then the locus of P is called a conic section.

(i) **Focus :** The fixed point is called the focus of conic section.

(ii) **Directrix :** The fixed straight line is called the directrix of the conic.

(iii) **Eccentricity :** The constant ratio is called the eccentricity of the conic.

(iv) **Axis :** The straight line passing through the focus and perpendicular to the directrix is called the axis of the conic section.

(v) **Vertex :** The point of intersection of conic section and the axis is called vertex.

(vi) **Centre :** The point which bisects the chord passing through it is called the centre of conic section.

(vii) **Latus rectum :** The latus rectum of a conic is the chord passing through focus and parallel to the directrix.

For e = 1, the conic section is called **Parabola**.

For e < 1, the conic section is called **Ellipse**.

For e > 1, the conic section is called **Hyperbola**.

For e = 0, the conic section is called **Circle** which is a special type of ellipse

For e = ∞, the conic section is called **Pair of straight line**.

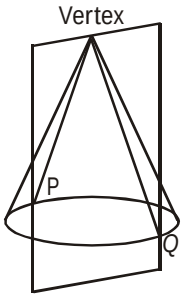
Let $S(\alpha, \beta)$ be the focus, e the eccentricity and $ax + by + c = 0$, the directex of a conic section, then the general equation of the conic section is given by $(x-\alpha)^2 + (y-\beta)^2 = e^2 . \frac{(ax+by+c)^2}{a^2+b^2}$.

2. CONIC SECTION

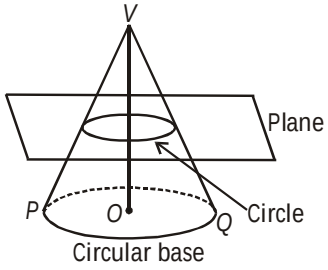
Considering various plane sections of a right circular cone Euclid, the father of creative geometry, found many curves known as conic sections.

Section of a Right Circular Cone by Different Planes

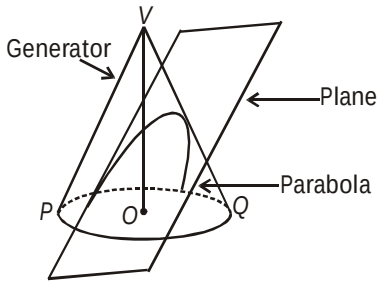
(i) Section of a right circular cone by a plane passing through its vertex is a pair of straight lines passing through the vertex as shown in the figure.



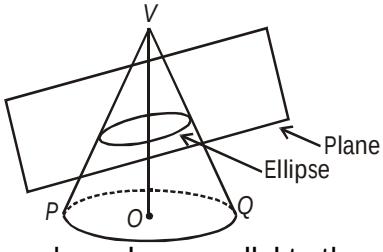
(ii) Section of a right circular cone by a plane parallel to its base is a circle.



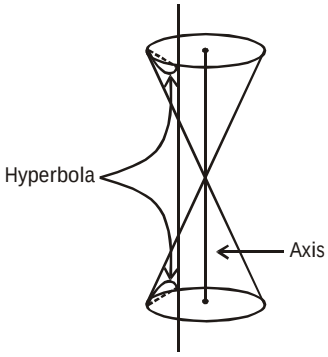
(iii) Section of a right circular cone by a plane parallel to a generator of the cone is a parabola.



(iv) Section of a right circular cone by a plane not parallel to any generator of the cone and not perpendicular or parallel to the axis of the cone is an ellipse.



(v) Section of a right circular cone by a plane parallel to the axis of the cone is a hyperbola.



From the general equation of the conic section, we observe that it is of second degree equation in x and y Let the general equation of second degree in two independent variables x and y be

$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ....(i)

where $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

where $\Delta = 0$, the equation (i) represents the Degenerate conic whose nature is specified as follows

Conditions	**Nature of conic**
$\Delta = 0$ and $h^2 - ab = 0$	A pair of coincident straight lines
$\Delta = 0$ and $h^2 - ab \neq 0$	A pair of intersecting straight lines
$\Delta = 0$ and $h^2 - ab > 0$	Real and distinct pair of straight lines
$\Delta \neq 0$, and $h^2 - ab < 0$, $2fg = ch$	Point

where $\Delta \neq 0$, the equation (i) represents the Non-degenerate conic whose nature is specified as follows.

Conditions	**Nature of conic**
$\Delta \neq 0$, $h = 0$, $a = b \neq 0$, $f^2 + g^2 - ac \geq 0$	Circle
$\Delta \neq 0$, $h^2 - ab = 0$	Parabola
$\Delta \neq 0$, $h^2 - ab < 0$	Ellipse
$\Delta \neq 0$, $h^2 - ab > 0$	Hyperbola
$\Delta \neq 0$, $h^2 - ab > 0$ and $a + b = 0$	Rectangular hyperbola

The centre of the conics can be obtained by solving simultaneously the equations

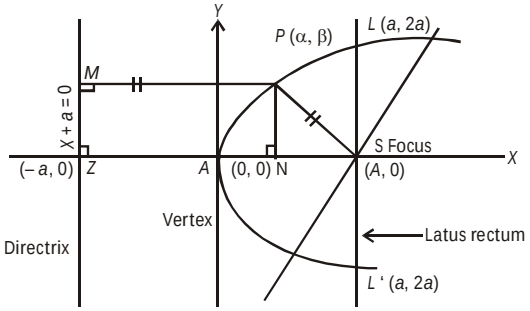
$\frac{\partial f}{\partial x} = 0 = ax + hy + g$ and $\frac{\partial f}{\partial y} = 0 = hx + by + f$ which is $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$

3. PARABOLA

Standard Equation of Parabola

The locus of a point which moves in a plane such that its distance from a fixed point is equal to its distance from a fixed straight line not containing the fixed point is called **Parabola**.

The fixed point is called Focus. The fixed straight line is called Directrix and the straight line perpendicular from the focus to its directrix is called the axis of the parabola. The point of intersection of the parabola with the axis of the parabola is called vertex of the parabola.



Let the focus and directrix of the parabola be given by $S(a, 0)$ and $x + a = 0$ respectively then from the definition of the parabola.

$SP = PM$

$\Rightarrow (\alpha - a)^2 + \beta^2 = (\alpha + a)^2$

$\Rightarrow \beta^2 = (\alpha + a)^2 - (\alpha - a)^2 = 4a\alpha$

$\Rightarrow$ Locus of $P(\alpha, \beta)$ is

$y^2 = 4ax$

Thus $y^2 = 4ax$ represents a parabola with following specifications :

Focus is $(a, 0)$

Vertex is $(0, 0)$

Directrix is given by $x + a = 0$

Axis of the parabola is $y = 0$

The equation of the tangent at the vertex is $x = 0$

The ends of Latus rectum are $(a, 2a)$ and $(a, -2a)$

Length of latus rectum = $4a$

Equation of the latus rectum is $x = a \Rightarrow x - a = 0$ and $(y-\beta)^2 = 4a(x-\alpha)$ is $x = \alpha + at^2,\ y = \beta + 2at$

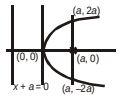
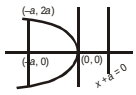
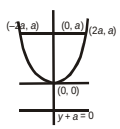
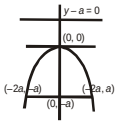
Parametric equations are $x = at^2$, $y = 2at$

Thus the point on the parabola $y^2 = 4ax$ may be taken as $(at^2, 2at)$.

Any chord passing through the focus $S(a, 0)$ is called focal chord of the parabola.

Focal distance of a point $P(x, y) = x + a$ ($\because PS = PM = ZA + AN = a + x$).

PARABOLAS WITH LATUS RECTUM 4a

Equation of Parabola	Vertex	Focus	Equation of the Axis	Equation of the tangents at the vertex	Length of the Latus rectum	Ends of Latus rectum	Equation of the Latus rectum	Parametric coordinates of a point on the Parabola	Equation of the Directrix	Figure	Focal distance of a point (x, y)
(i) $y^2 = 4ax$, $a > 0$	(0, 0)	$(a, 0)$	$y = 0$	$x = 0$	$4a$	$(a, 2a)$ and $(a, -2a)$	$x - a = 0$	$(at^2, 2at)$	$x + a = 0$		$a + x$
(ii) $y^2 = -4ax$, $a > 0$	(0, 0)	$(-a, 0)$	$y = 0$	$x = 0$	$4a$	$(-a, 2a)$ and $(-a, -2a)$	$x + a = 0$	$(-at^2, 2at)$	$x - a = 0$		$a - x$
(iii) $x^2 = 4ay$, $a > 0$	(0, 0)	$(0, a)$	$x = 0$	$y = 0$	$4a$	$(2a, -a)$ and $(-2a, a)$	$y - a = 0$	$(2at, at^2)$	$y + a = 0$		$a + y$
(iii) $x^2 = -4ay$, $a > 0$	(0, 0)	$(0, -a)$	$x = 0$	$y = 0$	$4a$	$(2a, -a)$ and $(-2a, -a)$	$y + a = 0$	$(2at, -at^2)$	$y - a = 0$		$a - y$

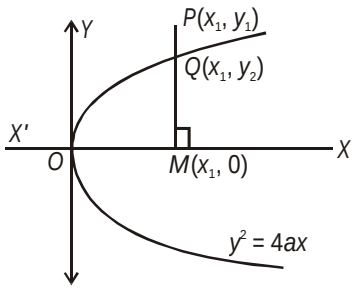
Student Task

1. Find the focus and the equation of the directrix of the parabola $2x^2 + 3x - 2y - 1 = 0$
2. Find the value of λ for which the equation $(10x - 5)^2 + (10y - 7)^2 = \lambda^2(5x + 12y + 7)^2$ represents the equation of parabola.
3. A double ordinate of the parabola $y^2 = 8ax$ is of length $16a$. Then find the angle subtended by it at the vertex of the parabola.
4. Show that the locus of mid-point of segment of a line between the focus and a moving point on the parabola $x^2 = 4(y - 1)$ is another parabola. Also find the equation of its directrix.

Ans. 1. $\left(\frac{-3}{4}, \frac{-13}{16}\right)$; $16y + 21 = 0$; **2.** $\lambda = \pm\frac{10}{13}$; **3.** $\frac{\pi}{2}$; **4.** $y - 1 = 0$

Position of a Point Relative to a Parabola

Let $P(x_1, y_1)$ be a point.



Let us draw perpendicular PM from P on the axis OX of the parabola $y^2 = 4ax$.

Then $y_2^2 = 4ax_1$

Now, P will be outside, on or inside the parabola $y^2 = 4ax$ according as

$\Rightarrow PM >/< QM$

$\Rightarrow PM^2 >/< QM^2$

$\Rightarrow y_1^2 >/< y_2^2$

$\Rightarrow y_1^2 >/< 4ax_1$

$\Rightarrow y_1^2 - 4ax_1 >/< 0$

☺ **REMARK :**

1. The point (x_1, y_1) lies inside, on or outside $y^2 = -4ax$ according as $y_1^2 + 4ax_1 \lessgtr 0$.
2. The point (x_1, y_1) lies inside, on or outsidr $x^2 = 4ay$ according as $x_1^2 - 4ay_1 \lessgtr 0$.
3. The point (x_1, y_1) lies inside, on or outside $x^2 = -4ay$ according as $x_1^2 + 4ay_1 \lessgtr 0$.

Position of a line with respect to a Parabola

Let the line is $y = mx + c$ ... (i)

Parabola is $y^2 = 4ax$ ....(ii)

By (i) and (ii) $(mx + c^2) = 4ax$

$\Rightarrow m^2x^2 + 2x(mc - 2a) + c^2 = 0$

Discriminant $D = 4(mc - 2a)^2 - 4m^2c^2$

$D = 16a(a - mc)$

(i) If the line is chord, then $D > 0 \Rightarrow a > mc$

(ii) If the line is tangent, then $Q = 0 \Rightarrow a = mc \Rightarrow c = \frac{a}{m}$

(iii) If the line is outside the parabola, then $D < 0 \Rightarrow a < mc$

Thus the line $y = mx + c$ touches the parabola $y^2 + 4ax$ if $c = \frac{a}{m}$. Consequently the equation of the tangent to the parabola $y^2 = 4ax$ in slope form can be put in the form $y = mx + \frac{a}{m}$

If the line $lx + my + n = 0$ touches $y^2 = 4ax$, then $ln = am^2$

Equation of Director Circle

The director circle of the parabola is the locus of point of intersection of perpendicular tangents.

Let us take a tangent of $y^2 = 4ax$ is $y = mx + \frac{a}{m}$

$mk = m^2h + a$

$m^2h - mk + a = 0$

This equation has two roots m_1, m_2 which represents the slopes of tangents drawn from (h, k).

Hence $m_1 + m_2 = \frac{k}{h}$ .... (i)

$m_1 m_2 = \frac{a}{h}$ .... (ii)

If tangents are perpendicular, then $m_1 m_2 = -1 \Rightarrow h = -a$
locus of (h, k) is $x + a = 0$
Hence the director circle of any parabola is directrix.

Equation of Tangent in Different From

(i) Point form : The equation of tangents at (x_1, y_1) is $yy_1 - 2a(x + x_1) = 0$

(ii) Parametric form : At $(at^2, 2at)$ is $yt = x + at^2$ ...(i)

(iii) Slope form : $y = mx + \frac{a}{m}$ at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ .... (ii)

For better understanding we can make a chord.

Parabola	Tangent in point form	Tangent in parametric form	Tangent in slope form
$y^2 = 4ax$	$yy_1 - 2a(x + x_1) = 0$ at (x_1, y_1)	$yt = x + at^2$ at $(at^2, 2at)$	$y = mx + \frac{a}{m}$ at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$
$y^2 = -4ax$	$yy_1 + 2a(x + x_1) = 0$ at (x_1, y_1)	$yt = -x + at^2$ at $(-at^2, 2at)$	$y = mx - \frac{a}{m}$ at $\left(-\frac{a}{m^2}, \frac{2a}{m}\right)$
$x^2 = 4ax$	$xx_1 + 2a(y + y_1) = 0$ at (x_1, y_1)	$xt = y + at^2$ at $(2at^2, 2at^2)$	$y = mx - am^2$ at $(2am, am^2)$
$x^2 = -4ax$	$xx_1 + -2a(y + y_1)$ at (x_1, y_1)	$xt = -y + at^2$ at $(2at, -at^2)$	$y = mx + am^2$ at $(-2am, -am^2)$

Point of Intersection of Tangents at t_1, t_2

At $(at_1^2, 2at_1)$ tangents is $yt_1 = x + at_1^2$ ...(i)
At $(at_2^2, 2at_2)$ tangents $yt_2 = x + at_2^2$ ....(ii)
By (i) and (ii) $x = at_1t_2$
$y = a(t_1 + t_2)$
Here we find that x coordinate of point of intersection is the geometric mean of at_1^2 and at_2^2 while y coordinate is the arithmetic mean of $2at_1$ and $2at_2$.

Student Task

1. Two straight lines are at right angles to one another and one of them touches $y^2 = 4a(x + a)$ and the other $y^2 = 4b(x + b)$. Prove that the point of intersection of the lines lies on the line $x + a + b = 0$.
2. TP and TQ are any two tangents to a parabola and the tangents at a third point R cuts them in P' and Q'. Prove that $\frac{TP'}{TP} + \frac{TQ'}{TQ} = 1$
3. Find the eqution(s) of the common tangent(s) touching the curves $y^2 - 6y - 4x + 9 = 0$ and $x^2 + y^2 - 6x - 6y + 9 = 0$

Ans. 3. $x = 0$, $\sqrt{3}y = x + 3 + 3\sqrt{3}$, $x + \sqrt{3}y + 3 - 3\sqrt{3} = 0$

NORMAL

(i) Point form

The eqution of normal at (x_1, y_1) is

$y - y_1 = \frac{y_1}{2a}(x - x_1)$ .....(i)

(ii) Parametric form

By putting $x_1 = at^2$, $y_1 = 2at$ in (i) we get

$$y - 2at = -\frac{2at}{2a}(x - at^2)$$

$\Rightarrow y + xt = 2at + at^3$ ....(ii)

(iii) Slope form

Let the slope of (ii) is m, then $m = -t \Rightarrow t = -m$

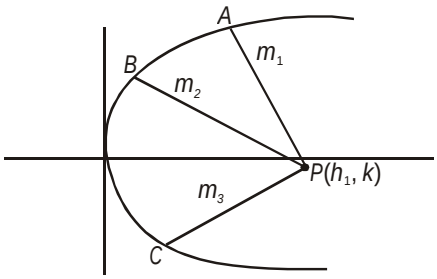
Hence equation (ii) becomes

$y = mx - 2am - am^3$, at $(am^2, -2am)$ ....(iii)

from (iii) we may conclude that, if $y = mx + c$ is normal to $y^2 = 4ax$, then $c = -2am - am^3$.

(iv) Conormal points

If normals at some points with respect to parabola are concurrent, then the points are called conormal points.



Let $y = mx - 2am - am^3$, which passes through (h, k)

$\Rightarrow k = mh - 2am - am^3$

$am^3 + 2am - mh + k = 0$

This equation is cubic in m hence maximum three normals (atleast one) can be drawn from any point to the parabola.

The roots of (i) represent the slopes of normals, hence

$m_1 + m_2 + m_3 = 0$ ....(i)

$$m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a - h}{a} \quad \text{....(iii)}$$

$$m_1m_2m_3 = \frac{k}{a} \quad \text{....(iv)}$$

(v) Properties of Conormal Points

(i) The algebraic sum of the slopes of three conormal points is zero.

(ii) The algebraic sum of ordinates of feet of three normals drawn to a parabola is zero.

(iii) If $h > 2a$, then three normals can be drawn from (h, k) to $y^2 = 4ax$.

(iv) The centroid of a triangle formed by conormal points lies on the axis of the parabola.

(vi) Intersection of Normals

Let $y + xt_1 = 2at_1 + at_1^3$ ....(i)

$y + xt_2 = 2at_2 + at_2^3$ ....(ii)

are two normals at the points t_1 and t_2. By solving (i) and (ii) we get

$x = 2a + a(t_1^2 + t_2^2 + t_1t_2)$

$y = -at_1t_2(t_1 + t_2)$

4. SOME IMPORTANT POINTS

(i) Equation of the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ is $y(t_1 + t_2) = 2(x + at_1t_2)$

(ii) Length of focal chord joining t_1 and t_2 is $a(t_1 - t_2)^2$.

(iii) If the chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ passes through focus $(a, 0)$, then $t_1t_2 = -1$.

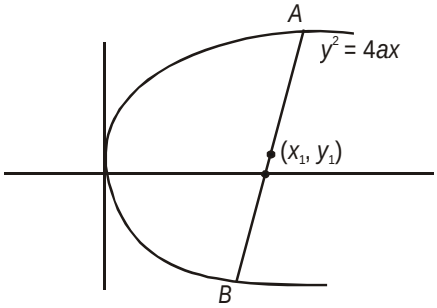
i.e. if one end of a focal chord of $y^2 = 4ax$ is $(at^2, 2at)$, then other end will be $\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$, $t \neq 0$.

(iv) Let the chord which is normal at $(at_1^2, 2at_1)$ meet the parabola at $(at_2^2, 2at_2)$, then $t_2 = -t_1 - \frac{2}{t_1}$.

(v) Semi-latus rectum is the harmonic mean of focal radii if l_1, l_2 are focal raddii, then semi-latus rectum = $\frac{2l_1l_2}{l_1 + l_2}$.

(vi) The equation of chord of contact of tangents drawn from (x_1, y_1) to the parabola $y^2 = 4ax$ is $yy_1 = 2a(x + x_1)$.

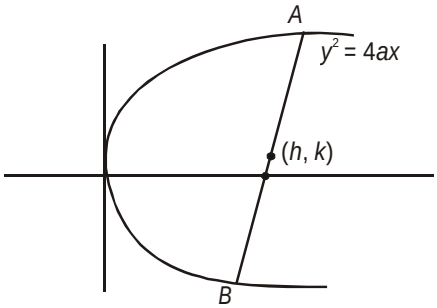
Equation of Chord with Middle Point (x_1, y_1)



Equation of AB is $yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$
or in short $T = S_1$

Equation of Diameter

Diameter of a parabola is the locus of middle point of parabola chord. Let us take the slope of chord = m.



The equation of AB is $yk - 2a(x + h) = k^2 - 4ah$

But its slope = $m = \Rightarrow \frac{2a}{k} = m$

$\Rightarrow k = \frac{2a}{m}$, hence locus of (h, k) is $y = \frac{2a}{m}$.

Hence diameter of parabola is always parallel to axis of the parabola.

Equation of Pair of Tangent

The equation of tangents from (x_1, y_1) to the parabola $y^2 = 4ax$ is given by $T^2 = SS_1$
Where $T = yy_1 - 2a(x + x_1)$
$S = y^2 - 4ax$

$S_1 = y_1^2 - 4ax_1$

Pole and Polar

The equation of polar (x_1, y_1) with respect to the respect to the parabola $y^2 = 4ax$ is given by $yy_1 = 2a(x + x_1)$.

The point (x_1, y_1) is called the pole of the polar.

Student Task

1. If the line $x - y + 2 = 0$ is a normal to the parabola $y^2 - 6y - 4x + k = 0$, then find k.
2. Prove that the normal chord to a parabola at the point whose ordinate is equal to the abscissa subtends a right angle at the focus.
3. Prove that the circle drawn on any focal chord of a parabola as diameter touches the directrix.
4. Find the locus of the point $P(h, k)$; $h \neq 1$, the point of intersection of those normals to the parabola $x^2 - 4y - 2x + 5 = 0$, which are at right angles.
5. Find the locus of mid-points of the chords to the parabola $y^2 = 12x$ which pass through (2, 3)

Ans. 1. 1, **4.** $(x - 1)^2 = y - 4$, **5.** $-y^2 - 6x - 3y + 12 = 0$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic :1 Definiton of the Conic

Q.1 The curve is given by $x = \cos 2t$, $y = \sin t$ represents
(a) A parabola
(b) Circle
(c) Part of a parabola
(d) A pair of straight lines

Q.2 The curve described parametrically by $x = t^2 + t + 1, y = t^2 - t + 1$ represents
(a) a pair of straight lines
(b) an ellipse
(c) a parabola
(d) a hyperbola

Topic : 2 Standard Equation of Parabola

Q.3 If b and c are the lengths of the segements of any focal length of the parabola then $y^2 = 8x$ semi-latus rectum is

(a) $\frac{b+c}{2}$ (b) $\frac{bc}{b+c}$

(c) $\frac{2bc}{b+c}$ (d) $[\sqrt{bc}]$

Q.4 Let $P(1, 0)$ and Q any point on $y^2 = 8x$. The locus of mid-points of PQ must be
(a) $y^2 + 2 = 4x$ (b) $y^2 + 4x + 2 = 0$
(c) $x^2 + 2 = 4y$ (d) $x^2 + 4y + 2 = 0$

Q.5 If the vertex and the focus of the parabola are (–1, 1) and (2, 3) respectively, then the equation of the directrix is
(a) $3x + 2y + 14 = 0$ (b) $3x + 2y - 25 = 0$
(c) $2x - 3y + 10 = 0$ (d) $x - y + 5 = 0$

Q.6 If the parabola $y^2 = 4ax$ passes through (3, 2), then the length of its latus rectum is

(a) $\frac{2}{3}$ (b) $\frac{4}{3}$ (c) $\frac{1}{3}$ (d)4

Q.7 The vertex of the parabola $y^2 = (a - b)(x - a)$ is
(a)(b, a) (b)(a, b) (c)$(a, 0)$ (d)$(a, 0)$

Q.8 The length of a focal chord of the parabola $y^2 = 4ax$ at a distance b from the vertex is c. Then
(a) $a^2 = bc$ (b) $a^3 = b^2c$
(c) $b^2 = ac$ (d) $b^2c = 4a^3$

Q.9 The number of points with integral co-ordinates that lie in the interior of the region common to the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 4x$ is
(a)8 (b)10 (c)18 (d)20

Q.10 The point $(a, 2a)$ is an interior point of the region bounded by the parabola $y^2 = 16x$ and the double ordinate through the focus. Then
(a) $a < 4$ (b) $0 < a < 4$
(c) $0 < a < 2$ (d) $a > 4$

Q.11 The vertex of the parabola $y^2 = 8x$ is at the centre of a circle and the parabola cuts the circle of the ends of the latus rectum. Then the equation of the circle

(a) $x^2 + y^2 = 4$ (b) $x^2 + y^2 = 20$
(c) $x^2 + y^2 = 80$ (d) $x^2 + y^2 = 1$

Q.12 The particles moves on a parabolic path $y^2 = 4ax$. If its distance from the focus is minimum, then x is
(a)–1 (b)0 (c)–a (d)a

Q.13 The mirror image of the focus of the parabola $y^2 = 4(x + y)$ w.r.t. the directrix is
(a) (0, 2) (b) (2, 2)
(c) (–4, 2) (d) (–2, 2)

Q.14 A point on the curve $y^2 = 4x$, which is nearest to the point (2, 1) is
(a) (1, –2) (b) (–2, 1)
(c) $(1, 2\sqrt{2})$ (d) (1, 2)

Q.15 The length of the latus rectum of the parabola $x^2 - 6x + 5y = 0$ is
(a) 3 (b) 5 (c) 7 (d) 1

Q.16 Length of the chord intercepted by the parabola $y = x^2 + 3x$ on the line $x + y = 5$ is
(a) $3\sqrt{26}$ (b) $2\sqrt{26}$
(c) $\sqrt{26}$ (d) $6\sqrt{2}$

Topic : 3 Director circle, Tangent of Parabola

Q.17 Two common tangents to the circle $x^2 + y^2 = \frac{a^2}{2}$ and the parabola $y^2 = 4ax$ are
(a) $x = \pm(y + 2a)$ (b) $y = \pm(x + 2a)$
(c) $x = \pm(y + a)$ (d) $y = \pm(x + a)$

Q.18 If the line $y - \sqrt{3}x + 3 = 0$ cuts the parabola $y^2 = x + 2$ at A and B. If $P(\sqrt{3}, 0)$ then $PA.PB$ is
(a) $\frac{2}{3}(\sqrt{3}+2)$ (b) $2\sqrt{3}$
(c) $\frac{4}{3}(2-\sqrt{3})$ (d) $\frac{4}{3}(\sqrt{3}+2)$

Q.19 If tangent at A and B on the parabola intersect at point C, then the ordinates of A, C and B are in
(a) A.P. (b) G.P. (c) H.P. (d) A.G.P.

Q.20 If the parabola $y = (a - b)x^2 + (x - b)x + (c - a)$ touches the x-axis in the interval (0, 1) then the line $ax + by + c = 0$ always passes through a fixed point is
(a) (1, 2) (b)(–2, 1) (c) (2, 1) (d)(2, –1)

Q.21 A tangent to the parabola $y^2 = 8x$ makes an angle of 45° with the straight line $y = 3x + 5$. The equation of tangent is
(a) $x + 2y + 1 = 0$ (b) $2x + y + 1 = 0$
(c) $x + y + 2 = 0$ (d) $x + y + 1 = 0$

Q.22 Let m_1, m_2 are slopes of two tangents that are drawn from (2, 3) to the parabola $y^2 = 4x$ and α is the harmonic mean of m_1 and m_2. If β is the value of $(1 + \tan 23^o)(1 + \tan 22^o)$, then $\frac{3}{2}\alpha + \beta$ is
(a) 2 (b) 3 (c) 1 (d) 5

Q.23 The equation of the tangent to the parabola $y^2 = 9x$, which passes through the point (4, 10) is
(a) $x + 4y + 1 = 0$ (b) $9x + 4y + 4 = 0$
(c) $x - 4y + 36 = 0$ (d) $19x - 14y + 4 = 0$

Q.24 The coordinates of the point at which the line $x \cos\alpha + y \sin\alpha + a \sin\alpha \tan\alpha = 0$ touches the parabola $y^2 = 4ax$ are
(a) $(a \tan\alpha, 2a \tan\alpha)$
(b) $(2a \tan\alpha, a \tan\alpha)$
(c) $(a \tan^2\alpha, -2a \tan\alpha)$
(d) $(a \cos\alpha, 2a \cos\alpha)$

Q.25 If the segment intercepted by the parabola $y^2 = 4ax$ with the line $\ell x + my + n = 0$ subtends a right angle at the vertex, then
(a) $4a\ell + n = 0$ (b) $4a\ell + 4am + n = 0$
(c) $4am + n = 0$ (d) $a\ell + n = 0$

Topic : 4 Tangent and Normal in diffrent form

Q.26 The locus of the point of intersection of the normals at the ends of a focal chord of the parabola $y^2 = 4ax$ is
(a) $y^2 = 4a(x - 3a)$ (b) $y^2 = 2a(x - 3a)$
(c) $y^2 = a(x - 3a)$ (d) $y^2 = 16a(x - 3a)$

Q.27 Three normals are drawn to the curve $y^2 = x$ from a point $(c, 0)$. Out of three, one is always the x-axis. If two other normals are perpendicular to each other, 'c' is
(a)$\frac{3}{4}$ (b)$\frac{1}{2}$ (c)$\frac{3}{2}$ (d)2

Q.28 If $y + b = m_1(x + a)$ and $y + b = m_2(x + a)$ are two tangents to the parabola $y^2 = 4ax$, then
(a) $m_1 + m_2 = 0$ (b) $m_1m_2 = 1$
(c) $m_1m_2 = -1$ (d) $m_1 = m_2$

Q.29 The line $y = a - x$ touches the parabola $y = x - x^2$ and $f(x) = \sin^2 x + \sin^2\left(x + \frac{\pi}{3}\right) + \cos x \cdot \cos\left(x + \frac{\pi}{3}\right) g\left(\frac{5}{4}\right) = 1$ and $b = gof$, then
(a) $a = b$ (b) $a = 2b$
(c) $a = -b$ (d) $2a = b$

Q.30 The co-ordiantes of the points on the parabola $y^2 = 8x$, which is at minimum distance

from the circle $x^2 + (y + 6)^2 = 1$ are
(a) (2, 4) (b) (–2, 4)
(c) (2, –4) (d) (–2, –4)

Q.31 If a circle intersects the parabola $y^2 = 4ax$ at points $A(at_1^2, 2at_1)$, $B(at_2^2, 2at_2)$, $C(at_3^2, 2at_3)$, $D(at_4^2, 2at_4)$, then $t_1 + t_2 + t_3 + t_4$ is
(a)1 (b)–1 (c)0 (d)2

Q.32 The number of distinct normals that can be drawn from (–2, 1) to the parabola $y^2 = –4x – 2y – 3 = 0$ is
(a) 1 (b) 2 (c) 3 (d) 0

Q.33 A ray of light moving parabola to x-axis gets refelcted from a parabolic mirror whose equation is $4(x + y) – y^2 = 0$. After reflection the ray pass through the $pt(a, b)$. Then the value of $a + b$ is
(a) 2 (b) 1 (c) –2 (d) –1

Q.34 The length of the chord of the parabola $y^2 = 12x$ passing through the vertex and making an angle of 60° with the axis of x is
(a) 8/3 (b) 8 (c) 16/3 (d) 4

Q.35 For the parabola $y^2 = 4ax$, the length of the chord passing through the vertex and inclined to the x-axis at angle θ, is

(a) $\frac{4a\cos\theta}{\sin^2\theta}$ (b) $\frac{4a\sin\theta}{\cos^2\theta}$

(c) $a\sec^2\theta$ (d) $a\,\text{cosec}^2\theta$

Q.36 If the normals drawn at the points t_1 and t_2 on the parabola meets the parabola again at its point t_3, then t_1t_2 equals

(a)2 (b)–1 (c)–2 (d) $t_3 – \frac{2}{t_3}$

Q.37 The line $4x – 7y + 10 = 0$ intersects the parabola, $y^2 = 4x$ at the points P and Q. The coordinates of the point of intersection of the tangents of the points P and Q are

(a) $\left(-\frac{7}{2}, -\frac{5}{2}\right)$ (b) $\left(\frac{5}{2}, \frac{7}{2}\right)$

(c) $\left(-\frac{5}{2}, -\frac{7}{2}\right)$ (d) $\left(\frac{7}{2}, \frac{5}{2}\right)$

Q.38 The length of the normal chord to the parabola $y^2 = 4x$, which subtends a right angle at the vertex is
(a)$3\sqrt{3}$ (b)$6\sqrt{3}$ (c)2 (d)1

Section-B
(Assertion-Reason)

Q.1 **Statement – 1 :** If normal at ends of double ordinate $x = 4$ of parabola $y^2 = 4x$ meet the curve again at P and P' respectively, then $PP' = 12$ units.
Statement – 2 : If normal at t_1 of $y^2 = 4ax$ meet parabola again at t_2, then $t_2 = –t_1 – \frac{2}{t_1}$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1:** If the parabola $y = (a – b)x^2 + (b – c)x + (c – a)$ touches the x-axis in the interval (0, 1) then the line $ax + by + c = 0$ always passes through a fixed point.
Statement-2: The equation $L_1 + \lambda L_2 =$ or $\mu L_1 + \nu L_2 = 0$ represents a line passing through intersection of lines $L_1 = 0$ and $L_2 = 0$ which is a fixed point, when λ, μ, ν are constants.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1:** The lines from vertex to the two extremities of a focal chord of parabola $y^2 = 4ax$ are at an angle $\pi/4$.
Statements-2: If extremities of focal chord of parabola are $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ then $t_1t_2 = –1$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** The length of latus rectum of the parabola $(x – y + 2)^2 = 8\sqrt{2}\{x + y – 6\}$ is $8\sqrt{2}$.

Statement–2 : The length of latus rectum of parabola $(y-a)^2 = 8\sqrt{2}(x-b)$ is $8\sqrt{2}$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Year Question)

Q.1 Let P be the point on the parabola, $y^2 = 8x$, which is at a minimum distance from the centre C of the circle, $x^2 + (y+6)^2 = 1$. Then, the equation of the circle, passing through C and having its centre at P is

(a) $x^2 + y^2 - 4x + 8y + 12 = 0$

(b) $x^2 + y^2 - x + 4y - 12 = 0$

(c) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$

(d) $x^2 + y^2 - 4x + 9y + 18 = 0$

Q.2 Let O be the vertex Q be any point on the paralbola $x_2=8y$. If the point P divides the line segment OQ internally in the ratio 1 : 3, then the locus of P is

(a) $x^2 = y$ (b) $y^2 = x$

(c) $y^2 = 2x$ (d) $x^2 = 2y$

Q.3 Let (x, y) be any point on the parabola $y^2 = 4x$. Let P be the point that divides the line segment from (0, 0) to (x,y) in the ratio 1 : 3. Then, the locus of P is

(a) $x^2 = y$ (b) $y^2 = 2x$

(c) $y^2 = x$ (d) $x^2 = 2y$

Q.4 Axis of a parabola is y = x and vertex and focus are at a distance $\sqrt{2}$ and $2\sqrt{2}$ respectively from the origin. Then, equation of the parabola is

(a) $(x + y)^2 = 8 (x + y - 2)$

(b) $(x + y)^2 = 2(x + y - 2)$

(c) $(x- y)^2 = 4 (x + y - 2)$

(d) $(x + y)^2 = 2 (x- y + 2)$

Q.5 The locus of the mid-point of the line segment joining the focus to a moving point on the parabola $y^2 = 4\alpha x$ is another parabola with directrix

(a) $x = -\alpha$ (b) $x = -\frac{\alpha}{2}$

(c) $x = 0$ (d) $x = \frac{\alpha}{2}$

Q.6 The equation of the directrix of the parabola $y^2 + 4y + 4x + 2 = 0$ is

(a) x= -1 (b) x= 1

(c) $x = -3/2$ (d) $x = 3/2$

Q.7 If the line x-1=0 is the directrix of the parabola $y^2 - kx + 8 = 0$, then one of the values of k is

(a) $\frac{1}{8}$ (b) 8 (c) 4 (d) $\frac{1}{4}$

Q.8 The curve described parametrically by $x = t^2 + t + 1$, $y = t^2 - t + 1$, represents

(a) a pair of straight lines (b) an ellipse

(c) a parabola (d) a hyperbola

Q.9 The radius of a circle having minimum area, which touches the curve $y = 4 - x^2$ and the lines $y = |x|$, is

(a) $2\left(\sqrt{5}+1\right)$ (b) $2\left(\sqrt{5}-1\right)$

(c) $4\left(\sqrt{2}-1\right)$ (d) $4\left(\sqrt{2}+1\right)$

Q.10 The slope of the line touching both the parabolas $y^2 = 4x$ and $x^2 = -32y$ is

(a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) $\frac{1}{8}$ (d) $\frac{2}{3}$

Q.11 The tangent at (1,7) to the curves $x^2 = y - 6x$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at

(a) (6, 7) (b) (-6, 7) (c) (6, -7) (d) (-6, -7)

Q.12 The angle between the tangents drawn from the point (1,4) to the parabola $y^2 = 4x$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

Q.13 The focal chord to $y^2 = 16x$ is tangent to $(x-6)^2 + y^2 = 2$, then the possible values of the slope of this chord are

(a) {-1, 1} (b) {-2,2}
(c) {-2, 1/2} (d){2, -1/2}

Q.14 The equation of the common tangent to the curves $y^2 = 8x$ and $xy = -1$ is

(a) $3y = 9x + 2$ (b) $y = 2x + 1$
(c) $2y = x + 8$ (d) $y = x + 2$

Q.15 The equation of the common tangent touching the circle $(x-3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the X-axis is

(a) $\sqrt{3}y = 3x + 1$ (b) $\sqrt{3}y = -(x+3)$
(c) $\sqrt{3}y = x + 3$ (d) $\sqrt{3}y = -(3x+1)$

Q.16 If $x + y = k$ is normal to $y^2 = 12x$, then k is
(a) 3 (b) 9 (c) -9 (d) -3

Q.17 If a chord, which is not a tangent of the parabola $y^2 = 16x$, has the equation $2x + y = p$, and mid-point (h, k), then which of the following is (are) posible value (s) of p, h and k?
(a) $p = -1, h = 1, k = -3$ (b) $p = 2, h = 3, k = -4$
(c) $p = -2, h = 2, k = -4$ (d) $p = 5, h = 4, k = -3$

Q.18 Let A and B be two distinct points on the parabola $y^2 = 4x$. If the axis of the parabola touches a circle of radus r having AB as its diameter, then the slope of the line joining A and B can be

(a) $-\frac{1}{r}$ (b) $\frac{1}{r}$ (c) $\frac{2}{r}$ (d) $-\frac{2}{r}$

Section-D
(Board pattern)

Q.1 If y_1, y_2, y_3 be the ordinates of a vertex of the triangle inscribed in a parabola $y^2 = 4ax$. Then show that the area of the triangle is $\frac{1}{8a}|(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$.

Q.2 Find the equation of a parabola whose vertex is (1, 3) and directrix is the line $x - y + 1 = 0$.

Q.3 Find the vertex, focus, equation of directrix and axis and length of latus rectum of the parabola $y^2 - x - 2y + 2 = 0$.

Q.4 If the parabola $y^2 = 4ax$ passes through the point (2, 3) then find the length of latus rectum.

Q.5 Find the equation of parabola whose axis is parallel to *y*-axis and which passes through the point (0, 4)(1, 9) and (–2, 6).

Q.6 Find the point of intersection of parabola $y^2 = 8x$ and line $x + y = 2$ and also find length of intercept made by line on the parabola.

Q.7 Find the mid-point of the chord $2x + y - 4 = 0$ of the parabola $y^2 = 4x$.

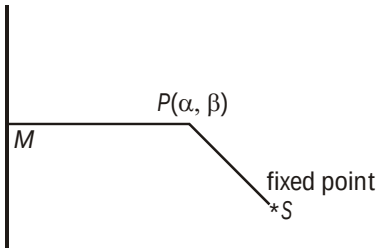
CHAPTER 3

ELLIPSE

1. STANDARD EQUATION OF ELLIPSE

The locus of a point which moves in a plane such that the ratio of its distances from a fixed point to a fixed straight line is always constant and is less than one is called an ellipse.

i.e., $\frac{SP}{PM} = e < 1$, then locus of S is an ellipse.



The fixed point is called focus and the fixed straight line is called the directrix of the ellipse.
An ellipse also the locus of a point moving in a plane such that the sum of its distances from two fixed points $(ae, 0)$ remains always $2a$.
Let $P(\alpha, \beta)$ be a moving point and $S(ae, 0)$ and $S'(-ae, 0)$ be the fixed points such that
$PS + PS' = 2a$

$\Rightarrow \sqrt{(\alpha - ae)^2 + \beta} + \sqrt{(\alpha + ae)^2 + \beta^2} = 2a$ $\quad \frac{\Delta_1}{\Delta_2}$

$\Rightarrow (\alpha + ae)^2 + \beta^2 = 4a^2 - 4a\sqrt{(\alpha - ae)^2 + \beta^2} + (\alpha - ae)^2 + \beta^2$

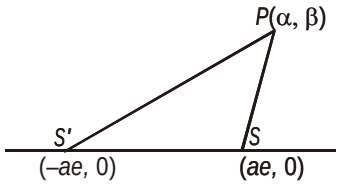
$\Rightarrow 4ae\alpha - 4a^2 = -4a\sqrt{(\alpha - ae)^2 + \beta^2}$

$\Rightarrow e^2\alpha^2 + a^2 - 2ae\alpha = \alpha^2 + a^2e^2 - 2ae\alpha + \beta^2$

$\Rightarrow a^2(1 - e^2) = \alpha^2(1 - e^2) + \beta^2$

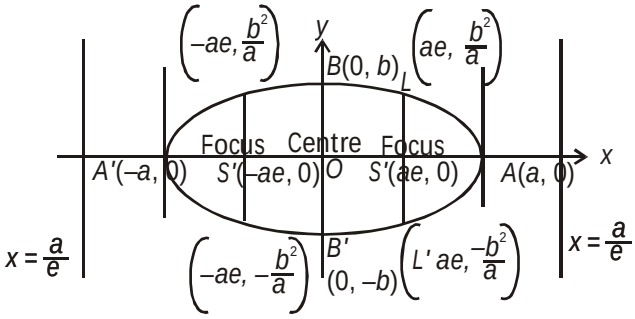
$\Rightarrow \frac{\alpha^2}{a^2} + \frac{\beta^2}{a^2(1 - e^2)} = 1$

$\Rightarrow \frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} = 1$, $b^2 = a^2(1 - e^2)$



Locus of (α, β) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $b^2 = a^2(1 - e^2)$

Thus $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $(a > h)$ represents an ellipse with following specifications :



(i) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a > b$

where $b^2 = a^2(1 - e^2)$

Centre O : (0, 0)

Vertices A and A' : $(\pm a, 0)$

Foci S and S' : $(\pm ae, 0)$

Equations of the directrices : $x = \pm \frac{a}{e}$

Equation of the major axis $y = 0$

Equation of the minor axis $x = 0$

Length of the major axis = $2a$

Length of the minor axis = $2b$

End points of the minor axis B and B' = $(0, \pm b)$

Equation of the latus rectum : $x = \pm ae$

End points of the latus rectum : $\left(ae, \pm\frac{b^2}{a}\right), \left(-ae, \pm\frac{b^2}{a}\right)$

Length of the latus rectum = $\frac{2b^2}{a}$

Focal distance of the point $(x, y) = a \pm ex$

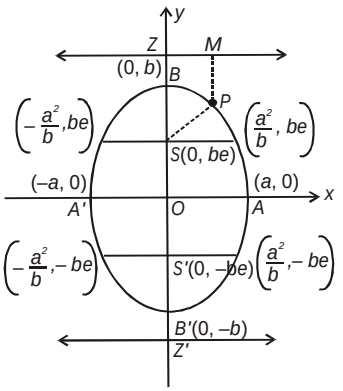
(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $a < b$

where $a^2 = b^2(1 - e^2)$

Centre O : (0, 0)

Vertices B and B' : $(0, \pm b)$

Foci S and S' : $(0, \pm be)$



Equation of the directrices : $y = \pm \frac{b}{e}$

Equation of the major axis : $x = 0$

Equation of the minor axis : $y = 0$

Length of the major axis = $2b$

Length of the minor axis = 2a

End points of the minor axis A and A' : $(\pm a, 0)$

Equation of the latus rectum : $y = \pm be$

End points of the latus rectum : $\left(\pm\frac{a^2}{b}, be\right), \left(\pm\frac{a^2}{b}, -be\right)$

Length of the latus rectum = $\frac{2a^2}{b}$

Focal distance of a point $(x, y) = b \pm ey$

2. PARAMETRIC FORM AND AUXILIARY CIRCLE

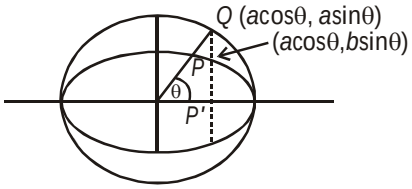
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ... (i)

$x = a\cos\theta$, $y = b\sin\theta$ is the parametric form of the ellipse (i)

Let a circle is being drawn on the major axis of the ellipse as its diameter, therefore its equation is $x^2 + y^2 = a^2$

Let there be a point on this circle $Q(a\cos\theta, a\sin\theta)$.

A perpendicular is drawn from Q to major axis which intersects ellipse at P and P'.



$\therefore$ Co-ordinates of P are $(a\cos\theta, k)$. It lies on ellipse

$\therefore \frac{a^2\cos^2\theta}{a^2} + \frac{k^2}{b^2} = 1$

$\therefore k = \pm b\sin\theta$

Hence a general point on ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is described as $(a\cos\theta, b\sin\theta)$.

Circle $x^2 + y^2 = a^2$ is called auxiliary circle and θ is called eccentric angle of the ellipse.

Student Task

1. Find the eccentricity, length of latus rectum, foci and the equations of directrices of the ellipse $25x^2 + 9y^2 - 50x - 36y - 164 = 0$
2. Let the line segment joining foci of an ellipse subtend a right angle at one end of the minor axis. Find its eccentricity.
3. Find the length of the latus rectum of the ellipse whose foci are (–2, –1) and (1, 2) and one of the directrices is $x + y = 5$

Ans. 1. $e = \frac{4}{5}$, latus rectum = $\frac{18}{5}$, foci are (1, 6) and (1, 2), equations of directions are $4y - 33 =$

0, $4y + 17 = 0$ **2.** $\frac{1}{\sqrt{2}}$ **3.** $2\sqrt{\frac{6}{5}}$ units

Position of Point

Let $S_1 = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$

Point (x_1, y_1) will be outside, or inside the circle respectively, if $S_1 > 0, = 0, < 0$.

3. DIRECTOR CIRCLE OF AN ELLIPSE

The locus of the point of intersection of mutually perpendicular tangents to an ellipse is a circle called the director circle of the ellipse.

The equation of any tangent of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

The slope form may be written as

$y - mx = \sqrt{a^2m^2 + b^2}$ ... (i)

and the equation of tangent perpendicular to (i) is

$x + my = \sqrt{a^2 + b^2m^2}$ (replacing m by $-\frac{1}{m}$ in equation (i)) ....(ii)

The required locus of the point of intersection is obtained by eliminating m between (i) and (ii)

Squaring (i) and (ii) and then adding we get

$(1 + m)^2 (x^2 + y^2) = (1 + m^2)a^2 = (1 + m^2)b^2$

$x^2 + y^2 = a^2 + b^2$

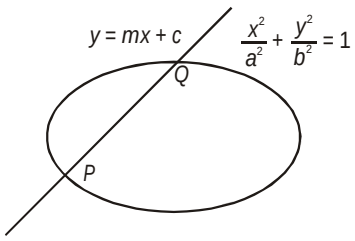
which is the equation of the director circle of given ellipse.

Point of intersection of a Straight Line and an Ellipse

Let $y = mx + c$ be the equation of a given straight line and

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be the ellipse.

The points of intersection of the line and the ellipse can be found by solving the two equations simultaneously.



Thus $\frac{x^2}{a^2} + \frac{(mx + c)^2}{b^2} = 1$

i.e., $(b^2 + a^2m^2)x^2 + 2a^2mcx + a^2(c^2 - b^2) = 0$

Which being quadratic equation in x gives abscissa of the point of intersection. The given straight will intersect of touch or neither touch nor cut the given ellipse according as the discriminant of the above equation is greater or equal or less than zero.

i.e., $4a^4m^2c^2 - 4(a^2)(c^2 - b^2)(b^2 + a^2m^2) >$ or $= < 0$

$a^2m^2c^2 - b^2c^2 + b^4 - a^2m^2c^2 + a^2m^2b^2 >/< 0$

$c^2 >/< a^2m^2 + b^2$

Thus the straight line y = mx + c will touch the given ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

iff $c^2 = a^2m^2 + b^2$

i.e., $c = \pm \sqrt{a^2m^2+b^2}$

Hence the equation of the tangents to the ellipse in slope form are given by $y = mx \pm \sqrt{a^2m^2+b^2}$

4. EQUATION OF TANGENT AND NORMAL

Equation of Tangent

(i) Point form : The equation of the tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at (x_1, y_1) is

$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$.

(ii) Parametric form : The equation of the tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at ($a\cos\theta$, $b\sin\theta$) is $\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta=1$.

Equation of the Normal

(i) Point form : Normal (x_1, y_1) is $\frac{a^2x}{x_1}-\frac{b^2y}{y_1}=a^2-b^2$

(ii) Parametric form : Normal at ($a\cos\theta$, $b\sin\theta$) is $ax\sec\theta - by\,\mathrm{cosec}\theta = a^2 - b^2$

i.e. by $\tan^4\frac{\theta}{2}+2(ax+a-b)\tan^3\frac{\theta}{2}+2(ax-a+b)\tan\frac{\theta}{2}-by=0$, which will be a biquadratic equation in $\tan\frac{\theta}{2}$ for fixed x and y. Hence from a fixed point atmost four real normals can be drawn to the ellipse.

Student Task

1. Let A and B be the corresponding points on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the auxilliary circle respectively. The normal at A meets that the line joining centre of the ellipse to B at M. Prove that the distance of the centre from M is $a + b$.
2. Prove that the line px + qy + r = 0 touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ if $a^2p^2 + b^2q^2 = r^2$.
3. Prove that the product of perpendicular drawn from foci to any tangent of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is b^2.

Equation of Chord passing through ($a\cos\alpha$, $b\sin\alpha$) and ($a\cos\beta$, $b\sin\beta$)

Equation of chord is

$$\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$$

If chord passes through foci then the chord is major axis of the ellipse

Equation of the Chord with Middle Point (x_1, y_1)

$$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1 \text{ or } T = S_1.$$

Equation of Pair of Tangents from External Point (x_1, y_1)

Equation is $T^2 = SS_1$

$$\text{or } \left(\frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1\right)^2=\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}-1\right)\left(\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1\right)$$

5. DIAMETER AND CONJUGATE DIAMETER OF AN ELLIPSE

The locus of the middle points of a system of parallel chords of an ellipse is called a diameter of the ellipse and two diameters of an ellipse are said to be conjugate diameters if each bisects the chords parallel to the other.

Let the mid-point of a chord parallel to a given chord y = mx, be (h, k).

$\therefore$ Equation of the chord is

$T = S_1$

$$\Rightarrow \frac{xh}{a^2}+\frac{yk}{b^2}-1=\frac{h^2}{a^2}+\frac{k^2}{b^2}-1$$

Slope of this chord = $-\frac{h}{a^2}.\frac{b^2}{k}=m$ $\quad\therefore$ Locus of (h, k) is $y = -\frac{b^2}{a^2m}x$

Now if $y = m_1x$ and $y = m_2x$ are the conjugate diameters then

$$m_2=-\frac{b^2}{a^2m_1} \qquad \Rightarrow m_1m_2=\frac{-b^2}{a^2}$$

Properties of conjugate diameters :

(i) The eccentric angles of ends of the semi-conjugate diameters differ by $\frac{\pi}{2}$

Slope of $CP = \frac{b\sin\theta}{a\cos\theta}=m_1$ (say)

Slope of $CQ = \frac{b\sin\phi}{a\cos\phi}=m_2$ (say)

$\because\ m_1m_2 = \frac{b^2}{a^2}$ $\qquad\therefore \frac{b\sin\theta}{a\cos\theta}.\frac{b\sin\phi}{a\cos\phi}=-\frac{b^2}{a^2}$

$\therefore \cos(\phi-\theta) = 0 \quad \Rightarrow \phi-\theta = \frac{\pi}{2}$

$\therefore$ If end point of one semi-conjugate diameter is $(a\cos\theta, b\sin\theta)$ then end point of the other semi conjugate diameter is $(-a\sin\theta, b\cos\theta)$.

(ii) Sum of the squares of the lengths of semi-conjugate diameters is always constant

$CP^2 + CQ^2 = (a^2\cos^2\theta + b^2\sin^2\theta) + (a^2\sin^2\theta + b^2\cos^2\theta)$

$= a^2 + b^2$

= constant

Pole and Polar

The equation of the polar of a point (x_1, y_1) with respect to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ the point (x_1, y_1) is called the pole of this polar.

Equation of the Chord of Contact

The equation of the chord of contact of the tangents drawn from the point (x_1, y_1) to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$.

Student Task

1. Find the locus of the point of intersection of the tangents at the end points of the semi conjugate diameters of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$
2. Let a normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at $P(a\cos\theta, b\sin\theta)$ passes through the focus. Find the eccentricity of the ellipse in terms of θ.
3. Let a variable line drawn from $A(h, k)$ intersects the ellipse $x^2 + 2y^2 = 1$ at P and Q. The tangents at P and Q intersect at R and the locus of R is $\lambda x + (\lambda + 1)y = 1$. Find the values of λ such that the point A lies inside the ellipse.

Ans. 1. $\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}=2\right)$, **2.** $\left(1-\frac{b^2}{a^2}\right)\cos\theta$, **3.** $\lambda\in\left(-1,\frac{1}{3}\right)$

MISCELLANEOUS EXAMPLES

Illustration 1: Find the equation of the ellipse whose focus is at (1, –1) directrix is the line $x + y = 3$ eccentricity is $\frac{1}{3}$.

Solution : If (x, y) is any point on the ellipse then

$$\sqrt{(x-1)^2+(y+1)^2}=\frac{1}{3}\left(\frac{x+y-3}{\sqrt{2}}\right)$$

or $18[x – 1)^2 + (y + 1)^2] = (x + y – 3)^2$

or $17x^2 + 17y^2 – 2xy – 30x + 42y + 27 = 0$

Illustration 2 : Find the equation of the ellipse whose two foci are (4, 0) and (–4, 0) and whose eccentricity is $\frac{1}{3}$.

Solution : Since the distance between foci = 8 we must have

Also $b^2 = a^2 (1 – e^2) = 144 (1 – \frac{1}{9}) = 128$

Now since the midpoint of the line joining the foci is centre of the ellipse and the point of line joining (4, 0) and (–4, 0) is origin.

$\Rightarrow$ Origin is centre of the ellipse and x-axis is the major axis.

We also conclude that y-axis is the major axis. Thus the equation of the ellipse is

$$\frac{x^2}{y^2}+\frac{y^2}{b^2}=1\Rightarrow\frac{x^2}{144}+\frac{y^2}{128}=1$$

Illustration 3: Find the eccentricity of the ellispe if its latus rectum be equal to half of the minor axis.

Solution : We must have $2\frac{b^2}{a} = \frac{1}{2}b \Rightarrow a = 4b$

As $b^2 = a^2(1 - e^2)$, we have $b^2 = 16b^2(1 - e^2)$

$\Rightarrow e = \frac{\sqrt{15}}{4}$

Illustration 4: Find focal distance of the point $(4\sqrt{3}, 5)$ on the ellipse $25x^2 + 16y^2 = 1600$.

Solution : The ellipse is $\frac{x^2}{64} + \frac{y^2}{100} = 1 \Rightarrow a^2 = 64, b^2 = 100$

$a^2 = b^2(1 - e^2)$ (since $a < b$)

$\Rightarrow 64 = 100(1 - e^2) \Rightarrow e = \frac{3}{5}$

The focal distances of any point (x_1, y_1) on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $(a < b)$ must be $b - ey_1$, $b + ey_1$ which, in this question are $10 - \frac{3}{5} \times 5$, $10 + \frac{3}{5} \times 5$ or 7 and 13.

Illustration 5: Show that the locus of a point which moves such that the sum of its distances from two fixed points $(ae, 0)$ and $(-ae, 0)$ is always a constant equal to $2a$ is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution : Let P be (x, y) then from the condition given we have

$\Rightarrow \sqrt{(x - ae)^2 + y^2} + \sqrt{(x + ae)^2 + y^2} = 2a$

$\Rightarrow \sqrt{x^2 + y^2 + a^2e^2 - 2aex} + \sqrt{x^2 + y^2 + a^2e^2 + aex} = 2a$

$\Rightarrow \sqrt{\lambda - 2aex} + \sqrt{\lambda + 2aex} = 2a$ (where $\lambda = x^2 + y^2 + a^2e^2$)

On squaring the last relation we get

$\lambda - 2aex + \lambda + 2aex + \sqrt{\lambda^2 - 4a^2e^2x^2} = 4a^2 \Rightarrow 2\sqrt{\lambda^2 + 4a^2e^2x^2} = 4a^2 - 2\lambda$

On squaring again we get

$\lambda^2 - 4a^2e^2x^2 = 4a^4 + \lambda^2 - 4a^2\lambda \Rightarrow -4a^2e^2x^2 = 4a^4 - 2a^2(x^2 + y^2 + a^2e^2)$

$\Rightarrow x^2 + y^2 + a^2e^2 - e^2x^2 = a^2 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2(1 - e^2)} = 1$

Thus locus of a point P the sum of whose distances from two fixed points $S(ae, 0)$, $S'(-ae, 0)$ is a constant $(= 2a)$ is an ellipse. Note that this constant is equal to length of the major axis of the ellipse.

Illustration 6 : Reduce the equation of the ellipse $4x^2 + 3y^2 + 8x + 12y = 1$ to the standard form and hence determine/locate centre, focus, directrix of the ellipse.

Solution : Grouping terms we get

$4x^2 + 8x + 3y^2 + 12y = 1$

$\Rightarrow 4(x^2 + 2x) + 3(y^2 + 4y) = 1 \Rightarrow 4(x^2 + 2x + 1) + 3(y^2 + 4y + 4) = 1 + 4 + 12$

$\Rightarrow 4(x + 1)^2 + 3(y + 2)^2 = 17 \Rightarrow \frac{(x + 1)^2}{17/4} + \frac{(y + 2)^2}{17/3} = 1$

Put X = x + 1, Y = y + 2 we get $\frac{X^2}{17/4}+\frac{Y^2}{17/3}=1$

$\Rightarrow a^2 = \frac{17}{4}$, $b^2 = \frac{17}{3}$ since $b > a$ the major axis of the ellipse must be along y-axis.

On which foci will lie. from $a^2 = b^2(1 - e^2)$ we get $\frac{17}{4} = \frac{17}{3}(1 - e^2) \Rightarrow e = \frac{1}{2}$

Thus the eccentricity of the given ellipse is $\frac{1}{2}$, the lengths of major and minor axes and 2b and 2a or $2\sqrt{\frac{17}{3}}, \sqrt{17}$ respectively.

Now observe the following table for various locations :

Location	***New axes***	***Old axes***
Centre	(0, 0)	(1, 2)
	(On putting $X = 0$, $Y = 0$ in $X = x + 1$, $Y = y + 1$)	
Foci	$\left(0, \sqrt{\frac{17}{3}}.\frac{1}{2}\right)$	$\left(-1, \sqrt{\frac{17}{3}}.\frac{1}{2}-1\right)$
	$\left(0, -\sqrt{\frac{17}{3}}.\frac{1}{2}\right)$	$\left(-1, -\sqrt{\frac{17}{3}}.\frac{1}{2}-1\right)$
Directrix	$X = \pm\frac{\sqrt{17/3}}{1/2}$	$x+1 = \pm\frac{\sqrt{17/3}}{1/2}$

Thus student must note that the eccentricity, length of major, minor axes, length of latus rectum are invariants. They remain same in all frame of reference. In addition you must note that if the general equation of second degree represents an ellipse then

(1) eccentricity = $\sqrt{1-\frac{\text{coeff. of } y^2}{\text{coeff. of } x^2}}$, if coeff. of y^2 < coeff. of x^2.

Or

eccentricity = $\sqrt{1-\frac{\text{coeff. of } y^2}{\text{coeff. of } x^2}}$, if coeff . of y^2 > coeff. of x^2.

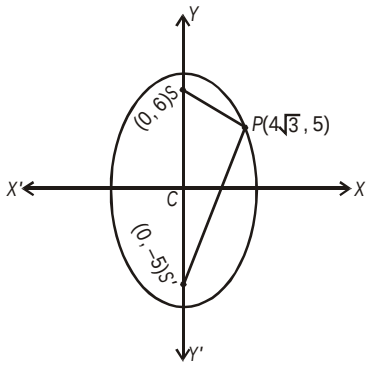
(2) If the general equation of second degree $f(x, y) = 0$ represents an ellipse and xy term is abscent then centre of the ellipse is given by $\frac{\partial f}{\partial x} = 0, \frac{\partial f}{\partial y} = 0$

where $\frac{\partial f}{\partial x}$ denotes the derivative of ($f(x, y)$ with respect to x, keeping y constant.

Illustration 7: Find the lengths and equations of the focal radii drawn from the point $(4\sqrt{3}, 5)$ on the ellipse $25x^2 + 16y^2 = 1600$.

Solution : The equation of the ellipse is

$25x^2 + 16y^2 = 1600$ or $\frac{x^2}{64}+\frac{y^2}{100}=1$



Here $b > a$

$a^2 = 64$, $b^2 = 100$

$a^2 = b^2 (1 - e^2)$

$\therefore\ 64 = 100(1 - e^2)$

$\Rightarrow e = 3/5$

Let $P(x_1, y_1) \equiv (4\sqrt{3}, 5)$

be a point on the ellipse then SP and $S'P'$ are the focal radii

$\therefore\ SP = b - ey_1$ and $S'P = b + ey_1$

$\therefore\ SP = 10 - \frac{3}{5} \times 5$ and $S'P = 10 + \frac{3}{5} \times 5$

$\Rightarrow SP = 7$ and $S'P = 13$

i.e., $\left(0, 10 \times \frac{3}{5}\right)$ *i.e.*, (0, 6)

and S' is $(0, -be)$

Illustration 8: Find the eccentric angles of the extremities of latus-recta of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution : The coordinates of the end-points of latus recta of the given ellipse are $\left(\pm a\, e, \pm \frac{b^2}{a}\right)$

Let θ be the eccentric angle of an end of a latus-rectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Then, its coordinates are $(a \cos\theta, b \sin\theta)$ of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

$\therefore\ a \cos\theta = a\, e$ and $b \sin\theta = \frac{b^2}{a}$

$\Rightarrow \frac{b\sin\theta}{a\cos\theta} = \frac{b^2}{a^2 e} \Rightarrow \tan\theta = \frac{b}{ae} \Rightarrow \tan^{-1}\left(\frac{b}{ae}\right)$

Similarly, the eccentric angles of other ends of latus recta are given by $\theta = \tan^{-1}(-b/ae)$.

Hence, the eccentric angles of the extremities of the latus-recta of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are given by $\theta = \tan^{-1}(\pm b/ae)$.

Illustration 9: Show that the line $lx + my + n = 0$ will cut the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in points whose eccentric angles differ by $\pi/2$, if $a^2 l^2 + b^2 m^2 = 2n^2$.

Solution : Suppose the line $lx + my + n = 0$ cuts the ellipse at $P(a\cos\theta, b\sin\theta)$ and $Q(a\cos(\pi/2+\theta), b\sin(\pi/2+\theta))$. Then these two points lie on the line.

$\therefore la\cos\theta + mb\sin\theta + n = 0$

and $-la\sin\theta + mb\cos\theta + n = 0 \quad \Rightarrow la\cos\theta + mb\sin\theta = -n$

and, $-la\sin\theta + mb\cos\theta = -n$

$\Rightarrow (la\cos\theta + mb\sin\theta)^2 + (-la\sin\theta + mb\cos\theta)^2 = n^2 + n^2$

$\Rightarrow l^2a^2 + m^2b^2 = 2n^2$.

Illustration 10: Show that the line $x\cos\alpha + y\sin\alpha = p$ touches the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ if $a^2\cos^2\alpha + b^2\sin^2\alpha = p^2$ and that points of contact is $\left(\frac{a^2\cos\alpha}{p}, \frac{b^2\sin\alpha}{p}\right)$.

Solution : The given line is $x\cos\alpha + y\sin\alpha = p$

$y = -x\cot\alpha + p\,\text{coses}\,\alpha$

Comparing this line with $y = mx + c$

$\therefore m = -\cot\alpha$ and $c = p\,\text{coses}\,\alpha$

Hence the given line touches the ellipse then

$c^2 = a^2m^2 + b^2$

$\Rightarrow p^2\,\text{coses}^2\alpha = a^2\cot^2\alpha + b^2$

$\Rightarrow p^2 = a^2\cos^2\alpha + b^2\sin^2\alpha$

and point of contact is $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$

i.e., $\left(-\frac{a^2(-\cot\alpha)}{p\,\text{cosec}\alpha}, \frac{b^2}{p\,\text{cosec}\alpha}\right)$

i.e., $\left(\frac{a^2\cos\alpha}{p}, \frac{b^2\sin\alpha}{p}\right)$

Illustration 11: Find the equations of the tangents to the ellipse $3x^2 + 4y^2 = 12$ which are perpendicular to the line $y + 2x = 4$.

Solution : Let m be the slope of the tangent, since the tangent is perpendicular to the line $y + 2x = 4$.

$\therefore m \times (-2) = -1 \Rightarrow m = \frac{1}{2}$

Since $3x^2 + 4y^2 = 12$

or $\frac{x^2}{4}+\frac{y^2}{3}=1$

Comparing this with $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

$\therefore a^2 = 4$ and $b^2 = 3$

So the equation of the tangnets are

$y=\frac{1}{2}x \pm \sqrt{4\times\frac{1}{4}+3}$

$\Rightarrow y=\frac{1}{2}x \pm 2$ or $x - 2y \pm 4 = 0$

Illustration 12: Find the condition that the line $lx + my = n$ may touch the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Also, find the point of contact.

Solution : The equation of the line and the ellipse are

$lx + my = n$ .....(i)

and $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ .....(ii)

respectively.

Let the line (i) touch the ellipse (ii) at (x_1, y_1). Then the equation of the tangnet at (x_1, y_1) is

$\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ .....(iii)

Now, (i) and (ii) will represent the same line if

$$\frac{x_1 / a^2}{l}=\frac{y_1 / b^2}{m}=\frac{1}{n}\Rightarrow x_1=\frac{a^2 l}{n}, y_1=\frac{b^2 m}{n}$$

Thus, $(x_1, y_1) = \left(\frac{a^2 l}{n}, \frac{b^2 m}{n}\right)$ is the point of contact.

Since (x_1, y_1) lies on (ii), therefore

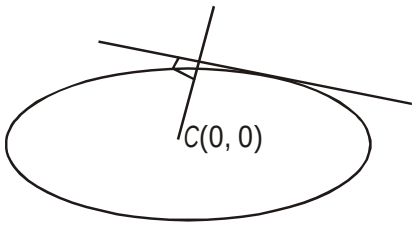
$$\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}=1\Rightarrow\frac{a^2 l^2}{n^2}+\frac{b^2 m^2}{n^2}=1$$

$\Rightarrow a^2l^2 + b^2m^2 = n^2$

This is the required condition of tangency.

Illustration 13: Find the locus of the foot of the perpendicular drawn form centre upon any tangnet to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

Solution : $y = mx + \sqrt{(a^2m^2 + b^2)}$ ....(1)



Equation of the line perpendicular to(1) and passing through (0, 0) is

$y=\frac{1}{m}x$ or $m=-\frac{x}{y}$

Substituting the value of m from (2) in (1) then

$$y=-\frac{x^2}{y}+\sqrt{\left(a^2\frac{x^2}{y^2}+b^2\right)}$$

$\Rightarrow (x^2 + y^2)^2 = a^2x^2 + b^2y^2$

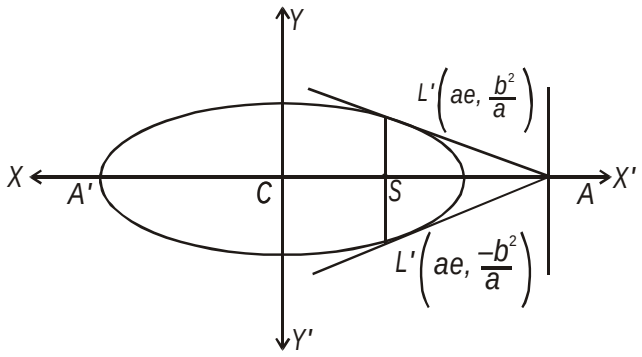
or changing to polars by putting $x = r\cos\theta$, $y = r\sin\theta$ it becomes

$r^2 = a^2\cos^2\theta + b^2\sin^2\theta$

Illustration 14: Prove that the tangents at the extremities of latus rectum of an ellipse intersect at the corresponding directrix.

Solution : The *LSL'* be a latus rectum of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

The co-ordinates of *L* and *L'* are $\left(ae,\frac{b^2}{a}\right)$ and $\left(ae,-\frac{b^2}{a}\right)$ respectively



$\therefore$ Equation of tangent at $L\left(ae,\frac{b^2}{a}\right)$ is

$\Rightarrow \frac{x(ae)}{a^2}+\frac{y\left(\frac{b^2}{a}\right)}{b^2}=1$

$\Rightarrow ex + y = a$ .....(2)

Solving (1) and (2) we get $x=\frac{a}{e}$ and $y = 0$

Thus the tangents at *L* and *L'* intersect at $(a/e, 0)$ which is a point lying on the corresponding directrix *i.e.*, $x=\frac{a}{e}$ and $y = 0$

Thus the tangents at *L* and *L'* intersect at $(a/e, 0)$ which is a point lying on the corresponding directrix *i.e.*, $x=\frac{a}{e}$.

Illustration 15: Prove that the product of the perpendicular from the foci upon any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is b^2.

Solution : The equation of any tangnet to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is $y = mx + \sqrt{a^2m^2+b^2}$

$\Rightarrow mx - y + \sqrt{a^2m^2+b^2} = 0$ .....(i)

The two foci of the given ellipse are $S(ae, 0)$ and $S'(-ae, 0)$. Let p_1 and p_2 be the lengths of perpendicular from *S* and *S'* respectively on (i). Then,

p_1 = Length of $\perp$ from $S(ae, 0)$ on (i) = $\frac{mae-0+\sqrt{a^2m^2+b^2}}{\sqrt{m^2+(-1)^2}}$

$$= \frac{mae + \sqrt{a^2m^2 + b^2}}{\sqrt{m^2 + 1}}$$

and p_2 = Length of $\perp$ from S' $(-ae, 0)$ on (i)

$$= \frac{-mae - 0 + \sqrt{a^2m^2 + b^2}}{\sqrt{m^2 + (-1)^2}}$$

$$= \frac{-mae + \sqrt{a^2m^2 + b^2}}{\sqrt{m^2 + 1}}$$

$$p_1p_2 = \left(\frac{mae + \sqrt{a^2m^2 + b^2}}{\sqrt{m^2 + 1}}\right)\left(\frac{-mae + \sqrt{a^2m^2 + b^2}}{\sqrt{m^2 + 1}}\right)$$

$$= \frac{-m^2a^2e^2 + a^2m^2 + b^2}{m^2 + 1}$$

$$= \frac{a^2m^2(1 - e^2) + b^2}{1 + m^2} = \frac{m^2\{a^2(1 - e^2)\} + b^2}{1 + m^2}$$

$$= \frac{m^2b^2 + b^2}{1 + m^2} \quad [\because\ b^2 = a^2\,(1 - e^2)]$$

$$= \frac{b^2(m^2 + 1)}{(m^2 + 1)} = b^2$$

Illustration 16: Show that the point of intersection of the tangents at two points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose eccentric angles differ by a right angle lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$.

Solution : Let P $(a\cos\theta, b\sin\theta)$ and Q $(a\cos\phi, b\sin\phi)$ be two points on the ellipse such that $\theta - \phi = \pi/2$.

The equations of tangents at P and Q are

$$\frac{x}{a}\cos\theta + \frac{y}{b}\sin\theta = 1 \quad .....(i)$$

and, $\frac{x}{a}\cos\phi + \frac{y}{b}\sin\phi = 1$ ....(ii)

respectively.

Since $\theta - \phi = \pi/2$, so (i) can be written as

$$-\frac{x}{a}\sin\phi + \frac{y}{b}\cos\phi = 1 \quad ....(iii)$$

Squaring (ii) and (iii) and then adding, we get $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$.

Hence, the points of intersection of tangents at P and Q lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$.

Illustration 17 : Prove that the locus of the mid-points of the portion of the tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepted between the axes is $a^2y^2 + b^2x^2 = 4x^2y^2$.

Solution : Let (x_1, y_1) be any point on the ellipse

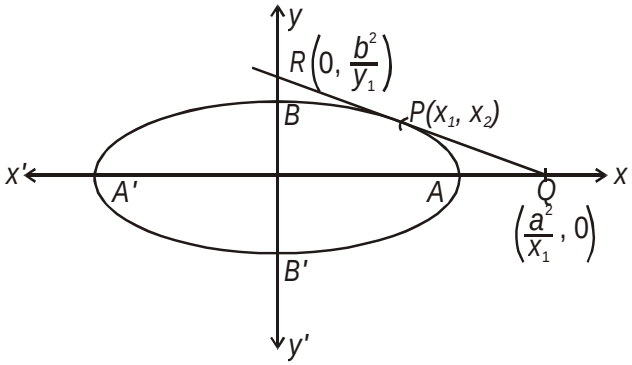
$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \text{.....(i)}$$

The equation of the tangent at (x_1, y_1) to (i) is $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$

This meets the coordinate axes at $Q\left(\frac{a^2}{x_1},0\right)$ and $R\left(0,\frac{b^2}{y_1}\right)$.

Let $L(h, k)$ be the mid-point of QR. Then,

$$h=\frac{a^2}{2x_1}, k=\frac{b^2}{2y_1} \Rightarrow x_1=\frac{a^2}{2h}, y_1=\frac{b^2}{2k} \quad \text{.....(ii)}$$



Since (x_1, y_1) lies on the ellipse (i),

$$\therefore \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}=1 \Rightarrow \frac{a^4}{4h^2a^2}+\frac{b^4}{4k^2b^2}=1 \Rightarrow \frac{a^2}{4h^2}+\frac{b^2}{4k^2}=1$$

Hence, the locus of (h, k) is $\frac{a^2}{4x^2}+\frac{b^2}{4y^2}=1$

$\Rightarrow a^2y^2 + b^2x^2 = 4x^2y^2$.

Illustration 18: Prove that the straight line lx + my + n = 0 is a normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ if

$$\frac{a^2}{l^2}+\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}.$$

Solution : The equation of any normal to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is

$ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$

The straight line $lx + my + n = 0$ will be a normal to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, if

$ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$

Since $lx + my + n = 0$ and (i) represent the same line.

$$\therefore \frac{a\sec\theta}{1}=\frac{b\operatorname{cosec}\theta}{m}=\frac{a^2-b^2}{-n}$$

$$\Rightarrow \cos\theta = -\frac{an}{l(a^2-b^2)} + \frac{b^2n^2}{m(a^2-b^2)^2}$$

$$\therefore \cos^2\theta + \sin^2\theta = \frac{a^2n^2}{l^2(a^2-b^2)^2} + \frac{b^2n^2}{m(a^2-b^2)^2}$$

$$\Rightarrow \frac{(a^2-b^2)^2}{n^2} = \frac{a^2}{l^2} + \frac{b^2}{m^2}$$

Illustration 19: If the normal at an end of a latus rectum of an ellipse passes through one extremity of the minor axis, show that the eccentricity of the ellipse is given by $e^4 + e^2 - 1 = 0$.

Solution : Let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be the ellipse. The coordinates of an end of the latusrectum are $(ae, b^2/a)$.

The equation of the normal at $(ae, b^2/a)$ is

$$\frac{a^2x}{ae} - \frac{b^2y}{b^2/a} = a^2 - b^2 \text{ or } \frac{ax}{e} - ay = a^2 - b^2$$

It passes through one extremity of the minor axis whose coordinates are $(0, -b)$.

$\therefore ab = a^2 - b^2 \Rightarrow a^2b^2 = (a^2 - b^2)^2$

$\Rightarrow a^2 \times a^2 (1 - e^2) = (a^2 e^2)^2 \Rightarrow 1 - e^2 = e^4 \Rightarrow e^4 + e^2 - 1 = 0$

Illustration 20: Any ordinate MP of an ellipse meets the auxiliary circle in Q. Prove that the locus of the point of intersection of the normals at P and Q is the circle $x^2 + y^2 = (a + b)^2$.

Solution : Let $P(a\cos\theta, b\sin\theta)$ be any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and let $Q(a\cos\theta, a\sin\theta)$ be the corresponding point on the auxiliary circle $x^2 + y^2 = a^2$. The equation of the normal at $P(a\cos\theta, b\sin\theta)$ to the ellipse is

$ax\sec\theta - by\,\text{cosec}\,\theta = a^2 - b^2$ .....(i)

The equation of the normal at $Q(a\cos\theta, a\sin\theta)$ to the circle $x^2 + y^2 = a^2$ is $y = x\tan\theta$

Let (h, k) be the point of intersection of (i) and (ii). Then,

$ah\sec\theta - bk\,\text{cosec}\,\theta = a^2 - b^2$ .....(iii)

and, $k = h\tan\theta$ .....(iv)

Eliminating θ from (iii) and (iv), we get

$$ah\sqrt{1+\frac{k^2}{h^2}} - bk\sqrt{1+\frac{h^2}{k^2}} = a^2 - b^2$$

$\Rightarrow (a - b)\sqrt{h^2 + k^2} = a^2 - b^2 \Rightarrow h^2 + k^2 = (a + b)^2$

Hence, the locus of (h, k) is $x^2 + y^2 = (a + b)$

Illustration 21: If α, β, γ are the eccentric angles of three points on the ellipse, the normals at which are concurrent, then $\sin(\alpha + \beta) + \sin(\beta + \gamma) + \sin(\gamma + \alpha) = 0$.

Solution : We known the equation by. $t^4 + 2t^3(ax + a^2e^2) + 2t(ax - a^2e^2) - by = 0$

$\Sigma t_1 t_2 = 0$ ....(i)

and $t_1, t_2\, t_3\, t_4 = -1$ .....(ii)

Now, $\Sigma t_1\, t_2 = 0$

$\Rightarrow t_1\, t_2 + t_2\, t_3 + t_3\, t_1 = -t_4(t_1 + t_2 + t_3)$

$$\Rightarrow t_1\, t_2 + t_2\, t_3 + t_3\, t_1 = \frac{t_1 + t_2 + t_3}{t_1 t_2 t_3} \qquad \left[\begin{array}{l} \because t_1 t_2 t_3 t_4 = -1 \\ \therefore -t_4 = \dfrac{1}{t_1 t_2 t_3} \end{array}\right]$$

$$\Rightarrow t_1 t_2 + t_2 t_3 + t_3 t_1 = \frac{1}{t_2 t_3} + \frac{1}{t_1 t_3} + \frac{1}{t_1 t_2}$$

$$\Rightarrow \tan\frac{\alpha}{2}\tan\frac{\beta}{2} + \tan\frac{\beta}{2}\tan\frac{\gamma}{2} + \tan\frac{\alpha}{2}\tan\frac{\gamma}{2} = \cot\frac{\beta}{2}\cot\frac{\gamma}{2} + \cot\frac{\alpha}{2}\cot\frac{\gamma}{2} + \cot\frac{\alpha}{2}\cot\frac{\beta}{2}$$

$$\Rightarrow \sum\left(\tan\frac{\alpha}{2}\tan\frac{\beta}{2} - \cot\frac{\alpha}{2}\cot\frac{\beta}{2}\right) = 0$$

$$\Rightarrow \qquad \Sigma\left(\frac{\sin^2\frac{\alpha}{2}\sin^2\frac{\beta}{2} - \cos^2\frac{\alpha}{2}\cos^2\frac{\beta}{2}}{\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}\sin\frac{\beta}{2}\cos\frac{\beta}{2}}\right) = 0$$

$$\Rightarrow \sum -4\left(\frac{\left(\cos\frac{\alpha}{2}\cos\frac{\beta}{2} + \sin\frac{\alpha}{2}\sin\frac{\beta}{2}\right)\left(\cos\frac{\alpha}{2}\cos\frac{\beta}{2} - \sin\frac{\alpha}{2}\sin\frac{\beta}{2}\right)}{\sin\alpha\sin\beta}\right) = 0$$

$$\Rightarrow \frac{\sum -4\cos\frac{(\alpha + b)}{2}\cos\frac{(\alpha - \beta)}{2}}{\sin\alpha\sin\beta} = 0$$

$$\Rightarrow \frac{\sum 2(\cos\alpha + \cos\beta)}{\sin\alpha\sin\beta} = 0 \Rightarrow \frac{\sum 2\sin\gamma(\cos\theta + \cos\beta)}{\sin\alpha\sin\beta} = 0$$

$\Rightarrow \sin\gamma\,(\cos\alpha + \cos\alpha)\sin\beta\,(\cos\alpha + \cos\gamma) + \sin\alpha\,(\cos\beta + \cos\gamma) = 0$
$\Rightarrow \sin(\alpha + \beta) + \sin(\beta + \gamma) + \sin(\gamma + \alpha) = 0.$
Note : You can remember this as a property.

Illustration 22: Find the locus of the points of the intersection of tangents to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which make an angle θ.

Solution : Given ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \text{.....(1)}$$

Equation of any tangent to ellipse (1) in terms of slope (m) is

$$y = mx + \sqrt{(a^2m^2 + b^2)}$$

Since its passes through $P(\alpha, \beta)$ then

$$\beta = m\alpha + \sqrt{(a^2m^2 + b^2)} \qquad \Rightarrow (\beta - m\alpha) = \sqrt{(a^2m^2 + b^2)}$$

$\Rightarrow (\beta - m\alpha)^2 = a^2m^2 + b^2$
$\Rightarrow m^2(a^2 - \alpha^2) + 2\alpha\beta m + (b^2 - \beta^2) = 0$ .....(2)
(2) being a quadratic equation in m.
Let roots of equation (2) be m_1 and m_2 then

$$\therefore\ m_1 + m_2 = -\frac{2\alpha\beta}{(a^2 - \alpha^2)}, m_1m_2 = \frac{b^2 - \beta^2}{a^2 - \alpha^2}$$

$$\therefore\ (m_1 - m_2) = \sqrt{(m_1 + m_2)^2 - 4m_1m_2}$$

$$= \sqrt{\frac{4\alpha^2\beta^2}{(a^2-\alpha^2)^2} - \frac{4(b^2-\beta^2)}{(a^2-\alpha^2)}}$$

$$= \sqrt{\frac{4\alpha^2\beta^2 - 4(b^2-\beta^2)(a^2-\alpha^2)}{(a^2-\alpha^2)^2}}$$

$$= \sqrt{\frac{4\{\alpha^2\beta^2 - a^2b^2 + b^2\alpha^2 + a^2\beta^2 - \alpha^2\beta^2\}}{(a^2-\alpha^2)^2}}$$

$$= \frac{2}{|(a^2-\alpha^2)|}\sqrt{(a^2\beta^2 + b^2\alpha^2 - a^2b^2)}$$

$\because$ θ be the angle between these two tangents, then

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$= \left|\frac{\frac{2}{(a^2-\alpha^2)}\sqrt{(a^2\beta^2 + b^2\alpha^2 - a^2b^2)}}{\left(1 + \frac{b^2-\beta^2}{a^2-\alpha^2}\right)}\right|$$

$$\tan\theta = \left|\frac{2\sqrt{(a^2\beta^2 + b^2\alpha^2 - a^2b^2)}}{a^2 + b^2 - \alpha^2 - \beta^2}\right|$$

or $(a^2 + b^2 - \alpha^2 - \beta^2)^2 \tan^2\theta = 4(a^2\beta^2 - b^2\alpha^2 - a^2b^2)$

$\Rightarrow (\alpha^2 + \beta^2 - a^2 - b^2)^2 \tan^2\theta = 4(b^2\alpha^2 + a^2\beta^2 - a^2b^2)$

$\therefore$ Locus of $P(\alpha, \beta)$ is

$(x^2 + y^2 - a^2 - b^2)^2 \tan^2\theta = 4(b^2x^2 + a^2y^2 - a^2b^2)$

Illustration 23: Prove that the chord of contact of tangents drawn from the points (h, k) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ will subtends a right at the centre, if $\frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{a^2} + \frac{1}{b^2}$. Also, find the locus of (h, k).

Solution : The equation of chord of contact of tangents drawn from P(h, k) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1 \quad ....(1)$$

The equation of the straight lines *CA* and *CB* is obtained by making homogeneous ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with the help of (1)

$$\therefore \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = \left(\frac{hx}{a^2} + \frac{ky}{b^2}\right)^2$$

$$\Rightarrow \left(\frac{k^2}{a^4} - \frac{1}{a^2}\right)x^2 + \left(\frac{k^2}{b^4} - \frac{1}{b^2}\right)y^2 + \frac{2hk}{a^2b^2}xy = 0$$

But given $\angle ACB = 90^o$

$\therefore$ co-efficient of x^2 + co-efficient of $y^2 = 0$

$$\Rightarrow \frac{h^2}{a^4}-\frac{1}{a^2}+\frac{k^2}{b^4}-\frac{1}{b^2}=0$$

$$\Rightarrow \frac{h^2}{a^4}+\frac{k^2}{b^4}=\frac{1}{a^2}+\frac{1}{b^2}$$

Hence locus of (h, k) is

$$\frac{x^2}{a^4}+\frac{y^2}{b^4}=\frac{1}{a^2}+\frac{1}{b^2}$$

Illustration 24: Show that the locus of the middle points of chord of an ellipse, which pass through a fixed point, is another ellipse.

Solution : Let $P(x_1, y_1)$ be the middle point of any chord AB of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ then equation of chord AB is $T = S_1$.

$$\Rightarrow \frac{xx_1}{a^2}+\frac{yy_1}{b^2}-1=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}-1$$

$$\Rightarrow \frac{xx_1}{a^2}+\frac{yy_1}{b^2}=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}$$

But it passes through a fixed point $Q(h, k)$, its coordinates must satisfy equation (i),

$$\therefore \frac{hx_1}{a^2}+\frac{ky_1}{b^2}=\frac{x_1^2}{a^2}+\frac{y_1^2}{b^2}$$

this can be re-written as

$$\frac{\left(x_1-\frac{h}{2}\right)^2}{a^2}+\frac{\left(y_1-\frac{k}{2}\right)^2}{b^2}=\frac{1}{4}\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)$$

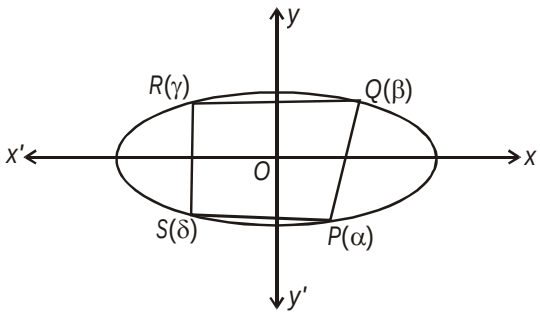
Hence locus of $P(x_1, y_1)$ is

$$\frac{\left(x-\frac{h}{2}\right)^2}{a^2}+\frac{\left(y-\frac{k}{2}\right)^2}{b^2}=\frac{1}{4}\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)$$

which is obviously an ellipse with centre at $\left(\frac{h}{2},\frac{k}{2}\right)$ and axes parallel to coordinate axes.

Illustration 25: If three of the sides of a quadrillateral inscribed in an ellipse are parallel respectively to three given straight lines. Show that fourth side will also be parallel to a fixed straight line.

Solution : Let $PQRS$ be a quadrilateral inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.



Let *PQ*, *QR* and *RS* be the three sides parallel to the given lines.
Equation of *PQ* is

$$\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right)+\frac{y}{\beta}\sin\left(\frac{a+\beta}{2}\right)=\cos\left(\frac{\alpha-\beta}{2}\right)$$

Its slope is $-\frac{b}{a}\cot\left(\frac{\alpha+\beta}{2}\right)$ which is constant by hypothesis

$\therefore\ \alpha + \beta$ = constant = $2\lambda_1$ (say)
Similary $\beta + \gamma$ = constant = $2\lambda_2$ (say)
and $\gamma + \delta$ = constant = $2\lambda_3$ (say)
Now the equation of *SP* is

$$\frac{x}{a}\cos\left(\frac{\alpha+\delta}{2}\right)+\frac{y}{b}\sin\left(\frac{\alpha+\delta}{2}\right)=\cos\left(\frac{\alpha-\delta}{2}\right)$$

Its slope $m=-\frac{b}{a}\cot\left(\frac{\alpha+\delta}{2}\right)$

But $\alpha + \delta = (\alpha + \beta) - (\beta + \gamma) + (\gamma + \delta)$
$= 2\lambda_1 - 2\lambda_2 + 2\lambda_3$ = constant
Hence, the slope of the fourth side *PS* is constant. Hence the fourth side is also parallel to a fixed straight line.

Illustration 26: Find the locus of the mid-point of a normal chord of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

Solution : Let (h, k) be the mid-point of a normal chord of the given ellipse. Then, its equation is

$$\frac{hx}{a^2}+\frac{ky}{b^2}-1=\frac{h^2}{a^2}+\frac{k^2}{b^2}-1 \text{ [Using } T = S\text{]}$$

or $\frac{hx}{a^2}+\frac{ky}{b^2}=\frac{h^2}{a^2}+\frac{k^2}{b^2}$ .....(i)

If (i) is a normal chord, then it must be of the form
$ax \sec\theta - by \operatorname{cosec}\theta = a^2 - b^2$ .....(ii)

$$\therefore \frac{h}{a^3\sec\theta}=\frac{h}{-b^3\operatorname{cosec}\theta}=\frac{\frac{h^2}{a^2}+\frac{k^2}{b^2}}{a^2-b^2}$$

$$\Rightarrow \cos\theta=\frac{a^3}{h(a^2-b^2)}\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right),\sin\theta=\frac{-b^3}{k(a^2-b^2)}\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)$$

Eliminating θ from the above relations, we get

$$\frac{a^6}{h^2(a^2-b^2)}\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)+\frac{b^6}{k^2(a^2-b^2)^2}\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)=1$$

$$\Rightarrow\left(\frac{a^6}{h^2}+\frac{b^6}{k^2}\right)\left(\frac{h^2}{a^2}+\frac{k^2}{b^2}\right)^2=a^2-b^2$$

Hence, the locus of (h, k)

$$\Rightarrow\left(\frac{a^6}{x^2}+\frac{b^6}{y^2}\right)\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)^2=(a^2-b^2)^2$$

Illustration 27: A ray emanating from the point (–3, 0) is incident on the ellipse $16x^2 + 25y^2 = 400$ at the point P with ordinate 4. Find the equation of the reflected ray after first reflection.

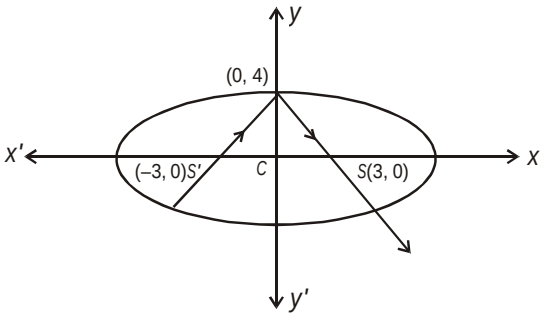
Solution : For point P, y-co-ordinate = 4

$\because$ given ellipse is

$16x^2 + 25y^2 = 400$

$16x^2 + 25(4)^2 = 400$

co-ordinate of P is (0, 4)



$$e^2=1-\frac{16}{25}=\frac{9}{25}$$

$$e=\frac{3}{5}$$

foci $(\pm ae, 0)$ *i.e.*, $(\pm 3, 0)$

Equation of reflected ray (*i.e.*, PS) is

$$\frac{x}{3}+\frac{y}{4}=1 \text{ or } 4x + 3y = 12.$$

Illustration 28: Let P be a point on the ellipse whose foci are S' and S. If normal at P meets the axes at G and G' then

(i) $S'G = e\,S'P$

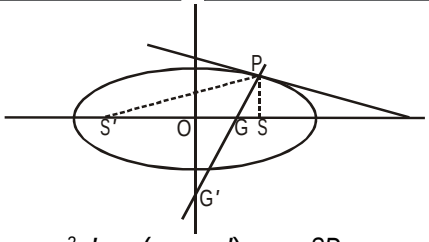
(ii) $S'G : SG = SP : SP$

(iii) PG bisects the angle $S'PS$

Solution : If P be (x', y') then equation of PGG' is $ax \sec\phi = by \operatorname{cosec}\phi = c^2 - b^2$

On putting $y = 0$ we get $OG = \dfrac{a^2-b^2}{a\sec\phi}=\dfrac{a^2-b^2}{a^2}a\cos\phi=-e^2x^1$

Now $S'G = S'O + OG = ae + e^2x' = e(a + ex') = e(S'P)$



Also $SG = OS - OG = ae - e^2x' = e(a - ex') = e \cdot SP$

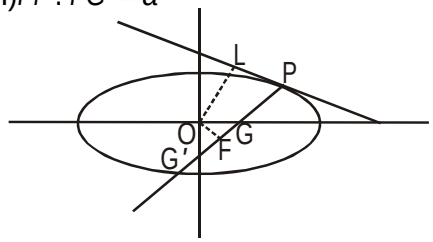
Hence $S'G : SG = S'P : SO$ (*)

The last relation clearly implies $\angle S'PG = \angle GPS$

Thus normal at any point P bisects the angle between $S'P$ and SP where S' and S are two foci.

Illustration 29: If the normal at any point P meets the major axes in G and G' and if OF be the perpendicular upon this normal then prove that

(i) $PF \cdot PG = b^2$ (ii) $PF \cdot PG' = a^2$



Solution : Let P be $(a\cos\phi, b\sin\varphi)$ then equation of tanget at P is $\frac{x}{a}\cos\varphi + \frac{y}{b}\sin\varphi = 1$.

Now $PF = OL$ = distance of origin from the tangent

$$= \frac{1}{\sqrt{\frac{\cos^2\varphi}{a^2} + \frac{\sin^2\varphi}{b^2}}} = \frac{1}{\sqrt{a^2\sin\varphi + b^2\cos^2\varphi}}$$

Also the normal at P ia $ax \sec \frac{\tan\frac{\varphi_1}{2}}{\tan\frac{\varphi_2}{2}} = \left(\frac{e-1}{e+2}\right)^2$ – by cosec $\varphi = a^2 - b^2 \Rightarrow OG = \frac{a^2-b^2}{a}\cos\varphi$

$\Rightarrow G$ is $\left(\frac{a^2-b^2}{a}\cos\varphi, 0\right)$

$$\Rightarrow PG^2 = \left(a\cos\varphi - \frac{a^2-b^2}{a}\cos\varphi\right)^2 + b^2\sin^2\varphi = \frac{b^2}{a^2}(a^2\sin^2\varphi + b^2\cos^2\varphi)$$

$\Rightarrow PG = \frac{a}{b}\sqrt{b^2\cos^2\varphi + a^2\sin^2\varphi}$. Whence it follows that $PF \cdot PG = b^2$

Again on putting $x = 0$ in the equation of the normal we get G' as $\left(0, \frac{a^2-b^2}{a}\sin\varphi\right)$

$$\Rightarrow PG'^2 = a^2\cos^2\varphi + \left(b\sin\varphi + \frac{a^2-b^2}{a}\sin\varphi\right)^2 = \frac{a^2}{b^2}(b^2\cos^2\varphi + a^2\sin^2\varphi)$$

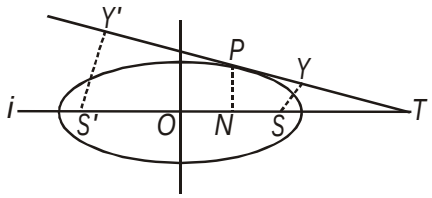
$\Rightarrow PG' = \frac{a}{b}\sqrt{b^2\cos^2\varphi + a^2\sin^2\varphi}$ whence it follows that $PF \cdot PG' = a^2$.

Illustration 30: If *SY* and *S' Y'* be the perpendicular from the foci *S* and *S'* upon the tangent at any point *P* of the ellipse then show that
(a)*Y* and *Y'* lie on auxiliary circle
(b)*SY* . *S' Y'* = b^2
(c)*OY* and *S'P* are parallel.

Solution :

Let *P* be ($a\cos\varphi$, $b\sin\varphi$)

then equation of tangent at *P* is $\frac{x}{a}\cos\varphi+\frac{y}{b}\sin\varphi=1$



Slope of tangent = $-\frac{\frac{\cos\varphi}{a}}{\frac{\sin\varphi}{b}}(x-ae)$

Equation of *SY* must be $y-0=-\frac{\frac{\sin\varphi}{b}}{\frac{\cos\varphi}{a}}(x-ae)$

Rewriting equation (i) and (ii) as
$bx\cos\varphi+ay\sin\varphi-ab=0$ and $ax\sin\varphi-by\cos\varphi-a^2e\sin\varphi=0$
On solving for *x* and *y* we get the co-ordinates of *Y* as

$$x=\frac{a(a^2e\sin^2\varphi+b^2\cos\varphi)}{a^2\sin^2\varphi+b^2\cos^2\varphi},y=\frac{a^2b\sin\varphi+(1-e\cos\varphi)}{a^2\sin^2\varphi+b^2\cos^2\varphi}$$

The *x*-coordinate can be simplified to

$$x=\frac{a^3\left(e\sin^2\varphi+\frac{b^2}{a^2}\cos\varphi\right)}{a^2\left(\sin^2\varphi+\frac{b^2}{a^2}\cos^2\varphi\right)}$$

$$=\frac{a(e-e\cos^2\varphi+(1-e^2)\cos\varphi)}{a^2(1-\cos^2\varphi+\cos^2\varphi-e^2\cos^2\varphi)}\qquad\left(\because\frac{b^2}{a^2}=1-e^2\right)$$

$$=\frac{a(1-e\cos\varphi)(e+\cos\varphi)}{(1-e^2\cos^2\varphi)}=\frac{a(e+\cos\varphi)}{1+e\cos\varphi}\text{ (i)}$$

They *y*-coordinate can be simplified to

$$y=\frac{a^2b\sin\varphi(1-e\cos\varphi)}{a^2\left(\sin^2\varphi+\frac{b^2}{a^2}\cos^2\varphi\right)}=\frac{b\sin\varphi}{1-e\cos\varphi}\text{ (ii)}$$

Note carefully that co-ordinates of Y' will be obtained by simply replacing e by $-e$.*i.e.* Y' must $x^2 + y^2 = a^2$. Indeed (for Y)

$$x^2 + y^2 = \frac{a^2(e+\cos\varphi)^2 + b^2\sin^2\varphi}{(1+e\cos\varphi)^2}$$

$$= \frac{a^2e^2 + a^2\cos^2\varphi + 2a^2e\cos\varphi + a^2(1-e^2)\sin^2\varphi}{(1+e\cos\varphi)^2}$$

$$= \frac{a^2[e^2\cos^2\varphi + 1 + 2e\cos\varphi]}{(1+e\cos\varphi)^2} = a^2$$

As similar proof can be given on show that Y' also satisfies $x^2 + y^2 = a^2$.

(b)Now $SY = \perp$ distance of S from tangent $= \dfrac{|e\cos\varphi - 1|}{\sqrt{\dfrac{\cos^2\varphi}{a^2} + \dfrac{\sin^2\varphi}{b^2}}}$

$S'Y' = \dfrac{|-e\cos\varphi - 1|}{\sqrt{\dfrac{\cos^2\varphi}{a^2} + \dfrac{\sin^2\varphi}{b^2}}}$ (replacing e by $-e$)

$$\Rightarrow SY\,S'Y' = \frac{1 - e^2\cos^2\varphi}{\dfrac{\cos^2\varphi}{a^2} + \dfrac{\sin^2\varphi}{b^2}} = \frac{a^2b^2(1-e^2\cos^2\varphi)}{b^2\cos^2\varphi + a^2\sin^2\varphi}$$

$$= \frac{b^2(1-e^2\cos^2\varphi)}{\dfrac{b^2}{a^2}\cos^2\varphi + \sin^2\varphi} \qquad \left(\because \frac{b^2}{a^2} = 1 - e^2\right)$$

(c)We have slope $OY = \dfrac{b\sin\varphi}{a(e+\cos\varphi)}$

Slope $S'P = \dfrac{b\sin\varphi - 0}{a\cos\varphi + ae} = \dfrac{b\sin\varphi}{a(e+\cos\varphi)}$

= slope $OY \Rightarrow OY \parallel S'P$.

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic : 1 Ellipse

Q.1 The eccentricity of the ellipse, which passes through the points (2, –2) and (–3, 1) is

(a) $\frac{1}{\sqrt{5}}$ (b) $\sqrt{\frac{2}{5}}$ (c) $\sqrt{\frac{3}{5}}$ (d) $\sqrt{\frac{4}{5}}$

Q.2 A rod given length moves such that its extremities lie on two fixed perpendicular lines. Any point on the rod describes a

(a) Circle (b) Parabola
(c) Ellipse (d) Straight line

Q.3 The centre of the ellipse $4x^2 + 9y^2 + 16x - 18y - 11 = 0$ is

(a) (–2, –1) (b) (–2, 1)
(c) (2, –1) (d) (2, 1)

Q.4 The length of the latus rectum of the ellipse $2x^2 + 3y^2 - 4x - 6y - 13 = 0$ is

(a) 5 (b) 4
(c) 8 (d) 12

Q.5 The co-ordinates of foci of an ellipse $3x^2 + 4y^2 + 12x + 16y - 8 = 0$ is

(a) $(\pm\sqrt{3} - 2, -2)$ (b) (–2, –2)

(c) $(2 - \sqrt{3}, \pm 2)$ (d) $(2 + \sqrt{3}, -2)$

Q.6 If the line joining foci subtends an angle of 90° at an extremity of minor axis, then the eccentricity e is

(a) $\frac{1}{\sqrt{6}}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{2\sqrt{2}}$

Q.7 In an ellipse the distance between the foci is 8 and the distance between the directrices is 25, then the ratio of the length of major and minor axis is

(a) $\frac{5}{\sqrt{17}}$ (b) $\frac{3}{\sqrt{17}}$ (c) $\frac{4}{\sqrt{17}}$ (d) $\frac{6}{\sqrt{17}}$

Q.8 The locus of the foot of perpendicular from the centre on any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

(a) A circle
(b) A pair of straight lines
(c) Another ellipse
(d) Curve of degree 4

Q.9 If the equation $\frac{x^2}{2-r} + \frac{y^2}{r-5} + 1 = 0$ represents an ellipse, then r lie in

(a) $(-\infty, 2) \cup (5, \infty)$ (b) (2, 5)
(c) (5, 10) (d) $(-\infty, 1)$

Q.10 If the focal distance of an end of minor axis of any ellipse, (whose axes along the x and y axes respectively is k and the distance between the foci is $2h$. Then the equation of the ellipse is

(a) $\frac{x^2}{h^2} + \frac{y^2}{k^2} = 1$ (b) $\frac{x^2}{k^2} + \frac{y^2}{k^2 - h^2} = 1$

(c) $x^2 + \frac{y^2}{k^2} = 1$ (d) $\frac{x^2}{h^2} + y^2 = 1$

Q.11 Distance between the foci of the curve represented by the equation $x = 3 + 4\cos\theta$, $y = 2 + 3\sin\theta$, is

(a) $3\sqrt{7}$ (b) $2\sqrt{7}$
(c) $\sqrt{7}$ (d) $\sqrt{7}/2$

Topic : 2 Parametric form and auxiliary

Q.12 The parametric coordinates of a point on the ellipse, whose foci are (–3, 0) and (9, 0) and eccentricity 1/3, are

(a) $(-3 + 9\cos\theta, 9\sin\theta)$
(b) $(4 - 3\cos\theta, 4 + 9\sin\theta)$
(c) $(3 + 18\cos\theta, 4 + 9\sin\theta)$
(d) $(3 + 18\cos\theta, 12\sqrt{2}\sin\theta)$

Q.13 The sum of the squares of the perpendicular on any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ from two points on the minor axis each at the distance $\sqrt{a^2 - b^2}$ from the centre is

(a) a^2 (b) b^2 (c) $2a^2$ (d) $2b^2$

Q.14 The locus of the foot of the perpendicular from any focus upon any tangent to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

(a) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (b) $x^2 + y^2 = a^2 + b^2$

(c) $x^2 + y^2 = a^2$ (d) None of these

Q.15 If the tangent at a point ($a\cos\alpha, b\sin\alpha$) on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ meets the auxiliary circle in two points and the chord joining them subtends a right angle at the centre, then the eccentricity of the ellipse is given byby

(a) $1+\cos^2\alpha$ (b) $\frac{1}{\sqrt{1+\sin^2\alpha}}$

(c) $\frac{1}{\sqrt{1+\cos^2\alpha}}$ (d) $1+\sin^2\alpha$

Q.16 If CD is a diameter of an ellipse $x^2 + 9y^2 = 25$ and the eccentric angle of C is $\pi/6$ then the eccentric angle of D is

(a) $\frac{5\pi}{6}$ (b) $-\frac{5\pi}{6}$

(c) $-\frac{2\pi}{3}$ (d) $\frac{2\pi}{3}$

Q.17 Number of points on the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ from which pair of perpendicular tangents are drawn to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$, is

(a)2 (b)3
(c)4 (d)1

Q.18 The area of the region bounded by the ellipse $\frac{x^2}{16}+\frac{y^2}{9}\le 1$ and the circle $x^2 + y^2 \le 9$ is

(a)12π (b)3π (c)9π (d)6π

Q.19 The radius of the circle passing through the point of intersection of ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2 - y^2 = 0$ is

(a)$\frac{12}{5}$ (b)$\frac{12\sqrt{2}}{5}$ (c)$\frac{6\sqrt{2}}{5}$ (d)$\frac{9\sqrt{2}}{5}$

Q.20 Area of the region bounded by the curve $\left\{(x,y):\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1\le\frac{x}{a}+\frac{y}{b}\right\}$ is

(a) $\left(\frac{\pi}{4}-\frac{1}{2}\right)ab$ (b) $\left(\frac{\pi}{4}+\frac{1}{2}\right)ab$

(c) $\left(\frac{\pi}{4}-\frac{1}{3}\right)ab$ (d) $\frac{\pi}{4}ab$

Q.21 The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the co-ordinate axes which is turn in inscribed in another ellipse that passes through the (4, 0). Then the equation of the ellipse is

(a) $x^2 + 12y^2 = 16$ (b) $4x^2 + 48y^2 = 48$
(c) $4x^2 + 64y^2 = 48$ (d) $x^2 + 16y^2 = 16$

Q.22 P is any variable point on the ellipse $4x^2 + 9y^2 = 36$ and F_1, F_2 are its foci maximum area of ΔPF_1F_2 is (e is eccentricity of ellipse)

(a)9e (b)4e (c)6e (d)10e

Q.23 The curve described parametrically by $x = t^2 + t + 1$, $y = t^2 – t + 1$ represents

(a) A pair of straight lines
(b) An ellipse
(c) A hyperbola
(d) A parabola

Topic: 3 Director Circle of an Ellipse

Q.24 The radius of the circle passing through the foci of $\frac{x^2}{16}+\frac{y^2}{9}=1$, and having centre (0, 3) is

(a) 4 (b) 3
(c) $\sqrt{12}$ (d) 7/2

Q.25 The locus of the feet of the perpendiculars drawn from the centre to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ on any tangent to it is

(a) $(x^2 + y^2)^2 = 9x^2 + 4y^2$
(b) $(x^2 + y^2)^2 = 4x^2 + 9y^2$
(c) $(x^2 + y^2)^2 = 3x^2 + 2y^2$
(d) $(x^2 + y^2)^2 = 2x^2 + 3y^2$

Q.26 The minimum area of the triangle formed by the tangent to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and the co-ordinte axes is

(a)16 (b)9
(c)12 (d)144

Q.27 An ellipse has eccentricity 1/2, and one focus at the point $P\left(\frac{1}{2},1\right)$. Its one directrix is the common tangent, nearer to the point P, to $x^2 + y^2 = 1$ and the hyperbola $x^2 – y^2 = 1$. The equation of the ellipse is

(a) $\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}}+\frac{(y-1)^2}{\frac{1}{12}}=1$

(b) $\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{8}}+\frac{(y-1)^2}{\frac{1}{12}}=1$

(c) $\frac{\left(x-\frac{1}{3}\right)^2}{\frac{1}{9}}+\frac{(y-1)^2}{\frac{1}{8}}=1$

(d) None of these

Topic : 4 Tangent and Normal of an Ellipse

Q.28 The line $2x + y = 3$ cuts the ellipse $4x^2 + y^2 = 5$ at P and Q. If θ be the angle between the normals at these points, then $\tan\theta$ is equal to
(a)1/2 (b)3/4
(c)3/5 (d)5

Q.29 The number of values of c, such that $y = 4x + c$ touches $\frac{x^2}{4} + y^2 = 1$, is
(a)0 (b)1
(c)2 (d)Infinite

Q.30 Two common tangents to $x^2 + y^2 = 2a^2$ and $y^2 = 8ax$ are
(a) $x = \pm(y + 2a)$ (b) $x = \pm(y + a)$
(c) $y = \pm(x + 2a)$ (d) $y = \pm(x + a)$

Q.31 The line $\ell x + my + n = 0$ is a normal to $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, provided

(a) $\frac{a^2}{m^2}+\frac{b^2}{\ell^2}=\frac{(a^2-b^2)^2}{n^2}$

(b) $\frac{a^2}{\ell^2}+\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$

(c) $\frac{a^2}{\ell^2}-\frac{b^2}{m^2}=\frac{(a^2-b^2)^2}{n^2}$

(d) None of these

Q.32 From a point P, two tangents are drawn to the parabola $y^2 = 4ax$. If the slope of one tangent is twice the slope of other, the locus of P is
(a) Circle (b) Straight line
(c) Parabola (d) Ellipse

Q.33 Tangents are drawn to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ at ends of latus rectum. The area of quadrilateral so formed is
(a) 27 (b) 27/2 (c) 27/4 (d) 27/55

Q.34 The tangent at any point on the ellipse $16x^2 + 25y^2 = 400$ meets the tangents at the ends of the major axis at T_1 and T_2. The circle on T_1T_2 as diameter passes through
(a) (3, 0) (b) (0, 0) (c) (0, 3) (d) (4, 0)

Q.35 If $\theta_1, \theta_2, \theta_3, \theta_4$ be eccentric angles of the four concyclic points of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, then $\theta_1 + \theta_2 + \theta_3 + \theta_4$ is (where $n \in I$)

(a) $(2n + 1)\frac{\pi}{2}$ (b) $(2n + 1)\pi$
(c) $2n\pi$ (d) $n\pi$

Q.36 Let the eccentric angles of three point A, B and C or the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ are θ_1, $\frac{\pi}{2}$ + θ_1, $\pi + \theta_1$. A circle through A, B and C cuts the ellipse again at D. Then the eccentric angle of D is

(a) $\pi - 3\theta_1$ (b) $\frac{3\pi}{2} - 3\theta_1$

(c) $\frac{\pi}{2} - 3\theta_1$ (d) $3\theta_1 - \frac{\pi}{2}$

Q.37 The number of values of k such that $y = 4x + k$ touches the curve $\frac{x^2}{4} + y^2 = 1$ is
(a)0 (b)1
(c)2 (d)4

Q.38 The number of common tangents to the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and the circle $x^2 + y^2 = 4$ is
(a) 2 (b) 1
(c) 0 (d) 4

Q.39 A common tangent to $9x^2 + 16y^2 = 144$, $y^2 = x - 4$ and $x^2 + y^2 - 12x + 32 = 0$ is
(a) $y = 3$ (b) $y = -3$ (c) $x = 4$ (d) $x = -4$

Q.40 The equation of a common tangent to the curves $y^2 = 8x$ and $xy = -1$ is
(a) $y = x + 2$ (b) $y = 2x + 1$
(c) $y = \frac{x}{2} + 4$ (d) $y = 3x + \frac{2}{3}$

Topic : 5 Different form of Chord of an Ellipse

Q.41 The locus of the mid points of the portion of the tangents to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intercepted between the axes is

(a) $\frac{a^2}{x^2}+\frac{b^2}{y^2}=4$ (b) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=4$

(c) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=16$ (d) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=25$

Q.42 The equation of the chord of the ellipse $x^2 + 4y^2 = 4$ having the middle point at $\left(-2, \frac{1}{2}\right)$ is

(a) $2x - 2y + 5 = 0$ (b) $x - 2y = 0$
(c) $3x - 2y + 4 = 0$ (d) $2x - 2y + 5 = 0$

Q.43 If tangents are drawn to the ellipse $2x^2 + 3y^2 = 6$, then the locus of the mid-point of the intercept made by the tangents between the co-ordinate axes is

(a) $\frac{1}{4x^2}+\frac{1}{2y^2}=1$ (b) $\frac{3}{4x^2}+\frac{1}{2y^2}=1$

(c) $\frac{1}{x^2}+\frac{1}{y^2}=1$ (d) $\frac{1}{x^2}+\frac{3}{4y^2}=1$

Topic : 6 Different form of Chord of an Ellipse

Q.44 The locus of the middle points of the chords of hyperbola $\frac{x^2}{9}-\frac{y^2}{b^2}=1$, which pass through the fixed point (1, 2) is a hyperbola whose eccentricity is

(a) $\frac{3}{2}$ (b) $\frac{\sqrt{7}}{2}$

(c) $\frac{\sqrt{13}}{2}$ (d) $\frac{\sqrt{15}}{2}$

Q.45 If angle between the asymptotes of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ is

(a) $\tan^{-1}\left(\frac{3}{4}\right)$ (b) $2\tan^{-1}\left(\frac{3}{4}\right)$

(c) $\tan^{-1}\left(\frac{4}{3}\right)$ (d) $2\tan^{-1}\left(\frac{4}{3}\right)$

Q.46 Find the length of the chord of the hyperbola $x^2 - y^2 = 2a^2$ if (2a, a) is the mid-point of the chord

(a) $\frac{20\sqrt{2}}{3}a$ (b) $\frac{10\sqrt{2}}{3}a$

(c) $\frac{20}{3}a$ (d) None of these

Section-B
(Assertion-Reason)

Q.1 **Statement–1 :** Two tangents are drawn from the point $(\sqrt{24}, 1)$ to $\frac{x^2}{16}+\frac{y^2}{9}=1$ then they must be perpendicular.
Statement–2 : The equation of the director circle to $\frac{x^2}{16}+\frac{y^2}{9}=1$ is given by $x^2 + y^2 = 25$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.2 Let C_1 and C_2 be two given circles and C be a moving circle which touches both the circles.
Statement–1 : The locus of centre of circle C must be an ellipse.
Statement–2 : The locus of a moving point whose sum of distances from two given points is always constant is called an ellipse.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.3 Let $S_1 : x^2 + y^2 = 25$ and $S_2 : x^2 + y^2 - 2x - 2y - 14 = 0$ be two circles.
Statement–1 : S_1 and S_2 have exactly two common tangents.
Statement–2 : If two circles touches each other internally they have one common tangent.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** The points on the auxilary circle corresponding to P and subtends a right angle at the centre.

Statement–2 : If the eccentric angles of the ends, *P* and *D* of a pair of conjugate diameters be ϕ and ϕ' differ by a right angle.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement–1 :** If *P* and *D* be the ends of conjugate diameter, then the locus of intersection of tangents at *P* and *D* is an elipse.

Statement–2 : If *P* and *D* be the ends of conjugate diameter, then the locus of intersection of tangents *P* and *D* is an ellipse.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.6 **Statement–1 :** The tangent at the extremity of any diameter is parallel to the chords which it bisects.

Statement–2 : Chords of the $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ always touches the concentric and coaxial ellipse $\frac{x^2}{\alpha^2}+\frac{y^2}{\beta^2}=1$, then the locus of their poles is an ellipse.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Year Question)

Q.1 The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having centre at (0, 8) is

(a) $x^2+y^2-6y-7=0$

(b) $x^2+y^2-6y+7=0$

(c) $x^2+y^2-6y-5=0$

(d) $x^2+y^2-6y+5=0$

Q.2 The ellipse $E_1:\frac{x^2}{9}+\frac{y^2}{4}=1$ is inscribed in a reactangle R whose sides are parallel to the coordinate axes. Another ellipse E_2 passing through the point (0, 4) circumscribes the rectangle R, The eccentricity of the ellipse E_2 is

(a) $\frac{\sqrt{2}}{2}$ (b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$ (d) $\frac{3}{2}$

Q.3 If $P=(x,y), F_1=(3,0), F_2=(-3,0)$ and $16x^2+25y^2=400$, then PF_1+PF_2 equals

(a) 8 (b) 6

(c) 10 (d) 12

Q.4 The eccentricity of an ellipse whose centre is at the origin is 1/2. If one of its directrices is x = -4, then the equation of the normal to it at $\left(1+,\frac{3}{2}\right)$ is

(a) $2y-x=2$ (b) $4x-2y=1$

(c) $4x+2y=7$ (d) $x+2y=4$

Q.5 The area (in sq units) of the quadrilateral formed by the tangents at the end points of the latusrectum recta to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ is

(a) $\frac{27}{4}$ (b) 18

(c) $\frac{27}{2}$ (d) 27

Q.6 The locus of the foot of perpendicualr drawn from the centre of the ellipse $x^2+3y^2=6$ on any tangent to it is

(a) $(x^2-y^2)^2=6x^2+2y^2$

(b) $(x^2-y^2)^2=6x^2-2y^2$

(c) $(x^2+y^2)^2=6x^2+2y^2$

(d) $(x^2+y^2)^2=6x^2-2y^2$

Q.7 The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the X-axis at Q. M is the mid-point of the line segment PQ, then the locus of M intersects the latusrectum of the given ellipse at the points

(a) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$ (b) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$

(c) $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$ (d) $\left(\pm 2\sqrt{3}, \pm\frac{4\sqrt{3}}{7}\right)$

Q.8 The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse Then, the area (insqunits) of the triangle with vertices at A, M and the origin O is

(a) $\frac{31}{10}$ (b) $\frac{29}{10}$

(c) $\frac{21}{10}$ (d) $\frac{27}{10}$

Q.9 Tangents are drawn to the ellipse $x^2 + 2y^2 = 2$, then the locus of the mid-point of the intercept made by the tangents between the coordinate axes is

(a) $\frac{1}{2x^2} + \frac{1}{4y^2} = 1$ (b) $\frac{1}{4x^2} + \frac{1}{2y^2} = 1$

(c) $\frac{x^2}{2} + \frac{y^2}{4} = 1$ (d) $\frac{x^2}{4} + \frac{y^2}{2} = 1$

Q.10 Tangent is drawn to ellipse $\frac{x^2}{27} + y^2 = 1$ at $\left(3\sqrt{3}\cos\theta, \sin\theta\right)$ (where, $\theta \in (0, \pi/2)$. Then, the value of θ such that the sum of intercepts on axes made by this tangent is minimum, is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$

(c) $\frac{\pi}{8}$ (d) $\frac{\pi}{4}$

Q.11 If $\alpha > 2b > 0$, then positive value of m for which $y = mx - b\sqrt{1+m^2}$ is a common tangent to $x^2 + y^2 = b^2$ and $(x-\alpha)^2 + y^2 = b^2$ is

(a) $\frac{2b}{\sqrt{a^2 - 4b^2}}$ (b) $\frac{\sqrt{a^2 - 4b^2}}{2b}$

(c) $\frac{2b}{a-2b}$ (d) $\frac{b}{a-2b}$

Q.12 The number of values of c such that the straight line $y = 4x + c$ touches the curve $\frac{x^2}{4} + y^2 = 1$ is

(a) 0 (b) 2 (c) 1 (d) ∞

Section-D
(Board pattern)

Q.1 Find the co-ordinates of the foci, the vertices, the length of major axis, the minor axis, the eccentricity and the length of the latus rectum of the ellipse, $x^2 + 4y^2 = 16$.

Q.2 Find the equation for the ellipse, whose vertices are (± 4, 0) and foci(± 3, 0).

Q.3 Find the eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} =$ 1, whose latus rectum is half of its minor axis.

Q.4 Find the equation of the ellipse passing through (4, 2) and having eccentricity 1/2.

Q.5 Find the eccentric angle of a point on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ whose distance from the centre of the ellipse is 2.

Q.6 Find the equation of the circle radius 3 units and whose centre is coincide with the focus of the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$.

Q.7 Find the equation of the ellipse whose foci coincide with focus of the parabola $y^2 = 12x$ and whose length of minor axis is 8.

Q.8 Find the location of the point (2, 3) with respect to the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$.

CHAPTER 4

HYPERBOLA

1. HYPERBOLA

Standard Equation of the Hyperbola

The locus of a point which moves in a plane such that the ratio of its distance from a fixed point and a fixed straight line is always constant and is greater then unity is called Hyperbola. The fixed point is called focus, the fixed straight line is called the directrix and the fixed ratio is called the eccentricity of the hyperbola.

A hyperbola is also the locus of a point which moves in a plane such that the difference of its distances from two fixed points (ae, 0) and ($-ae$, 0) remains always $2a$.

Let $P(\alpha, \beta)$ be a moving point. Then by hypothesis.

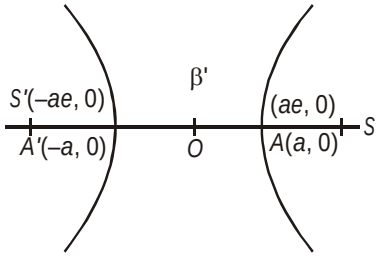
$$\sqrt{(\alpha+ae)^2+\beta^2}-\sqrt{(\alpha-ae)^2+\beta^2}=2a$$

$\Rightarrow$ $(\alpha + ae)^2 + \beta^2 = 4a^2 - 4a\sqrt{(\alpha-ae)^2+\beta^2} + (\alpha - ae)^2 + b^2$

$\Rightarrow$ $4ae\alpha - 4a^2 = -4a^2\sqrt{(\alpha-ae)^2+\beta^2}$

$\Rightarrow$ $e^2\alpha^2 + a^2 - 2ae\alpha = \alpha^2 + a^2e^2 - 2ae\alpha + \beta^2$

$\Rightarrow$ $(e^2 - 1)\alpha^2 - \beta^2 = a^2(e^2 - 1)$



$\Rightarrow$ $\frac{\alpha^2}{a^2}-\frac{\beta^2}{a^2(e^2-1)}=1$ $\Rightarrow \frac{\alpha^2}{a^2}-\frac{\beta^2}{b^2}=1$ $b^2 = a^2(e^2 - 1)$, say,

Locus of $P(\alpha, \beta)$ is

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

Thus $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, $b^2 = a^2(e^2 - 1)$ represents a hyperbola with following specifications.

Its centre is at O (0, 0), foci are S (ae, 0) and S' ($-ae$, 0), directrices are $x = \frac{a}{e}$ and $x = -\frac{a}{e}$, vertices are $A(a, 0)$ $A'(-a, 0)$ and AA' is called the transverse axis of the hyperbola.

The hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ does not meet y-axis. However we take two points B and B' on y-axis on opposite sides of the centre O of the hyperbola such that

$OB = OB' = b$

$B'B$ is called the conjugate axis of the parabola. The ends of the latera are

$\left(ae, \frac{b^2}{a}\right), \left(ae, -\frac{b^2}{a}\right), \left(-ae, \frac{b^2}{a}\right)$ and $\left(-ae, -\frac{b^2}{a}\right)$ and the length of latus rectum is $\frac{2b^2}{a}$. The equations of latera recta are $x = ae$ and $x = -ae$.

The difference of the focal distances of any point on a hyperbola is a constant and is equal to the length of the transverse axis of the hyperbola.

When the transverse and conjugate axes of one hyperbola coincide respectively with the conjugate and transverse axes of another hyperbola, then the hyperbolas are called conjugate, to one another.

The equation of the hyperbola conjugate to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$$

or $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Thus, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$

represents a hyperbola with the following specifications

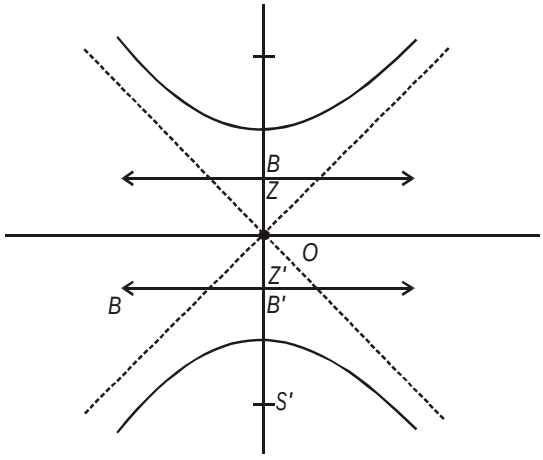
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$

where $a^2 = b^2(e^2 - 1)$

Centre O : (0, 0)

Vertices B and B' : $(0, \pm b)$

Foci S and S' $(0, \pm be)$



Equations of the directrices : $y = \pm \frac{b}{e}$

Equation of the transverse axis : $x = 0$

Equation of the conjugate axis : $y = 0$

Length of the transverse axis = $2b$

Length of the conjugate axis = $2a$

Equation of the latus rectum : $y = \pm be$

End points of the latus rectum : $\left(\pm\frac{a^2}{b}, be\right), \left(\pm\frac{a^2}{b}, -be\right)$

Length of the latus rectum = $\frac{2a^2}{b}$

2. PARAMETRIC FORM AND AUXILIARY CIRCLE

Parametric equations of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $x = a\sec\theta$, $y = b\tan\theta$.

The circle described on the transverse axis of the hyperbola as diameter is called auxiliary circle of the hyperbola. Its equation is $x^2 + y^2 = a^2$.

Student Task

1. Find the equation of the hyperbola whose one focus is at (1, 2) and the equation of the corresponding directrix is $x + y = 5$, whose eccentricity is $\sqrt{\frac{3}{2}}$.
2. If foci of the ellipse $\frac{(x-1)^2}{4} + \frac{(y-1)^2}{1} = 1$ and the hyperbola $\frac{(x-1)^2}{a^2} + \frac{(y-1)^2}{2} = 1$ coincide, then find a.
3. Find the foci and the length of the latus rectum of the hyperbola $9x^2 - 16y^2 - 72x + 96y - 144 = 0$

Ans. 1. $x^2 - 6xy + y^2 + 22x + 14y - 55 = 0$, **2.** 1, **3.** (9, 3), (–1, 3), 9/2

Point of Intersection of a Straight Line and a Hyperbola

Let the given straight line be $y = mx + c$ which intersects the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

For the point of intersection, we have

$$\frac{x^2}{a^2} - \frac{(mx+c)^2}{b^2} = 1$$

$\Rightarrow (b^2 - a^2m^2)x^2 - 2a^2mcx - a^2(b^2 + c^2) = 0$ ....(i)

Which is a quadratic equation in x giving two values of abscissae of the points of intersection

i.e., $x_1 + x_2 = \frac{2a^2mc}{b^2 - a^2m^2}$ and $x_1x_2 = \frac{a^2(b^2 + c^2)}{b^2 - a^2m^2}$

and the corresponding ordinates y_1, y_2 can be obtained by substituting x_1, x_2 in $mx + c$. The straight line will be tangent to the given hyperbola if the roots of (i) are equal,

i.e., if $4a^4m^2c^2 + 4(b^2 + c^2)a^2(b^2 - a^2m^2) = 0$

$\Rightarrow b^2c^2 + b^4 - a^2b^2m^2 = 0$

$\Rightarrow c^2 = a^2m^2 - b^2$

$\Rightarrow c = \pm\sqrt{a^2m^2 - b^2}$

Thus the equation of the tangent in slope form to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ can be written as

$y = mx \pm \sqrt{a^2m^2 - b^2}$

Equation of Tangent

(i) The equation of the tangent at (x_1, y_1) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is given by

$$\frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1$$

(ii) The equation in parametric form i.e., the equation of the tangent at $(a\sec\theta, b\tan\theta)$ is

$$\frac{x}{a}\sec\theta-\frac{y}{b}\tan\theta=1$$

Equation of Normal

(i) The equation of the normal at (x_1, y_1) to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

$$\frac{x-x_1}{\frac{x_1}{a^2}}=\frac{y-y_1}{\frac{y_1}{b^2}}$$

(ii) The equation of the normal in parametric form *i.e.*, at $(a\sec\theta, b\tan\theta)$ is given by $ax\cos\theta + by\cot\theta = a^2 + b^2$

Director Circle of a Hyperbola

The locus of the points of intersection of the mutually perpendicular tangents to a hyperbola is a circle called director circle of the hyperbola.

The equation of any tangents to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

$$y - mx = \sqrt{a^2m^2 - b^2}$$

Let it passes through (h, k)

$\therefore\ (k - mh)^2 = a^2m^2 - b^2$

$\Rightarrow m^2.(h^2 - a^2) - 2khm + (k^2 + b^2) = 0$

which is quadratic equation in m

Let roots of this equation be m_1, m_2.

$$m_1m_2 = \frac{k^2+b^2}{h^2-a^2} \quad \text{... (i)}$$

$\because$ Tangents are perpendicular

$\therefore\ m_1m_2 = -1$

From (i) & (ii)

$$\frac{k^2+b^2}{h^2-a^2} = -1$$

$\Rightarrow h^2 + k^2 = a^2 - b^2$

$\therefore$ The locus of (h, k) is $x^2 + y^2 = a^2 - b^2$

☺ **CONCEPTS**

In hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, if $a > b$ only the real director circle can be drawn. If $a < b$ then angle between tangents is always acute.

Equation of a Chord whose Middle Point is (x_1, y_1) :

$$\frac{xx_1}{a^2}-\frac{yy_1}{b^2}-1=\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1$$

$\Rightarrow T = S_1$

Equation of the Pair of Tangents From (x_1, y_1) :

$SS_1 = T^2$

Or $\left(\frac{x^2}{a^2}-\frac{y^2}{b^2}-1\right)\left(\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1\right)^2=\left(\frac{xx_1}{a^2}-\frac{yy_1}{b^2}-1\right)^2$

Diameter and Conjugate Diameter of a Hyperbola

The locus of the middle points of a system of parallel chords of a hyperbola is called a diameter of the hyperbola and two diameters are said to be conjugate diameters of each

bisects the chords parallel to the other $y = \frac{b^2}{a^2m}x$ is the equation a diameter whose m is the slope of the chord of system of parallel chords.

Two diameters $y = m_1x$ and $y = m_2x$ are conjugate diameters w.r.t hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ if $m_1m_2 = b^2/a^2$.

Student Task

1. If $x = 4$ is the chord of contant of tangents drawn from a point to the hyperbola $x^2 - y^2 = 4$, then find the equation of the corresponding pair of tangents.
2. Find the equations of common tangents to the hyperbola $3x^2 - 4y^2 = 12$ and the parabola $y^2 = 4x$.

Ans. 1. $4x^2 - 3y^2 - 8x + 4 = 0$, **2.** $x + y + 1 = 0$ and $x - y + 1 = 0$

3. RECTANGULAR HYPERBOLA

(i) Asymptote : A line which approaches to become tangent at infinity is called asymptote of the curve. In hyperbola, equation of hyperbola and combines equation of asymptotes differ by a constant term only and asymptotes pass through the centre.

Let the hyperbola be $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

$\therefore$ Combined equation of asymptotes is

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=0\Rightarrow y=\pm\frac{b}{a}x$$

(ii) **Rectangular hyperbola :** If the asymptotes of a hyperbola are perpendicular to each other then hyperbola is called rectangular hyperbola

i.e. $\frac{b}{a}x-\frac{b}{a}=-1\Rightarrow b = a$

$\therefore x^2 - y^2 = a^2$ is the equation of rectangular hyperbola whose asymptotes are $y = \pm x$.

Eccentricity of a rectangular hyperbola is always $\sqrt{2}$.

$\because e=\sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+1}$ ($\because b = a$) $\therefore$ e $=\sqrt{2}$

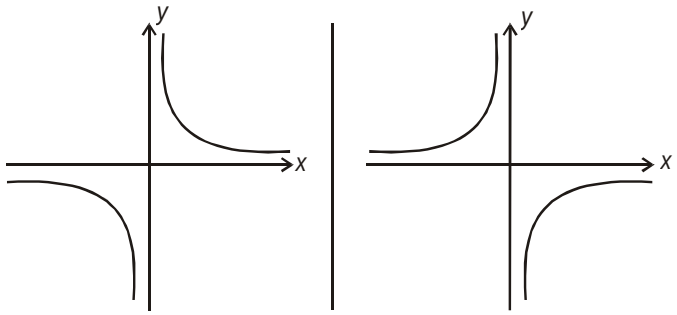
Note :- All results of $x^2 - y^2 = a^2$ are obtained by putting b = a is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(iii) Rectangular hyperbola whose asymptotes are the co-ordinate axes :

Equations of x-axis and y-axis are $y = 0$ and $x = 0$ respectively

$\therefore$ Combined equation of asymptotes is $xy = 0$

$\therefore$ Equation of rectangular hyperbola whose asymptotes are the co-ordinates axes is $xy = c^2$ or $xy = -c^2$



$x^2 - y^2 = a^2$ is a rectangular hyperbola whose asymptotes are $y = \pm x$.

Let axes are rotated about origin through an angle of 45° in clockwise direction

Putting $x = \frac{x+y}{\sqrt{2}}, y = \frac{-x+y}{\sqrt{2}}$, we get

$$\left(\frac{x+y}{\sqrt{2}}\right)^2 - \left(\frac{-x+y}{\sqrt{2}}\right)^2 = a^2$$

$4xy = 2a^2$

$\Rightarrow xy = \frac{a^2}{2}$ $\Rightarrow xy = c^2$ whose $c^2 = \frac{a^2}{2}$

(iv) Parametric form : The parametric equations of the hyperbola $xy = c^2$ are $x = ct$ and $y = \frac{c}{t}$.

$\therefore$ A general point of rectangular hyperbola $xy = c^2$ is $\left(ct, \frac{c}{t}\right)$

(v) Equation of the tangent :

(a) The equation of the tangent at (x_1, y_1) is $xy_1 + x_1y = 2c^2$ or $\frac{x}{x_1} + \frac{y}{y_1} = 2$

(b) The equation of the tangent at $\left(ct, \frac{c}{t}\right)$ is $\frac{x}{t} + ty = 2c$

(vi) Equation of the normal :

(a) The equation of the normal at (x_1, y_1) is $xx_1 - yy_1 = x_1^2 - y_1^2$

(b) The equation of the normal at $\left(ct, \frac{c}{t}\right)$ is $t^3x - ty - ct^4 + c = 0$

Student Task

1. A circle cuts two perpendicular lines so that each intercept is of given length. Prove that the locus of the center of circle is rectangular hyperbola.
2. A variable straight line with slope 2 intersects the hyperbola $xy = 1$ at two points. Find the locus of the mid-points of the line segment.
3. Find the locus of the mid-points of the chords of the hyperbola $x^2 - y^2 = 1$ which touch the parabola $y^2 = 4x$.

Ans. 2. $2x + y = 0$, **3.** $y^2(x - 1) = x^3$

MISCELLANEOUS EXAMPLES

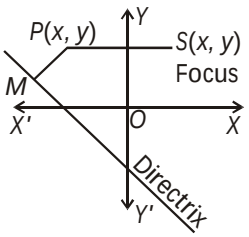
HYPERBOLA

Illustration 1: Find the equation of the hyperbola whose focus is (1, 2) directrix is the line $x + y + 1 = 0$ and eccentricity $\frac{3}{2}$.

Solution : Let S (1, 2) be the focus and let $P(x, y)$ be a point on the hyperbola. Draw perpendicular PM from P on the directrix $x + y + 1 = 0$. Then

$SP = e\,PM$

$$\Rightarrow \sqrt{(x-1)^2+(y-2)^2} = \frac{3}{2}\left(\frac{x+y+1}{\sqrt{1^2+1^2}}\right) \qquad \Rightarrow (x-1)+(y-2)^2 = \frac{9}{4}\left[\frac{(x+y+1)^2}{2}\right]$$



$\Rightarrow 8[(x-1)^2 + (y-2)^2] = 9[(x+y+1)^2]$

$= 8x^2 + 8y^2 - 16x - 32y + 40$

$= 9x^2 + 9y^2 + 9 + 18xy + 18x + 18y$

$\Rightarrow x^2 + y^2 + 18xy + 34x + 50y - 31 = 0$

This is the equation of the required hyperbola.

Illustration 2: Find the eccentricity of the conic represented by $x^2 - y^2 - 4x + 4y + 16 = 0$

Solution : We have $x^2 + y^2 - 4x + 4y + 16 = 0$

$\Rightarrow (x^2 - 4x) - (y^2 - 4y) = -16 \Rightarrow (x^2 - 4x + 4) - (y^2 - 4y + 4) = -16$

$$\Rightarrow (x-2)^2 - (y-2)^2 = -16 \Rightarrow \frac{(x-2)^2}{4^2} - \frac{(y-2)^2}{4^2} = -1$$

Shifting the origin at(2, 2), we obtain $\frac{X^2}{4^2} - \frac{Y^2}{4^2} = -1$, where $x = X + 2$, $y = Y + 2$

This is rectangular hyperbola, whose eccentricity is always $\sqrt{2}$

Illustration 3: Find the centre, eccentricity, foci, directrices and the lengths of the transverse and conjugate axes of the hyperbola, whose equation is $(x-1)^2 - 2(y-2)^2 + 6 = 0$

Solution : The equation of the hyperbola can be written as $(x-1)^2 - 2(y-2)^2 + 6 = 0$

or $-\frac{(x-1)^2}{\sqrt{(6)^2}} + \frac{(y-2)^2}{\left(\sqrt{3}\right)^2} = 1$ or $\frac{Y^2}{\left(\sqrt{3}\right)^2} - \frac{X^2}{\left(\sqrt{6}\right)^2} = 1$

Where $Y = (y-2)$ and $X = (x-1)$ .....(1)

$\therefore$ centre : $X = 0$, $Y = 0$, *i.e.*, $(x-1) = 0$, $x = 1$ & $(y-2) = 0$, $y = 2$.

So, $a = \sqrt{3}$ and $b = \sqrt{6}$ so transverse axis = $2\sqrt{3}$, and conjugate axis = $2\sqrt{6}$.

Also $b^2 = a^2(e^2 - 1) \Rightarrow 6 = 3(e^2 - 1)$ *i.e.*, $e = \sqrt{3}$

In (X, Y) coordinates, foci are $(0, \pm ae)$

i.e., $(0, \pm 3)$ $\therefore$ foci are $(1, 2 \pm 3)$ *i.e.*, (1, 5) and (1, –1)

Equations of directrices, $Y = \pm a/e$

$\therefore$ directrices $y - 2 = \pm \sqrt{3} / \sqrt{3} = \pm 1$ or $y = 3$, $y = 1$.

Illustration 4: Find the equation of the hyperbola whose foci are (8, 3), (0, 3) and eccentricity = $\frac{4}{3}$.

Solution : The centre of the hyperbola is the mid-point of the line joining the two foci. So the coordinates of the centre are $\left(\frac{8+0}{2}, \frac{3+3}{2}\right)$ i.e. (4, 3).

Let $2a$ and $2b$ be the length of transverse and conjugate axes and let e be the eccentricity. Then the equation of the hyperbola is

$$\frac{(x-4)^2}{a^2} - \frac{(y-3)^2}{b^2} = 1 \qquad \text{....(i)}$$

Now, distance between the two foci = $2ae$

$$\Rightarrow \sqrt{(8-0)^2 + (3-3)^2} = 2ae \Rightarrow ae = 4 \Rightarrow a = 3 \left(\because e = \frac{4}{3}\right)$$

Now, $b^2 = a^2(e^2 - 1) \Rightarrow b^2 = 9\left(-1 + \frac{16}{9}\right) = 7$

Thus, the equation of the hyperbola is

$\frac{(x-4)^2}{9} - \frac{(y-3)^2}{7} = 1$ [putting the values of a and b in (i)]

$\Rightarrow 7x^2 - 9y^2 - 56x + 54y - 32 = 0$

Illustration 5: Show that the equation $x^2 - 2y^2 - 2x + 8y - 1 = 0$ represents a hyperbola. Find the coordinates of the centre, lengths of the axes, eccentricity, latus rectum, coordinates of the foci and vertices and equations of directrices of the hyperbola.

Solution : $x^2 - 2y^2 - 2x + 8y - 1 = 0$

$\Rightarrow (x^2 - 2x) - 2(y^2 - 4y) = 1$

$\Rightarrow (x^2 - 2x + 1) - 2(y^2 - 4y + 4) = -6$

$\Rightarrow (x-1)^2 - 2(y-2)^2 = -6$

$$\Rightarrow \frac{(x-1)^2}{\left(\sqrt{6}\right)^2} - \frac{(y-2)^2}{\left(\sqrt{3}\right)} = -1 \quad \text{....(i)}$$

Shifting the origin at (1, 2) without rotating the coordinates axes and denoting the new coordinates with respect to these axes by X and Y, we have

$x = X + 1$ and $y = Y + 2$ .....(ii)

Using these relations, equation (i) reduces to

$$\frac{X^2}{\left(\sqrt{6}\right)^2} - \frac{Y^2}{\left(\sqrt{3}\right)^2} = -1 \quad \text{.....(iii)}$$

This equation is of the form $\frac{X^2}{a^2} - \frac{Y^2}{b^2} = -1$ where $a^2 = \left(\sqrt{6}\right)^2$ and $b^2 = \left(\sqrt{3}\right)^2$. So we have :

Centre The coordinates of the centre with respect to the new are ($X = 0$, $Y = 0$ in So, the coordinates of the centre with respect to the old axes are (1, 2) [Putting $X = 0$, $Y = 0$ in (ii)]

Lengths of the Axes :

Since the transverce axis of the hyperbola is along new Y-axis.

$\therefore$ Transverse axis = $2b = 2\sqrt{3}$, Conjugate axis = $2a = 2\sqrt{6}$.

Eccentricity :

The eccentricity e is given by $e = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{6}{3}} = \sqrt{3}$

Latus rectum :

Length of the latus rectum = $\frac{2a^2}{b} = \frac{12}{\sqrt{3}} = 4\sqrt{3}$

Foci :

The coordinates of foci with respect to the new axes are

($X = 0$, $Y = \pm be$) *i.e.*, ($X = 0$, $Y = \pm 3$).

So, the coordinates of the vertices with respect to the old axes are (1, 2 ± 3) i.e. (1, 5) and (1, –1) [Putting $X = 0$, $Y = \pm 3$ in (ii)]

Vertices :

The coordinates of the vertices with respect to the new axes are $X = 0$, $Y = \pm b$ *i.e.*, ($X = 0$, $Y = \pm \sqrt{3}$)

So the coordinates of the vertices w.r.t. to the old axes are

(1, 2 ± $\sqrt{3}$) *i.e.* (1, 2 + $\sqrt{3}$) and (1, 2 – $\sqrt{3}$) [Putting $X = 0$, $Y = \pm \sqrt{3}$ in (ii)]

Directrices :

The equations of the directrices with respect to the new axes are $Y = \pm b/e$ *i.e.* $Y = \pm 1$

So the equations of the directrices with respect to the old axes are

$y = 2 \pm 1$ *i.e.* $y = 1$ and $y = 3$ [Putting $Y = \pm 2$ in (ii)]

Illustration 6: If e and e' be the eccentricities of a hyperbola and its conjugate, prove that $\frac{1}{e^2} + \frac{1}{e'^2} = 1$.

Solution : Let the equation of the hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ .....(i)

Then the equation of the hyperbola conjugate to (i) is $\frac{x^2}{b^2} - \frac{y^2}{a^2} = 1$ .....(ii)

e = Eccentricity of (i) = $\sqrt{1 + \left(\frac{\text{conjugate axis}}{\text{Transverse axis}}\right)^2}$

$$\Rightarrow e = \sqrt{1+\left(\frac{2b}{2a}\right)^2} \Rightarrow e^2 = 1+\frac{b^2}{a^2} \Rightarrow e^2 = \frac{a^2+b^2}{a^2}$$

And, e' = Eccentricity of (ii) = $\sqrt{1+\left(\frac{\text{conjugate axis}}{\text{Transverse axis}}\right)^2}$

$$\Rightarrow e' = \sqrt{1+\left(\frac{2a}{2b}\right)^2} \Rightarrow e'^2 = 1+\frac{a^2}{b^2} \Rightarrow e'^2 = \frac{a^2+b^2}{b^2}$$

From (iii) and (iv), we have

$$\frac{1}{e^2}+\frac{1}{e'^2.} = \frac{a^2}{a^2+b^2}+\frac{b^2}{a^2+b^2} = \frac{a^2+b^2}{a^2+b^2} \Rightarrow \frac{1}{e^2}+\frac{1}{e'^2.} = 1$$

Illustration 7: Find the condition for the line $x\cos\alpha + y\sin\alpha = p$ to be a tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.

Solution : We have,

$x\cos\alpha + y\sin\alpha = p \Rightarrow y = -x\cot\alpha - p\,\text{cosec}\,\alpha$

$\Rightarrow y = (-\cot\alpha)\,x + (-p\,\text{cosec}\,\alpha)$.....(i)

This will touch $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ if

$(-p\,\text{coses}\,\alpha)^2 = a^2(-\cot\alpha)^2 - b^2$ [Using $c^2 = a^2m^2 - b^2$]

$\Rightarrow p^2\,\text{cosec}^2\alpha = a^2\cot^2\alpha - b^2 \Rightarrow p^2 = a^2\cos^2\alpha - b^2\sin^2\alpha$

Illustration 8: Prove that the product of the lengths of the perpendiculars drawn from foci on any tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is b^2.

Solution : The equation of any tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $y = mx + \sqrt{a^2m^2-b^2}$ .....(i)

Let $S(ae, 0)$, $S'(-ae, 0)$ be two foci of the hyperbola.

Let P and P' be the lengths of perpendicular from $S(ae, 0)$ and $S'(-ae, 0)$ on (i). Then,

p = length of the $\perp$ from $S(ae, 0)$ on (i)

$$= \left|\frac{mae-0+\sqrt{a^2m^2-b^2}}{\sqrt{m^2+1}}\right| \left|\frac{mae+\sqrt{a^2m^2-b^2}}{\sqrt{1+m^2}}\right|$$

p' = Length of the $\perp$ from $S'(-ae, 0)$ on (i)

$$= \left|\frac{-mae+\sqrt{a^2m^2-b^2}}{\sqrt{m^2+1}}\right| = \left|\frac{-mae+\sqrt{a^2m^2-b^2}}{\sqrt{1+m^2}}\right|$$

$$\text{Now, } p.p' = \left|\frac{mae+\sqrt{a^2m^2-b^2}}{\sqrt{1+m^2}}\right| . \left|\frac{-mae+\sqrt{a^2m^2-b^2}}{\sqrt{1+m^2}}\right|$$

$$= \left|\frac{-m^2a^2e^2+a^2m^2-b^2}{1+m^2}\right| = \left|\frac{-m^2a^2(e^2-1)-b^2}{m^2+1}\right| \qquad = \left|\frac{m^2b-b^2}{m^2+1}\right| = \frac{(m^2+1)b^2}{(m^2+1)} = b^2$$

Illustration 9: If the tangent at the point (p, q) on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ cuts the auxiliary circle in points whose ordinates are y_1 and y_2 then q is Harmonic mean of y_1 and y_2.

Solution : Hence we choose the tangent as

$$\frac{xp}{a^2}-\frac{yq}{b^2}=1 \text{ where } \frac{p^2}{a^2}-\frac{q^2}{b^2}=1 \quad ....(1)$$

Its intersection with $x^2 + y^2 = a^2$ is given by eliminating x as we are concerned with ordinates

$$\left(1+\frac{yq}{b^2}\right).\frac{a^4}{p^2}+y^2=a^2$$

or $(b^2 + yq)^2 a^4 + b^4y^2p^2 = b^2$

or $y^2 (a^4q^2 + b^4p^2) + 2yqb^2a^4 – a^2p^2b^4 = 0$ .....(2)

Above is a quadratic in y. We have to prove that q is H.M. 'H' of y_1 and y_2

$$\text{Now H} = \frac{2y_1y_2}{y_1+y_2}\ \frac{(a^4b^4-a^2p^2b^4)}{-2qb^2a^4} \text{ by (2)}$$

$$= \frac{a^4b^4(1-p^2/a^2)}{-2qb^2.a^4}=\frac{b^2}{-q}\left[\frac{q^2}{b^2}\right]=q \text{, by (1).}$$

$\therefore$ q is H.M of y_1 and y_2.

Illustration 10: Determine the equation of common tangents to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and

$$\frac{x^2}{b^2}-\frac{y^2}{a^2}=1.$$

Solution : Tangent to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

$$y = m_1x \pm \sqrt{(a^2m_1^2-b^2)} \quad ....(1)$$

The other hyperbola is $\frac{x^2}{(-b^2)}-\frac{y^2}{(-a^2)}=1$

Any tangnet to it is $y = m_2x \pm \sqrt{(-b^2)m_2^2-(a^2)}$ .....(2)

If (1) and (2) are same, then

$m_1 = m_2$ and $a^2m_1^2 – b^2 = – b^2m_2^2 + a^2$

or $a^2m_1^2 – b^2 = a^2 – b^2m_1^2$

or $(a^2 + b^2)m_1^2 = a^2 + b^2$

$\therefore$ $m_1^2 = 1$ $\therefore$ $m_1 = \pm 1$

Hence the common tangnets are

$$y = \pm x \pm \sqrt{a^2-b^2}$$

Illustration 11: Prove that the line lx + my + n = 0 will be a normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ if

$$\frac{a^2}{l^2}-\frac{b^2}{m^2}=\frac{(a^2+b^2)^2}{n^2}$$

Solution : The equation of the normal at (a sec θ, b tan θ) to the hyperbola
$a\,x \sin\theta + b\,y = (a^2 + b^2)\tan\theta$ .....(i)
and the equation of the line is
$lx + my + n = 0$ .....(ii)
Equations (i) and (ii) will represent the same line. If

$$\frac{a\sin\theta}{l} = \frac{b}{m} = \frac{(a^2+b^2)\tan\theta}{-n}$$

$$\Rightarrow \text{cosec}\theta = \frac{am}{bl} \text{ and } \cot\theta = -\frac{(a^2+b^2)m}{b^2}$$

$$\therefore \text{cosec}^2\theta - \cot^2\theta = 1 \Rightarrow \frac{a^2m^2}{b^2l^2} - \frac{(a^2+b^2)m^2}{b^2n^2} = 1$$

$$\Rightarrow \frac{a^2}{l^2} - \frac{(a^2+b^2)^2}{n^2} = \frac{b^2}{m^2} \Rightarrow \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2+b^2)^2}{n^2}$$

Illustration 12: If the normal at 'θ' on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the transverse axis at G, prove that $AG.A'G = a^2\,(e^4 \sec^2\theta - 1)$
Where A and A' are the vertices of the hyperbola.

Solution : The equation of the normal at (a secθ, b sinθ) to the given hyperbola is
$ax \cos\theta + by \cot\theta = (a^2 + b^2)$
This meets the transverse axis *i.e.* x-axis at G. So, the coordinates of G are

$$\left(\frac{a^2+b^2}{a}\sec\theta, 0\right)$$

The coordinates of the vertices A and A' are A (a, 0) and A' ($-a$, 0) respectively

$$\therefore AG.A'G = \left(-a + \frac{a^2+b^2}{a}\sec\theta\right)\left(a + \frac{a^2+b^2}{a}\sec\theta\right)$$

$= (-a + a\,e^2 \sec\theta)\,(a + e^2 \sec\theta)$
$= a^2\,(e^4 \sec^2\theta - 1)$

Illustration 13: Show that the locus of the middle points of the normal chords of the rectangular hyperbola $x^2 - y^2 = a^2$ is $(y^2 - x^2)^3 = 4a^2x^2y^2$.

Solution : If (h, k) be the mid-point of the chord of the hyperbola $x^2 - y^2 = a^2$ then its equation by $T = S_1$ is $hx - ky = h^2 - k^2$ .....(1)
But since (1) is normal to the hyperbola its equation is
$x \cos\theta + y \cot\theta = 2a$ .....(2)
Putting $b = a$
Comparing (1) and (2), we get

$$\frac{h}{\cos\theta} = -\frac{k}{\cos\theta} = \frac{h^2-k^2}{2a}$$

$$\therefore \sec\theta = \frac{h^2-k^2}{2a} \text{ and } \tan\theta = \frac{h^2-k^2}{-2ak}$$

Put in $\sec^2\theta - \tan^2\theta = 1$

$$\therefore \frac{\left(h^2-k^2\right)^2}{4a^2}\left(\frac{1}{h^2} - \frac{1}{k^2}\right) = 1$$

Hence the focus of the mid-point (h, k) is
$(x^2 - y^2)^2 (y^2 - x^2) = 4a^2 x^2y^2$
or $(y^2 - x^2)^3 = 4a^2x^2y^2$

Illustration 14: A normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the axes in Q and R, and lines QL and RL are drawn at right angles to the axes and meet at L. [In other words the rectangle $QORL$ is completed].
Prove that locus of the point L is the hyperbola
$a^2x^2 - b^2y^2 = (a^2 + b^2)^2$
Prove that further that the locus of the middle point of QR is $4(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$

Solution : Any normal to the hyperbola is
$ax \cos\theta + by \cot\theta = a^2 + b^2$ .....(1)
Putting $y = 0$ and then $x = 0$, we get the points Q and R where it meets the axes as

$$Q\left(\frac{a^2+b^2}{a}\sec\theta, 0\right) \text{ and } R\left(0, \frac{a^2+b^2}{b}\tan\theta\right)$$

Line through Q perpendicular to x-axis is $x = \frac{a^2+b^2}{a}\sec\theta$

Line through R perpendicular to y-axis is $y = \frac{a^2+b^2}{b}\tan\theta$

Both these lines meet at the point L and in order to find its locus, we have to eliminate θ between their equations by the help of relation $\sec^2\theta - \tan^2\theta = 1$.

or $\frac{a^2x^2}{(a^2+b^2)} - \frac{b^2y^2}{(a^2+b^2)^2} = 1$ or $a^2x^2 - b^2y^2 = (a^2 + b^2)^2$

2nd part : if (h, k) be the mid-point of QR, then

$$2h = \frac{a^2+b^2}{a}\sec\theta, 2k = \frac{a^2+b^2}{a}\tan\theta$$

$4(a^2h^2 - b^2k^2) = (a^2 + b^2)^2 (\sec^2\theta - \tan^2\theta)$
or $(a^2x^2 - b^2y^2) = (a^2 + b^2)^2$
is the required locus.

Chord with A Given Mid-Point

The equation of chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with $P(x_1, y_1)$ as its middle point is given by $T = S_1$ where $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

Chord of Contact

The equation of chord of contact of tangent drawn from a point $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $T = 0$ where $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

The equation of the chord joining the points $P \equiv (a\sec\theta_1, b\tan\theta_1)$ and $Q \equiv (a\sec\theta_2, \tan\theta_2)$ is

$$\frac{x}{a}\cos\left(\frac{\theta_1-\theta_2}{2}\right) - \frac{y}{b}\sin\left(\frac{\theta_1+\theta_2}{2}\right) = \cos\left(\frac{\theta_1+\theta_2}{2}\right) \text{ or } \begin{vmatrix} x & y & 1 \\ a\sec\theta_1 & b\tan\theta_1 & 1 \\ a\sec\theta_2 & b\tan\theta_2 & 1 \end{vmatrix} = 0$$

Equation of A Diameter of A Hyperbola

The equation of the diameter bisecting chords of slope m of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $y=\frac{b^2}{a^2m}$

Conjugate Diameters

Two diameters of a hyperbola are said to be conjugate diameters if each bisects the chord parallel to the other. If m_1 and m_2 be the slopes of the conjugate diameters of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, then $m_1m_2 = \frac{b^2}{a^2}$

Illustration 15: Chords of the circle $x^2 + y^2 = a^2$ touch the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. Prove that locus of their middle point is the curve $(x^2 + y^2) = a^2x^2 – b^2y^2$.

Solution : Let (h, k) be the mid-point of the chord of the circle $x^2 + y^2 = a^2$, so that its equation by $T = S_1$ is

$hx + ky = h^2 + k^2$ or $y=-\frac{h}{k}x+\frac{h^2+k^2}{k}$ *i.e.* of the form

$y = mx + c$

$\therefore \left(\frac{h^2+k^2}{k}\right)=a^2\left(-\frac{h}{k}\right)^2-b^2$ or $(h^2 + k^2)^2 = a^2h^2 – b^2k^2$

Generalising, the locus of mid-point (h, k) is
$(x^2+ y^2)^2 = a^2x^2 – b^2y^2$

Illustration 16: Prove that the locus of the middle points of the chord of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ which pass through a fixed point (α, β) is a hyperbola whose centre is $(\alpha/2, \beta/2)$.

Solution : Equation of the chord of the hyperbola whose mid-point is (h, k) by $T = S_1$ is

$\frac{hx}{a^2}-\frac{ky}{b^2}=\frac{h^2}{a^2}-\frac{k^2}{b^2}$

It passes through the point (α, β)

$\therefore \frac{h\alpha}{a^2}-\frac{k\beta}{b^2}=\frac{h^2}{a^2}-\frac{k^2}{b^2}$ $\quad\quad$ $\therefore$ Locus of (h, k) is $\frac{x^2-\alpha x}{a^2}-\frac{y^2-\beta y}{b^2}=0$

or $\frac{(x-\alpha/2)^2}{a^2}-\frac{(y-\beta/2)^2}{b^2}=\frac{1}{4}\left(\frac{\alpha^2}{a^2}-\frac{\beta^2}{b^2}\right)=k^2$,

Above equaiton represents a hyperbola whose centre is $(\alpha/2, \beta/2)$.

Illustration 17: From points on circle $x^2 + y^2 = a^2$ tangents are drawn to the hyperbola $x^2 – y^2 = a^2$. Prove that the locus of the middle points of the chords of contact is the curve $(x^2 – y^2) = a^2 (x^2 + y^2)$.

Solution : Any point on the curve $x^2 + y^2 = a^2$ is $(a\cos\theta, a\sin\theta)$. Chord of contact of this point w.r.t. the hyperbola $x^2 – y^2 = a^2$ is

$x (a\cos\theta) – y (a\sin\theta) = a^2$

or $x\cos\theta – y\sin\theta = a$ .....(1)

If its mid-point be (h, k), then it is same as $T = S_1$.

or $hx - ky = h^2 - k^2$ .....(2)

Comparing (1) and (2), we get

$$\frac{\cos\theta}{h} = \frac{\sin\theta}{k} = \frac{a}{h^2 - k^2}$$

But $\cos^2\theta + \sin^2\theta = 1$

$$\therefore \left(\frac{ah}{h^2 - k^2}\right)^2 + \left(\frac{ak}{h^2 - k^2}\right)^2 = 1$$

Hence the locus of the mid points (h, k) is

$a^2(x^2 + y^2) = (x^2 - y^2)^2$

Illustration 18: For any real t, $x = \frac{e^t + e^{-t}}{2}, y = \frac{e^t - e^{-t}}{2}$ is a point on the hyperbola $x^2 - y^2 = 1$. Show that the area bounded by the hyperbola and the lines joining its centre to the points corresponding to t_1 and $-t_1$ is t_1.

Solution : Substituting the point in the equation of hyperbola, we get

$$\frac{1}{4}[(e^t + e^{-t})^2 - (e^t - e^{-t})^2] = \frac{1}{4} 4e^t e^{-t} = 1$$

Above is true for all values of t. Hence the point lies on the hyperbola $x^2 - y^2 = 1$.

Let $P(x_1, y_1)$ be the point corresponding to t_1 then

$$x_1 = \frac{1}{2}(e^t + e^{-t}), y_1 = \frac{1}{2}(e^t - e^{-t})$$

If $Q(x_2, y_2)$ be the point corresponding to $-t_1$ then

$$x_2 = \frac{1}{2}(e^{-t} + e^t) = x_1,$$

$$y_2 = \frac{1}{2}(e^{-t} + e^t) = y_1,$$

$\therefore$ The point Q is (x_1, y_1). Hence PQ is a double ordinate.

The vertex A (1, 0) correxponding to $t = 0$

$$\Delta CPQ = \left|\frac{1}{2}(x_1 y_2 - x_2 y_1)\right| = \left|\frac{1}{2}(-x_1 y_1 - x_1 y_1\right|$$

$= |-x_1 y_1| = x_1 y_1$

Area of $APMQA$ = 2 area $APMA$ = $2\int_0^{t_1} \frac{dx}{dt} dt$

$$= 2\int_0^{t_1} \frac{1}{2}(e^t - e^{-t}).\frac{1}{2}(e^t - e^{-t})dt \quad = \frac{1}{2}\int_0^{t_1}(e^{2t} - e^{-2t} - 2)dt \quad = \frac{1}{2}\left[\frac{e^{2t}}{2} - \frac{e^{-2t}}{2} - 2t\right]_0^{t_1}$$

$$= \frac{1}{2}\left[\frac{e^{2t_1}}{2} - \frac{e^{-2t_1}}{2} - 2t_1\right] \quad = \frac{1}{2}(e^{t_1} + e^{-t_1}).\frac{1}{2}(e^{t_2} - e^{-t_1}) - t_1$$

$= x_1 y_1 - t_1$

Hence the required shaded area is

$(x_1 y_1) - (x_1 y_1 - t_1) = t_1$

Alternative : Required area = $2\left[\Delta - \int_1^{x1} y dx\right]$ .....(1)

$\Delta = \frac{1}{2}x_1y_1$ and $\int_1^{x1} ydx = \int_1^{x1} \sqrt{(x^2-1)}dx$

$= \left[\frac{x}{2}\sqrt{x^2-1} - \frac{1}{2}\log\left\{x+\sqrt{x^2-1}\right\}\right]_1^{x1} = \frac{x1}{2}\sqrt{x_1^2-1} - \frac{1}{2}\log\left\{x_1+\sqrt{x_1^2-1}\right\} - 0$

$= \frac{1}{2}x_1y_1 - \frac{1}{2}\log(x_1+y_1) = \Delta\log e^{t_1} = \Delta - \frac{1}{2}t_1$

Putting in (1), we get Area = $2\Delta - 2\Delta + 2.\frac{1}{2}t_1 = t_1$

Illustration 19: Find the equation of the asymptotes of the hyperbola $2x^2 - 5xy - 3y^2 - 5x - 3y - 21 = 0$.

Solution : Since the equation of the pair of asymptotes of the hyperbola differ by the hyperbola by a constant only. So, let the equation of the pair of asymptotes be

$2x^2 - 5xy - 3y^2 - 5x - 3y + \lambda = 0$

This equation represents a pair of straight lines

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

Here : $a = 2$, $b = -3$, $h = -5/2$, $g = -5/2f^2 = -3/2$ and $c = \lambda$

$\therefore -6\lambda - \frac{75}{4} - \frac{9}{2} + \frac{75}{2} - \frac{25}{4}\lambda = 0 \Rightarrow -49\lambda - 75 - 18 + 75 = 0 \Rightarrow \lambda = -18/49.$

Putting the value of λ in (i), we get the required equation

$2x^2 - 5xy - 3y^2 - 5x - 3y -$

Illustration 20: A series of chords of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ touch the circle on the line joining the foci as diameter. Show that the locus of the poles of these chords with respect to the hyperbola is $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2+b^2}$.

Solution : The foci of the hyperbola are the points S (ae, 0) and S'($-ae$, 0). Circle on SS' as diameter is $(x - ae)(x + ae) + y^2 = 0$

or $x^2 + y^2 = a^2e^2$ .....(1)

If (h, k) be the pole of the chord which touches (1), then its equation is the polar of (h, k) w.r.t. hyperbola

i.e. $\frac{hx}{a^2} - \frac{ky}{b^2} = 1$ .....(2)

Since the line (2) touches the circle (1) therefore perpendicular from centre (0, 0) should be equation to radius ae.

$\therefore \quad \frac{1}{\sqrt{\left(\frac{h^2}{a^4} + \frac{k^2}{b^2}\right)}} = ae$

or $\frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{a^2+b^2}$ $\quad \therefore \quad b^2 = a^2(e^2-1)$

Generalising, the locus of the pole (h, k) is the ellipse $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2+b^2}$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic : 1 Hyperbola

Q.1 If the chord of contact of tangent from a point P to the parabola $y^2 = 4ax$ touches the parabola $x^2 = 4by$, then the locus of P is
(a) A circle
(b) A parabola
(c) A pair of straight lines
(d) A hyperbola

Q.2 If LM is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that OLM is an equilateral triangle, O being the centre of the hyperbola, then the eccentricity e of the hyperbola, satisfies
(a) $e > \frac{2}{\sqrt{3}}$ (b) $e = \frac{2}{\sqrt{3}}$
(c) $e < \frac{2}{\sqrt{3}}$ (d) $1 < e < \frac{2}{\sqrt{3}}$

Q.3 The equation of the hyperbola, whose foci are (6, 4) and (–4, 4) and eccentricity is 2, is
(a) $12(x-1)^2 - 4(y-4)^2 = 75$
(b) $12(x+1)^2 - 4(y+4)^2 = 75$
(c) $4(x-1)^2 - 12(y-4)^2 = 75$
(d) $4(x+1)^2 - 12(y+4)^2 = 75$

Q.4 The equation $\frac{x^2}{1-r} - \frac{y^2}{1+r} = 1$, $r > 1$ represents
(a) An ellipse (b) A hyperbola
(c) A circle (d) None of these

Q.5 The equation of the hyperbola with centre at (0, 0) and co-ordinate axes as its axes, distance between the directrices being $\frac{4}{\sqrt{3}}$ and passing through the point (2, 1), is
(a) $3x^2 - 2y^2 = 10$ (b) $3x^2 - 2y^2 = 2$
(c) $2x^2 - 3y^2 = 10$ (d) $x^2 - y^2 = 3$

Q.6 If t is a parameter, then $x = a\left(t + \frac{1}{t}\right)$ and $y = b\left(t - \frac{1}{t}\right)$ represents
(a) An ellipse (b) A parabola
(c) A hyperbola (d) A circle

Q.7 If the equation $\frac{x^2}{10-\lambda} + \frac{y^2}{6-\lambda} = 1$ represents a hyperbola, then λ is
(a) $\lambda > 10$ (b) $6 < \lambda < 10$
(c) $\lambda < 6$ (d) $(-\infty, 6] \cup [10, \infty)$

Q.8 If $3x^2 - 5y^2 - 6x + 20y - 32 = 0$ represents a hyperbola, then the co-ordinates of foci are
(a) $(\pm 2\sqrt{2}, 0)$ (b) $(1 \pm 2\sqrt{2}, 2)$
(c) $(0, \pm 2\sqrt{2})$ (d) (1, 2)

Q.9 If the foci of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{15}$ coincide, then the value of b^2 is
(a) $\frac{101}{25}$ (b) $\frac{144}{25}$ (c) $\frac{81}{25}$ (d) $\frac{44}{25}$

Q.10 If $3x^2 - 3y^2 - 18x + 12y + 2 = 0$ represents a hyperbola, then the equation of directrices are
(a) $x = 3 \pm \sqrt{\frac{6}{13}}$ (b) $x = 3 \pm \sqrt{\frac{13}{6}}$
(c) $2 \pm \sqrt{\frac{13}{6}}$ (d) $2 \pm \sqrt{\frac{6}{13}}$

Q.11 If the locus of the point of intersection of two perpendicular tangents to a hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ is a circle with centre (0, 0), then the radius of a circle is
(a) 5 (b) 4 (c) 3 (d) 7

Topic : 2 Tangent and Normal of Hyperbola

Q.12 A normal to the parabola $y^2 = 4px$ with slope m touches the rectangular hyperbola $x^2 - y^2 = p^2$, if
(a) $m^6 - 4m^4 - 3m^2 + 1 = 0$
(b) $m^6 + 4m^4 + 3m^2 + 1 = 0$
(c) $m^6 + 4m^4 - 3m^2 + 1 = 0$
(d) $m^6 - 4m^4 + 3m^2 - 1 = 0$

Q.13 The equation of the tangent to the hyperbola $3x^2 - 4y^2 = 12$, which makes equal intercepts on the axes is
(a) $x - y + 1 = 0$ (b) $x + y + 1 = 0$
(c) $x + y - 1 = 0$ (d) All are correct

Q.14 The equation of the tangent to the hyperbola $3x^2 - 8y^2 = 24$ and perpendicular to the line $3x - 2y = 4$ is
(a) $3x + 2y \pm \sqrt{5} = 0$ (b) $2x + 3y \pm \sqrt{7} = 0$
(c) $2x + 3y \pm 2 = 0$ (d) $2x + 3y \pm \sqrt{5} = 0$

Q.15 The locus of a point $P(\alpha, \beta)$, such that the line $y = ax + b$ is a tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, which pass through the fixed point (1, 2) is a hyperbola whose eccentricity is

(a) $\frac{3}{2}$ (b) $\frac{\sqrt{7}}{2}$ (c) $\frac{\sqrt{13}}{2}$ (d) $\frac{\sqrt{15}}{2}$

Q.16 if (h, k) is the point of intersection of the normals at P and Q, then K is equal to

(a) $\frac{a^2 + b^2}{a}$ (b) $-\left(\frac{a^2 + b^2}{a}\right)$

(c) $\frac{a^2 + b^2}{b}$ (d) $-\left(\frac{a^2 + b^2}{b}\right)$

Q.17 If the line $y = mx + \sqrt{a^2m^2 - b^2}$, $m = \frac{1}{2}$ touches the hyperbola $\frac{x^2}{16} - \frac{y^2}{3} = 1$ at the point $(4 \sec \theta, \sqrt{3} \tan \theta)$ then θ is
(a) $\pi/2$ (b) $\pi/4$ (c) $2\pi/3$ (d) $\pi/6$

Q.18 A common tangent to $9x^2 - 16y^2 = 144$ and $x^2 + y^2 = 9$ is
(a) $y\sqrt{7} = \sqrt{2}x + 15$ (b) $y\sqrt{7} = 3\sqrt{2}x + 15$
(c) $y = 3\sqrt{2}x + 15$ (d) $y = 3x + 15$

Topic : 3 Different form of Chord of an Hyperbola

Q.19 The locus of the middle points of the chords of hyperbola $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$, which pass through the fixed point (1, 2) is a hyperbola whose eccentricity is

(a) $\frac{3}{2}$ (b) $\frac{\sqrt{7}}{2}$ (c) $\frac{\sqrt{13}}{2}$ (d) $\frac{\sqrt{15}}{2}$

Q.20 If angle between the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

(a) $\tan^{-1}\left(\frac{3}{4}\right)$ (b) $2\tan^{-1}\left(\frac{3}{4}\right)$

(c) $\tan^{-1}\left(\frac{4}{3}\right)$ (d) $2\tan^{-1}\left(\frac{4}{3}\right)$

Q.21 Find the length of the chord of the hyperbola $x^2 - y^2 = 2a^2$ if (2a, a) is the mid-point of the chord

(a) $\frac{20\sqrt{2}}{3}$a (b) $\frac{10\sqrt{2}}{3}$a

(c) $\frac{20}{3}$a (d) None of these

Topic : 4 Rectangular Hyperbola

Q.22 The normal at any point $P(t^2, 2t)$ on the parabola $y^2 = 4x$ meets the curve again at Q, then the $ar(\Delta POQ)$ in the form of $\frac{k}{|t|}(1 + t^2)(2 + t^2)$. The value of k is
(a)$k > 2$ (b)$k = 2$ (c)$k < 2$ (d)$k = 1$

Q.23 If the line $\frac{x}{a} + \frac{y}{b} = \sqrt{2}$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepted between the axes is
(a)0 (b)90° (c)45° (d)60°

Q.24 If the polars of (x_1, y_1) and (x_2, y_2) w.r.t. the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are at right angles, then the value of $\frac{x_1x_2}{y_1y_2}$ is

(a) $\frac{a^4}{b^4}$ (b) $-\frac{a^4}{b^4}$ (c) $\frac{b^4}{a^4}$ (d) $-\frac{b^4}{a^4}$

Q.25 Shortest distance between the curves $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{x^2}{4} + \frac{y^2}{4} = 1$ is
(a)1 (b)2 (c)3 (d)4

Q.26 The centre of a rectangular hyperbola lies on the line $y = 3x$. If one of the asymptotes is $x + y + 2 = 0$, then the other asymptotes is
(a) $x + y + 1 = 0$ (b) $x - y + 1 = 0$
(c) $x - y + 2 = 0$ (d) $x + y + 3 = 0$

Q.27 Area of the triangle formed by any arbitrary tangents of the hyperbola $xy = 4$, with the co-ordinate axes is
(a)2 (b)4 (c)6 (d)8

Q.28 If the normal to the rectangular hyperbola $xy = 4$ at the point $\left(2t_1, \frac{2}{t_1}\right)$ meets the curve again at $\left(2t_2, \frac{2}{t_2}\right)$, then

(a) $t_1^3 y_2 = 1$ (b) $t_1^3 t_2 = -1$
(c) $t_2^3 t_1 = 1$ (d) $t_1 t_2^3 = -1$

Q.29 If the circle $x^2 + y^2 = 4$ intersect the hyperbola $xy = 4$ in four point (x_i, y_i), $i = 1, 2, 3, 4$ then
(a) $x_1x_2x_3x_4 = 4$ (b) $x_1x_2x_3x_4 = 16$
(c) $x_1x_2x_3x_4 = 8$ (d) $x_1x_2x_3x_4 = 32$

Q.30 PQ and Rs are two perpendicular chords of the reactangualr hyperbola $xy = c^2$. If C is the centre of the reactangular hyperbola, then the product of the slopes of CP, CQ, CR and CS is
(a) 0 (b) -1
(c) 1 (d) None of these

Q.31 If A, B, and C be thre points on the hyperbola $xy = c^2$ such that AB subtends a right anlge at C, then AB is parallel to .....to hyperbola at point C.
(a) Normal (b) Tangent
(c) Any line (d) None of these

Section-B
(Assertion-Reason)

Q.1 **Statement – 1 :** From any point, at most four tangents can be drawn to a hyperbola.
Statement – 2 : From the centre of a hyperbola, four tangents can be drawn which are known as asymptotes.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1:** The line $y = \frac{b}{a}x$ will not meet the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $(a > b > 0)$.
Statement-2: The line $y = \frac{b}{a}x$k is an asymptote to the hyperbola.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1:** Lines $3x - 4y + 7 = 0$ and $4x + 3y + 8 = 0$ are the asymptotes of rectangular hyperbola.
Statements-2: In rectangular hyperbola, asymptotes intersect at right angle.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** Foci at the hyperbola $f(x, y)$, $7y^2 - 9x^2 + 54x - 28y - 116 = 0$ are (3, 6) and (3, –2).
Statement–2 : Foci is the solution of $\frac{df(x,y)}{dx} = 0$ and $\frac{df(x,y)}{dy} = 0$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement–1 :** Equation of director circle of the hyperbola $xy = c^2$ is a point circle.
Statement–2 : For rectangular hyperbola, $a = b$, i.e. $x^2 + y^2 = 0$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.6 **Statement–1 :** If eccentricity of a hyperbola is 2, then eccentricity of its conjugate hyperbola is $\frac{2}{\sqrt{3}}$.
Statement–2 : If e and e' are the eccentricity of hyperbola and its conjugate hyperbola, then $\frac{1}{e^2} + \frac{1}{e'^2} = 1$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True;

Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.7 **Statement–1 :** A hyperbola and its conjugate hyperbola have the same asymptotes.
Statement–2 : In a second degree curve, equation of asymptotes, if exists differ by constant only.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.8 **Statement–1 :** The line $3x + 4y = 5$ intersects the hyperbola $9x^2 - 16y^2 = 144$ only at one point.
Statement–2 : Given line is parallel to an asymptotes of the hyperbola.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.9 **Statement–1 :** If lines $y = m_1x$ and $y = m_2x$ are the conjugate diameter of the hyperbola $xy = c^2$, then $m_1 + m_2 = 0$.
Statement–2 : Two lines are called conjugate diameter of hyperbola if they bisect the chords parallel to each other.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.10 **Statement–1 :** Tangents at any point $P(x_1, y_1)$ on the hyperbola $xy = c^2$ meets the co-ordinate axes at points Q and R, the circumcentre of ΔOQR has co-ordinate (x_1, y_1).
Statement–2 : Equation of tangent at point (x_1, y_1) to the curve $xy = c^2$ is $\frac{x}{x_1} + \frac{y}{y_1} = 2$.
(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Year Question)

Q.1 The eccentricity of the hyperbola whose length of the latusrectum is equal to 8 and the length of its conjugate axis is equal to half of the distance between its foci, is

(a) $\frac{4}{3}$ (b) $\frac{4}{\sqrt{3}}$

(c) $\frac{2}{\sqrt{3}}$ (d) $\sqrt{3}$

Q.2 Consider a branch of the hyperbola $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A. Let B be one of the end points of its latusrectum. If C is the focus of the hyperbola nearest to the point A, Then the area of the ΔABC is

(a) $1 - \sqrt{2/3}$ squnit (b) $1\sqrt{3/2} - 1$ squnit

(c) $1 + \sqrt{2/3}$ squnit (d) $\sqrt{3/2} + 1$ squnit

Q.3 A hyperbola, having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then, its equation is

(a) $x^2\,\mathrm{cosec}^2\theta - y^2\sec^2\theta = 1$

(b) $x^2\sec^2\theta - y^2\,\mathrm{cosec}^2\theta = 1$

(c) $x^2\sin^2\theta - y^2\cos^2\theta = 1$

(d) $x^2\cos^2\theta - y^2\sin^2\theta = 1$

Q.4 If e_1 is the eccentricity if the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ and e_2 is the eccentricity of the hyperbola passing through the foci of the ellipse and $e_1e_2 = 1$, then equation of the hyperbola is

(a) $\frac{x^2}{9} - \frac{y^2}{16} = 1$ (b) $\frac{x^2}{16} - \frac{y^2}{9} = -1$

(c) $\frac{x^2}{9} - \frac{y^2}{25} = 1$ (d) None of these

Q5 For hyperbola $\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$, which of the following remains constant with change in 'α'?
(a) Abscissae of vertices
(b) Abscissae of foci

(c) Eccentricity
(d) Directrix

Q.6 The equation $\frac{x^2}{1-r}-\frac{y^2}{1+r}=1, |r|<1$ represents
(a) an ellipse
(b) a hyperbola
(c) a circle
(d) None of the above

Q.7 If hyperbola passes through the point $P\left(\sqrt{2},\sqrt{3}\right)$ and foci at $(\pm 2,0)$, then the tangent to this hyperbola at P also passes through the point

(a) $\left(3\sqrt{2},2\sqrt{3}\right)$ (b) $\left(2\sqrt{2},3\sqrt{3}\right)$

(c) $\left(\sqrt{3},\sqrt{2}\right)$ (d) $\left(-\sqrt{2},-\sqrt{3}\right)$

Q.8 Let P(6, 3) be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If the normal at the point P intersects the X-axis at (9, 0), then the eccentricity of the hyperbola is

(a) $\sqrt{\frac{5}{2}}$ (b) $\sqrt{\frac{3}{2}}$

(c) $\sqrt{2}$ (d) $\sqrt{3}$

Q.9 If the line $2x+\sqrt{6}y=2$ touches the hyperbola $x^2-2y^2=4$, then the point of contact is

(a) $\left(-2,\sqrt{6}\right)$ (b) $\left(-5,2\sqrt{6}\right)$

(c) $\left(\frac{1}{2},\frac{1}{\sqrt{6}}\right)$ (d) $\left(4,-\sqrt{6}\right)$

Q.10 Let P $(a\sec\theta, b\tan\theta)$ and $Q(a\sec\phi, b\tan\phi)$ Q $(\alpha\sec\phi, b\tan\phi)$, where $\theta+\phi=\frac{\pi}{2}$, be two points on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If (h, k) is the point of the intersection of the normals at P and Q, then k is equal to

(a) $\frac{a^2+b^2}{a}$ (b) $-\left(\frac{a^2+b^2}{a}\right)$

(c) $\frac{a^2+b^2}{b}$ (d) $-\left(\frac{a^2+b^2}{b}\right)$

Q.11 If x= 9 is the chord of contact of the hyperbola $x^2-y^2=9$, then the equation of the corresponding pair of tangents is

(a) $9x^2-8y^2+18x-9=0$

(b) $9x^2-8y^2-18x+9=0$

(c) $9x^2-8y^2-18x-9=0$

(d) $9x^2-8y^2+18x+9=0$

Section-D
(Board pattern)

Q.1 Find the co-ordinate of the foci and the vertices, the eccentricity and the length of the latus rectum of the hyperbola $4y^2-9x^2=36$.

Q.2 Find the equation of the hyperbola whose vertices are (±4, 0) and foci are (±5, 0).

Q.3 Find the equation of the hyperbola one of whose foci coincide with the centre of the circle $x^2+y^2-10y+9=0$ and length of semi-conjugate axis is 3.

Q.4 Find the eccentricity of the hyperbola with latus-rectum 6 and semi-conjugate axis is $\sqrt{3}$.

Q.5 Find the the equation of focal chord of the hyperbola $\frac{x^2}{8}-\frac{y^2}{1}=1$ which is inclined 30° with the positive *x*-axis.

Q.6 For what value of *k* the equation $\frac{x^2}{7-k}+\frac{y^2}{4-k}=1$ represents a hyperbola.

Q.7 If the foci of ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ coincide with the foci of the hyperbola $\frac{x^2}{b^2}+\frac{y^2}{12}=1$ then find the value of *b*.

Q.8 Find the equation of the hyperbola whose length of conjugate axis is 4 and one of the focal chord is *x* + *y* = 3 and centre (0, 0).

Q.9 For what value of *k* parabola $x^2=\frac{y}{4}$ and $\frac{x^2}{a^2}-\frac{y^2}{16}=1$ intersect each other?

Q.10 If e_1 and e_2 are the eccentricities of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}$ = 1 and its conjugate hyperbola, then find the value of (e_2-e_1).

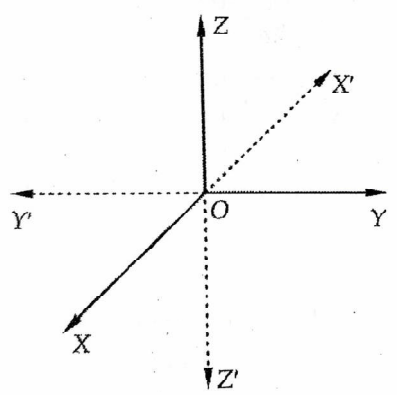
CHAPTER 4 ANALYTICAL GEOMETRY [THREE DIMENSION]

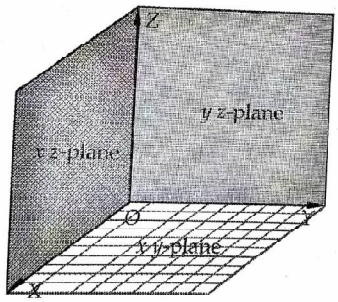
1. RECAPITULATION

COORDINATES OF POINT IN SPACE

In this section, we will recapitulate various concepts learnt in class XI . We have learnt in class XI tha three mutually perpendicualr lines in space define three mutually perpendicualr planes which in turn divide the space into eight parts known as octants and the lines are known as the coordiante axes.

Let X'OX,Y'OY and Z'OZ be three mutually perpendiucalr lines intersecting at O such that two of them viz. Y'OY and Z'OZ lie in the plane of the paper and the third X'OX is perpendicualr to the plane of the paper and is projecting out from the plane of the paper





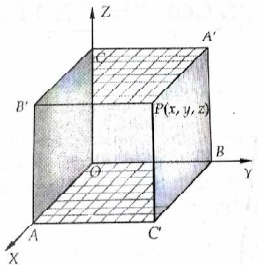
Let O be the origin and the lines X'OX, Y'OY and Z'OZ be x-axis, y-axis and z-axis. respectively. These three lines are also called the rectangualr axes of coordinates. The Planes containing the lines X'OX, Y'OY and Z'OZ in pairs, determine three mutually perpendicualr planes XOY, YOX and ZOZ or simply XY,YZ and ZX which are called rectangular coordinate planes as shown in figure.

Let P be point in space figure . Through P draw three planes parallel to the coordinate planes to meet the axes in A, B and C respectively. Let OA = x, OB = y and OC = z. These three real numbers taken in this order determined by the point P are called the coordiantes of the point P, written as (x, y, z), x, y,z are positive or negative according as they are measured along positive or neagitive directions of the coordinate axes.

Conversely, given an ordered triad (x,y,z) of real numbers we can always find the point whose coordinates are (x,y,) in the following manner.

(i) Measure OA, OB, OC along x-axis, y-axis and z-axis respectively.

(ii) Through the points A, B,C draw planes parallel to the coordinate planes YoZ, ZOX and XOY respectively. The point of intersection of these planes is the required point P.



To give another explanation about the coordiates of a point P we draw three planes through P parallel to the coordiante planes. These thre planes determine a rectangular parallelopiped which has three pairs of rectangualr faces, viz. PB' AC', OCA'B; PA'BC', OAB'C; PA'CB'; OAC'B as shown in Figure Clearly,

$x = OA = CB' = PA' =$ perpendicualr distance from P on the YOZ plane;

$y = OB = A'C = PB' =$ perpendicualr distance from P on the ZOX plane;

$z = OC = A'B = PC' =$ perpendicualr distance from P on the XOY plane.

Thus, the coordinates of the point P are the perpendicular distances from P on the three mutually rectangular coordinate planes YOZ, ZOX and XOY respectivley.

Further, since the line PA lies in the plane PB'AC' which is perpendicular to the line OA, we have PA perpendicualr to OA. Similarly, PB perpendicular to OB and PC perpendicualr to OC.

Thus, the coordinates of a point are the distances from the origin of the feet of the perpendiulars from the point on the respective coordinate axes.

SIGNS OF COORDINATES OF A POINT

To determine the signs of the coordiantes of a point in three dimension we follow the sign convention analogus to the sign convention in two dimensional geometry that all distances measured along or parallel to OX, OY, OZ will be possitive and distances moved along or parallel to OX', OY', OZ' will be negative. As discussed in previous article that three mutually perpendiular lines X'OX, Y'OY and Z'OZ determine three mutually perpendicular coordinate planes which in turn divide the space into eight compartments known as octants. The octant haivng OX, OY and OZ as its edges is denoted by OXYZ. Similarly, the other octants are denoted by OX'YZ, OXY'Z, OX'Y'Z, OXYZ',OX'YZ'OXY'Z" OX'Y'Z'. The signs of the coordiates of a point depend upon hte octant in which it lies. LEt P be a point and Let A, B, C be the feet of the perpendiculars drawn from P on X'OX, Y'OY and Z'OZ respectively. IF P lies in octant OXYZ, then clearly A, B, C lie on OX, OY and OZ respectively. Therefore, by our sign convention OA, OB and OC are positive. Thus all the three coordiantes of P are Positive. IF P lies in octant OX'YZ, then A,B and C lie on OX', OY and OZ respectively. Therefore, x-coordiante of P is negative and y and z coordiantes are positive.

The following table shows the signs of coordiantes of points in various octants:

Octant / Coordiante	OXYZ	OX'YZ	OX'Y'Z	OX'YZ'	OXYZ'	OX'YZ'	OXY'Z'	OX'Y'Z'
x	+	−	+	−	+	−	+	−
y	+	+	−	−	+	+	−	−
z	+	+	+	+	−	−	−	−

REMARKS 1 If If a point P lies in x y-plane, then by the definition of coordiantes of a point, z-coordinate of P is zero. therefore, the coordiates of a point on xy-plane are of the form (x, y, 0) and we may take the equation of xy-plane as z =0. Similarly the coordinates of any point in yz and zx-planes are of the forms (0, y, z) and (x, 0, z) respectively and their equations may be taken as x =0 and y = 0 respectively.

REMARKS 2 If a point lies on the x-axis, then its y and z-coordiantes are both zero. therefore, the coordinates of a point on x-axis are of the form (x, 0, 0) and we may take the equation of x-axis as y = 0, z = 0. Similarly, the coordinates of a point on y and z-axes are of the the form (0, y,)) and (0, 0, z) respectively and their equations may be taken as x = 0, z =0 and x = 0, y=0 respectively.

2. DISTANCE FORMULA

THEOREM Prove that the distance between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

PROOF Let O be the origin and let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be two given points. Then,

$\overrightarrow{OP} = x_1\hat{i} + y_2\hat{j} + z_1\hat{k}, \overrightarrow{OQ} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$

Now,

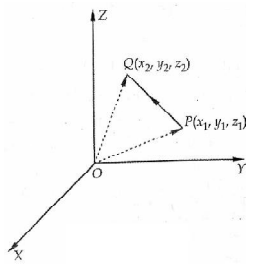
$\overrightarrow{PQ}$ = Position vector of Q-Position vector of P

$\Rightarrow \overrightarrow{PQ} = \left(x_2\hat{i} + y_2\hat{j} + z_2\hat{k}\right) - \left(x_1\hat{i} + y_1\hat{j} + z_1\right)\hat{k}$

$\Rightarrow \overrightarrow{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

$\therefore \quad PQ = |\overrightarrow{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

Hence, $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$



EXAMPLE 1 Find the distance between the points P(-2,4,1) and Q(1,2,-5)

SOLUTION We have, $PQ = \sqrt{(1-(-2))^2 + (2-4)^2 + (-5-1)^2} = \sqrt{9+4+36} = 7$ units

EXAMPLE 2 Prove by using distance formula tha the points P (1,2,3), Q(-1,-1,-1) and R(3, 5, 7) are collinear.

SOLUTION We have, $PQ = \sqrt{(-1-1)^2 + (-1-2)^2 + (-1-3)^2} = \sqrt{4+9+16} = \sqrt{29}$,

$QR = \sqrt{(3+1)^2 + (5+1)^2 + (7+1)^2} = \sqrt{16+36+64} = \sqrt{116} = 2\sqrt{29}$

and $PR = \sqrt{(3-1)^2 + (5-1)^2 + (7-3)^2} = \sqrt{4+9+16} = \sqrt{116} = \sqrt{29}$

Since $QR = PQ + PR$. Therefore, the given points are collinear.

EXAMPLE 3 Determine the point in XY-plane which is equidistant from three points A(2,0,3), B(0, 3, 2) and C (0, 0,1).

SOLUTION We know that z-coordiante of every point on xy-plane is zero. So, let P(x, y,0) be a point on xy plane such that PA=PB=PC.

Now, PA = PB

$\Rightarrow \quad PA^2 = PB^2$

$\Rightarrow \quad (x-2)^2 + (y-0)^2 + (0-3)^2 = (x-0)^2 + (y-3)^2 + (0-2)^2$

$\Rightarrow \quad 4x - 6y = 0 \Rightarrow 2x - 3y = 0$

$PB^2 = PC^2$

$\Rightarrow \quad (x-0)^2 + (y-3)^2 + (0-2)^2 = (x-0)^2 + (y-0)^2 + (0-1)^2$

$\Rightarrow \quad -6y + 12 = 0 \Rightarrow y = 2$

Putting y = 2 in (i), we obtain x = 3.

Hence, the required point is (3, 2, 0).

EXAMPLE 4 Show taht the points A (0,1,2), B(2-1,3) and C (1,-3,1) are vertices of an isosceles right - angled triangle.

SOLUTION We have,

$$AB=\sqrt{(2-0)^2+(-1-1)^2+(3-2)^2}=\sqrt{4+4+1}=3,$$

$$BC=\sqrt{(1-2)^2+(-3+1)^2+(1-3)^2}=\sqrt{1+4+4}=3$$

and $CA=\sqrt{(1-0)^2+(-3-1)^2+(1-2)^2}=\sqrt{1+16+1}=3\sqrt{2}$

Clearly, $AB=BC$ and $AB^2+BC^2=AC^2$.

Hence, triangle ABC is a an isosceles right-angled triangle.

EXAMPLE 5 Find the lopcus of the point which is equidistant from the points A(0,2,3)and B(2,-2,1)

SOLUTION Let P (x, y, z) be any point which is equidistant from A(0,2,3) and B(2,-2,1). Then,

PA= PB

$$\Rightarrow \quad PA^2=PB^2$$

$$\Rightarrow \quad \sqrt{(x-0)^2+(y-2)^2+(z-3)^2}=\sqrt{(x-2)^2+(y+2)^2+(z-1)^2}$$

$$\Rightarrow 4x-8y-4z+4=0\Rightarrow x-2y-z+1=0$$

Hence, the required locus is $x-2y-z+1=0$

EXAMPLE 6 Find the coordinates of a point equidistant from the four points O(0,0,0), A(a, 0,0),B(0,b,0) and C(0, 0, c).

SOLUTION Let $P(x,y,z)$ be the required point. Then, OP=PA =PB=PC.

Now, OP = PA

$\Rightarrow OP^2=PA^2$

$$\Rightarrow x^2+y^2+z^2=(x-a)^2+(y-0)^2+(z-0)^2$$

$$\Rightarrow 0=-2ax+a^2\Rightarrow x = a/2$$

Similarly, $OP=PB\Rightarrow y=b/2$ and OP = PC $\Rightarrow$ z=c / 2.

Hence, the coordinates of the required point are $\left(\frac{a}{2},\frac{b}{2},\frac{c}{2}\right)$

EXAMPLE 7 Using vector method: Prove that the points A(3,-2,4),B(1,1,1) and C(-1,4,-2) are collinear.

SOLUTION We have,

$\overrightarrow{AB}=$ position vector of B- Position vector of A

$$\Rightarrow \quad \overrightarrow{AB}=\left(\hat{i}+\hat{j}+\hat{k}\right)-\left(3\hat{i}-2\hat{j}+4\hat{k}\right)=-2\hat{i}+3\hat{j}-3\hat{k}$$

and , $\overrightarrow{BC}=\left(-\hat{i}+4\hat{j}-2\hat{k}\right)-\left(\hat{i}+\hat{j}+\hat{k}\right)=-2\hat{i}+3\hat{j}-3\hat{k}$

Clearly, $\overrightarrow{AB}=\overrightarrow{BC}$

This shows that AB is parallel to BC. But, B is common to AB and BC.

Hence, A, B, C are collinear.

EXAMPLE 8 Find the distance between the points A and B with positiion vectors $\hat{i}-\hat{j}$ and $2\hat{i}+\hat{j}+2\hat{k}$.

SOLUTION We have,

$\overrightarrow{AB}=$ Position vector of B-Position vector of A

$$\Rightarrow \overrightarrow{AB}\left(2\hat{i}+\hat{j}+2\hat{k}\right)-\left(\hat{i}-\hat{j}+0\hat{k}\right)=\hat{i}+2\hat{j}+2\hat{k}$$

$$\therefore \quad AB=|\overrightarrow{AB}|=\sqrt{1+4+4}=3$$

3. SECTION FORMULAS

THEOREM 1 (Internal Division) Let P (x_1, y_1, z_1) and Q (x_2, y_2, z_2) be two points. Let R be a point on the line segment joining P and Q such that it divides the join of P and Q internally in the ratio $m_1 : m_2$. Then, the coordiantes of R are

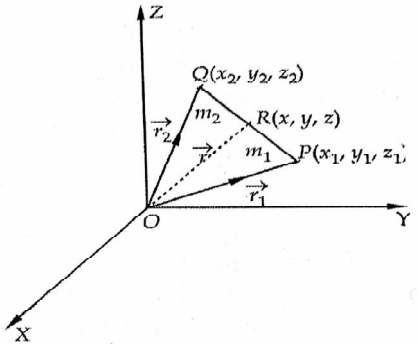
$$\left(\frac{m_1x_1+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}, \frac{m_1z_2+m_2z_1}{m_1+m_2}\right)$$

PROOF Let the coordinates of R be (x, y, z). Let $\vec{r_1}, \vec{r_2}$ and $\vec{r}$ be the position vectors of P, Q and R respectively. Then,

$$\vec{r_1} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}, \vec{r_2} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k} \text{ and, } \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

Since R divides PQ internally in the ratio $m_1 : m_2$.

$\therefore$ Position vector $\vec{r}$ of point R is given by $\vec{r} = \frac{m_1\vec{r_2} + m_2\vec{r_1}}{m_1 + m_2}$



$$\Rightarrow x\hat{i} + y\hat{j} + z\hat{k} = \frac{m_1\left(x_2\hat{i} + y_2\hat{j} + z_2\hat{k}\right) + m_2\left(x_1\hat{i} + y_1\hat{j} + z_1\hat{k}\right)}{m_1 + m_2}$$

$$\Rightarrow x\hat{i} + y\hat{j} + z\hat{k} = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}\right)\hat{i} + \left(\frac{m_1y_2 + m_2y_1}{m_1 + m_2}\right)\hat{j} + \left(\frac{m_1z_2 + m_2z_1}{m_1 + m_2}\right)\hat{k}$$

$$\Rightarrow x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}, y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}, z = \frac{m_1z_2 + m_2z_1}{m_1 + m_2}$$

Hence, the coordinates of R are

$$\left(\frac{m_1x_2+m_2x_1}{m_1+m_2}, \frac{m_1y_2+m_2y_1}{m_1+m_2}, \frac{m_1z_2+m_2z_1}{m_1+m_2}\right)$$

COROLLARY If R is the mid-point of the segment joining P (x_1, y_1, z_1) and Q (x_2, y_2, z_2), then

$$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

THEOREM 2 (External Division) Let P (x_1, y_1, z_1) and Q $(x_2, y_2 z_2)$ be two points, and let R be a point on PQ produced dividing it externally in the ratio $m_1 : m_2$ $(m_1 \neq m_2)$. Then, the coordinates of R are

$$\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}, \frac{m_1z_2 - m_2z_1}{m_1 - m_2}\right)$$

PROOF Let the coordinates of R be (x,y,z). Let $\vec{r_1}, \vec{r_2}$ and $\vec{r}$ be the position vectors of P, Q and R respectively. Then,

$\vec{r_1} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}, \vec{r_2} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Since R divides PQ externally in the ratio m_1: m_2. Therefore, position vector $\vec{r}$ of point R is given by

$$\vec{r} = \frac{m_1\vec{r_2} - m_2\vec{r_1}}{m_1 - m_2}$$

$$\Rightarrow \quad x\hat{i} + y\hat{j} + z\hat{k} = \frac{m_1\left(x_2\hat{i} + y_2\hat{j} + z_2\hat{k}\right)}{m_1 - m_2}$$

$$\Rightarrow \quad x\hat{i} + y\hat{j} + z\hat{k} = \left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}\right)\hat{i} + \left(\frac{m_1y_2 - m_2y_1}{m_1 - m_2}\right)\hat{j} + \left(\frac{m_1z_2 - m_2z_1}{m_1 - m_2}\right)\hat{k}$$

$$\Rightarrow \quad x = \frac{m_1x_2 - m_2x_1}{m_1 - m_2}, y = \frac{m_1y_2 - m_2y_1}{m_1 - m_2}, z = \frac{m_1z_2 - m_2z_1}{m_1 - m_2}$$

Hence, the coordinates of R are given by

$$\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}, \frac{m_1z_2 - m_2z_1}{m_1 - m_2}\right)$$

EXAMPLE 1 Find the coordinates of the point with divides the join of P(2, -1, 4) and Q(4, 3,2) in the ratio 2 :3 (i) internally (ii) externally.

SOLUTION Let R(x,y,z)be the required point. Then,

(i) $x = \frac{2\times4 + 3\times2}{2+3}, y = \frac{2\times3 + 3\times-1}{2+3}, z = \frac{2\times2 + 3\times4}{2+3}$

$\Rightarrow x = \frac{14}{5}, y = \frac{3}{5}, z = \frac{16}{5}$

So, coordinates of the required point are $R\left(\frac{14}{5}, \frac{3}{5}, \frac{16}{5}\right)$

(ii) $x = \frac{2\times4 - 3\times2}{2-3}, y = \frac{2\times3 - 3\times-1}{2-3}, z = \frac{2\times2 - 3\times4}{2-3}$

$\Rightarrow x = -2, y = -9, z = 8$

EXAMPLE 2 Find the ratio in which the line joining the points (1,2,3) and (-3, 4,-5) is divided by the xy-plane. also find the coordinates of the point of division.

SOLUTION Suppose the line joinig the points P (1,2,3) and Q(-3,4-5)is divided by the xy-plane at a point R in the ratio $\lambda:1$. Then, the coordiantes of R are

$$\left(\frac{-3\lambda + 1}{\lambda + 1}, \frac{4\lambda + 2}{\lambda + 1}, \frac{-5\lambda + 3}{\lambda + 1}\right)$$

Since R lies on xy-plane i.e. z=0. Therefore,

$$\frac{-5\lambda + 3}{\lambda + 1} = 0 \Rightarrow \lambda = \frac{3}{5}$$

So, the required ratio is $\frac{3}{5}:1$ or, $3:5$

Putting $\lambda = \frac{3}{5}$ in (i), we obtain the coordinates of R as (-1/ 2,11/ 4,0).

EXAMPLE 3 Find the ratio in which join the A (2, 1, 5) and B (3, 4, 3) is divided by the plane $2x+2y-2z=1$. Also, find the coordinates of the point of division.

SOLUTION Suppose the plane $2x+2y-2z=1$ divides the line joinig the points A(2,1,5) and B(3,4,3) at a point C in the ratio $\lambda:1$. Then, the coordinates of C are

$$\left(\frac{3\lambda+2}{\lambda+1},\frac{4\lambda+1}{\lambda+1},\frac{3\lambda+5}{\lambda+1}\right)$$

Since point C lies on the plane $2x+2y-2z=1$. Therefore, coordinates of C must satisfy the equation of the plane

i.e. $$2\left(\frac{3\lambda+2}{\lambda+1}\right)+2\left(\frac{4\lambda+1}{\lambda+1}\right)-2\left(\frac{3\lambda+5}{\lambda+1}\right)=1$$

$\Rightarrow$ $$8\lambda-4=\lambda+1\Rightarrow\lambda=\frac{5}{7}$$

So, the required ratio is $\frac{5}{7}:1$ or $5:7$.

Putting $\lambda=\frac{5}{7}$ in (i), the coordinates of the point of division C are (29 / 12 ,9/4, 25/6).

EXAMPLE 4 Using section formula, prove that the three points A(-2,3,5), B(1,2,3)and C (7, 0 , -1)are collinear

SOLUTION Suppose the given point are collinear and C divides AB in the ratio $\lambda:1$. Then, coordinates of C are

$$\left(\frac{\lambda-2}{\lambda+1},\frac{2\lambda+3}{\lambda+1},\frac{3\lambda+5}{\lambda+1}\right)$$

But , coordinates of C are (7, 0,-1).

$\therefore$ $$\frac{\lambda-2}{\lambda+1}=7,\frac{2\lambda+3}{\lambda+1}=0 \text{ and } \frac{3\lambda+5}{\lambda+1}=-1$$

From each of these equations, we get $\lambda=-\frac{3}{2}$

since each of these equations give the same value of λ. Therefore, the given points are collinear and C divides AB extrnally in the ratio 3:2.

EXAMPLE 5 Give that P(3,2,-4),Q(5,4,-6) and R(9,8,-10)are collinear. find the ratio in which Q divides PR.

SOLUTION Suppose Q divides PR in the ratio $\lambda:1$. Then, coordinates of Q are

$$\left(\frac{9\lambda+3}{\lambda+1},\frac{8\lambda+2}{\lambda+1},\frac{-10\lambda-4}{\lambda+1}\right)$$

But, coordinates of Q are (5,4,-6).

$\therefore$ $$\frac{9\lambda+3}{\lambda+1}=5,\frac{8\lambda+2}{\lambda+1}=4,\frac{-10\lambda-4}{\lambda+1}=6$$

These three equations give $\lambda=\frac{1}{2}$. So, Q divides PR in the ratio $\frac{1}{2}:1$ or 1 :2.

MISCELLANEOUS EXAMPLES

Illustration 1: Find the locus of the point which is equidistant from the points (1, 3, 5) and (–2, 1, 4).

Solution : Let $P(x, y, z)$ be any point equidistant from the point $A(1, 3, 5)$ and $B(-2, 1, 4)$, then

$PA = PB \Rightarrow PA^2 = PB^2$

$\Rightarrow (x-1)^2 + (y-3)^2 + (z-5)^2 = (x+2)^2 + (y-1)^2 + (z-4)^2$

$\Rightarrow x^2 - 2x + 1 - 6y + 9 + z^2 - 10z + 25 = x^2 + 4x + y^2 - 2y + 1 + z^2 - 8z + 16$

$\Rightarrow 6x + 4y + 2z - 14 = 0$

$\Rightarrow 3x + 2y + z - 7 = 0$

which is the required locus.

Illustration 2 : Find the ratio in which the join of (2, –1, 4) and (4, 3, 2) is divided by the point (–2, –9, 8).

Solution : Let the ratio be $\lambda : 1$

The point which divides the join of (2, –4, 4) and (4, 3, 2) in ratio $\lambda : 1$ is

$$\left(\frac{4\lambda+2}{\lambda+1}, \frac{3\lambda-1}{\lambda+1}, \frac{2\lambda+4}{\lambda+1}\right)$$

But the given point is (–2, –9, 8) $\therefore$

$$\frac{4\lambda+2}{\lambda+1} = -2$$

$\Rightarrow 4\lambda + 2 = -2\lambda - 2 \quad \Rightarrow \quad 6\lambda = -4$

$$\Rightarrow \lambda = -\frac{2}{3}$$

$\therefore$ Required ratio is $-\lambda : 1$ i.e., $-2 : 3$.

Thus, (–2, –9, 8) divide the join of (2, –1, 4) and (4, 3, 2) extrenally in ratio 2 : 3.

Illustration 3: Find the ratio in which the line joining the points (1, 2, 3) and (–3, 4, –5) is divided by the *xy*-plane. Also find the coordinates of the point of division.

Solution : Let the ratio be $\lambda : 1$.

The point which divides the join of (1, 2, 3) and (–3, 4, –5) in ratio $\lambda : 1$ is

$$\left(\frac{-3\lambda+1}{\lambda+1}, \frac{4\lambda+2}{\lambda+1}, \frac{-5\lambda+3}{\lambda+1}\right)$$

....(i)

But this point lies on *xy*-plane, therefore z-coordinate is zero,

i.e., $\frac{-5\lambda+3}{\lambda+1} = 0 \Rightarrow = \frac{3}{5}$

Therefore, required ratio is $\frac{3}{5} : 1$, i.e. 3 : 5.

The required point is $\left(\frac{3(-3)+5(1)}{3+5}, \frac{3(4)+5(2)}{3+5}, \frac{3(-5)+5(3)}{3+5}\right)$

i.e. $\left(-\frac{4}{8}, \frac{22}{8}, 0\right)$ i.e. $\left(-\frac{1}{2}, \frac{11}{4}, 0\right)$

Illustration 4: Two vertices of a triangle *ABC* are $A(2, -4, 3)$ are and $B(3, -1, 2)$ and its centroid is (1, 0, 3). Find its third vertex *C*.

Solution : Let the third vertex be (x_3, y_3, z_3), then centroid is

$$\left(\frac{2+3+x_3}{3}, \frac{-4-1+y_3}{3}, \frac{3-2+z_3}{3}\right)$$

i.e., $\left(\frac{5+x_3}{3}, \frac{-5+y_3}{3}, \frac{1+z_3}{3}\right)$

But centroid is given to be (1, 0, 3). $\therefore \frac{5+x_3}{3}=1, \frac{-5+y_3}{3}=0$ and $\frac{1+z_3}{3}=3$

$\Rightarrow x_3 = -2$, $y_3 = 5$ and $z_3 = 8$
Thus, the third vertex is (–2, 5, 8).

Illustration 5: If origin is the centroid of triangle *ABC* with vertices *A*(*a*, 1, 3), *B*(–2, *b*, 5) and *C*(4, 7, *c*), find the values of *a*, *b*, and *c*.

Solution : Centroid of triangle *ABC* with vertices *A*(*a*, 1, 3), *B*(–2, *b*, –5) and *C*(4, 7, *c*) is

$$\left(\frac{a-2+4}{3}, \frac{1+b+7}{3}, \frac{3-5+c}{3}\right)$$

i.e. $\left(\frac{a+2}{3}, \frac{8+b}{3}, \frac{-2+c}{3}\right)$

But, centroid is given to be (0, 0, 0)

$$\therefore \frac{a+2}{3}=0, \frac{8+b}{3}=0, \frac{-2+c}{3}=0$$

$\Rightarrow$ a = –2, b = –8 and c = 2.

Illustration 6: The mid-points of the sides of a triangle are (1, 5, –1), (0, 4, –2) and (2, 3, 4). Find its vertices.

Solution : Let $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$ and $C(x_3, y_3, z_3)$ be the vertices of the triangle and let *D*(1, 5, –1), *E*(0, 4, –2) and *F*(2, 3, 4) be the mid-points of the sides *BC*, *CA* and *AB* respectively. Then

$$\frac{x_2+x_3}{2}=1, \frac{y_2+y_3}{2}=5, \frac{z_2+z_3}{2}=1;$$

$$\frac{x_3+x_1}{2}=0 \quad \frac{y_3+y_1}{2}=4 \quad \frac{z_3+z_1}{2}=-2$$

$$\frac{x_1+x_2}{2}=2 \quad \frac{y_1+y_2}{2}=3 \quad \frac{z_1+z_2}{2}=4$$

Thus $x_2+x_3=2 \quad x_3+x_1=0 \quad x_1+x_2=4$

Adding them
$(x_1 + x_2 + x_3) = 3$
so from the above relations we get
$x_1 = 1, \quad x_2 = 3 \quad x_3 = -1$
Similary we get the values
$y_1 = 2 \qquad y_2 = 4 \quad y_3 = 6$
$z_1 = 3 \qquad z_2 = 5 \quad z_3 = -7$
Hence vertices of triangle are
A(1, 2, 3) *B*(3, 4, 5) & *C*(–1, 6, –7)

Illustration 7: If $\frac{2}{3}, -\frac{2}{3}, \frac{1}{3}$ are direction cosines of a vector $\vec{r}$, then its direction ratios are 2, –2, 1 or

–2, 2, –1 or 4, –4, 2, because

Solution : $\frac{2/3}{2}=\frac{-2/3}{2}=\frac{1/3}{1},\frac{2/3}{-2}=\frac{-2/3}{2}$

$\Rightarrow \quad \frac{1/3}{-1},\frac{2/3}{4}=\frac{-2/3}{-4}=\frac{1/3}{2}$

Illustration 8: A vector $\overrightarrow{OP}$ is inclined to *OX* at 45 and *OY* at 60. Find the angle at which $\overrightarrow{OP}$ is inclined to *OZ*.

Solution : Suppose $\overrightarrow{OP}$ is inclined at an angle γ to *OZ*. Let ℓ, *m*, *n* be the direction cosines of $\overrightarrow{OP}$. Then $\ell = \cos 45^o$, $m = \cos 60^o$, $n = \cos\gamma$.

$\Rightarrow \ell = \frac{1}{\sqrt{2}}, m=\frac{1}{2}, n=\cos\gamma$

Now, $\ell^2 + m^2 + n^2 = 1$

$\Rightarrow \quad \frac{1}{2}+\frac{1}{4}+n^2=1$

$\Rightarrow \quad n^2 = \frac{1}{2} \Rightarrow n = \pm\frac{1}{2}$

$\Rightarrow \quad \cos\gamma = \pm\frac{1}{2} \Rightarrow \gamma = 60^o$ or 120°

Hence, $\overrightarrow{OP}$ is inclined to *OZ* either at 60° or at 120°.

Illustration 9: If a vector makes angles α, β, γ with *OX, OY* and *OZ* respectively, prove that : $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Solution : Let $\ell^2 + m^2 + n^2 = 1$

$\Rightarrow \quad \cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow \quad (1 - \sin^2\alpha) + (1 - \sin^2\beta) + (1 - \sin^2\gamma) = 1$

$\Rightarrow \quad \sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Illustration 10: Find the direction cosines of a vector $\vec{r}$ which is equally inclined with *OX*, *OY* and *OZ*. If $|\vec{r}|$ is given, find the total number of such vectors.

Solution : Let ℓ, *m, n* be the direction cosines of $\vec{r}$. Since $\vec{r}$ is equally inclined with *OX*, *OY* and *OZ* Therefore,

$\ell = m = n \quad [\because \alpha = \beta = \gamma \ \therefore \cos\alpha = \cos\beta = \cos\gamma]$

now, $\ell^2 + m^2 + n^2 = 1 \quad \Rightarrow 3\ell^2 = 1 \Rightarrow \ell = \pm\frac{1}{\sqrt{3}}$

Hence, direction cosines of $\vec{r}$ are $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$.

Now, $\vec{r} = |\vec{r}|(\ell\hat{i} + m\hat{j} + n\hat{k})$

$\Rightarrow \quad \vec{r} = |\vec{r}|\left(\pm\frac{1}{\sqrt{3}}\hat{i} \pm \frac{1}{2}\hat{j} \pm \frac{1}{\sqrt{3}}\hat{k}\right)$

Since + and –signs can be arranged at three places in 2 × 2 × 2 = 8 ways. Therefore, there are eight vectors of given magnitude which are equally inclined with the coordinate axes.

Illustration 11: A vector $\vec{r}$ has length 21 and direction ratios 2, –3, 6. Find the direction cosines and components of $\vec{r}$, given that $\vec{r}$ makes an acute angle with *x*-axis.

Solution : Recall that if *a*, *b*, *c* direction ratios of a vector, then its direction cosines are

$$\pm\frac{a}{\sqrt{a^2+b^2+c^2}},\pm\frac{b}{\sqrt{a^2+b^2+c^2}},\pm\frac{c}{\sqrt{a^2+b^2+c^2}}$$

Therefore direction cosines of $\vec{r}$ are

$$\pm\frac{2}{\sqrt{2^2+(-3)^2+6^2}},\frac{-3}{\sqrt{2^2+(-3)^2+6^2}},\frac{6}{\sqrt{2^2+(-3)^2+6^2}}$$

Since $\vec{r}$ makes an acute angle with *x*-axis, therefore $\alpha > 0$ i.e. $\ell > 0$.

So, direction $\vec{r}$ are $\frac{2}{7},-\frac{3}{7},\frac{6}{7}$

$\therefore \quad \vec{r}=21\left(\frac{2}{7}\hat{i}-\frac{3}{7}\hat{j}+\frac{6}{7}\hat{k}\right)$ [Using $\vec{r}=|\vec{r}|(\ell\hat{i}+m\hat{j}+n\hat{k})$]

or $\vec{r}=6\hat{i}-9\hat{k}+18\hat{k}$

So, coomponents of $\vec{r}$ along *OX, OY* and *OZ* are $6\hat{i},-8\hat{j}$ and $18\hat{k}$ respectively.

Illustration 12: Find the angles at which the vector $2\hat{i}-\hat{j}+2\hat{k}$ is inclined to each of the coordiantes axes.

Solution : Let $\vec{r}$ be the given vector, and let it make angles α, β, γ with *OX*, *OY* and *OZ* respectively. Then its direction cosine are $\cos\alpha, \cos\beta, \cos\gamma$.

We have $\vec{r}=2\hat{i}-\hat{j}+2\hat{k}$ so, direction ratios of $\vec{r}$ are 2, –1, 2.

$\therefore$ Direction cosines of $\vec{r}$ are

$$\frac{2}{\sqrt{2^2+(-1)^2+2^2}},\frac{-1}{\sqrt{2^2+(-1)^2+2^2}},\frac{2}{\sqrt{2^2+(-1)^2+2^2}},$$

i.e. $\frac{2}{3},-\frac{1}{3},\frac{2}{3}$

$$\therefore \cos\alpha=\frac{2}{3},\cos\beta=-\frac{1}{3},\cos\gamma=\frac{2}{3} \quad \Rightarrow \alpha=\cos^{-1}\left(\frac{2}{3}\right),\beta=\cos^{-1}\left(-\frac{1}{3}\right),\gamma=\cos^{-1}\left(\frac{2}{3}\right)$$

$$\Rightarrow \alpha=\cos^{-1}\left(\frac{2}{3}\right),\beta=\pi-\cos^{-1}\left(-\frac{1}{3}\right),\gamma=\cos^{-1}\left(\frac{2}{3}\right)$$

Illustration 13: Find the angle between the lines whose direction-cosines are $\frac{-3}{7},\frac{6}{7},\frac{-2}{7}$ and $\frac{-2}{3},\frac{2}{3},\frac{1}{3}$.

Solution : Let θ be the angle between the two lines whose direction-cosines given, then

$$\cos\theta=\left(-\frac{3}{7}\right)\left(-\frac{2}{3}\right)+\left(\frac{6}{7}\right)\left(\frac{2}{3}\right)+\left(-\frac{2}{7}\right)\left(\frac{1}{3}\right)$$

$$=\frac{6}{21}+\frac{12}{21}-\frac{2}{21}=\frac{16}{21}$$

$$\Rightarrow \theta=\cos^{-1}\left(\frac{6}{21}\right)$$

Illustration 14: Find the the coordinates of point P such that $\overrightarrow{OP}$ is inclined to x-axis at 45° and to y-axis at 60° and $\left|\overrightarrow{OP}\right| = 2$.

Solution : $\ell = \cos 45^o = \frac{1}{\sqrt{2}}$

$m = \cos 60^o = \frac{1}{2}$ and $n = \cos\gamma$

Now, $\ell^2 + m^2 + n^2 = 1 \Rightarrow \frac{1}{2}+\frac{1}{4}+n^2 = 1$

$n^2 = \frac{1}{4} \Rightarrow n = \pm\frac{1}{2}$ $\Rightarrow \cos\gamma = \pm\frac{1}{2} \Rightarrow \gamma = 60^o$ *or* 120°

Now, $\widehat{OP} = \frac{1}{\sqrt{2}}\hat{i}+\frac{1}{2}\hat{j}\pm\frac{1}{\sqrt{2}}\hat{k}$

$\overrightarrow{OP} = \left|\overrightarrow{OP}\right|\widehat{OP} = 2\left(\frac{1}{\sqrt{2}}\hat{i}+\frac{1}{2}\hat{j}\pm\frac{1}{\sqrt{2}}\hat{k}\right) = \frac{2}{\sqrt{2}}\hat{i}+\hat{j}\pm\hat{k} = \sqrt{2}\hat{i}+\hat{j}\pm\hat{k}$

$\therefore$ coordinates of point P are $\left(\sqrt{2},1,\pm1\right)$

Illustration 15: The projections of a line segment on the coordinates axes are 6, 2, 3. Find the length of the line and its direction-cosines.

Solution : Let ℓ, m, n be the direction-cosines of the line segment and r be its length
Then,
Projection on x-axis = $r\ell = 6$
Projection on y-axis = $rm = 2$
and Projection on z-axis = $rn = 3$
$\Rightarrow (r\ell)^2 + (rm)^2 + (rn)^2 = (6)^2 + (2)^2 + (3)^2$
$\Rightarrow r^2\ell^2 + r^2m^2 + r^2n^2 = 36 + 4 + 9$
$\Rightarrow r^2(\ell^2 + m^2 + n^2) = 49$ $[\because \lambda^2 + m^2 + n^2 = 1]$ $\Rightarrow r^2 = 49 \Rightarrow r = 7$

$\therefore rl = 6 \Rightarrow \ell = \frac{6}{7}$

$rm = 2 \Rightarrow m = \frac{2}{7}$ and $rn = 3 \Rightarrow n = \frac{3}{7}$

Illustration 16: Find the direction-ratios of a line pependicular to the two lines having direction-ratios 1, 3, –2 and –2, 2, 4 respectively.

Solution : Let a, b, c be the required direction-ratios.
Since, this line is perpendicular to the two lines having direction-ratios 1, 3, –2 and –2, 2, 4.
$\therefore a + 3b - 2c = 0$
$-2a + 2b + 4c = 0$

$\therefore \frac{a}{3\ \ -2} = \frac{-b}{1\ \ -2} = \frac{c}{1\ \ 3}$ $\quad$ *i.e.*, $\frac{a}{16} = \frac{b}{0} = \frac{c}{8}$ *i.e.*, $\frac{a}{2} = \frac{b}{0} = \frac{c}{1}$
$\quad 2\ \ 4\ \ -2\ \ 4\ \ -2\ \ 2$

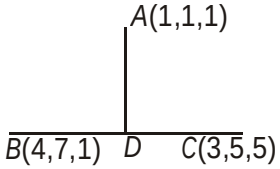
Thus, direction-ratios are 2, 0, 1.
Note : Do not get confuse with O in denominator here.

Illustration 17: Find the foot of the perpendicular from the point (1, 1, 1) to the joint of the points (4, 7, 1) and (3, 6, 3).

Solution : Let $A(1, 1, 1)$, $B(4, 7, 1)$ and $C(3, 5, 3)$ be three points and D be the foot of perpendicular from A on BC

Let D divide BC in ratio $\lambda : 1$, then coordinate of D are

$$\left(\frac{3\lambda+4}{\lambda+1}, \frac{5\lambda+7}{\lambda+1}, \frac{3\lambda+1}{\lambda+1}\right)$$



Direction-ratios of AD are

$$\frac{3\lambda+4}{\lambda+1}-1, \frac{5\lambda+7}{\lambda+1}-1, \frac{3\lambda+1}{\lambda+1}-1$$

i.e. $\frac{2\lambda+3}{\lambda+1}, \frac{4\lambda+6}{\lambda+1}, \frac{2\lambda}{\lambda+1}$

Direction-ratios of BC are

$3 – 4, 5 – 7, 3 – 1$ *i.e.* $–1, –2, 2$

Since $AD \perp BC$

$$\therefore\ -1\left(\frac{2\lambda+3}{\lambda+1}\right)-2\left(\frac{4\lambda+6}{\lambda+1}\right)+2\left(\frac{2\lambda}{\lambda+1}\right)=0$$

$$\Rightarrow \frac{-2\lambda-3-8\lambda-12+4\lambda}{\lambda+1}=0 \qquad \Rightarrow \frac{-6\lambda-15}{\lambda+1}=0 \Rightarrow -6\lambda = 15\ \lambda=-\frac{5}{2}$$

$\therefore$ D divides BC in ratio 5 : 2 externally. $\therefore$ Coordinates of point D are

$$\left(\frac{15-8}{3}, \frac{25-14}{3}, \frac{15-2}{3}\right) \quad \textit{i.e.} \quad \left(\frac{7}{3}, \frac{11}{3}, \frac{13}{3}\right)$$

Illustration 18: If the edges of a rectangular parallelopiped are a, b, c prove that the angle between the four diagonals are given by $\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}\right)$.

Solution : Let OA, OB, OC be the coterminous edges of the paralleopiped taken along the coordinateaxes and let

$OA = a$, $OB = b$ and $OC = c$

The coordinates of the vertices are

$O(0,0,0)$, $A(a,0,0,)$, $B(0, b, 0)$, $C(0, 0, c)$

$D(a,b,0)$, $E(a,0,c)$, $F(0,b,c)$, $G(a,b,c)$

The direction-ratios of the diagonal

OG are $a – 0, b – 0, c – 0$ *i.e.* a,b,c

AF are $0 – a, b – a, c – 0$, *i.e.* $–a,b,c$

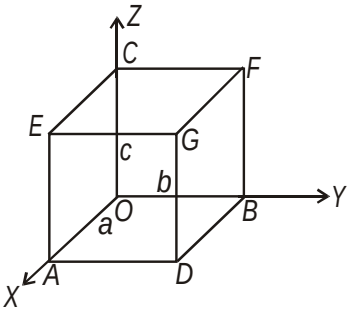
BE are $a – 0, 0 – b, c – 0$, *i.e.* $a,b,–c$

and CD are $a – 0, b – 0, 0 – c$, *i.e.*, $a,b,–c$

Thus, direction-cosines of OG, AF, BE and CD are

$$\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$

$$\frac{-a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$



$$\frac{a}{\sqrt{a^2+b^2+c^2}},\frac{-b}{\sqrt{a^2+b^2+c^2}},\frac{c}{\sqrt{a^2+b^2+c^2}}$$

$$\frac{a}{\sqrt{a^2+b^2+c^2}},\frac{b}{\sqrt{a^2+b^2+c^2}},\frac{-c}{\sqrt{a^2+b^2+c^2}}$$ respectively.

If θ_1 is the angle between *OG* and *AF*, then

$$\cos\theta_1 = \frac{-a^2+b^2+c^2}{a^2+b^2+c^2} \qquad \Rightarrow \theta_1 = \cos^{-1}\left(\frac{-a^2+b^2+c^2}{a^2+b^2+c^2}\right)$$

Again, if θ_2 is the angle between *OG* and *BE*, then

$$\cos\theta_2 = \frac{a^2-b^2+c^2}{a^2+b^2+c^2} \qquad \Rightarrow \theta_2 = \cos^{-1}\left(\frac{a^2-b^2+c^2}{a^2+b^2+c^2}\right)$$

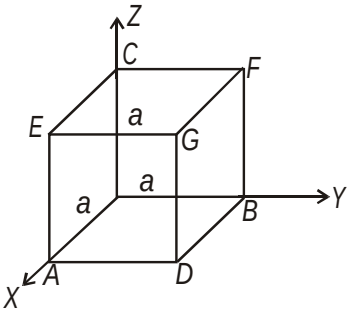
Similary, the angle between the other pairs of diagonals can be. obtained.
Thus, the angles between the four diagonals may be given by

$$\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}\right)$$

Illustration 19: Show that the angle between any two diagonals of a cube is $\cos^{-1}(1/3)$.

Solution : Let OA, OB, OC be the coterminous edges of a cube, taken along the coordinates axes and let

OA = OB = OC = a



The coordinates of the vertices are
O(0,0,0,), *A*(*a*,0,0), *B*(0,*a*,0), *C*(0,0,*c*)
D(*a*,*a*,0), *E*(*a*,0,*a*), *F*(0,*a*,0), *G*(*a*,*a*,*a*)
The direction-ratios of the diagonal OG are
a,*a*,*a*; *AF* are –*a*,–*a*,*a* and *CD* are *a*,*a*,–*a*
Thus, direction-cosines of *OG*, *AF*, *BE* and *CD* are

$$\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}};\ \frac{-1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}};$$

$$\frac{1}{\sqrt{3}},\frac{-1}{\sqrt{3}},\frac{1}{\sqrt{3}} \text{ and } \frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{-1}{\sqrt{3}}; \text{ respectively.}$$

If θ_1 be the angle between *OG* and *AF*, then

$$\cos\theta_1 = \left(\frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{3}}\right)+\left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right)+\left(\frac{1}{\sqrt{3}}\right)\left(\frac{1}{\sqrt{3}}\right)$$

$$= -\frac{1}{3}+\frac{1}{3}+\frac{1}{3} \qquad \Rightarrow \cos\theta_1 = \frac{1}{3} \Rightarrow \theta_1 = \cos^{-1}\left(\frac{1}{3}\right)$$

Similarly, the angle between any two diagonals of the cube is $\cos^{-1}(1/3)$.

Illustration 20: A line makes angle $\alpha,\beta,\gamma,\delta$ with the four diagonals of a cube, then prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$.

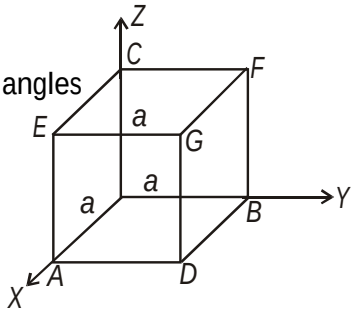
Solution : Let OA, OB, OC be the coterminous edges of a cube, taken along the coordinates axes and let *OA* = *OB* = *OC* = *a* (Figure)
The coordinates of the vertices are
O(0,0,0), *A*(*a*,0,0), *B*(0,*a*,0), *C*(0,0,*a*)
D(*a*,*a*,0), *E*(*a*,0,*a*), *F*(0,*a*,*a*), *G*(*a*,*a*,*a*)
The direction-ratios of the diagonals

OG are *a*,*a*,*a*; *AF* are –*a*,*a*,*a*; *BE* are *a*,–*a*,*a* and *CD* are *a*,*a*,–*a*. *BE* and *CD* are

$$\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}};-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}$$

$$\frac{1}{\sqrt{3}},\frac{-1}{\sqrt{3}},\frac{1}{\sqrt{3}} \text{ and } \frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{-1}{\sqrt{3}}$$

respectively.



Let ℓ,m,n be the direction cosines of a line which makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of the cube. Then

$$\cos\alpha = \frac{\ell}{\sqrt{3}},\frac{m}{\sqrt{3}},\frac{n}{\sqrt{3}} = \frac{\ell+m+n}{\sqrt{3}}$$

$$\cos\beta = -\frac{\ell}{\sqrt{3}}+\frac{m}{\sqrt{3}}+\frac{n}{\sqrt{3}} = \frac{-\ell+m+n}{\sqrt{3}}$$

$$\cos\gamma = \frac{\ell}{\sqrt{3}}-\frac{m}{\sqrt{3}}+\frac{n}{\sqrt{3}} = \frac{\ell+m+n}{\sqrt{3}}$$

$$\cos\delta = \frac{\ell}{\sqrt{3}}+\frac{m}{\sqrt{3}}-\frac{n}{\sqrt{3}} = \frac{\ell+m-n}{\sqrt{3}}$$

On squaring and adding, we get

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta$

$$= \left(\frac{\ell+m+n}{\sqrt{3}}\right)^2+\left(\frac{-\ell+m+n}{\sqrt{3}}\right)^2+\left(\frac{\ell-m+n}{\sqrt{3}}\right)^2+\left(\frac{\ell+m-n}{\sqrt{3}}\right)^2$$

$$= \frac{1}{3}[\ell^2 + m^2 + n^2 + 2\ell m + 2mn + 2n\ell + \ell^2 + m^2 + n^2 - 2lm - 2mn - 2nl$$

$$+ \ell^2 + m^2 + n^2 - 2\ell m - 2mn + 2n\ell + \ell^2 + m^2 + n^2 - 2\ell m - 2mn - 2n\ell]$$

$$= \frac{1}{3}[4\ell^2 + m^2 + n^2)] = \frac{4}{3}$$

Illustration 21 If ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 be the direction-cosines of two mutually perpendiuclar lines, show that the direction-cosines of the line perpendicular to both of then are $m_1n_1 - m_2n_1, n_1\ell_2 - n_2\ell_1, \ell_1m_2 - \ell_2m_1$.

Solution : Let ℓ, m, n be the direction-cosines of the line perpendicular to each one of the given lines. Then

$\ell\ell_1 + mm_1 + nn_1 = 0$

$\ell\ell_2 + mm_2 + nn_2 = 0$

$$\therefore \frac{\ell}{\begin{vmatrix} m_1 & n_1 \\ m_2 & n_2 \end{vmatrix}} = \frac{-m}{\begin{vmatrix} \ell_1 & n_1 \\ \ell_2 & n_2 \end{vmatrix}} = \frac{n}{\begin{vmatrix} \ell_1 & m_1 \\ \ell_2 & m_2 \end{vmatrix}}$$

$$\Rightarrow \frac{\ell}{m_1n_2 - m_2n_1} = \frac{m}{n_1\ell_2 - n_2\ell_1} = \frac{n}{\ell_1m_2 - m_1\ell_2}$$

$$\Rightarrow \frac{\ell}{m_1n_2 - m_2n_1} = \frac{m}{n_1\ell_2 - n_2\ell_1} = \frac{n}{\ell_1m_2 - m_1\ell_2} = \frac{\sqrt{\ell^2+m^2+n^2}}{\sqrt{\sum(m_1n_2 - m_2n_1)^2}}$$ [Using langranges identity]

$$\Rightarrow \frac{\ell}{m_1n_2 - m_2n_1} = \frac{m}{n_1\ell_2 - n_2\ell_1} = \frac{n}{\ell_1m_2 - m_1\ell_2} = \frac{1}{\sin\theta}$$

where θ is the angle between the given lines.

Since, $\theta = \pi/2$, $\therefore \sin\theta = 1$

Thus, $\ell = m_1n_2 - m_2n_1$, $m = n_1\ell_2 - n_2\ell_1$ and $n = \ell_1m_2 - \ell_1m_1$

Hence, the direction-cosines of the required line are

$m_1n_2 - m_2n_1$, $n_1\ell_2 - n_2\ell_1$, $\ell_1m_2 - \ell_2m_1$.

Illustration 22: Verify that $\frac{\ell_1+\ell_2+\ell_3}{\sqrt{3}}, \frac{m_1+m_2+m_3}{\sqrt{3}}, \frac{n_1+n_2+n_3}{\sqrt{3}}$ can be taken as the direction-cosines of a line equally inclined to three mutually perpendicular lines with direction-cosines ℓ_1,m_1,n_1 : ℓ_2,m_2,n_2 and ℓ_3,m_3,n_3.

Solution : $\ell_1,m_1,n_1;\ell_2,m_2,n_2$ and ℓ_3,m_3,n_3 be the direction-cosines of three mutually perpendicular lines.

$$\therefore \ \ell_1^2 + m_1^2 + n_1^2 = 1, \ell_2^2 + m_2^2 + n_2^2 = 1 \text{and} \ell_3^2 + m_3^2 + n_3^2 = 1$$

and $\ell_1\ell_2 + m_1m_2 + n_1n_2 = 0$

$\ell_2\ell_3 + m_2m_3 + n_2n_3 = 0$

and $\ell_3\ell_1 + m_3m_1 + n_3n_1 = 0$

Let θ_1 be the angle between the line whose direction-cosines are

$$\frac{\ell_1+\ell_2+\ell_3}{\sqrt{3}}, \frac{m_1+m_2+m_3}{\sqrt{3}}, \frac{n_1+n_2+n_3}{\sqrt{3}}$$

and ℓ_1, m_1, n_1 then

$$\cos\theta_1 = \ell_1\left(\frac{\ell_1+\ell_2+\ell_3}{\sqrt{3}}\right) + m_1\left(\frac{m_1+m_2+m_3}{\sqrt{3}}\right) + n_1\left(\frac{n_1+n_2+n_3}{\sqrt{3}}\right)$$

$$= \frac{1}{\sqrt{3}}[(\ell_1^2 + m_1^2 + n_1^2) + (\ell_1\ell_2 + m_1m_2 + n_1n_2) + (\ell_1\ell_3 + m_1m_3 + n_1n_3)$$

$$= \frac{1}{\sqrt{3}}[1+0+0] = \frac{1}{\sqrt{3}} \qquad \therefore \ \theta_1 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Similarly, we can show that the angle between the other pairs *i.e.*, between the line with direction-cosines.

$$\frac{\ell_1+\ell_2+\ell_3}{\sqrt{3}}, \frac{m_1+m_2+m_3}{\sqrt{3}}, \frac{n_1+n_2+n_3}{\sqrt{3}} \text{ and } \ell_2,m_2,n_2$$

and $\frac{\ell_1+\ell_2+\ell_3}{\sqrt{3}}, \frac{m_1+m_2+m_3}{\sqrt{3}}, \frac{n_1+n_2+n_3}{\sqrt{3}}$ and ℓ_3,m_3,n_3 is also $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$.

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic:1 Distance between two points

Q.1 The points A (1, –1, – 5), B (3, 1, 3) and C(9, 1, –3) are the vertices of -
(a) an equilateral triangle
(b) an isosceles triangle
(c) a right angled triangle
(d) none of these

Q.2 Distance of the point (x, y, z) from y-axis is-
(a) y (b) $\sqrt{x^2+y^2}$
(c) $\sqrt{y^2+z^2}$ (d) $\sqrt{z^2+x^2}$

Q.3 The distance of a point P (x, y, z) from yz plane is –
(a) x (b) y
(c) z (d) x + y + z

Q.4 The co-ordinates of the point which are lie equally distance from the point (0 , 0 , 0) ; (a, 0, 0) ; (0, b, 0) and (0, 0, c)
(a) (a/2, b/2, c/2)
(b) (–a/2, b/2, c/2)
(c) (–a/2, –b/2, c/2)
(d) (a/2, –b/2, –c/2)

Q.5 Distance of the point (a, b, c) from z- axis is -
(a) $\sqrt{a^2+b^2}$ (b) $\sqrt{b^2+c^2}$
(c) $\sqrt{c^2+a^2}$ (d) c

Q.6 The point on xy-plane which is equidistant from the points (2, 0, 3), (0, 3, 2), (0, 0, 1) is
(a) (2, 3, 0) (b) (3, 0, 2)
(c) (3, 2, 0) (d) (2, 3, 1)

Q.7 The point which lie on z -axis has the following condition -
(a) z - coordinate are zero
(b) Both x and y coordinate are zero
(c) Both y and z coordinate are zero
(d) Both x and z coordinate are zero

Q.8 The distance of the point (1, 2, 3) from x-axis is -
(a) $\sqrt{13}$ (b) $\sqrt{5}$
(c) $\sqrt{10}$ (d) None of these

Q.9 If P ≡ (0, 5, 6), Q ≡ (2, 1, 2), R ≡ (a, 3, 4) and PQ = QR then 'a' equal to -
(a) 1 (b) 2
(c) 3 (d) None

Q.10 Points (1, 2, 3); (3, 5, 7) and (–1, –1, –1) are-
(a) Vertices of a equilateral triangle
(b) Vertices of a right angle triangle
(c) Vertices of a isosceles triangle
(d) Collinear

Q.11 If the vertices of points A,B,C of a tetrahedron ABCD are respectively (1, 2, 3) ; (–1, 2, 3),(1, –2, 3) and his centroid is (0, 0, 3/2) then co-ordinate of point D are
(a) (1, 2, –3) (b) (–1, –2, 3)
(c) (–1, –2, –3) (d) (0, 0, 0)

Q.12 The distance of point (1, 2, 3) from coordinate axis are -
(a) 1, 2 , 3 (b) $\sqrt{5},\sqrt{10},\sqrt{13}$
(c) $\sqrt{10},\sqrt{13},\sqrt{5}$ (d) $\sqrt{13},\sqrt{10},\sqrt{5}$

Q.13 The coordinates of the points A and B are (–2, 2, 3) and (13, –3, 13) respectively. A point P moves so that 3PA = 2 PB, then locus of P is –
(a) $x^2 + y^2 + z^2 + 28x – 12y + 10z – 247 = 0$
(b) $x^2 + y^2 + z^2 + 28x – 12y + 10z + 247 = 0$
(c) $x^2 + y^2 + z^2 – 28x + 12y – 10z – 247 = 0$
(d) None of these

Q.14 A point which lie in yz plane, the sum of co-ordinate is 3, if distance of point from xz plane is twice the distance of point from xy plane, then co-ordinates are-
(a) (1, 2, 0) (b) (0, 1, 2)
(c) (0, 2, 1) (d) (2, 0, 1)

Q.15 A point located in space is moves in such a way that sum of distance from xy and yz plane is equal to distance from zx plane the locus of the point are -

(a) $x - y + z = 2$ (b) $x + y - z = 0$
(c) $x + y - z = 2$ (d) $x - y + z = 0$

Q.16 A (1, 3, 5) and B (– 2, 3, – 4) are two points, A point P moves such that $PA^2 - PB^2 = 6c$, then locus of P is -
(a) $x + 3z + 1 - c = 0$
(b) $x + 3z - 1 + c = 0$
(c) $2x + 3z + 1 - c = 0$
(d) $2x + 3z - 1 + c = 0$

Q.17 The locus of the point which moves such that its distance from (1, –2, 2) is unity, is
(a) $x^2 + y^2 + z^2 - 2x + 4y + 4z + 8 = 0$
(b) $x^2 + y^2 + z^2 - 2x - 4y - 4z + 8 = 0$
(c) $x^2 + y^2 + z^2 + 2x + 4y + 4z + 8 = 0$
(d) $x^2 + y^2 + z^2 - 2x + 4y - 4z + 8 = 0$

Q.18 If distance of any point from z - axis is thrice its distance from xy-plane, then its locus is-
(a) $x^2 + y^2 - 9z^2 = 0$ (b) $y^2 + z^2 - 9x^2 = 0$
(c) $x^2 - 9y^2 + z^2 = 0$ (d) $x^2 + y^2 + z^2 = 0$

Q.19 The points (1, 2, 3), (– 1, – 2, – 1), (2, 3, 2) and (4, 7, 6) form a -
(a) rectangle (b) square
(c) parallelogram (d) rhombus

Q.20 If distance of any point from z-axis is thrice its distance from xy-plane, then its locus is-
(a) $x^2 + y^2 - 9z^2 = 0$ (b) $y^2 + z^2 - 9x^2 = 0$
(c) $x^2 - 9y^2 + z^2 = 0$ (d) $x^2 + y^2 + z^2 = 0$

Topic: 2 Coordinates of division point

Q.21 Find the ratio in which the segment joining the points (2, 4, 5), (3, 5, –4) is divided by the yz-plane.
(a) 3 : 1 (b) – 2 : 3
(c) – 1 : 3 (d) 1 : 2

Q.22 Find the ratio in which the segment joining (1, 2, –1) and (4, –5, 2) is divided by the plane $2x - 3y + z = 4$.
(a) 2 : 1 (b) 3 : 2
(c) 3 : 7 (d) 1 : 2

Q.23 If points A (3, 2, –4) ; B(5,4, –6) and C(9, 8,–10) are collinear then B divides AC in the ratio-
(a) 2 : 1 (b) 1 : 2
(c) 2 : 3 (d) 3 : 2

Q.24 If zx plane divides the line joining the points (1, –1, 5) and (2, 3, 4) in the ratio l :1 then equals to –
(a) 1/3 (b) 3
(c) –3 (d) –1/3

Q.25 OABC is a tetrahedron whose vertices are O (0, 0, 0) ; A (a, 2, 3); B (1, b, 2) and C (2, 1, c) if its centroid is (1, 2, –1) then distance of point (a, b, c) from origin are
(a) $\sqrt{14}$ (b) $\sqrt{107}$
(c) $\sqrt{107/14}$ (d) None of these

Q.26 If A(1, 2, –1) and B (–1, 0, 1) are two points then co-ordinate of points which divide AB externally in the ratio of 1 : 2
(a) (3, 4, –3) (b) $\frac{1}{3}(3, 4, -3)$
(c) $\frac{1}{3}(1, 4, -1)$ (d) None of these

Q.27 The ratio in which the yz-plane divides the join of the points (–2, 4, 7) and (3, –5, 8) is -
(a) 2 : 3 (b) 3 : 2
(c) –2 : 3 (d) 4 : –3

Q.28 A (3, 2, 0), B (5, 3, 2) and C (–9, 6, –3) are vertices of a triangle ABC. If the bisector of ĐA meets BC at D, then its coordinates are–
(a) $\left(\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$ (b) $\left(-\frac{19}{8}, \frac{57}{16}, \frac{17}{16}\right)$
(c) $\left(\frac{19}{8}, \frac{57}{16}, -\frac{17}{16}\right)$ (d) $\left(-\frac{19}{8}, -\frac{57}{16}, \frac{17}{16}\right)$

Q.29 If origin is the centroid of the triangle ABC with vertices A (a, 1, 3), B (–2, b, –5) and C (4, 7, c) then values of a, b, c are respectively–
(a) 2, 8, 2 (b) 0, 2, 2
(c) –2, –8, 2 (d) None of these

Q.30 The line joining the points (2, –3, 1) and (3, –4, –5) and cuts the plane $2x + y + z = 7$ in those points, the point are –
(a) (1, 2, 7) (b) (–1, 2, 7)
(c) (1, –2, 7) (d) (1, –2, –7)

Q.31 The vertices of a triangle ABC are A (4, 3, –2), B(3, 0, 1) and C(2, –1 , 3), the length of the median drawn from point 'A' –

(a) $\frac{1}{2}\sqrt{122}$ (b) $\sqrt{122}$

(c) $\frac{1}{3}\sqrt{122}$ (d) None of these

Q.32 The orthocentre of the triangle with vertices (2, 3, 4), (3, 4, 2) and (4, 2, 3) is

(a) (1, 1, 1) (b) (2, 2, 2)

(c) (3, 3, 3) (d) None of these

Q.33 The z-coordinates of a point R is 3, which is lie on a line meets the point P(2, 7, 1) and Q(3, 10, 11) then coordinates of R is

(a) (2, 7, 3) (b) (3, 10, 3)

(c) (11/5, 38/5, 3) (d) (38/5, 11/5, 3)

Q.34 If three consecutive vertices of a parallelogram are A (1, 2, 3), B (– 1, – 2, – 1) and C (2, 3, 2). Its fourth vertex is -

(a) (– 4, 5, 3) (b) (4, 7, 6)

(c) (3, – 5, 2) (d) (4, 5, 3)

Q.35 The points trisecting the line segment joining the points (0, 0, 0) and (6, 9, 12) are -

(a) (2, 3, 4), (4, 6, 8)

(b) (3, 4, 2), (6, 8, 4)

(c) (2, 3, 4), (4, 8, 6)

(d) none of these

Q.36 The point which divides the line joining the points (2, 4, 5) and (3, 5, – 4) in the ratio – 2 : 3 lieson -

(a) XOY plane (b) YOZ plane

(c) ZOX plane (d) none of these

Q.37 The line joining the points (0,0,0) and (1,– 2, – 5) is divided by plane x – y + z = 1 in the ratio -

(a) 1 : 1 (b) 1 : 2

(c) 1 : 3 (d) 3 : 1

Section-B
(Assertion-Reason)

Q.1 **Statement – 1 :** The triangle with vertices (1, 3, 5), (2, 4, 6) and (0, 5, 7) must be a right angle triangle.

Statement – 2 : If the dot product of two non-zero vectors is zero then they must be perpendicular

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement-1:** The points (2, 3, 5), (7, 5, 7) and (–3, 1, 3) are vertices of an equilateral triangle.

Statement-2: The triangle with equal sides is called an equilateral triangle.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statements-1:** If a line making an angle $\pi/4$ with x-axis, $\pi/4$ with y-axis then it must be perpendicular to z-axis.

Statements-2: If direction ratios of two lines are l_1, m_1, n_1 and l_2, m_2, n_2 then the angle between them is given by $\theta = \cos^{-1}(l_1l_2 + m_1m_2 + n_1n_2)$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement–1 :** The centroid of a tetrahedron with vertices (0, 0, 0), (4, 0, 0), (0, –8, 0), (0, 0, 12) is (1, –2, 3).

Statement–2 : The centroid of a triangle with vertices (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) is

$$\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right)$$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.

(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Years Question)

Q.1 The distance of the point (1, -5, 9) from the plane $x-y+z=5$ measured along the line $x=y=z$ is

(a) $3\sqrt{10}$ (b) $10\sqrt{3}$

(c) $\frac{10}{\sqrt{3}}$ (d) $\frac{20}{3}$

Q.2 The angle between the lines whose direction cosines satisfy the equations $l+m+n=0$ and $l^2+m^2+n^2$, is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

Q.3 If the line, $\frac{x-3}{2}=\frac{y+2}{-1}=\frac{z+4}{3}$ lines in the plane, $lx+my-z=9$, then l^2+m^2 is equal to

(a) 26 (b) 18
(c) 5 (d) 2

Q.4 If the lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and $\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar, then k can have

(a) any value

(b) exactly one value

(c) exactly two values

(d) exactly three values

Q.5 If the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-k}{2}=\frac{z}{1}$ intersect, then the value of k is

(a) $\frac{3}{2}$ (b) $\frac{9}{2}$

(c) $-\frac{2}{9}$ (d) $-\frac{3}{2}$

Q.6 If the image of the point P(1, -2, 3) in the plane $2x+3y-4z+22=0$ measured parallel to the line $\frac{x}{1}=\frac{y}{4}=\frac{z}{5}$ is Q, then PQ is equal to

(a) $3\sqrt{5}$ (b) $2\sqrt{42}$

(c) $\sqrt{42}$ (d) $6\sqrt{5}$

Q.7 The distance of the points(1, 3,-7) from the plane passing through the point (1, -1,-1) having normal perpendicular to both the lines $\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-4}{3}$ and $\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$, is

(a) $\frac{20}{\sqrt{74}}$ units (b) $\frac{10}{\sqrt{83}}$ units

(c) $\frac{5}{\sqrt{83}}$ units (d) $\frac{10}{\sqrt{74}}$ units

Q.8 The equation of the plane passing through the point (1, 1, 1) and perpendicular to the planes $2x+y-2z=5$ and $3x-6y-2z=7$ is

(a) $14x+2y-15z=1$

(b) $-14x+2y+15z=3$

(c) $14x-2y+15z=27$

(d) $14x+2y+15z=31$

Q.9 Consider a pyramid OPQRS located in the first octant $(x\geq 0, y\geq 0, z\geq 0)$ with O as origin, and OP and OR along the X-axis and the Y-axis, respectively. The base OPQR of the pyramid is a square with OP = 3. The point S is directly a above the mid-point T of diagonal OQ such that TS=3. Then,

(a) the acute angle between OQ and OS is $\frac{\pi}{3}$

(b) the equation of the plane containing the ΔOQS is $x-y=0$

(c) the length of the perpendicualr from P to the Plane containing the ΔOQS is $\frac{3}{\sqrt{2}}$

(d) the perpendicualr distance from O to the straight line containing RS is $\sqrt{\frac{15}{2}}$

Q.10 Let P be the image of the point (3, 1, 7) with respect to the plane $x-y+z=3$. Then, the equation of the plane passing through P and containig the straight line $\frac{x}{1}=\frac{y}{2}=\frac{z}{1}$ is

(a) $x+y-3z=0$ (b) $3x+z=0$

(c) $x-4y+7z=0$ (d) $2x-y=0$

Q.11 In R^3, consider the planes P_1 : y =0 and P_2 : x+z=1. Let P_3 be a plane, different from P_1 and P_2, which passes through the intersection of P_1 and P_2. If the distance of the point (0, 1, 0) from P_3 is 1 and the distance of a point (α,β,γ) from P_3 is 2, then which of the following relation(s) is/are true?

(a) $2\alpha+\beta+2\gamma+2=0$

(b) $2\alpha-\beta+2\gamma+4=0$

(c) $2\alpha+\beta-2\gamma-10=0$

(d) $2\alpha-\beta+2\gamma-8=0$

Q.12 In R^3, let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes P_1 : x+ 2y-z+1=0 and P_2 : 2x-y+z-1=0. Let M be the locus of the foot of the perpendiulars drawn from the points on L to the plane P_1. Which of the following points(s) on M?

(a) $\left(0,-\frac{5}{6},-\frac{2}{3}\right)$ (b) $\left(-\frac{1}{6},-\frac{1}{3},\frac{1}{6}\right)$

(c) $\left(-\frac{5}{6},0,\frac{1}{6}\right)$ (d) $\left(-\frac{1}{3},0,\frac{2}{3}\right)$

Q.13 The equation of the plane containing the lines $2x-5y+z=3, x+y+4z=5$ and parallel to the plane $x+3y+6z=1$ is

(a) $2x+6y+12z=13$

(b) $x+3y+6z=-7$

(c) $x+3y+6z=7$

(d) $2x+6y+12z=-13$

Q.14 The distance of the point (1, 0,2) from the point of intersection of the line $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ and the plane $x-y+z=16$ is

(a) $2\sqrt{14}$ (b) 8

(c) $3\sqrt{21}$ (d) 13

Q.15 The image of the line $\frac{x-1}{3}=\frac{y-3}{1}=\frac{z-4}{-5}$ in the plane $2x-y+z+3=0$ is the line

(a) $\frac{x+3}{3}=\frac{y-5}{1}=\frac{z-2}{-5}$

(b) $\frac{x+3}{-3}=\frac{y-5}{-1}=\frac{z+2}{5}$

(c) $\frac{x-3}{3}=\frac{y+5}{1}=\frac{z-2}{-5}$

(d) $\frac{x-3}{-3}=\frac{y+5}{-1}=\frac{z-2}{5}$

Q.16 Perpendicular are drawn from points on the line $\frac{x+2}{2}=\frac{y+1}{-1}=\frac{z}{3}$ to the plane $\frac{x+2}{2}=\frac{y+1x+y+z=3.}{-1}=\frac{z}{3}$ The feet of perpendiculars lie on the line

(a) $\frac{x}{5}=\frac{y-1}{8}=\frac{z-2}{-13}$ (b) $\frac{x}{2}=\frac{y-1}{3}=\frac{z-2}{-5}$

(c) $\frac{x}{4}=\frac{y-1}{3}=\frac{z-2}{-7}$ (d) $\frac{x}{2}=\frac{y-1}{-7}=\frac{z-2}{5}$

Q.17 Distance between two parallel planes $2x+y+2z=8$ and $4x+2y+4z+5=0$ is

(a) $\frac{3}{2}$ (b) $\frac{5}{2}$

(c) $\frac{7}{2}$ (d) $\frac{9}{2}$

Q.18 The equation of a plane passing through the line of intersection of the planes $x+2y+3z=2$ and $x-y+z=3$ and at a distance $\frac{2}{\sqrt{3}}$ from the point $(3,1,-1)$ is

(a) $5x-11y+z=17$ (b) $\sqrt{2}x+y=3\sqrt{2}-1$

(c) $x+y+z=\sqrt{3}$ (d) $x-\sqrt{2}y=1-\sqrt{2}$

Q.19 The point P is the intersection fo the straight line joining the points Q(2, 3, 5) and R(1, -1, 4) with the plane $5x-4y-z=1$. If S is the foot of the perpendicualr drawn from the point T (2, 1,4) to QR, then the length of the line segement PS is

(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$

(c) 2 (d) $2\sqrt{2}$

Q.20 If the distance of the point P(1, -2, 1) from the plane $x+2y-2z=\alpha$, where $\alpha>0$. is 5, then the foot of the perpendicular form P to the plane is

(a) $\left(\frac{8}{3},\frac{4}{3},-\frac{7}{3}\right)$ (b) $\left(\frac{4}{3},\frac{4}{3},\frac{1}{3}\right)$

(c) $\left(\frac{1}{3},\frac{2}{3},\frac{10}{3}\right)$ (d) $\left(\frac{2}{3},\frac{1}{3},\frac{5}{2}\right)$

Q.21 Equation of the plane containing the straight line $\frac{x}{2}=\frac{y}{3}=\frac{z}{4}$ and perpendicualr to the plane containing the straight lines $\frac{x}{3}=\frac{y}{4}=\frac{z}{2}$ and $\frac{x}{4}=\frac{y}{2}=\frac{z}{3}$ is

(a) $x+2y-2z=0$ (b) $3x+2y-2z=0$

(c) $x-2y+z=0$ (d) $5x+2y-4z=0$

Q.22 A line with positive direction cosines passes through the point P(2, -1, 2) and makes equal angles with the coordinate axes. The line meets the plane $2x+y+z=9$ at point Q. The length of the line segment PQ equals

(a) 1 (b) $\sqrt{2}$

(c) $\sqrt{3}$ (d) 2

Q.23 It P is (3,2,6) is a point in space and Q be a point on the line $\vec{r}=\left(\hat{i}-\hat{j}+2\hat{k}\right)+\mu\left(-3\hat{i}+\hat{j}+5\hat{k}\right)$. Then the value of μ for which the vector PQ is parallel to the plane $x-4y+z=1$ is

(a) $\frac{1}{4}$ (b) $-\frac{1}{4}$

(c) $\frac{1}{8}$ (d) $-\frac{1}{8}$

Q.24 A plane passes through (1,-2, 1) and is perpendicular to two planes $2x-2y+z=0$ and $x-y+2z=4$, then the distance of the plane from the point (1,2,2) is

(a) 0 (b) 1

(c) $\sqrt{2}$ (d) $2\sqrt{2}$

Q.25 A variable plane $\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$ at a unit distance from origin cuts the coordinate axes at A, B and C. Centroid (x, y, z) satisfies the equation $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=K$. The value of K is

(a) 9 (b) 3

(c) 1 / 9 (d) 1 / 3

Q.26 THe value of k such that $\frac{x-4}{1}=\frac{y-2}{1}=\frac{z=k}{2}$ lies in the plane $2x-4y+z=7$, is

(a) 7 (b) -7

(c) No real Value (d) 4

Section-D
(Board pattern)

Q.1 Find the direction cosines of the line which is perpendicular to the lines with direction ratios (1, –2, –2) and (0, 2, 1).

Q.2 Find angle between the lines whose direction ratio are (4, –3, 5) and (3, 4, 5).

Q.3 Direction ratio of two lines are (1, 2, –3) and (λ, –1, 2). If two lines are perpendicular then find λ.

Q.4 Find the ratio in which the join of the points *P*(2, 1, 5) and *Q*(3, 4, 3) is divided by the plane $x + y - z - \frac{1}{2} = 0$. Find the co-ordinates of the point of division.

Q.5 Find co-ordinates of the point which divides the join of *P*(2, –1, 4) and *Q*(4, 3, 2) in the ratio 2 : 3 (i) internally (ii) externally.

Q.6 If α, β, γ be angle made by a line with positive direction of *x*-axis then find value of $\cos2\alpha + \cos2\beta + \cos2\gamma$.

Q.7 Find direction cosine of line joining (1, –1, 1) and (2, –1, 2).

Q.8 Find distance of point (1, 3, –4) from *x*-axis.

Q.9 Find distance of point (1, 2, 3) from *x*-*y* plane.

Q.10 Find distance between points (1, 3, –2) and (2, –1, 1).

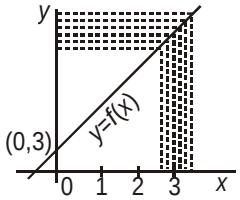
CHAPTER 6 LIMITS AND DERIVATIVES

1. LIMITS

Let us consider a function f defined as $f(x) = \dfrac{x^2-9}{x-3}$, $x \neq 3$

Let us study the behaviour of $f(x)$ in the neighbourhood of $x = 3$.

Here we observe that as x becomes closer and closer to 3, numerator $(x^2 - 9)$ as well as denominator $x - 3$ tends to 0 and $f(x)$ becomes closer to 6, which is clear from the following table and graph of the function.

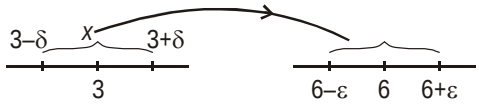


X	2.9	2.999	3.0001	3.01	3.0025	2.8579	3.012
$f(x)$	5.9	5.999	6.0001	6.01	6.0025	5.8579	6.012

From the above observation we may conclude that when x belongs to the immediate neighbourhood of 3, $f(x)$ belongs to the neighbourhood of 6. In other words, the difference between the values of $f(x)$ and 6 can be made as small as we please when x lies in the neigbhourhood of 3.

i.e., for a given $\varepsilon > 0$, however small, positive number δ such that

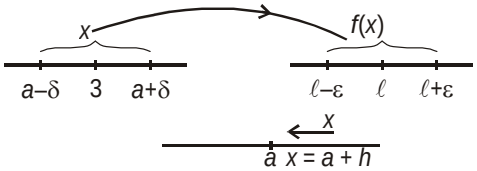
$x \in (3 - \delta, 3 + \delta) \Rightarrow f(x) \in (6 - \varepsilon, 6 + \varepsilon)$



In this case we say that $f(x)$ tends to 6 when x tends to 3. Mathematically $\lim\limits_{x\to 3} f(x) = 6$.

Thus a real valued function $f(x)$ is said to have limit ℓ and x tends to a, mathematically written as $\lim\limits_{x\to 3} f(x) = \ell$ if for a given $\varepsilon > 0$, however small, there exists a corresponding number $\delta(\varepsilon) > 0$ such that $|f(x) - l| < \varepsilon$ whenever $|x - a| < \delta$.

i.e., $x \in (a - \delta, a + \delta) \Rightarrow f(x) \in (\ell - \varepsilon, \ell + \varepsilon)$



Now x may approach to a from right hand side of a i.e., $x \to a^+$ and the corresponding limit

$\lim\limits_{x\to a^+} f(x) = \lim\limits_{h\to 0} f(a+h) = \ell$ is called Right hand limit of $f(x)$ at $x = a$.

Also x may approach to a from left hand side of a

i.e., $x \to a^-$ and the corresponding $\xrightarrow[x = a - h]{x} \overset{a}{|}$ limit $\lim\limits_{x\to a^-} f(x) = \lim\limits_{h\to 0} f(a-h) = \ell$ is called left-hand limit of $f(x)$ at $x = a$.

For the existence of limit $\lim\limits_{x\to a} f(x)$ we must have

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = \lim_{x\to a} f(x) = \ell$$

i.e., $\lim_{h\to 0} f(a-h) = \lim_{x\to 0} f(a+h) = \lim_{x\to a} f(x) = \ell$

Note :- Instead of lim (as the short form of limit) we can also use Lt. i.e. $\underset{x\to a}{Lt}\ f(x) = \underset{x\to a}{Lt}\ f(x)$

2. DETERMINATE AND INDETERMINATE FORMS

If a unique value be assigned to $f(a)$, then $f(x)$ is said to be determinate at $x = a$, otherwise $f(x)$ is said to be indeterminate at $x = a$.

For example,

for $f(x) = \dfrac{x^2-9}{x-3}$

$f(4) = 7$, which is unique

Hence $f(x)$ is said to be determinate at $x = 4$. But $f(3) = \dfrac{9-9}{3-3} = \left(\dfrac{0}{0}\right)$, which is indetermiate. Here we can not assign a unique value to $f(3)$ as

$\dfrac{0}{0} = k$, say

$\Rightarrow$ $0 = k.0$, which is true for any $k \in R$. There is no unique value of k which satisfies the above equation.

Hence, $\dfrac{0}{0}$ is an indeterminate quantity.

The other indeterminate form are $\dfrac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, 1^∞, 0^0, ∞^0 etc.

Justification for indeterminacy

(i) Let $\dfrac{\infty}{\infty} = k$

$\Rightarrow$ $\dfrac{1}{\infty} = \dfrac{1}{\infty}.k$

$\Rightarrow$ $0 = 0.k$

which is true for any $k \in R$. There is no unique value of k which satisfies the above equation.

Hence $\dfrac{\infty}{\infty}$ is an indeterminate quantity.

(ii) $0 \times \infty$

Let $0.\infty = k$

$\Rightarrow$ $0 = \dfrac{1}{\infty}.k$

$\Rightarrow$ $0 = 0.k$

which is true for any $k \in R$. There is no unique value of k, which satisfies the above equation. Hence $0 . \infty$ is an indeterminate quantity.

(iii) $\infty - \infty$

Let $\infty - \infty = k$

$\Rightarrow \infty = k + \infty$, which is true for any $k \in R$.

There is no unique value of k, which satisfies the above equation. Hence $\infty - \infty$ is an indeterminate quantity.

(iv) 1^{∞}

Let $1^{\infty} = k$

$\Rightarrow \quad \infty \log 1 = \log k$

$\Rightarrow \quad \log 1 = \frac{1}{\infty} \log k$

$\Rightarrow \quad 0 = 0.\ \log k$, which is true for all $k \in R^+$.

There is no unique value of k which satisfies the above equation. Consequently 1^{∞} is an indeterminate quantity.

(v) 0^0

Let $0^o = k$

$\Rightarrow \quad 0 \log_e = \log_e k$

$\Rightarrow \quad 0(-\infty) = \log_e k$

[$\log 0$ is not defined, but $\log_a 0 \to -\infty$ for $a > 1$ and ∞ for $0 < a < 1$]

$\Rightarrow \quad 0 = -0 \log_e k$ which is true for all $k \in R^+$.

There is no unique value of k which satisfies the above equation. Hence, 0^0 is an indeterminate quantity.

(vi) ∞^0

Let $\infty^0 = k$

$\Rightarrow \quad 0 \log \infty = \log k$

$\Rightarrow \quad 0 = \frac{1}{\log \infty} .\log k$

$\Rightarrow \quad 0 = 0.\log k$, which is true for all $k \in R^+$.

There is no unique value of k which satisfies the above equation. Hence, ∞^0 is an indeterminate quantity.

Evaluation of the limit of a function

In order to evaluate $\lim_{x \to a} f(x)$ we first put $x = a$ in $f(x)$ to get $f(a)$. Now two cases arise.

$\lim_{x \to a} f(x) \xrightarrow[\text{Put } x=a \text{ in } f(x)]{} f(a)$ — Determine $\lim_{x \to a} f(x) = f(a)$

Indeterminate

$\frac{0}{0}\left[\frac{\infty}{\infty}\right]$, $[0 \times \infty]$ $[\infty - \infty]$, 1^{∞}, 0^0, ∞^0 etc.

Case I :

When $f(a)$ is in determinate form, limit can be evaluated directly by using the defintion of limit or putting a in place of x if $x \to a$. No method is required in this case.

For example,

$$\lim_{x \to 1}(4x^2 - 5x + 2) = 1$$

$$\lim_{x \to 1} \frac{x^2 - 4}{x - 2} = 3$$

$$\lim_{x \to 1} \frac{\sin x}{x} = \sin 1$$

Case II

When $f(a)$ is of indeterminate form, then there are different methods to evaluate the limit in case

of different indeterminate forms $\frac{0}{0}, \frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, 1^{∞}, 0^0, ∞^0 etc. But in all cases our aim is to get rid of the indeterminate form and then to reduce the indeterminate form into determinate forms. After that we directly put the value to evaluate the limit.

For example,

Let $f(x) = \frac{x^3 - a^3}{x^2 - a^2}$ $x \neq a$ an we shall find $\lim_{x \to a} f(x)$.

Here we observe that the numerator as well as denominator become zero after putting $x = a$. Hence by factor theorem, it follows that $(x - a)$ is a factor of the numerator as well as denominator and this is the factor which makes the numerator and denominator zero. When $x \to a$, the expression $x - a$ tends to zero, but it is not zero.

Hence we can cancel the expression $(x - a)$ from the numerator and denominator and then the expression for the given function will reduce to determinate form. Hence the limit can be evaluated by putting directly $x = a$ in the determinate form.

This $\lim_{x \to a} f(x) = \lim_{x \to a} \frac{x^3 - a^3}{x^2 - a^2}$

$$= \lim_{x \to a} \frac{(x-a)(x^2 + xa + a^2)}{(x-a)(x+a)}$$

$$= \lim_{x \to a} \frac{x^2 + xa + a^2}{x + a}$$

$$= \frac{a^2 + a^2 + a^2}{a + a} = \frac{3a^2}{2a} = \frac{3}{2}a$$

3. IMPORTANT PROPERTIES OF LIMITS

(i) $\lim_{x \to a}(f(x) + g(x)) = \lim_{x \to a} f(x) + \lim_{x \to a} g(x)$

i.e., Limit of sum is equal to the sum of limits.

For example,

$$\lim_{x \to 3}(x^2 + \cos x) = \lim_{x \to 3} x^2 + \lim_{x \to 3} \cos x = 9 + \cos 3$$

(ii) $\lim_{x \to a} f((x) - g(x)) = \lim_{x \to a} f(x) - \lim_{x \to a} g(x)$

i.e., Limit of difference = Difference of Limits

For example,

$$\lim_{x \to 3}(\sin^2 x - 2^{\sin x}) = \sin^2 3 - 2^{\sin 3}$$

(iii) $\lim_{x \to a} f((x)g(x)) = \left(\lim_{x \to a} f(x)\right)\left(\lim_{x \to a}(g(x))\right)$

i.e., Limit of product = Product of limits

For example,

$$\lim_{x \to 2}(2^x(x^3)) = \lim_{x \to 2} 2^x . \lim_{x \to 2} x^3 = 2^2 . 2^3 = 32$$

(iv) $\lim_{x \to a}(kf(x)) = k \times \lim_{x \to a} f(x)$, k being a constant

i.e., Limit of k times = k times of limit

For example,

$$\lim_{x\to 3} 5\log_{10} x = 5\log_{10} 3$$

(v) $\lim_{x\to a}\left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$; provided $\lim_{x\to a} g(x) \neq 0$

i.e., Limit of quotient = Quotient of limits, provided limit of denominator is non-zero number.
For example,

$$\lim_{x\to 2}\frac{2^x}{\cos x} = \frac{\lim_{x\to 2} 2^x}{\lim_{x\to 2}\cos x} = \frac{4}{\cos 2}$$

(vi) $\lim_{x\to a}(f(x))^{\frac{m}{n}} = \left(\lim_{x\to a} f(x)\right)^{\frac{m}{n}}$

provided $\left(\frac{m}{n}\right)^{\text{th}}$ power of the limit of $f(x)$ is a real number.

i.e., Limit of power = Power of limit

(vii) $\lim_{x\to a}(f(x))^{g(x)} = \left(\lim_{x\to a} f(x)\right)^{\left(\lim_{x\to a} g(x)\right)}$

For example, $\lim_{x\to 2}(2x+4)^x = \left(\lim_{x\to 2}(2x+4)\right)^{\lim_{x\to 2} x} = 8^2 = 64$

(viii) If $f(x) < g(x)$, $\forall x$

$\Rightarrow \lim_{x\to a} f(x) \leq \lim_{x\to a} g(x)$

(ix) $\lim_{x\to a} |f(x)| = \left|\lim_{x\to a} f(x)\right|$

(x) $\lim_{x\to a} \ln f(x) = \ln\left(\lim_{x\to a} f(x)\right) = \ln(f(a))$ provided $f(a) > 0$

All these properties are used only when either both limits are definite or both are indeterminate, i.e., only in the common domain of the functions.

Working Rule for the evaluation of limit of [0/0] form

We factorize numerator and denominator and then cancel the common factor. The process is continued till all the common factors are cancelled and then the functional form will reduce to the determinate form when limit can be evaluated by putting directly the limiting point for the variable.

Student Task

Q.1 Evaluate $\lim_{x\to -\infty}\frac{\sqrt{x^2+x+1}}{\sqrt{x^2+x+2}}$

Q.2 $\lim_{x\to 2}\frac{x^4-16}{x^3-2x^2-x+2}$

Q.3 $\lim_{x\to\infty}\frac{\sqrt{x^2+1}-3\sqrt{x^3+1}}{4\sqrt{x^4+1}+5\sqrt{x^4+1}}$

Answer Key :

1. 1, 2. 32/3, 3. 0

For indeterminate forms $1^\infty, \infty^0, 0^0$, we take logarithm and proceed as above.

Important Results on Limit

1. $\lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}, n\in Q$ **, the set of rational numbers.**

Proof :

Let $x = a + h$ so that

$x \to a \Rightarrow h \to 0$

Thus $\lim_{x\to a}\frac{x^n-a^n}{x-a}=\lim_{h\to 0}\frac{(a+h)^n-a^n}{a+h-a}$

$$=\lim_{h\to 0}\frac{a^n\left[\left(1+\frac{h}{a}\right)^n-1\right]}{h}$$

$$=\lim_{h\to 0}\frac{a^n\left[\left(1+n.\frac{h}{a}+\frac{n(n-1)}{2!}.\frac{h^2}{a^2}+\frac{n(n-1)(n-2)}{3!}.\frac{h^3}{a^3}+....\right)-1\right]}{h}$$

$$=\lim_{h\to 0}a^n\left(\frac{n}{a}+\frac{n(n-1)}{2!}.\frac{h}{a^2}+\text{ terms containing higher powers of } h\right)$$

$$a^n.\frac{n}{a}=na^{n-1}$$

Illustration 2 :

2. Trigonometrical Limits

$\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1$ $\qquad$ $\lim_{\theta\to 0}\cos\theta=1$

$\lim_{\theta\to 0}\frac{\tan\theta}{\theta}=1$ $\qquad$ $\lim_{\theta\to 0}\cot\theta=1$

$\lim_{\theta\to 0}\text{cosec}\,\theta=1$ $\qquad$ $\lim_{\theta\to 0}\sec\theta=1$

Proof

Let PQ be an arc of a circle centred at 0 with radius r.

Let $\angle POQ = \theta$.

Let the tangent at P meets OQ produced at T.

From figure it is evident that area of ΔOQR < area of (Sector POQ) < area of (ΔOPT)

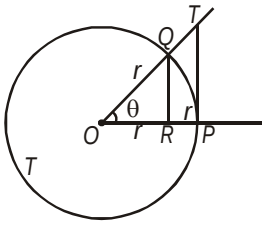
$$\Rightarrow \quad \frac{1}{2}.r.r\sin\theta<\frac{1}{2}r^2\theta<\frac{1}{2}r.r\tan\theta$$

$\Rightarrow$ $\sin\theta < \theta < \tan\theta$; θ being acute

$$\Rightarrow \quad 1<\frac{\theta}{\sin\theta}<\frac{1}{\cos\theta}$$

$$\Rightarrow \quad 1>\frac{\theta}{\sin\theta}>\cos\theta$$

Taking limit as $\theta \to 0$, we get



$$\lim_{\theta\to0}1 \geq \lim_{\theta\to0}\frac{\sin\theta}{\theta} \geq \lim_{\theta\to0}\cos\theta$$

$$\Rightarrow \quad 1 \geq \lim_{\theta\to0}\frac{\sin\theta}{\theta} \geq \lim_{\theta\to0}\cos\theta$$

$$\Rightarrow \quad 1 \geq \lim_{\theta\to0}\frac{\sin\theta}{\theta} \geq \lim_{\theta\to0}\cos\theta = 1$$

$$\Rightarrow \quad \lim_{\theta\to0}\frac{\sin\theta}{\theta} = 1, \text{ as } \lim_{\theta\to0}\cos\theta = 1$$

$$\lim_{\theta\to0}\frac{\tan\theta}{\theta} = \lim_{\theta\to0}\frac{\sin\theta}{\theta}.\frac{1}{\cos\theta}$$

$$= 1\times\frac{1}{1} = 1$$

$$\lim_{\theta\to0}\theta\cot\theta = \lim_{\theta\to0}\theta.\frac{\cos\theta}{\sin\theta} = \lim_{\theta\to0}\frac{\cos\theta}{\left(\frac{\sin\theta}{\theta}\right)} = \frac{1}{1} = 1$$

$$\lim_{\theta\to0}\theta\operatorname{cosec}\theta = \lim_{\theta\to0}\frac{\theta}{\sin\theta} = \lim_{\theta\to0}\frac{1}{\frac{\sin\theta}{\theta}} = \frac{1}{1} = 1$$

$$\lim_{\theta\to0}\sec\theta = \sec 0 = 1$$

☺ **CONCEPTS**

1. If $\theta < 0$, Let $\theta = -\phi$, where $\phi > 0$ and $\theta \to 0 \Rightarrow \phi \to 0$.

   Now $\lim_{\theta\to0}\frac{\sin\theta}{\theta} = \lim_{\varphi\to0}\frac{\sin(-\varphi)}{-\varphi} = \lim_{\varphi\to0}\frac{-\sin\varphi}{-\varphi} = \lim_{\varphi\to0}\frac{\sin\varphi}{\varphi} = 1$

2. $\lim_{x\to\infty}\sin x, \lim_{x\to\infty}\cos x, \lim_{x\to\infty}\tan x, \lim_{x\to\infty}\cot x, \lim_{x\to\infty}\sec x, \lim_{x\to\infty}\operatorname{cosec} x$ do not exist, but

   $$\lim_{x\to\infty} x\sin\frac{1}{x} = \lim_{x\to\infty}\frac{\sin\frac{1}{x}}{\frac{1}{x}} = \lim_{y\to0}\frac{\sin y}{y} = 1\left(\frac{1}{x} = y \text{ and } x\to\infty \Rightarrow y\to0\right)$$

   $$\lim_{x\to\infty}\cos\frac{1}{x} = 1$$

   $$\lim_{x\to\infty} x\tan\frac{1}{x} = 1$$

Illustration 1: **3. Proof**

When $a > 0$, $a^x = e^{x\log_e a} = e^{x\ln a}$

Now, $\lim_{x\to0}\frac{a^x - 1}{x} = \lim_{x\to0}\frac{e^{x\ln a} - 1}{x}$

$$= \lim_{x\to 0} \frac{1 + \frac{x\ln a}{1!} + \frac{x^2(\ln a)^2}{2!} + \frac{x^3(\ln a)^3}{3!} + ... - 1}{x}$$

$$= \lim_{x\to 0}\left[\ln a + \frac{x}{2!}(\ln a)^2 + \frac{x^2}{3!}(\ln a)^3 + ...\text{to}\ \infty \right]$$

$= \ln a$

For example,

$$\lim_{x\to 0} \frac{5^x - 1}{x} = \log_e 5$$

Corollary

(i) $\lim_{x\to 0} \frac{e^x - 1}{x} = \ln e = 1$

(ii) If $\lim_{x\to a} f(x) = 0$, then

$$\lim_{x\to a} \frac{e^{f(x)} - 1}{f(x)} = 1$$

$$\lim_{x\to a} \frac{b^{f(x)} - 1}{f(x)} = \ln b, \text{ if } b > 0$$

$$\lim_{x\to a} \frac{\sin f(x)}{f(x)} = 1, \lim_{x\to a} \frac{\tan(f(x))}{f(x)} = 1$$

Illustration 2: **4.** $\lim_{x\to 0} \frac{\log_a(1+x)}{x} = \log_a e, (a > 0, a \neq 1)$

Proof

We have

$$\lim_{x\to 0} \frac{\log_a(1+x)}{x} = \lim_{x\to 0} \frac{\log_e(1+x) \times \log_a e}{x}$$

$$= \lim_{x\to 0} \left[\frac{x - \frac{x^2}{2} + \frac{x^3}{3} - .....\text{to}\ \infty}{x} \right] \log_a e$$

$$= \lim_{x\to 0} \left[1 - \frac{x}{2} + \frac{x^2}{3} - .....\text{to}\ \infty \right] \log_a e$$

$= \log_a e$

Illustration 3: **5.** $\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = e^a = \lim_{h\to 0} (1 + ah)^{\frac{1}{h}}$

Proof

Let $\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = y \qquad \Rightarrow \ln y = \lim_{n\to\infty} n \ln\left(1 + \frac{a}{n}\right)$

$$= \lim_{n\to\infty} n\left[\frac{a}{n} - \frac{1}{2}.\frac{a^2}{n^2} + \frac{1}{3}.\frac{a^3}{n^3} - \frac{1}{4}\frac{a^4}{n^4} + ...\right]$$

$$= \lim_{n\to\infty} n\left[a - \frac{1}{2}.\frac{a^2}{n} + \frac{1}{3}.\frac{a^3}{n^2} - \frac{1}{4}\frac{a^4}{n^3} + ...\right] = a \qquad \Rightarrow y = e^a = \lim_{n\to\infty}\left(1+\frac{a}{n}\right)^n$$

Putting $\frac{1}{n} = h$ so that $n \to \infty \Rightarrow h \to 0$ thus the above limit reduces to

$$\lim_{n\to\infty}\left(1+\frac{a}{n}\right)^n = \lim_{h\to 0}(1+ah)^{\frac{1}{h}} = e^a$$

Short-cut method for 1^∞ form

CONCEPTS

$\lim_{x\to a} f(x)^{g(x)} = e^{\lim_{x\to a} g(x)[f(x)-1]}$ where $\lim_{x\to a} f(x) = 1$ and $\lim_{x\to a} g(x) = \infty$

Proof

We have, $\lim_{x\to a} f(x)^{g(x)} = \lim_{x\to a}\{1+(f(x)-1)\}^{g(x)}$

$$= \lim_{x\to a}\left[(1+(f(x)-1)^{\frac{1}{f(x)-1}}\right]^{(f(x)-1)g(x)}$$

$$= e^{\lim_{x\to a}(f(x)-1)g(x)}$$

De' L' Hospital's Rule

If $f(x)$ and $g(x)$ be two functions such that $f(a) = g(a) = 0$ and f and g are both differentiable everywhere in some neighbourhood of point of a except possibly at a.

i.e., if $\lim_{x\to a}\frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ then $\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{f'(x)}{g'(x)}$

We can repeat the above process if $\lim_{x\to a}\frac{f'(x)}{g'(x)}$ is also of the form $\frac{0}{0}$.

CONCEPTS

1. L' Hospital's Rule is applicable only for forms $\frac{0}{0}$ or $\frac{\infty}{\infty}$.
2. If we have any other indeterminate form other than $\frac{0}{0}$ or $\frac{\infty}{\infty}$ then we convert that form to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and then apply L' Hospital's Rule.

Note :- For using L' Hospital's Rule, we need to know how to find the derivative of a function which has been discussed in the next section of the chapter. However, we shall come across problems based on L' Hospital's Rule such problems may be left at this stage and should be attempted after having gone through the next section on derivative of a function.

Derivative of a function

The instantaneous rate of change of $f(x)$ at x mathematically written as $f'(x)$ or $\frac{df}{dx}$ is called the derivative of f at x. Geometrically, it represents the slope of the tangent at the point (x, y) on the curve $y = f(x)$.

Thus, $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ (derivative by first principle)

$\Rightarrow$ $f'(x) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ provided the derivatives exist.

Student Task

Using first principle find the derivatives of the following :

(i) $f(x) = \frac{x+1}{x+2}$ (ii) $f(x) = \sin 2x$ (iii) $f(x) = x^2 + 2x + 3$

4. DERIVATIVES OF STANDARD FUNCTIONS

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\text{cosec}^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\text{cosec } x) = -\text{cosec } x \cot x$$

$$\frac{d}{dx}(\text{constant}) = 0$$

$$\frac{d}{dx}(x^n) = nx^{n-1} n \in R$$

$$\frac{d}{dx}\left(\frac{1}{x^n}\right) = \frac{-n}{x^{n+1}}; x \neq 0$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(a^x) = a^x \ln a; a > 0$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}; x > 0 \left(\frac{d}{dx}(\ln|x|) = \frac{1}{x}, x \neq 0\right)$$

$$\frac{d}{dx}(\log_a x)=\frac{1}{x}\log_a e;\ x>0, a>0, a\neq 0$$

$$\frac{d}{dx}(|x|)=\frac{x}{|x|}\text{ or }\frac{|x|}{x};\ x\neq 0$$

Although the concept of inverse of a function is in the class XII, but the following derivatives help us in solving problems of limit with the help of L' Hospital's Rule.

$$\frac{d}{dx}(\sin^{-1}x)=\frac{1}{\sqrt{1-x^2}};\ -1<x<1$$

$$\frac{d}{dx}(\cos^{-1}x)=\frac{-1}{\sqrt{1-x^2}};\ -1<x<1$$

$$\frac{d}{dx}(\tan^{-1}x)=\frac{1}{1+x^2};\ x\in R$$

$$\frac{d}{dx}(\cot^{-1}x)=\frac{-1}{1+x^2};\ x\in R$$

$$\frac{d}{dx}(\sec^{-1}x)=\frac{1}{|x|\sqrt{x^2-1}};\ |x|>1$$

$$\frac{d}{dx}(\text{cosec}^{-1}x)=\frac{-1}{|x|\sqrt{x^2-1}};\ |x|>1$$

Basic Rules of Differentiation

The following rules will help us in finding the derivative of a function without first principle.

$$\frac{d}{dx}\{ku(x)\}=k\frac{du}{dx},\ k \text{ being a constant}$$

$$\frac{d}{dx}(u(x)\pm v(x))=\frac{du}{dx}\pm\frac{dv}{dx}$$

Generalisation

$$\frac{d}{dx}(u(x)\pm v(x)\pm w(x)\pm....)=\frac{du}{dx}\pm\frac{dv}{dx}\pm\frac{dw}{dx}\pm....$$

$$\frac{d}{dx}(u(x).v(x))=\frac{du}{dx}.v+u\frac{dv}{dx}$$

Generalisation

$$\frac{d}{dx}(u(x)v(x)w(x))=\frac{du}{dx}vw+u\frac{dv}{dx}w+uv\frac{dw}{dx}$$

$$\frac{d}{dx}(u(x)v(x)w(x)t(x)....)=\frac{du}{dx}(vwt...)+u\frac{dv}{dx}(wt...)+uv\frac{dw}{dx}(t....)+uvw\frac{dt}{dx}(...)+....$$

$$\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right)=\frac{v\frac{du}{dx}-u\frac{dv}{dx}}{v^2}$$

Student Task

Q.1 Let $f(x) = |x|$ then find $f'(1)$.

Q.2 If $y = f(x)^2$ and $f'(1) = \sin x$ then find the value of $\frac{dy}{dx}$.

Q.3 If $y = \sec^4 x - 2\sec^2 x \tan^2 x + \tan^4 x$ then find the value of $\frac{dy}{dx}$.

Answer Key :

1. 1, 2. $(2x)(\sin x^2)$, 3. 0

MISCELLANEOUS EXAMPLES

Example 1 : If $f(x) = \begin{cases} 5x-4, & 0 < x \le 1 \\ 4x^3 - 3x^2, & 1 < x < 2 \end{cases}$ show that $\lim_{x\to 1} f(x)$ exists.

Solution : We have,

LHL of $f(x)$ at $x = 1$

$$= \lim_{x\to 1^-} f(x) = \lim_{h\to 0} f(1-h)$$

$$= \lim_{h\to 0} 5(1-h) - 4 = \lim_{h\to 0} 1 - 5h = 1$$

RHL of $f(x)$ at $x = 1$

$$= \lim_{x\to 1^+} f(x) = \lim_{h\to 0} f(1+h)$$

$$= \lim_{h\to 0} 4(1+h)^3 - 3(1+h) = 4(1)^3 - 3(1) = 1$$

Thus RHL = LHL = 1. So $\lim_{x\to 1} f(x)$ exists and is equal to 1.

Example 2 : Evaluate the right hand limit and left hand limit of the function $f(x) = \begin{cases} \frac{|x-4|}{x-4}, & x \ne 4 \\ 0, & x = 4 \end{cases}$

Solution : RHL of $f(x)$ at $x = 4$

$$\lim_{x\to 4^+} f(x) = \lim_{h\to 0} f(4+h) = \lim_{h\to 0} = \frac{|4+h-4|}{4+h-4} = \lim_{h\to 0} \frac{|h|}{h} = 1$$

LHL of $f(x)$ at $x = 4$

$$\lim_{x\to 4^-} f(x) = \lim_{h\to 0} f(4-h) = \lim_{h\to 0} = \frac{|4-h-4|}{4-h-4} = \lim_{h\to 0} \frac{|h|}{-h} = \lim_{h\to 0} \frac{h}{-h} = -1$$

This RHL $\ne$ LHL. So, $\lim_{x\to 4} f(x)$ does not exist.

Example 3 : Show $\lim_{x\to 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$ does not exist.

Solution : Let $f(x) = \frac{e^{1/x} - 1}{e^{1/x} + 1}$. Then,

$$\text{LHL} = \lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0-h)$$

$$\lim_{h\to 0}\frac{e^{-1/h}-1}{e^{-1/h}+1}=\lim_{h\to 0}\frac{\left(\frac{1}{e^{1/h}}-1\right)}{\left(\frac{1}{e^{1/h}}+1\right)}=\frac{0-1}{0+1}=-1$$

$$\left[\text{as } h\to 0\Rightarrow\frac{1}{h}\to\infty\Rightarrow e^{1/h}\Rightarrow 1/e^{1/h}\to 0\right] \qquad \text{....(i)}$$

$$\text{RHL}=\lim_{x\to 0^+}f(x)=\lim_{x\to 0}f(0+h)$$

$$=\lim_{h\to 0}\frac{e^{1/h}-1}{e^{1/h}+1}=\lim_{h\to 0}\frac{(1-1/e^{1/h})}{(1+1/e^{1/h})}$$ [Dividing numerator and denominator by $e^{1/h}$]

$\frac{1-0}{1+0}=1$ LHL $\neq$ RHL, Hence limit does not exist.

Example 4 : Solve $\lim_{x\to 0}\left[\sin\frac{|x|}{x}\right]$, where [ ] denotes the greatest integer function.

Solution : Here $\lim_{x\to 0}\left[\sin\frac{|x|}{x}\right]$, since we have greatest integral function we must define function.

Now, RHL (put $x = 0 + h$)

$$\lim_{h\to 0}\left[\frac{\sin|0+h|}{0+h}\right],$$

we know $\lim_{h\to 0}\frac{\sin h}{g}\to 1$ as $h\to 0$ but less than 1 as $h > \sin h$

$$\therefore \lim_{h\to 0}0=0 \qquad \left\{\because\left[\frac{\sinh}{h}\right]=0 \text{ as } h\to 0\right\}$$

$\Rightarrow$RHL = 0

again (LHL (put $x = 0 - h$)

$$\lim_{h\to 0}\left[\sin\frac{|0-h|}{0-h}\right] \qquad \left\{\because\left[\frac{\sinh}{h}\right]=-1 \text{ as } h\to 0\right\}$$

we know $\frac{\sinh}{-h}\to -1$ as $h\to 0$ but greater than –1.

Example 5-I : (a) Find the $\underset{x\to\infty}{\text{Lt}}\frac{x^4+2x^3+3}{2x^4-x+2}$ (b) Find $\underset{n\to\infty}{\text{Lt}}\left(\frac{1^2}{n^3}+\frac{2^2}{n^3}+\frac{3^2}{n^3}+....+\frac{n^2}{n^3}\right)$

Solution : (a) $\underset{x\to\infty}{\text{Lt}}\frac{x^4+2x^3+3}{2x^4-x+2}=\underset{x\to\infty}{\text{Lt}}\frac{1+\frac{2}{x}+\frac{3}{x^4}}{-\frac{1}{x^3}+\frac{2}{x^4}}=\frac{1}{2}$

$$\left[\text{Since as } x\to\infty,\frac{2}{x}\to 0,\frac{3}{x^4}\to 0 \text{ and } \frac{2}{x^4}\to 0\right]$$

(b) $\underset{n\to\infty}{\text{Lt}}\left(\frac{1^2}{n^3}+\frac{2^2}{n^3}+\frac{3^2}{n^3}+....+\frac{n^2}{n^3}\right)$

$=\underset{n\to\infty}{\text{Lt}}\frac{1^2+2^2+3^2+...+n^2}{n^3}=\underset{n\to\infty}{\text{Lt}}\frac{n(n+1)(2n+1)}{6n^3}=\underset{n\to\infty}{\text{Lt}}\frac{(n+1)(2n+1)}{6n^2}$

$=\underset{n\to\infty}{\text{Lt}}\frac{(2n^2+3n+1)}{6n^2}=\underset{n\to\infty}{\text{Lt}}\frac{2+\frac{3}{n}+\frac{1}{n^2}}{6}=\frac{2}{6}=\frac{1}{3}$

Example 5-II : (a) Find $\underset{n\to\infty}{\text{Lt}}\sqrt{x}\left(\sqrt{x+c}-\sqrt{c}\right)$ (b) $\underset{n\to-\infty}{\text{Lt}}\left(\sqrt{x^2+4x}-\sqrt{x^2-4x}\right)$

(c) $\text{I}:\lim_{x\to\infty}\frac{\sqrt{x^2+1}-\sqrt[3]{x^2+1}}{\sqrt[4]{x^4+1}-\sqrt[5]{x^4+1}}$

Solution : (a) $\underset{x\to\infty}{\text{Lt}}\sqrt{x}\left(\sqrt{x+c}-\sqrt{x}\right)$

$=\underset{x\to\infty}{\text{Lt}}\frac{\sqrt{x}\left(\sqrt{x+c}-\sqrt{x}\right)\left(\sqrt{x+c}+\sqrt{x}\right)}{\left(\sqrt{x+c}+\sqrt{x}\right)}$

$=\underset{x\to\infty}{\text{Lt}}\frac{\sqrt{x}(x+c-x)}{\left(\sqrt{x+c}+\sqrt{x}\right)}=\underset{x\to\infty}{\text{Lt}}\frac{c}{\sqrt{x+c}+\sqrt{x}}$

$=\underset{x\to\infty}{\text{Lt}}\frac{c}{\frac{\sqrt{x+c}}{\sqrt{x}}+\frac{\sqrt{x}}{\sqrt{x}}}=\underset{x\to\infty}{\text{Lt}}\frac{c}{\sqrt{\frac{x+c}{x}}+1}$

$=\underset{x\to\infty}{\text{Lt}}\frac{c}{\sqrt{1+\frac{c}{x}}+1}=\frac{c}{1+1}=\frac{c}{2}$

(b) $\underset{x\to-\infty}{\text{Lt}}\left(\sqrt{x^2+4x}-\sqrt{x^2-4x}\right)$

$=\underset{x\to-\infty}{\text{Lt}}\frac{\left(\sqrt{x^2+4x}-\sqrt{x^2-4x}\right)\left(\sqrt{x^2+4x}+\sqrt{x^2-4x}\right)}{\left(\sqrt{x^2+4x}+\sqrt{x^2-4x}\right)}$

$=\underset{x\to-\infty}{\text{Lt}}\frac{(x^2+4x)-(x^2-4x)}{\sqrt{x^2+4x}+\sqrt{x^2-4x}}=\underset{x\to-\infty}{\text{Lt}}\frac{8x}{\sqrt{x^2+4x}+\sqrt{x^2-4x}}$

$=\underset{x\to-\infty}{\text{Lt}}\frac{8}{\frac{\sqrt{x^2+4x}+\sqrt{x^2-4x}}{x}}$

$=\underset{x\to-\infty}{\text{Lt}}\frac{8}{\sqrt{\frac{x^2+4x}{x^2}}+\sqrt{\frac{x^2-4x}{x^2}}}$

$\left[\text{Here } x<0 \because x=-\sqrt{x^2} \text{ for example } -4=-\sqrt{(-4)^2}=-\sqrt{16}\right]$

$$\lim_{x\to-\infty}\frac{8}{\sqrt{1+\frac{4}{x}}+\sqrt{1-\frac{4}{x}}}=\frac{8}{1+1}=\frac{8}{2}=4$$

(c)Dividing the numerator and denominator by x(which is the greatest power of x possible)

$$=\lim_{x\to\infty}\frac{\frac{\sqrt{x^2+1}}{x}-\frac{\sqrt[3]{x^2+1}}{x}}{\frac{\sqrt[4]{x^4+1}}{x}-\frac{\sqrt[5]{x^4+1}}{x}}$$

$$=\lim_{x\to\infty}\frac{\frac{\sqrt{x^2+1}}{x}-\frac{\sqrt[3]{x^2+1}}{x}}{\frac{\sqrt[4]{x^4+1}}{x}-\frac{\sqrt[5]{x^4+1}}{x}}$$

$$=\lim_{x\to\infty}\frac{\frac{(x^2+1)^{1/2}}{x}-\frac{(x^2+1)^{1/3}}{x}}{\frac{(x^4+1)^{1/4}}{x}-\frac{(x^4+1)^{1/5}}{x}}$$

$$=\lim_{x\to\infty}\frac{\left(\frac{x^2+1}{x^2}\right)^{1/2}-\left(\frac{x^2+1}{x^3}\right)^{1/3}}{\left(\frac{(x^4+1)}{x^4}\right)^{1/4}-\left(\frac{x^4+1}{x^5}\right)^{1/5}}=\lim_{x\to\infty}\frac{\left(1+\frac{1}{x^2}\right)^{1/2}-\left(\frac{1}{x}+\frac{1}{x^3}\right)^{1/3}}{\left(1+\frac{1}{x^4}\right)^{1/4}-\left(\frac{1}{x}+\frac{1}{x^5}\right)^{1/5}}$$

as $x\to\infty$ $\frac{1}{x^p}\to 0$ (for $p>1$)

$$=\frac{(1+0)-0}{(1+0)-0}=1$$

Example 6 : Illustration based on factorization :

(a) $\lim_{x\to a}\left(\frac{x^3-a^3}{x^2-ax}\right)$ (b) $\lim_{x\to 3}\frac{x^3-x^2-3x-9}{x^2-4x+3}$

(c) $\lim_{x\to 4}\frac{x^3-2x^2-9x+4}{x^2-2x-8}$ (d) $\lim_{x\to 1}\frac{x^7-2x^5+1}{x^3-3x^2+2}$

Solution : (a)Given $\lim_{x\to a}\left(\frac{x^3-a^3}{x^2-ax}\right)$

we know $x^3-a^3=(x-a)(x^2+ax+a^2)$

$$\therefore \lim_{x\to a}\frac{(x-a)(x^2+ax+a^2)}{x(x-a)}=\lim_{x\to a}\frac{(x^2+ax+a^2)}{x}=\frac{3a^2}{a}=3a$$

(b) Given $\lim_{x\to 3}\frac{x^3-x^2-3x-9}{x^2-4x+3}$

if we put $x = 3$ in numerator and denominator we get 0 in both, i.e. $(x-3)$ is a factor of both numerator and deniominator.

$\therefore$ limit becomes, $\lim_{x\to 3}\frac{(x-3)(x^2+2x+3)}{(x-3)(x-1)}$

now we can put $x = 3$

$$=\frac{9+6+3}{2}=\frac{18}{2}=9$$

(c) [When $x = 4$ numerator and denominator become zero]

$$=\underset{x\to 4}{\text{Lt}}\frac{x^3-4x^2-2x^2-8x-x+4}{x^2-4x+2x-8}=\underset{x\to 4}{\text{Lt}}\frac{(x-4)(x^2+2x-1)}{(x-4)(x+2)}$$

$$=\underset{x\to 4}{\text{Lt}}\frac{x^2-2x-1}{x+2}=\frac{23}{6}$$

Second Method : $\underset{x\to 4}{\text{Lt}}\frac{x^3-2x^2-9x+4}{x^2-2x-8}$

$$=\underset{x\to 4}{\text{Lt}}\frac{\left(\frac{x^3-4^3}{x-4}\right)(x-4)-2\left(\frac{x^2-4^2}{x-4}\right)(x-4)\left(\frac{x-4}{x-4}\right)(x-4)}{\left(\frac{x^2-4^2}{x-4}\right)(x-4)-\left(\frac{x-4}{x-4}\right)(x-4)}$$

$$=\underset{x\to 4}{\text{Lt}}\frac{\frac{x^3-4^3}{x-4}-2\left(\frac{x^2-4^2}{x-4}\right)-9\left(\frac{x-4}{x-4}\right)}{\frac{x^2-4^2}{x-4}-2\left(\frac{x-4}{x-4}\right)}$$

$$=\frac{3.4^2-2.2.4^1-9}{2.4^1-2}=\frac{23}{6}$$

(d) When $x = 1$ numerator and denominator both become zero and hence $(x-1)$ is a factor of both

Now $\underset{x\to 1}{\text{Lt}}\frac{x^7-2x^5+1}{x^3-3x^2+2}$

$$=\underset{x\to 1}{\text{Lt}}\frac{x^7-x^6+x^6-x^5-x^5+x^4-x^4+x^3-x^3+x^2-x^2+x-x+1}{x^3-x^2-2x^2+2x+2}$$

$$=\underset{x\to 1}{\text{Lt}}\frac{x^6(x-1)+x^5(x-1)-x^4(x-1)-x^3(x-1)-x^2(x-1)-x(x-1)-(x-1)}{x^2(x-1)-2x(x-1)-2(x-1)}$$

$$=\underset{x\to 1}{\text{Lt}}\frac{(x^6+x^5-x^4-x^3-x^2-x-1)}{x^2-2x-2}=\frac{-3}{-3}=1$$

Second Method : $\underset{x\to1}{\text{Lt}}\dfrac{x^7-2x^5+1}{x^3-3x^2+2}$

$$=\underset{x\to1}{\text{Lt}}\dfrac{\left(\dfrac{x^7-1^7}{x-1}\right)(x-1)-2\left(\dfrac{x^5-1^5}{x-1}\right)(x-1)}{\left(\dfrac{x^3-1^3}{x-1}\right)(x-1)-3\left(\dfrac{x^2-1^2}{x-1}\right)(x-1)}$$

$$=\underset{x\to1}{\text{Lt}}\dfrac{7(1)^6-2(5(1)^4)}{3(1)^2-3(2(1))^2}=\underset{x\to1}{\text{Lt}}\dfrac{7-10}{3-6}$$

$$=\underset{x\to1}{\text{Lt}}=\dfrac{-3}{-3}=1$$

Question based on formula

Example 7 : (a) $\lim\limits_{x\to5}\dfrac{x^4-625}{x^3-125}$ (b) $\lim\limits_{x\to64}\dfrac{x^{1/6}-2}{x^{1/3}-4}$ (c) $\lim\limits_{x\to0}\dfrac{(x-h)^{1/n}-x^{1/n}}{h}$ (d) $\lim\limits_{x\to0}\dfrac{(1+x)^5-1}{3x-5x^2}$

Solution : (a)Given $\lim\limits_{x\to0}\dfrac{x^4-625}{x^3-125}$

if we write it like $\lim\limits_{x\to5}\dfrac{x^4-5^4}{x^3-5^3}$

$$=\lim_{x\to5}\dfrac{\dfrac{x^4-5^4}{x-5}}{\dfrac{x^3-5^3}{x-5}}$$

and now we can use the formula = $\dfrac{4.5^3}{3.5^3}=\dfrac{20}{3}$

(b) $\lim\limits_{x\to64}\dfrac{x^{1/6}-2}{x^{1/3}-4}$

$$\lim_{x\to64}\dfrac{x^{1/6}-2}{x^{1/3}-4}=\lim_{x\to64}\dfrac{x^{1/6}-(64)^{1/6}}{x^{1/3}-(64)^{1/3}} \qquad \left[\because (64)^{1/6}=2,(64)^{1/3}=4\right]$$

$$=\lim_{x\to64}\dfrac{\dfrac{x^{1/6}-(64)^{1/6}}{x-64}}{\dfrac{x^{1/3}-(64)^{1/3}}{x-64}}=\dfrac{\frac{1}{6}(64)^{1/6-1}}{\frac{1}{3}(64)^{1/3-1}}=\dfrac{1}{2}(64)^{-\frac{5}{6}+\frac{2}{3}}$$

$$=\lim_{x\to64}\dfrac{1}{2}(64)^{-1/6}=\dfrac{1}{2}(2^6)^{-1/6}=\dfrac{1}{2}.\dfrac{1}{2}=\dfrac{1}{4}$$

(c) $\underset{h\to0}{\text{Lt}}\dfrac{(x+h)^{1/n}-x^{1/n}}{h}$

$$=\underset{(x+h)\to x}{\text{Lt}}\dfrac{(x+h)^{1/n}-x^{1/n}}{(x+h)-x} \qquad [\therefore \text{ when } h\to0,\ x+h\to x]$$

$$= \frac{1}{n} x^{\frac{1}{n}-1} = \frac{1}{n} x^{\frac{1-n}{n}}$$

Note : Here h is variable and x is a constant.

(d) $\underset{x\to 0}{\text{Lt}} \dfrac{(1+x)^5 - 1}{3x + 5x^2}$

$$= \underset{x\to 0}{\text{Lt}} \frac{(1+x)^5 - 1^5}{\dfrac{(1+x)-1}{x(3+5x)}}.x = \underset{x\to 0}{\text{Lt}} \frac{(1+x)^5 - 1^5}{\dfrac{(1+x)-1}{3+5x}} = \frac{5.1^4}{3} = \frac{5}{3}$$

Second Method : $\underset{x\to 0}{\text{Lt}} \dfrac{(1+x)^5 - 1}{3x + 5x^2}$

$$= \underset{x\to 0}{\text{Lt}} \frac{(1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5) - 1}{x(3+5x)}$$

$$= \underset{x\to 0}{\text{Lt}} \frac{(5x + 10x^2 + 10x^3 + 5x^4 + x^5)}{x(3+5x)}$$

$$= \underset{x\to 0}{\text{Lt}} \frac{(5x + 10x + 10x^2 + 5x^3 + x^4).x}{x(3+5x)} = \frac{5}{3}$$

Example 8 : Example based on rationalization :

(a) $\lim\limits_{x\to 0} \dfrac{\sqrt{1+x^2} - \sqrt{1+x}}{x}$ (b) $\lim\limits_{x\to 2} \dfrac{x-2}{\sqrt{x^2-4} + \sqrt{x-2}}$

(c) $\lim\limits_{x\to a} \dfrac{\sqrt{a+2x} - \sqrt{3x}}{\sqrt{3a+x} - 2\sqrt{x}}, a \neq 0$

Solution : (a)[Given function $\sqrt{1+x^2} - \sqrt{1+x}$ is of the form $\sqrt{a} - \sqrt{b}$]

$$\underset{x\to 0}{\text{Lt}} \frac{\sqrt{1+x^2} - \sqrt{1+x}}{x}$$

$$= \underset{x\to 0}{\text{Lt}} \left(\frac{\sqrt{1+x^2} - \sqrt{1+x}}{x} \right) \left(\frac{\sqrt{1+x^2} + \sqrt{1+x}}{\sqrt{1+x^2} + \sqrt{1+x}} \right)$$

$$= \underset{x\to 0}{\text{Lt}} \frac{1+x^2 - (1+x)}{x\left(\sqrt{1+x^2} + \sqrt{1+x}\right)} = \underset{x\to 0}{\text{Lt}} \frac{x^2 - x}{\left(\sqrt{1+x^2} + \sqrt{1+x}\right)x}$$

$$= \underset{x\to 0}{\text{Lt}} \frac{x(x-1)}{x\left(\sqrt{1+x^2} + \sqrt{1+x}\right)} = \underset{x\to 0}{\text{Lt}} \frac{x-1}{\left(\sqrt{1+x^2} + \sqrt{1+x}\right)}$$

$$= \frac{1}{\sqrt{1} + \sqrt{1}} = -\frac{1}{2}$$

(b) $\underset{x\to 2}{\text{Lt}} \dfrac{x-2}{\sqrt{x^2-4} + \sqrt{x-2}} \left[\dfrac{0}{0} \text{ form}\right]$

$$= \underset{x\to 2}{\text{Lt}} \frac{\left(\frac{x^1-2^1}{x-2}\right)(x-2)}{\sqrt{\left(\frac{x^2-2^2}{x-2}\right)(x-2)}+\sqrt{\frac{x^1-2^1}{x-2}(x-2)}} = \underset{x\to 2}{\text{Lt}} \frac{\left(\frac{x^1-2^1}{x-2}\right).(x-2)}{\left(\sqrt{\frac{x^2-2^2}{x-2}}+\sqrt{\frac{x^1-2^1}{x-2}}\right).\sqrt{x-2}}$$

$$= \underset{x\to 2}{\text{Lt}} \frac{\left(\frac{x^1-2^1}{x-2}\right).\sqrt{x-2}}{\left(\sqrt{\frac{x^2-2^2}{x-2}}+\sqrt{\frac{x^1-2^1}{x-2}}\right)} = \frac{1.0}{\sqrt{2.2^1}+1} = \frac{0}{3} = 0$$

Second Method :

$$= \underset{x\to 2}{\text{Lt}} \frac{x-2}{\sqrt{x^2-4}+\sqrt{x-2}} = \underset{x\to 2}{\text{Lt}} \frac{x-2}{\sqrt{(x-2)(x+2)}+\sqrt{x-2}}$$

$$= \underset{x\to 2}{\text{Lt}} \frac{x-2}{\sqrt{(x-2)}+\sqrt{x+2}+1} \underset{x\to 2}{\text{Lt}} \frac{\sqrt{x-2}}{\sqrt{x+2}+1} = \frac{0}{2+1} = 0$$

(c) Required limit

$$= \underset{x\to a}{\text{Lt}} \frac{\frac{\sqrt{a+2x}-\sqrt{3a}}{(a+2x)-3a}.2(x-a)-\frac{\sqrt{3x}-\sqrt{3a}}{3x-3a}.3(x-a)}{\frac{\sqrt{3a+x}-\sqrt{4a}}{(3a+x)-4a}.(x-a)-2\left(\frac{\sqrt{x}-\sqrt{a}}{x-a}\right)(x-a)}$$

$$= \underset{x\to a}{\text{Lt}} \frac{\frac{(a+2x)^{1/2}-(3a)^{1/2}}{a+2x-3a}.2-\frac{(3x)^{1/2}-(3a)^{1/2}}{3x-3a}.3}{\frac{(3a+x)^{1/2}-(4a)^{1/2}}{(3a+x)-4}.-2\left(\frac{x^{1/2}-a^{1/2}}{x-a}\right)}$$

Example 9 : (a)Find $\underset{\theta\to 0}{\text{Lt}} \frac{\sin a\theta}{\tan b\theta}$ (b) $\underset{x\to 0}{\text{Lt}} \frac{\tan x^o}{\tan x}$

Solution : (a) $\underset{\theta\to 0}{\text{Lt}} \frac{\sin a\theta}{\tan\beta} = \underset{\theta\to 0}{\text{Lt}} \frac{\left(\frac{\sin a\theta}{a\theta}\right)a\theta}{\left(\frac{\tan b\theta}{b\theta}\right)b\theta}$

$$= \underset{\theta\to 0}{\text{Lt}} \frac{\left(\frac{\sin a\theta}{a\theta}\right)a}{\left(\frac{\tan b\theta}{b\theta}\right)b} = \frac{1.a}{1.b} = \frac{a}{b}$$

(b) $x^o = \frac{\pi}{180}x$ radian

Now $$\lim_{x\to 0}\frac{\tan x^{o}}{x}=\lim_{x\to 0}\frac{\tan\frac{\pi x}{180}}{x}=\lim_{x\to 0}\frac{\frac{\tan\frac{\pi x}{180}}{\frac{\pi x}{180}}\cdot\frac{\pi x}{180}}{x}=\frac{\pi}{180}$$

Example 10 : (a) Find $\lim_{x\to 0}\frac{x(\cos x+\cos 2x)}{\sin x}$ (b) $\lim_{x\to 0}\frac{\cos x-\cos 3x}{x(\sin 3x-\sin x)}$

(c) $\lim_{x\to 0}\frac{\tan x-\sin x}{1-\cos x}$ (d) $\lim_{x\to \pi/2}\frac{\sin(\cos x)\cos x}{\sin x-\operatorname{cosec} x}$

Solution : (a) $=\lim_{x\to 0}\frac{x(\cos x+\cos 2x)}{\sin x}=\lim_{x\to 0}\frac{x(\cos x+\cos 2x)}{\frac{\sin x}{x}.x}$

[Here factor (cos *x* + cos 2*x*), does not tend to zero, hence it is not necessary to simplify it]

$$=\lim_{x\to 0}\frac{\cos x+\cos 2x}{\frac{\sin x}{x}}=\frac{1+1}{1}=2$$

(b) $$\lim_{x\to 0}\frac{\cos x-\cos 3x}{x(\sin 3x-\sin x)}=\lim_{x\to 0}\frac{2\sin\frac{x+3x}{2}\sin\frac{3x-x}{2}}{x.2\cos\frac{x+3x}{2}\sin\frac{3x-x}{2}}$$

$$=\lim_{x\to 0}\frac{2\sin 2x\sin x}{2x\cos 2x\sin x}=\lim_{x\to 0}\frac{\cos x-\cos 3x}{x(\sin 3x-\sin x)}=\lim_{x\to 0}\frac{\sin 2x}{x\cos 2x}$$

$$=\lim_{x\to 0}\frac{\frac{\sin 2x}{2x}.2x}{x\cos 2x}=\lim_{x\to 0}\frac{\frac{\sin 2x}{2x}.2}{\cos 2x}=\frac{2}{1}=2$$

(c) $$\lim_{x\to 0}\frac{\tan x-\sin x}{1-\cos x}=\lim_{x\to 0}\frac{\frac{\sin x}{\cos x}-\sin x}{1-\cos x}$$

$$=\lim_{x\to 0}\frac{\sin x-\sin x\cos x}{\cos x(1-\cos x)}=\lim_{x\to 0}\frac{\sin x(1-\cos x)}{\cos x(1-\cos x)}$$

$=\lim_{x\to 0}\tan x=0$ [By the definition of limit because the form is not indeterminate]

(d) $$\lim_{x\to \pi/2}\frac{\sin(\cos x)\cos x}{\sin x-\operatorname{cosec} x}$$

$$=\lim_{x\to \pi/2}\frac{\sin(\cos x)\cos x\sin x}{\sin^2 x-1}$$

$$=\lim_{x\to \pi/2}\frac{\sin(\cos x)\cos x\sin x}{\sin^2 x-1}$$

$$=\lim_{x\to \pi/2}-\frac{[\sin(\cos x)\sin x]}{\cos x}=-\lim_{x\to \pi/2}\frac{\sin(\cos x)\sin x}{\cos x}$$

as $x \to \frac{\pi}{2}$ $\cos x \to 0$,

$\therefore$ we can use $\frac{\sin x}{x}$ rule

$= -\lim_{x \to \pi/2} \sin x = -1$

Example 11 : $\left(\text{Using } \lim_{x \to 0} \frac{\sin x}{x} = 1\right)$ Evaluate the following limits

(a) $\lim_{x \to \pi/3} \frac{\tan x - \sqrt{3}}{9x^2 - \pi^2}$ (b) $\underset{x \to 0}{\text{Lt}} \frac{\tan x - \sin x}{x^3}$

(c) $\underset{x \to y}{\text{Lt}} \frac{\tan x - \tan y}{x - y}$ (d) $\underset{x \to a}{\text{Lt}} \frac{a \sin x - x \sin a}{ax^2 - a^2 x}$

Solution : (a) $\lim_{x \to \pi/3} \frac{\tan x - \sqrt{3}}{9x^2 - \pi^2} = \lim_{x \to \pi/3} \frac{\tan x - \tan \frac{\pi}{3}}{9x^2 - \pi^2}$

Using $\tan A - \tan B = \frac{\sin(A - B)}{\cos A \cos B}$ we get,

$$\lim_{x \to \pi/3} \frac{\sin\left(x - \frac{\pi}{3}\right)}{\cos x \cos \frac{\pi}{3} (3x - \pi)(3x + \pi)}$$

$$= \frac{1}{3} \frac{1}{\cos \frac{\pi}{3} \cos \frac{\pi}{3} (\pi + \pi)} \left(\text{Using } \lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1\right) = \frac{2}{3\pi}$$

(b) $\underset{x \to 0}{\text{Lt}} \frac{\tan x - \sin x}{x^3} = \underset{x \to 0}{\text{Lt}} \frac{\frac{\sin x}{\cos x} - \sin x}{x^3}$

$$= \underset{x \to 0}{\text{Lt}} \frac{\sin x (1 - \cos x)}{x^3 \cos x} = \underset{x \to 0}{\text{Lt}} \frac{\sin x . 2 \sin^2 \frac{x}{2}}{x^3 \cos x}$$

$$= \underset{x \to 0}{\text{Lt}} \frac{2\left(\frac{\sin x}{x}\right) x . \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}}\right) \left(\frac{x}{2}\right)^2}{x^3 \cos x} = \underset{x \to 0}{\text{Lt}} \frac{2\left(\frac{\sin x}{x}\right) x . \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}}\right)^2 . \frac{1}{4}}{\cos x} = \frac{2.1.1^2 \frac{1}{4}}{1} = \frac{1}{2}$$

(c)[Here independent variable x is not tending to zero rather x is tending to y, hence put $x = y + h$]

Let $x = y + h$, then as $x \to y$, $h \to 0$

Now $\underset{x \to y}{\text{Lt}} \frac{\tan x - \tan y}{x - y} = \underset{h \to 0}{\text{Lt}} \frac{\tan(y + h) - \tan y}{y + h - y}$

$$= \underset{h\to 0}{\text{Lt}} \frac{1}{h}\left[\frac{\sin(y+h)}{\cos(y+h)} - \frac{\sin y}{\cos y}\right]$$

$$= \underset{h\to 0}{\text{Lt}} \frac{\sin(y+h)\cos y - \cos(y+h)\sin y}{h\cos(y+h)\cos y}$$

$$= \underset{h\to 0}{\text{Lt}} \frac{\sin(y+h-y)}{h\cos(y+h)\cos y}$$

$$= \underset{h\to 0}{\text{Lt}} \frac{\sin h}{h}.\frac{1}{\cos(y+h)\cos y} = 1\frac{1}{\cos^2 y} = \sec^2 y$$

(d) $\lim\limits_{x\to a} \dfrac{a\sin x - x\sin a}{ax^2 - a^2 x}$

$$= \lim_{x\to a} \frac{a\sin x - x\sin x + x\sin x - x\sin a}{ax(x-a)}$$

$$= \lim_{x\to a} \frac{(a-x)\sin x + x(\sin x - \sin a)}{ax(x-a)}$$

$$= \lim_{x\to a} \frac{(a-x)\sin x}{ax(x-a)} + \lim_{x\to a} \frac{\sin x - \sin a}{a(x-a)}$$

$$= -\frac{\sin a}{a^2} + \lim_{x\to a} \frac{2\cos\left(\frac{x+a}{2}\right)}{2a}\left[\frac{\sin\frac{(x-a)}{2}}{\frac{(x-a)}{2}}\right] = -\frac{\sin a}{a^2} + \frac{\cos a}{a}$$

Example 12 : (a)If $f(x) = \dfrac{\tan 2x - x}{3x - \sin x}$, Find $\underset{x\to 0}{\text{Lt}}\, f(x)$ (b) $\underset{x\to 0}{\text{Lt}} \dfrac{\sin(\alpha+\beta)x + \sin(\alpha-\beta)x + \sin 2\alpha x}{\cos 2\beta x - \cos 2\alpha x}.x$

(c) $\underset{h\to 0}{\text{Lt}} \dfrac{(a+h)^2\sin(a+h) - a^2\sin a}{h}$ (d) $\underset{x\to 0}{\text{Lt}} \dfrac{1-\cos x\cos 2x\cos 3x}{\sin^2 x}$

Solution : (a) $\underset{x\to 0}{\text{Lt}}\, f(x) = \underset{x\to 0}{\text{Lt}} \dfrac{\tan 2x - x}{3x - \sin x}$

$$= \underset{x\to 0}{\text{Lt}} \frac{\left(\frac{\tan 2x}{2x}\right)2x - x}{3x - \left(\frac{\sin x}{x}\right)x} = \underset{x\to 0}{\text{Lt}} \frac{\left(\frac{\tan 2x}{2x}\right)2 - 1}{3 - \frac{\sin x}{x}} = \frac{2-1}{3-1} = \frac{1}{2}$$

(b) $\underset{x\to 0}{\text{Lt}} \dfrac{\sin(\alpha+\beta)x + \sin(\alpha-\beta)x + \sin 2\alpha x}{\cos 2\beta x - \cos 2\alpha x}.x$

$$= \underset{x\to 0}{\text{Lt}}\left[\frac{\frac{\sin(\alpha+\beta)x}{(\alpha+\beta)x}.(\alpha+\beta)x + \frac{\sin(\alpha-\beta)x}{(\alpha-\beta)x}.(\alpha-\beta)x + \frac{\sin 2\alpha x}{2\alpha x}.2\alpha x}{2\frac{\sin(\alpha+\beta)x}{(\alpha+\beta)x}.(\alpha+\beta)x.\frac{\sin(\alpha-\beta)}{(\alpha-\beta)x}.(\alpha-\beta)x}\right]x$$

$$= \lim_{x\to 0} \frac{\frac{\sin(\alpha+\beta)x}{(\alpha+\beta)x}.(\alpha+\beta)+\frac{\sin(\alpha-\beta)x}{(\alpha-\beta)x}.(\alpha-\beta)x+\frac{\sin 2\alpha x}{2\alpha x}.2\alpha}{2\frac{\sin(\alpha+\beta)x}{(\alpha+\beta)x}.(\alpha+\beta).\frac{\sin(\alpha-\beta)x}{(\alpha-\beta)x}.(\alpha-\beta)}$$

$$= \frac{1.(\alpha+\beta)+1.(\alpha-\beta)+1.2\alpha}{2.1(\alpha+\beta)1.(\alpha-\beta)} = \frac{4\alpha}{2(\alpha^2-\beta^2)} = \frac{2\alpha}{\alpha^2-\beta^2}$$

$= \alpha \cos \alpha + \sin \alpha = \sin \alpha - \alpha \cos \alpha$

(c) $\lim_{h\to 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$

$$= \lim_{h\to 0} \frac{(a^2+2ah+h^2)\sin(a+h)-a^2\sin a}{h}$$

$$= \lim_{h\to 0}\left[\frac{a^2(\sin(a+h))-\sin a)}{h}\right]+\frac{(2ah+h^2)\sin(a+h)}{h}$$

$$= \lim_{h\to 0}\left[\frac{a^2.2\cos\frac{2a+h}{2}\sin\frac{h}{2}}{h}+(2a+h)\sin(a+h)\right]$$

$$= \lim_{h\to 0} \frac{2a^2\cos\frac{2a+h}{2}\sin\frac{h}{2}.\frac{h}{2}}{\frac{h}{2}\quad h} + \lim_{h\to 0}(2a+h)\sin(a+h) = a^2\cos a + 2a\sin a$$

(d)cos x cos $2x$ cos $3x$

$= \frac{1}{2}$ (2cos x cos $3x$ cos $2x$)

$= \frac{1}{2}$ [(cos $2x$ + cos $4x$) cos $2x$]

$= \frac{1}{4}$ [2 cos^2 $2x$ + 2 cos $4x$ cos $2x$]

$= \frac{1}{4}$ [1 + cos $4x$ + cos $2x$ + cos $6x$]

Now $\lim_{x\to 0} \frac{1-\cos x\cos 2x\cos 3x}{\sin^2 2x}$

$$= \lim_{x\to 0} \frac{1-\frac{1}{4}(1+\cos 2x+\cos 4x+\cos 6x)}{\sin^2 2x}$$

$$= \lim_{x\to 0} \frac{1-\cos 2x+1-\cos 4x+1-\cos 6x}{4\sin^2 2x}$$

$$= \underset{x\to 0}{\text{Lt}} \frac{2\sin^2 x + 2\sin^2 2x + 2\sin^2 3x}{4\sin^2 2x}$$

$$= \underset{x\to 0}{\text{Lt}} \frac{2\left(\frac{\sin x}{x}\right)^2 .x^2 + 2\left(\frac{\sin 2x}{2x}\right)^2 .4x^2 + 2\left(\frac{\sin 3x}{3x}\right)^2 .9x^2}{4\left(\frac{\sin 2x}{2x}\right)^2 .4x^2} = \frac{28}{16} = \frac{7}{4}$$

Example 13 : (a) $\underset{x\to\infty}{\text{Lt}}\left(\frac{\sin x}{x}\right)$ (b) $\underset{x\to\infty}{\text{Lt}}\ 2x\tan\left(\frac{1}{x}\right)$

Solution : (a)We have $\underset{x\to\infty}{\text{Lt}} \frac{\sin x}{x}$

we know $-1 \le \sin x \le 1$, for all values of $x \in R$ and as $x \to \infty$, $\frac{1}{x} \to 0$

$\therefore$ Limit becomes

$\underset{x\to\infty}{\text{Lt}} (\to 0)$ (a number between –1 and 1)

= 0

Note : You can though remember this limit.

(b) $\underset{x\to\infty}{\text{Lt}}\ 2x\tan\left(\frac{1}{x}\right)$

put $x = \frac{1}{h}$, so as $x \to \infty\ h \to 0$

$\underset{h\to 0}{\text{Lt}} \frac{2}{h}\tan(h)$

$= 2\left\{\text{as } \underset{h\to 0}{\text{Lt}} \frac{\tanh}{h} = 1\right\}$

Example 14 : Find the limit, $\underset{x\to\infty}{\text{Lt}}\left(\frac{x+\cos x}{x+\sin x}\right)$

Solution : Given Limit is $\underset{x\to\infty}{\text{Lt}} \frac{x+\cos x}{x+\sin x}$

$$= \underset{x\to\infty}{\text{Lt}} \frac{1+\frac{\cos x}{x}}{1+\frac{\sin x}{x}}$$

$$= \underset{x\to\infty}{\text{Lt}} \frac{1+0}{1+0} \qquad \left[\underset{x\to\infty}{\text{Lt}} \frac{\cos x}{x} = 0 \text{ using the same concept as } \lim_{x\to 0}\frac{\sin x}{x}\right]$$

= 1

Example 15 :(a) $\underset{x\to 1}{\text{Lt}}\left(\frac{\log x}{x-1}\right)$ (b) $\underset{x\to 0}{\text{Lt}} \frac{e^x - 1 - x}{x^2}$ (c) $\underset{h\to 0}{\text{Lt}}\left(\frac{\log(x+h) - \log x}{h}\right)$

Solution : (a)Given limit is $\underset{x\to 1}{\text{Lt}}\left(\frac{\log x}{x-1}\right)$

replacing x by $h + 1$, {limit also changes}

$$= \underset{h\to 0}{\text{Lt}}\frac{\log(1+h)}{1+h-1}$$

$$= \underset{h\to 0}{\text{Lt}}\frac{\log(1+h)}{h} = 1$$

(b) $\underset{x\to 0}{\text{Lt}}\frac{e^x - 1 - x}{x^2}$

in this question if we try to use the formula $\left(\frac{e^x-1}{x}\right)$, then it will not be solved, why?

Because we will get zero in numerator and denominator, which becomes unsolvable.

$$\underset{x\to 0}{\text{Lt}}\frac{\frac{e^x-1}{x}-\frac{1}{x}}{x} = \underset{x\to 0}{\text{Lt}}\frac{(\to 1)-1}{x} = \frac{0}{0} \text{ form}$$

so either we use L' hospitals or we go for expansion series.
Here we will go for expansion series

$$= \underset{x\to 0}{\text{Lt}}\frac{\left(1+\frac{x}{1}+\frac{x^2}{2!}+\frac{x^3}{3!}....\right)-1-x}{x^2}$$

$$= \underset{x\to 0}{\text{Lt}}\left(\frac{x^2}{2!}+\frac{x^2}{3!}+....\right) = \underset{x\to 0}{\text{Lt}}\left(\frac{1}{2}+\frac{x}{3!}+\frac{x^2}{4!}....\right)$$

putting $x = 0$ in the rest = $\frac{1}{2}$

(c)Given limit is,

This is a simple limit, just use log properties i.e. $\log a - \log b = \log \frac{a}{b}$

$$= \underset{h\to 0}{\text{Lt}}\frac{\log\left(\frac{x+h}{x}\right)}{h} \qquad = \underset{h\to 0}{\text{Lt}}\frac{\log\left(1+\frac{h}{x}\right)}{h} = \frac{\log\left(1+\frac{h}{x}\right)}{\frac{h}{x}\times x}$$

$$= \left(\frac{1}{x}\right) \qquad \left\{\because \underset{h\to 0}{\text{Lt}}\log\frac{(1+h)}{h} = 1\right\}$$

Example 16 : (a)Evaluate $\lim_{x\to 0}\frac{(ab)^x - a^x - b^x + 1}{x^2}$ (b) Evaluate : $\lim_{x\to 0}\frac{e^{\tan x} - e^x}{\tan x - x}$

(c) $\lim_{x\to 0}\frac{3^x - 5^x}{x}$ (d) $\lim_{x\to 0}\frac{6^x - 2^x - 3^x + 1}{\sin^2 x}$

Solution : (a) $\lim_{x\to 0}\frac{(ab)^x - a^x - b^x + 1}{x^2}$

$= \lim_{x\to 0}\frac{a^x b^x - a^x - b^x + 1}{x^2}$

$= \lim_{x\to 0} = \frac{a^x(b^x - 1) - (b^x - 1)}{x^2}$

$= \lim_{x\to 0} = \frac{(a^x - 1)\times(b^x - 1)}{x} = \lim_{x\to 0}\frac{a^x - 1}{x} \times \lim_{x\to 0}\frac{(b^x - 1)}{x} = \log a \times \log b$

(b) $\lim_{x\to 0}\frac{e^{\tan x} - e^x}{\tan x - x} = \lim_{x\to 0}\frac{e^x \times \left[e^{(\tan x - x)} - 1\right]}{(\tan x - x)}$

$= \lim_{x\to 0}\frac{e^x\{e^{\tan x - x} - 1\}}{(\tan x - x)} = e^0 \times 1$ [As $x \to 0$, $\tan x - x \to 0$]

$= 1 \times 1 = 1$

(c) $\lim_{x\to 0}\frac{3^x - 5^x}{x} = \lim_{x\to 0}\left[\frac{3^x - 1}{x} - \frac{5^x - 1}{x}\right]$

$= \log 3 - \log 5 = \log\frac{3}{5}$

(d) $\lim_{x\to 0}\frac{6^x - 2^x - 3^x + 1}{\sin^2 x}$

$= \lim_{x\to 0}\frac{(2^x - 1)(3^x - 1)}{x^2}\frac{x^2}{\sin^2 x} = \lim_{x\to 0}\frac{2^x - 1}{x}\lim_{x\to 0}\frac{3^x - 1}{x}\lim_{x\to 0}\left(\frac{x}{\sin x}\right)^2$

$= \log_e 3 \log_e 2$

Example 17 : (a) $\underset{x\to\infty}{\text{Lt}}\left(1 + \frac{a}{x}\right)^x$ (b) $\underset{x\to\infty}{\text{Lt}}\left(e^{1/x} - e^{-1/x}\right)$

Solution : We try to convert these questions to type 2 only so that we can use formulas.

(a)Given, $\underset{x\to\infty}{\text{Lt}}\left(1 + \frac{a}{x}\right)^x$

Nowhere it is a type II question only. Compare it with formula $\underset{x\to 0}{\text{Lt}}(1 + ax)^{1/x}$. If you put $x = \frac{1}{h}$, limit changes to

$= \underset{h\to 0}{\text{Lt}}(1 + ah)^{1/h}$

$= e^a\{\underset{x\to 0}{\text{Lt}}(1 + x)^{1/h} = 0\}$

(b) $\underset{x\to 0}{\text{Lt}}\, x\left(e^{1/x} - e^{-1/x}\right)$

again putting $x = \frac{1}{h}$

$= \underset{h\to 0}{\text{Lt}} \frac{1}{h}(e^h - e^{-h})$

$= \underset{h\to 0}{\text{Lt}} \left[\frac{e^h - 1}{h} - \frac{(e^{-h} - 1)}{h}\right] = \underset{h\to 0}{\text{Lt}} \left[\frac{e^h - 1}{h}\right] - \underset{h\to 0}{\text{Lt}} \left[\frac{e^{-h} - 1}{h}\right]$

in second limit put $h = -h$

$= 1 - \underset{h\to 0}{\text{Lt}} \left[\frac{e^h - 1}{-h}\right] = 1 + 1 \left\{\because \underset{h\to 0}{\text{Lt}} \frac{e^h - 1}{h} = 1\right\}$

= 2

Example 18 : (a) $\lim_{x\to 0}(1-2x)^{1/x}$ (b) $\lim_{x\to\infty}\left(1 - \frac{1}{x^2}\right)^x$ (c) $\lim_{x\to\infty}\left(\frac{x+1}{x-2}\right)^{2x-1}$ (d) $\lim_{x\to 1} x^{\cot 2x}$

Solution : (a) $\lim_{x\to 0}(1-2x)^{1/x}$

using formula

$= \lim_{x\to 0}(1-2x)^{-1/2x}\left[\frac{1}{x} - 2x\right]$

$\Rightarrow e^{-1\lim_{x\to 0}\frac{2x}{x}} = e^{-2}$

(b) $\lim_{x\to\infty}\left(1 + \frac{1}{x^2}\right)^x$

our formula is $\lim_{x\to 0}(1 + f(x))^{\frac{1}{f(x)}} = e$ where $f(x) \to 1$

here $f(x) = \frac{1}{x^2}$ which as $x \to \infty$, $f(x) \to 0$, hence we can apply the same formula.

$\Rightarrow \lim_{x\to\infty}\left(1 + \frac{1}{x^2}\right)^x = \lim_{x\to\infty}\left(1 + \frac{1}{x^2}\right)^{x^2\left[x \times \frac{1}{x^2}\right]}$

$= e^{\lim_{x\to\infty}\frac{x}{x^2}} = e^{\lim_{x\to\infty}\frac{1}{x}}$

$= e^0 = 1$

(c) $\lim_{x\to\infty}\left(\frac{x+1}{x-2}\right)^{2x-1}$

$= \lim_{x\to\infty}\left(1 + \frac{x+1}{x-2} - 1\right)^{2x-1} = \lim_{x\to\infty}\left(1 + \frac{3}{x-2}\right)^{2x-1}$

again we can apply the concept as used in previous question

$= e^{\lim_{x\to\infty}\frac{3}{x-2}\times 2x-1} = e^6$

(d) $\lim_{x\to 1} x^{\cot \pi x}$

$= \lim_{x\to 1}(1 + (x-1))^{\cot \pi x}$

$= e^{\lim_{x\to1}(x-1)\cot\pi x} = e^{\lim_{x\to1}\frac{x-1}{\tan\pi x}}$ or $e^{\lim_{x\to1}\frac{x-1}{\tan(\pi-\pi x)}}$

$= e^{\lim_{x\to1}\frac{x-1}{\tan\pi(1-x)}}$

$= e^{\lim_{x\to1}\frac{-1}{\pi}\frac{\pi(1-x)}{\tan\pi(1-x)}}$ $\left\{\text{as } \lim_{\theta\to0}\frac{\tan\theta}{\theta}=1\right\}$

$= e^{\frac{-1}{\pi}}$

Example 19 : (a) $\lim_{x\to0}\frac{|x|}{x}$ (b) $\lim_{x\to0}[x-3]$ (c) $\lim_{x\to0}\sin\frac{1}{x}$ (d) $\lim_{x\to0}x\sin\left(\frac{1}{x}\right)$

Solution : (a)For the limit to exist LHL and RHL should be equal let's take LHL first

$\lim_{x\to0}\frac{|x|}{x}$ $\left\{\begin{array}{l}|x| \text{ gives } (-x) \text{ for negative no. and since } x \text{ is}\\ \text{approaching from negative side, } |x| \text{ gives } (-x)\end{array}\right\}$

$= \lim_{x\to0}\frac{-x}{x} = -1$ ....(i)

Now, RHL

$\lim_{x\to0^+}\frac{|x|}{x}$ {as $x \to 0^+$ i.e. from positive side $|x|$ returns $+x$}

$= \lim_{x\to0}\frac{x}{x} = 1$ ....(ii)

from (i) and (ii) LHL $\neq$ RHL, hence limit does not exist.

(b) $\lim_{x\to0}[x-3]$ again we will find LHL and RHL for this question

LHL $\lim_{x\to0^-}[x-3]$

$= \lim_{h\to0}[0-h-3]$

$= \lim_{h\to0}\underbrace{[-h-3]}$

This will be a number between (–4, –3) and we know that for this the value of greatest integer function is –4.

$\therefore$ LHL = –4

RHL $\lim_{x\to0^+}[x-3]$

$= \lim_{h\to0}[0+h-3]$

$= \lim_{h\to0}\underbrace{[h-3]}$

this will be a number between (–3, –2) and hence greatest integer function returns –3.

$\therefore$ RHL = –3

Now, LHL $\neq$ RHL

Hence limit does not exist.

(c) $\lim_{x\to0}\sin\left(\frac{1}{x}\right)$

now as $x \to 0$ $\left(\frac{1}{x}\right) \to \infty$

but for $\sin\left(\frac{1}{x}\right)$ or $\sin(\infty)$ is not a finite value. In fact it is a oscillatory value between [–1, +1] because we dont know the value of ∞.

Note : Some of the students get confused in this, in fact some think that $\sin(\infty) \to \infty$ which is absolutely wrong as $\sin x$ can never return a value other than [–1, 1].

Since the limit is not finite, limit does not exists.

(d) $\lim_{x\to 0} x\sin\left(\frac{1}{x}\right)$

This is a very important limit. Let us solve it.

We already solved the part $\lim_{x\to 0}\sin\left(\frac{1}{x}\right)$ in the above question. It is a value between [–1, 1]

but $x \to 0$, x approaches 0.

Hence limit becomes

$= \to 0 \times$ (a number between [–1, 1])

$= 0$

Hence limit exists and is equal to 0.

(You can check by equating LHL and RHL).

Example 20 : Evaluate the following limits :

(a) $\underset{x\to 1}{\text{Lt}}(1-x)\tan\frac{\pi x}{2}$ (b) $\underset{x\to \pi/2}{\text{Lt}}\frac{1-\sin x}{\left(\frac{\pi}{2}-x\right)\cot x}$

Solution :

(a) These type of limits are solved by substituting the limit.

See now in the limit,

$\underset{x\to 1}{\text{Lt}}(1-x)\tan\frac{\pi x}{2}$, $\tan\frac{\pi x}{2}$ approaches ∞ so somehow we need to remove this. If it can be converted to $\cot\frac{\pi x}{2}$, then the limit will be solved as $\cot\frac{\pi x}{2}$ will approach 0.

And we know $\cot\frac{\pi x}{2} = \tan\left(\frac{\pi}{2}-\frac{\pi}{2}x\right)$ or $\tan\frac{\pi}{2}(1-x)$

now, do you see something putting x as $(1 - x)$ solves the question $(1 - x)$ becomes x and lim changes to

$$\lim_{x\to 0} x\cot\frac{\pi}{x}\lim_{x\to 0}\frac{x}{\sin\frac{\pi x}{2}} = \frac{2}{\pi} \qquad [\text{as } \cos\theta = 1]$$

(b) $\underset{x\to \pi/2}{\text{Lt}}\frac{1-\sin x}{\left(\frac{\pi}{2}-x\right)\cot x}$

Note : In these type of questions you will get the clue of what to substitute from the question itself.

Like in this question we will substitute $\frac{\pi}{2} - x$ for x, hence limit becomes

$$\therefore \lim_{x\to 0}\frac{1-\sin\left(\frac{\pi}{2}-x\right)}{x\cot\left(\frac{\pi}{2}-x\right)} = \lim_{x\to 0}\frac{1-\cos x}{x\tan x}$$

$$= \lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)\left(\frac{x}{\tan x}\right) = \frac{1}{2}\times 1 \qquad \left\{\text{as } \lim_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}\right\}$$

Example 21 : Evaluate $\underset{x\to\infty}{\text{Lt}}\ \frac{x^5}{e^x}$

Solution : $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$

putting this back in the limit

$$\underset{x\to\infty}{\text{Lt}}\ \frac{x^5}{1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\frac{x^5}{5!}+....}$$

dividing by x^5

$$\underset{x\to\infty}{\text{Lt}}\ \frac{1}{\frac{1}{x^5}+\frac{1}{x^4}+\frac{1}{2!x^3}+\frac{1}{3!x^2}+\frac{1}{4!x}+1+\frac{x}{6!}+...}$$

we can see that denominator is approaching ∞

Hence the limit becomes $\underset{x\to\infty}{\text{Lt}}\ \frac{1}{\to\infty}$ which is 0.

Example 22 : Find $\underset{x\to 0}{\text{Lt}}\ \frac{\log(1+x)}{3^x-1}$

Solution : Here we will apply expansion series of both $\log(1 + x)$ and a^x which is

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...$$

and $a^x = 1 + x(\log a) + \frac{x^2}{2!}(\log a)^2 + ...$

using these

$$\underset{x\to 0}{\text{Lt}}\ \frac{x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}}{1+x\log 3+\frac{x^2}{2!}(\log 3)^2....-1}$$

$$= \underset{x\to 0}{\text{Lt}}\ \frac{x\left(1-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}.....\right)}{x\log 3\left(1+\frac{x}{2}\log 3+...\right)}$$

Now we can put $x = 0$ in the limit also

Hence limit is $\frac{1}{\log 3}$.

Example 23 : Find $\lim\limits_{x\to 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

Solution : $\lim\limits_{x\to 1^-} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$ $\left[\frac{0}{0}\text{ form}\right]$

$= \lim\limits_{x\to 1} \frac{7x^6 - 10x^4 + 0}{3x^2 - 6x + 0}$ [by L Hospital's rule]

$= \frac{7-10}{3-6} = \frac{(-3)}{(-3)} = 1$

Example 24 : Find $\lim\limits_{x\to \alpha} \frac{x\sin\alpha - \alpha\sin x}{x - \alpha}$

[Here x is the variable and α is a constant, therefore we will have to differentiate w.r.t. to x]

Solution : $\lim\limits_{x\to \alpha} \frac{x\sin\alpha - \alpha\sin x}{x - \alpha}$ $\left[\frac{0}{0}\text{ form}\right]$

$= \lim\limits_{x\to \alpha} \frac{1.\sin\alpha - \alpha\cos x}{1 - 0} = \sin\alpha - \alpha\cos\alpha$

Example 25 : Find $\lim\limits_{x\to 0} \frac{\tan x - \sin x}{x^3}$

Solution : $\lim\limits_{x\to 0} \frac{\tan x - \sin x}{x^3}$ $\left[\frac{0}{0}\text{ form}\right]$

$= \lim\limits_{x\to 0} \frac{\sec^2 x - \cos x}{3x^2}$ $\left[\frac{0}{0}\text{ form}\right]$

$= \lim\limits_{x\to 0} \frac{2\sec x \sec x \tan x + \sin x}{6x}$ $\left[\frac{0}{0}\text{ form}\right]$

$= \lim\limits_{x\to 0} \frac{2\sec^2 x.\sec^2 x + 2\tan x.2\sec x \sec x \tan x + \cos x}{6}$

$= \frac{2+0+1}{6} = \frac{1}{6}$

Example 26 : Find $\lim\limits_{h\to 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$ [Here h is the variable]

Solution : $\lim\limits_{h\to 0} \frac{(a+h)^2 \sin(a+h) - a^2 \sin a}{h}$ $\left[\frac{0}{0}\text{ form}\right]$

$= \lim\limits_{h\to 0} \frac{(a+h)^2 \cos(a+h)(0+1) + \sin(a+h)2(a+h)(+1) - 0}{1}$

$= a^2 \cos a + 2a \sin a$ [by L' Hospital's rule]

Example 27 : Evaluate $\lim_{x\to 0^+}(\sin x)^x$

Solution : Let $A = \lim_{x\to 0^+}(\sin x)^x$

$\Rightarrow \log A = \lim_{x\to 0^+} x\log(\sin x)$

$\log A = \lim_{x\to 0^+}\frac{\log(\sin x)}{1/x}$ [By L' Hospital's rule]

$$= \lim_{x\to 0^+}\frac{\frac{1}{\sin x}.\cos x}{-\frac{1}{x^2}} = -\lim_{x\to 0^+} x^2\cot x$$

$$= -\lim_{x\to 0^+}\frac{x^2}{\tan x} = 0$$

$\Rightarrow A = 1$ or $\lim_{x\to 0^+}(\sin x)^x = 1$

Example 28 : Evaluate $\lim_{x\to 0}(\text{cosec } x)^x$

Solution : Let $A = \lim_{x\to 0}(\text{cosec } x)^x$ (∞^0 form)

$\log A = \lim_{x\to 0} x\log(\text{cosec } x)$

$$= \lim_{x\to 0}\frac{\log(\text{cosec } x)}{\frac{1}{x}} \quad \left(\frac{\infty}{\infty}\text{form}\right)$$

$$= \lim_{x\to 0}\frac{1}{\text{cosec } x}.\frac{(-\text{cosec } x\cot x)}{-\frac{1}{x^2}}$$ [By L'Hospital's rule]

$$= \lim_{x\to 0}\frac{x^2}{\tan x} = 0$$

$\therefore \log A = 0$ or $A = 1$

$\Rightarrow \lim_{x\to 0}(\text{cosec } x)^x = 1$

Example 29 : Evaluate $\lim_{x\to 0} e^{\frac{1}{x\log x}}$.

Solution : Let $A = \lim_{x\to 0} e^{\frac{1}{x\log x}}$

$\log A = \lim_{x\to 0} e^{\frac{1}{x\log x}.\log e} = \lim_{x\to 0}\frac{1/x}{\log x}.\log e \quad \left(\frac{\infty}{\infty}\text{from}\right)$ [By L' Hosptial's rule]

$$= \lim_{x\to 0}\frac{-1/x^2}{1/x} = -\infty$$

$\log_e A = -\infty \quad \Rightarrow A = e^{-\infty}$ or $\lim_{x\to 0}\frac{1}{e^{x\log x}} = 0$

Example 30 : Evaluate $\lim_{x\to 0}|x|^{\sin x}$.

Solution : $\lim_{x\to 0}|x|^{\sin x}=\lim_{x\to 0}e^{\sin x\log_e|x|}=e^{\lim_{x\to 0}\frac{\log_e|x|}{\operatorname{cosec} x}}$

$$=\lim_{e^x\to 0}\frac{1/x}{-\operatorname{cosec} x\cot x}$$

$$=\lim_{e^x\to 0}-\frac{\sin^2 x}{x\cos x}=e^{e^x\to 0\ -\left(\frac{\sin x}{x}\right)^2\cdot\left(\frac{x}{\cos x}\right)}$$

$= e^{-(1).(0)} = e^0 = 1$

Example 31 : Solve $\lim_{x\to 0^+}\log_{\sin x}\sin 2x$

Solution : Here, $\lim_{x\to 0^+}\log_{\sin x}\sin 2x$

$$=\lim_{x\to 0^+}\frac{\log\sin 2x}{\log\sin x}\qquad\left(\frac{-\infty}{-\infty}\text{form}\right)$$

$$=\lim_{x\to 0^+}\frac{1}{\frac{\sin 2x}{\frac{1}{\sin x}.\cos x}}.2\cos 2x=\lim_{x\to 0^+}\frac{\left(\frac{2x}{\sin(2x)}\right)\cos 2x}{\left(\frac{x}{\sin x}\right)\cos x}$$

[By L' Hospital's]

$$=\lim_{x\to 0^+}\frac{\cos 2x}{\cos x}=1$$

Example 32 : Solve $\lim_{x\to 0^+}(\sin x)^{\tan x}$

Solution : Here $\lim_{x\to 0^+}(\sin x)^{\tan x}$

(0^0 form)

Let $A = \lim_{x\to 0^+}(\sin x)^{\tan x}$

Taking log on both sides, we get

$$\log_e A = \lim_{x\to 0^+}\tan x\log(\sin x)$$

$$=\lim_{x\to 0^+}\frac{\log(\sin x)}{\cot x}\left(\frac{-\infty}{-\infty}\text{form}\right)$$ [By L' Hospitals]

Applying L' Hospital rule $=\lim_{x\to 0^+}\frac{1}{\frac{\sin x}{-\operatorname{cosec}^2 x}}.\cos x=\lim_{x\to 0^+}-\sin x.\cos x=0$

$\log_e A = 0$

$A = e^0 = 1 \Rightarrow A = 1$

Example 33 : Evaluate $\lim_{n\to\infty}(\pi n)^{2/n}$.

Solution : Here, $A = \lim_{n\to\infty}(\pi n)^{2/n}$ (∞^0 form)

$$\log A = \lim_{n\to\infty} \frac{2\log(\pi n)}{n}$$

$\left(\frac{\infty}{\infty}\text{form}\right)$

$$= \lim_{n\to\infty} \frac{2.\frac{1}{\pi n}.\pi}{1} = \lim_{n\to\infty} \frac{2}{n} = 0$$

$\log_e A = 0 \Rightarrow A = 1$

Example 34 : Evaluate $\lim_{n\to\infty}\left(\frac{e^n}{\pi}\right)^{1/n}$.

Solution : Here, $A = \lim_{n\to\infty}\left(\frac{e^n}{\pi}\right)^{1/n}$

$(\infty^0 \text{ form})$

$\therefore \quad \log A = \lim_{n\to\infty} \frac{1}{n}\log\left(\frac{e^n}{\pi}\right)$

$$= \lim_{n\to\infty} \frac{1}{n}\log\left(\frac{e^n}{\pi}\right)$$

$\left(\frac{\infty}{\infty}\text{form}\right)$

$$= \lim_{n\to\infty} \frac{n\log e - \log\pi}{n}$$

$$= \lim_{n\to\infty} \frac{\log e - 0}{1}$$

$= \log e$

$\Rightarrow A = e$

Example 35 : Using first principle, find the derivatives of the following functions

(i) x^n

(ii) a^x, $a > 0$

(iii) $\tan x$, $x \neq (2n+1)\frac{\pi}{2}$, $n \in Z$

(iv) $\sec x$, $x \neq (2n+1)\frac{\pi}{2}$, $n \in Z$

(v) $\sin^{-1} x$, $-1 \leq x \leq 1$

Solution : (i) We have $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{(x+h)^n - x^n}{h}$

$$= \lim_{h\to 0} \frac{x^n\left[\left(1+\frac{h}{x}\right)^n - 1\right]}{h}$$

$$= \lim_{h\to 0} \frac{x^n\left[\left(1+\frac{h}{x}\right)^n - 1\right]}{x\left(1+\frac{h}{x}-1\right)} = x^{n-1}\lim_{h\to 0}\frac{t^n-1}{t-1}; t = 1+\frac{h}{x} \quad \text{say}$$

$= x^{n-1} \cdot n = nx^{n-1}$

(ii) $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{a^{x+h}-a^x}{h}$

$$= \lim_{h\to 0} a^x\left(\frac{a^h-1}{h}\right) = a^x \lim_{h\to 0}\frac{a^{x+h}-a^x}{h}$$

(iii) $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{\tan(x+h)-\tan x}{h}$

$$= \lim_{h\to 0}\frac{\tan(x+h)-\tan x}{1+\tan(x+h).\tan x}.\frac{1+\tan(x+h)\tan x}{h}$$

$$= \lim_{h\to 0}\frac{\tan(x+h-x)}{h}[1+\tan(x+h)\tan x]$$

$$= \lim_{h\to 0}\left(\frac{\tan h}{h}\right) = \sec^2 x$$

$= 1 \times [1 + \tan^2 x] = \sec^2 x$

$$\Rightarrow \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

(iv) $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{\sec(x+h)-\sec x}{h}$

$$= \lim_{h\to 0}\frac{\cos x - \cos(x+h)}{h\cos x\cos(x+h)}$$

$$= \lim_{h\to 0} 2\sin\left(x+\frac{h}{2}\right)\frac{\sin\frac{h}{2}}{2.\frac{h}{2}}.\frac{1}{\cos x\cos(x+h)}$$

$$= \lim_{h\to 0}\sin x.1.\frac{1}{\cos x\cos x} = \tan x\sec x$$

$$\Rightarrow \quad \frac{d}{dx}(\sec x) = \sec x \tan x$$

(v) We have $y = f(x) = \sin^{-1} x$

$$\Rightarrow \quad x = \sin y \Rightarrow y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$= \frac{dx}{dy} = \lim_{h\to 0}\frac{\sin(y+h)-\sin y}{h} = \lim_{h\to 0}\frac{2\cos\left(y+\frac{h}{2}\right)\sin\frac{h}{2}}{2\frac{h}{2}}$$

$$= \lim_{h\to 0} \cos\left(y + \frac{h}{2}\right).\left(\frac{\sin\frac{h}{2}}{\frac{h}{2}}\right) = \cos y = \sqrt{1-\sin^2 y}$$

$$= \sqrt{1-x^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} = \frac{d}{dx}(\sin^{-1} x)$$

Example 36 : Find $\frac{dy}{dx}$ when

(i) $y = \frac{x}{\log_a x}$

(ii) $y = \frac{x}{x\log_a x}$

(iii) $y = \cos x \cos 2x \cos 4x \cos 8x$

(iv) $y = x^3 + \tan x - e^x + x \log_e x + (\sin x).a^x + \text{cosec } x - 2007$, $a > 0$

(v) $y = (1 + x)(1 + x^2)(1 + x^4)(1 + x^8).....(1 + x^{2^n})$

(vi) $y = (\cos x - i \sin x)(\cos 3x - i \sin 3x)(\cos 5x - i \sin 5x).....(\cos 2007x - i \sin 2007)$

(vii) $y = \sin x \cos x \tan x \text{ cosec } x \sec x \cot x$, if all the functions are well defined

(viii) If $y = f(x)$, where $f\left(x + \frac{1}{x}\right) = x^4 + x^{-4}$

Solution : (i) We have

$$y = \frac{x}{\log_a x} = \frac{x}{\log_e x \times \log_a x} = \frac{1}{\log_a x}\left(\frac{x}{\log_e x}\right)$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{\log_a e}\frac{d}{dx}\left(\frac{x}{\log_e x}\right) = \frac{1}{\log_a e}\left[\frac{\frac{dx}{dx}\times\log_e x - x\frac{d}{dx}(\log_e x)}{(\log_e x)^2}\right]$$

$$= \frac{(\log_e a)\left[\log_e x - x.\frac{1}{x}\right]}{(\log_e x)^2} = \frac{(\log_e a)\log_e\left(\frac{x}{e}\right)}{(\log_e x)^2}$$

(ii) $y = \frac{1}{x\log_a x}$

$$\frac{dy}{dx} = \frac{\frac{d(1)}{dx}\times x\log_a x - 1.\frac{d}{dx}(x\log_e x \times \log_a e)}{(x\log_a x)^2}$$

$$= \frac{-\times x\log_a e - 1\times\log_a e\left[\frac{dx}{dx}\times\log_e x + x\frac{d}{dx}(\log_e x)\right]}{(x\log_a x)^2}$$

$$= \frac{-\log_a e\left[1\times\log_e x + x.\frac{1}{x}\right]}{(x\log_a x)^2} = \frac{-(1+\log_e x)\log_a e}{(x\log_a x)^2}$$

(iii) $y = \cos x \cos 2x \cos 4x \cos 8x$

$$= \frac{2\sin x}{2\sin x}\cos x \cos 2x \cos 4x \cos 8x$$

$$= \frac{2\sin 2x\cos 2x\cos 4x\cos 8x}{2\times 2\sin x}$$

$$= \frac{2\sin 4x\cos 4x\cos 8x}{2\times 4\sin x}$$

$$= \frac{2\sin 8x\cos 8x}{2\times 8\sin x}$$

$$= \frac{\sin 16x}{16\sin x}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{d}{dx}\left(\frac{\sin 16x}{16\sin x}\right) = \frac{1}{16}.\frac{\sin x\frac{d}{dx}(\sin 16x) - \sin 16x\frac{d}{dx}(\sin x)}{\sin^2 x}$$

$$= \frac{1}{16}.\frac{\sin x.16\cos 16x - \sin 16x.\cos x}{\sin^2 x}$$

$$= \frac{1}{16}\left[\frac{16\sin x\cos 16x - \sin 16x\cos x}{\sin^2 x}\right]$$

(iv) $y = x^3 + \tan x - e^x + x\log_e x + (\sin x)a^x + \operatorname{cosec} x - 2007$

$$\Rightarrow \frac{dy}{dx} = \frac{dx^3}{dx} + \frac{d}{dx}(\tan x) - \frac{d}{dx}(e^x) + \frac{d}{dx}(x\log_e x) + \frac{d}{dx}(a^x\sin x) + \frac{d}{dx}(\operatorname{cosec} x) - \frac{d}{dx}(2007)$$

$$\Rightarrow$$

$$\frac{dy}{dx} = 3x^2 + \sec^2 x - e^x + \frac{dx}{dx}\times\log_e x + x\frac{d}{dx}(\log_e^x) + \frac{da^x}{dx}\times\sin x + a^x\frac{d\sin x}{dx} + (-\operatorname{cosec} x\cot x) - 0$$

$= 3x^2 + \sec^2 x - e^x + \log_e x + x\,.\,\frac{1}{x} + a^x \sin x \log_e x + a^x \cos x - \operatorname{cosec} x \cot x$

$= 3x^2 + \sec^2 x - \operatorname{cosec} x \cot x - e^x + \log_e x + a^x \sin x \log_e a + a^x \cos x + 1$

(v) $y = (1 + x)(1 + x^2)(1 + x^4)(1 + x^8).....(1 + x^{2^n})$

$$= \frac{(1-x)}{(1-x)}(1+x)(1+x^2)(1+x^4)(1+x^8).....(1+x^{2^n})$$

$$= \frac{(1-x^2)(1+x^4)(1+x^8)....(1+x^{2^n})}{(1-x)}$$

$$= \frac{(1-x^8)(1+x^8)....(1+x^{2^n})}{(1-x)}$$

$$y = \frac{(1-x^{2^n})(1+x^{2^n})}{1-x} = \frac{1-x^{2^{n+1}}}{1-x}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{(1-x)\frac{d}{dx}(1-x^{2n+1}) - (1-x^{2^{n+1}})\frac{d}{dx}(1-x)}{(1-x)^2}$$

$$= \frac{(1-x)(0-2^{n+1}.x^{2^{n+2}-1}) - (1-x^{2^{n+1}})(0-1)}{(1-x)^2}$$

$$= \frac{(1-x^{2^{n+1}}) - 2^{n+1}(1-x)(x^{2^{n+1}-1})}{(1-x)^2}$$

(vi) $y = (\cos x - i \sin x)(\cos 3x - i \sin 3x)(\cos 5x - i \sin 5x).....(\cos 2007x - i \sin 2007x)$

$= e^{-ix} \cdot e^{-3/x} \cdot e^{-5/x}.....e^{-i2007x}$, $[e^{-i\theta} = \cos\theta - i \sin\theta]$

$$= e^{-i(1004)^2 x} = \cos(1004)^2 x - i\sin(1004)^2 x$$

$$\frac{dy}{dx} = -\sin(1004)^2 x \times (1004)^2 - i \times (1004)^2 \cos(1004)^2 x$$

$= -(1004)^2[\sin(1004)^2x + i \cos(1004)^2x]$

(vii) $y = \sin x \cos x \tan x \operatorname{cosec} x \sec x \cot x$

$= (\sin x \operatorname{cosec} x)(\cos x \sec x)(\tan x \cot x)$

$= 1 \times 1 \times 1 = 1$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{d}{dx}(1) = 0$$

(viii) We have,

$$f\left(x + \frac{1}{x}\right) = x^4 + \frac{1}{x^4} = \left(x^2 + \frac{1}{x^2}\right)^2 - 2$$

$$= \left[\left(x + \frac{1}{x}\right)^2 - 2\right]^2 - 2$$

$\Rightarrow \quad f(x) = (x^2 - 2)^2 - 2 = x^4 - 4x^2 + 2 = y$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{d}{dx}(x^4) - \frac{d}{dx}(4x^2) + \frac{d}{dx}(2)$$

$= 4x^3 - 8x + 0$

$= 4x(x^2 - 2)$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic 1: Limit and definition

Q.1 $\underset{x\to 2}{Lt}\left(\frac{[x]^3}{3}-\left[\frac{x}{3}\right]^3\right)$ is, where $[x]$ represents the integral part of x

(a) 0 (b) $\frac{64}{27}$

(c) $\frac{8}{3}$ (d) $\frac{3}{8}$

Q.2 $\underset{x\to 2}{Lt}\{[x-2]+[2-x]-x\}=$ where [.] represent greater integral function

(a) 0 (b) 3

(c) –3 (d) –2

Q.3 If $\{x\}$ denotes the frational part of x, then $\lim_{x\to 0}\frac{\{x\}}{\tan\{x\}}$ is equal to

(a) 1 (b) 0

(c) –1 (d)Limit doesn't exist

Q.4 If $0<\alpha<\beta$ then $\lim_{n\to\infty}(\beta^n+\alpha^n)^{\frac{1}{n}}$ is equal to

(a) α (b) β

(c) $\alpha\beta$ (d) e

Q.5 $\lim_{x\to 5^+}\frac{x^2-9x+20}{x-[x]}$ is equal to

(a) 0 (b) 1

(c) 2 (d) Not defined

Q.6 $\lim_{x\to\frac{\pi^+}{2}}\frac{\cos(\tan^{-1}(\tan x^2))}{x-\frac{\pi}{2}}$

(a) 0 (b) 1

(c) 2 (d) 3

Q.7 $\lim_{x\to 0}\frac{1-\cos x}{x\sqrt{x^2}}$

(a) 1/2 (b) –1/2

(c) 0 (d) Does not exist

Q.8 Let $f(2) = 4$, $f'(2) = 4$. Then $\underset{x\to 2}{Lt}\frac{xf(2)-2f(x)}{x-2}$ is

(a) – 1/3 (b) –2

(c) –4 (d) 3

Q.9 If $f(4) = 4$, $f'(4) = 1$ then $\lim_{x\to 4}\frac{2-\sqrt{f(x)}}{2-\sqrt{x}}$ is equal to

(a) 0 (b) 1

(c) –1 (d) 2

Q.10 The value of $\lim_{x\to 0}\left(\frac{x^4\sin\left(\frac{1}{x}\right)+x^2}{1+|x|^3}\right)$ equals

(a) 1 (b) –1

(c) 2 (d) 0

Q.11 If $f(x)$ and $g(x)$ be differentiable functions and $f(1) = g(1) = 2$ them

$\underset{x\to 1}{Lt}\frac{f(1).g(x)-f(x).g(1)-f(1)+g(1)}{g(x)-f(x)}$ is equal to

(a) 0 (b) 1

(c) 2 (d) –1

Topic: 2 Determinante and indeterminate form

Q.12 If $\underset{x\to\infty}{Lt}\left(1+\frac{a}{x}+\frac{b}{x^2}\right)^{2x}=e^2$, then the values of a and b are

(a) $a\in R, b\in R$ (b) $a = 1, b\in R$

(c) $a\in R, b = 2$ (d) $a = 1, b = 2$

Q.13 If $\underset{x\to 0}{Lt}(\cos x+a\sin bx)^{1/x}=e^2$, then the values of a and b are

(a) $a = 1, b = -2$ (b) $a = 2\sqrt{2}, b = \sqrt{2}$

(c) $a = 2\sqrt{2}, b = \frac{1}{2}$ (d) $a = -2, b = 1$

Q.14 The integer n for which $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is

(a) 4 (b) 3

(c) 2 (d) 1

Q.15 Let α and β be the distinct roots of $ax^2 + bx + c = 0$, then $\underset{x\to\alpha}{Lt}\frac{1-\cos(ax^2+bx+c)}{(x-\alpha)^2}$ is equal to

(a) $\frac{1}{2}(\alpha-\beta)^2$ (b) $-\frac{a^2}{2}(\alpha-\beta)^2$

(c) 0 (d) $\frac{a^2}{2}(\alpha-\beta)^2$

Q.16 $\lim_{x\to 0}\frac{\sin 2x}{2-\sqrt{4-x}}$ is

(a) 2 (b) 4
(c) 8 (d) 0

Q.17 Let $\lim_{x\to 0}\frac{\sin 2x}{x}=a$ and $\lim_{x\to 0}\frac{3x}{\tan x}=b$, then $a + b$ equals

(a) 5 (b) 6
(c) 0 (d) 4

Q.18 $\lim_{x\to\infty}\frac{\sin x}{x}+\lim_{x\to\infty}\frac{\log x}{x}$ equals

(a) 0 (b) 1
(c) 3 (d) ∞

Q.19 $\lim_{x\to 0}((x\cot x)+(x\log x))$ equals

(a) 1 (b) 2
(c) 3 (d) 0

Q.20 $\lim_{x\to 0}(\sqrt{x+1}-\sqrt{x})$ equals

(a) ∞ (b) 0
(c) –1 (d) 1

Topic: 3 Properties of Limit

Q.21 Let $S_n = \frac{1}{1.4}+\frac{1}{4.7}+\frac{1}{7.10}+\ldots\ldots$ to n terms then $\lim_{n\to\infty} S_n$ is equal to

(a) 1/3 (b) 3
(c) 1/4 (d) ∞

Q.22 Let $t_r = \frac{r}{1+r^2+r^4}$, then $\lim_{n\to\infty}\sum_{r=1}^{n} t_r$ equals

(a) 1/2 (b) 1
(c) 1/4 (d) 1/3

Q.23 $\lim_{n\to\infty}\cos\frac{x}{2}\cos\frac{x}{4}\ldots\ldots\cos\frac{x}{2^n}$ equals

(a) 1 (b) $\frac{\sin x}{x}$

(c) $\frac{x}{\sin x}$ (d) $\cos x$

Q.24 $\lim_{n\to\infty}\left(\tan\theta+\frac{1}{2}\tan\frac{\theta}{2}+\ldots\ldots+\frac{1}{2^n}\tan\frac{\theta}{2^n}\right)$ equals

(a) $\frac{1}{\theta}$ (b) $\frac{1}{\theta}-2\cot 2\theta$
(c) $2\cot 2\theta$ (d) $2\tan^2\theta$

Q.25 $\lim_{x\to\infty} x^2\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\left(1-\cos\frac{1}{x}\right)\sqrt{\ldots\ldots to\ \infty}}}}$ equals

(a) $\frac{1}{2}$ (b) 2

(c) $\frac{1}{4}$ (d) 1

Q.26 $m, n \in I^+$, then $\lim_{x\to 0}\frac{\sin x^n}{(\sin x)^m}$ equals

(a) 1, if $n < m$ (b) 0, if $n = m$
(c) n/m (d) 0, if $n > m$

Topic: 4 Important Result

Q.27 If α, β are roots of $ax^2 + bx + c = 0$, then $\underset{x\to\alpha}{Lt}(1+ax^2+bx+c)^{\frac{1}{x-\alpha}}$ is

(a) $a(\alpha-\beta)$ (b) $\log e^{|a(\alpha-\beta)|}$
(c) $e^{a(\alpha-\beta)}$ (d) $e^{a(\alpha+\beta)}$

Q.28 Let α and β be real and distinct root of $ax^2 + bx + c = 0$ then $\lim_{x\to\alpha}[1+\sin(ax^2+bx+c)]^{\frac{1}{x-\alpha}}$ is

(a) $e^{\alpha-\beta}$ (b) $e^{\beta-\alpha}$
(c) $e^{a(\alpha-\beta)}$ (d) $e^{a(\beta-\alpha)}$

Q.29 $\underset{x\to 0}{Lt}\frac{\sqrt{1-\cos 2x}}{\sqrt{2x}}=$

(a) 1 (b) –1
(c) zero (d) Does not exist

Q.30 If α and β be the roots of the equation $ax^2 + bx + c = 0$ then $\lim_{x\to\frac{1}{\alpha}}\sqrt{\frac{1-\cos^2(cx^2+bx+a)}{4(1-\alpha x)^2}}$

(a) Does not exist

(b) Equals $\left|\frac{c}{2\alpha}\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)\right|$

(c) Equals $\left|\frac{c}{2\alpha}\left(\frac{1}{\alpha}-\frac{1}{\beta}\right)\right|$

(d) Equals $\left|\frac{c}{2}\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)\right|$

Q.31 $\lim_{n\to 1}\left(\sec\frac{\pi}{2^x}\right)$. ln x is equal to

(a) $\frac{\pi}{2}$ (b) $-\frac{2}{\pi \ln 2}$

(c) $\frac{2}{\pi \ln 2}$ (d) $\frac{\pi}{\ln 2}$

Q.32 Let $\lim_{x\to 0}\frac{\sin 2x}{\tan\left(\frac{x}{2}\right)} = L_1$ and $\lim_{x\to 0}\frac{e^{2x}-1}{x} = L_2$ then the value of L_1L_2 is

(a) 4 (b) 8
(c) 6 (d) 2

Q.33 $\lim_{x\to 0}\frac{2^x-1}{x} + \lim_{x\to 0}\frac{3^x-1}{x} - \lim_{x\to 0}\left(\frac{6^x-1}{x}\right)$ equals

(a) 1 (b) 2
(c) 0 (d) 4

Q.34 $\lim_{x\to 0}\frac{\log(1+2x)}{x} + \lim_{x\to 2}\frac{x^4-2^4}{x-2}$ equals

(a) 30 (b) 32
(c) 35 (d) 34

Q.35 $\lim_{x\to 0}\frac{x\tan 2x - 2x\tan x}{(1-\cos 2x)^2}$ equals

(a) 1 (b) 1/2
(c) –1/3 (d) 1/4

Q.36 $\underset{x\to 0}{Lt}\frac{2^x-1}{\sqrt{1+x}-1} =$

(a) 2 (b) $\log_e 2$

(c) $\frac{\log_e 2}{2}$ (d) $2\log_e 2$

Q.37 $\lim_{x\to 0}\frac{(1+x+x^2)}{(\ln x)^3}$ is equal to

(a) 2 (b) e^2
(c) e^{-2} (d) ∞

Q.38 For $x \in R$, $\lim_{x\to\infty}\left(\frac{x-3}{x+2}\right)^x$ equals

(a) e (b) e^{-1}
(c) e^{-5} (d) e^5

Q.39 $\lim_{x\to\infty}\left(\frac{1+\tan x}{1+\sin x}\right)^{\text{cosec}x}$ equals

(a) –1 (b) e
(c) 1 (d) e^{-1}

Q.40 $\lim_{x\to 0}\left(\frac{4^x+9^x}{2}\right)^{\frac{1}{x}}$ equals

(a) 2 (b) 6
(c) 16 (d) 112

Section-B
(Assertion-Reason)

Q.1 **Statement–1 :** $\lim_{x\to\infty}\left(\frac{x-1}{x+1}\right)^{x+2} = e^{-2}$

Statement–2 : $\lim_{x\to\infty}\frac{x^5}{(2008)^x} = 120$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.2 **Statement–1 :** $\lim_{x\to\infty}\left(1+\sin\frac{p}{x}\right)^x = e^p$

Statement–2 : $\lim_{x\to\infty}\left(1+\frac{p}{x}\right)^x = e^p$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.3 **Statement–1 :** $\lim_{x\to\infty}\frac{1-\cos x}{x(2^x-1)} = \frac{1}{2}\log_2 e$

Statement–2 : $\lim_{x\to 0}\frac{\sin x}{x} = 1$,

$\lim_{x\to 0}\frac{a^x-1}{x} = \log a,\ a>0$

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1

(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.4 **Statement – 1 :** $\lim_{x\to 0}\frac{\sqrt{1-\cos 2x}}{x}$ does not exist.

Statement – 2 : For $\frac{\sqrt{1-\cos 2x}}{x}$ at (x = 0). Right hand limit ≠ Left hand limit

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.5 **Statement-1:** $\lim_{x\to 0}\frac{\sqrt{1-\cos 2x}}{x}=\sqrt{2}$

Statement-2: $\lim_{x\to 0}\frac{\sin x}{x}=1$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Q.6 **Statements-1:** $\lim_{x\to 0}\frac{\tan x^o}{x}=1$, where x^o means x degrees.

Statements-2: $\lim_{x\to 0} f(x) = l$, $\lim_{x\to 0} g(x) = m$, then $\lim_{x\to 0}\{f(x)g(x)\} = lm$.

(a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(b) Statement-1 is True, Statement-2 is True; Statement-2 is not correct explanation for Statement-1
(c) Statement-1 is True, Statement-2 is False.
(d) Statement-1 is False, Statement-2 is True.

Section-C
(Previous Year Question)

Q.1 Let $\alpha(a)$ abd $\beta(a)$ be the roots of the equation

$\left(\sqrt[3]{1+a}-1\right)x^2+\left(\sqrt{1+a}-1\right)x+\left(\sqrt[6]{1+a}-1\right)=0$

where $a>-1$. Then $\lim_{a\to 0^+}\alpha(a)$ and $\lim_{a\to 0^+}\beta(a)$ are

(a) $-\frac{5}{2}$ and 1 (b) $-\frac{1}{2}$ and -1

(c) $-\frac{7}{2}$ and 2 (d) $-\frac{9}{2}$ and 3

Q.2 If $\lim_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$, then

(a) $a=1,b=4$ (b) $a=1,b=-4$

(c) $a=2,b=-3$ (d) $a=2,b=3$

Q.3 If $\lim_{x\to 0}\left[1+x\ln\left(1+b^2\right)\right]^{1/x}=2b\sin^2\theta, b>0$ and $\theta\in(-\pi,\pi]$, then the value of θ is

(a) $\pm\frac{\pi}{4}$ (b) $\pm\frac{\pi}{3}$

(c) $\pm\frac{\pi}{6}$ (d) $\pm\frac{\pi}{2}$

Q.4 The value of $\lim_{x\to 0}\frac{1}{x^3}\int_0^x\frac{t\ln(1+t)}{t^4+4}dt$ is

(a) 0 (b) $\frac{1}{12}$

(c) $\frac{1}{24}$ (d) $\frac{1}{64}$

Q.5 Let $g(x)=\frac{(x-1)^n}{\log\cos^m(x-1)}; 0<x<2, m$ and n are integers, $m\neq 0, n>0$, and let p be the left hand derivative of $|x-1|$ at $x=1$. If lim g(x)=p, then

(a) $n=1,m=1$ (b) $n=1,m=-1$

(c) $n=2,m=2$ (d) $n>2, m=n$

Q.6 The value of $\lim_{x\to 0}\left((\sin x)^{1/x}+(1/x)^{\sin x}\right)$, where x > 0 is

(a) 0 (b) -1
(c) 1 (d) 2

Q.7 $\lim_{h\to 0}\frac{f\left(2h+2+h^2\right)-f(2)}{f\left(h-h^2+1\right)-f(1)}$, given that f'(2)=6 and f'(1)=4

(a) does not exist (b) is equal to -3/2
(c) is equal to 3/2 (d) is equal to 3

Q.8 If $\lim_{x\to 0}\frac{((a-n)nx-\tan x)\sin nx}{x^2}=0$, where n is non zero real number, then a is equal to

(a) 0 (b) $\frac{n+1}{n}$

(c) n (d) $n+\frac{1}{n}$

Q.9 Let $f:R\to R$ be such that f(1) =3 and $f'(1)=6$, then $\lim_{x\to 0}\left(\frac{f(1+x)}{f(1)}\right)^{\frac{1}{x}}$ equals

(a) 1 (b) $e^{\frac{1}{2}}$

(c) e^2 (d) e^3

Q.10 The integer n, for which $\lim_{x\to 0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number is

(a) 1 (b) 2
(c) 3 (d) 4

Q.11 $\lim_{x\to 0}\frac{\sin\{\pi\cos^2 x\}}{x^2}$ equals

(a) $-\pi$ (b) π
(c) $\pi/2$ (d) 1

Q.12 For $x\in R$, $\lim_{x\to\infty}\left(\frac{x-3}{x+2}\right)^x=$

(a) e (b) e^{-1}
(c) e^{-5} (d) e^5

Q.13 $\lim_{x\to 0}\frac{x\tan 2x-2x\tan x}{(1-\cos 2x)^2}$ is

(a) 2 (b) -2
(c) 1 / 2 (d) -1 / 2

Q.14 $\lim_{x\to 1}\frac{\sqrt{1-\cos 2(x-1)}}{x-1}$

(a) exists and it equals $\sqrt{2}$

(b) exists and it equals $-\sqrt{2}$

(c) does not exists because $x-1\to 0$
(d) does not exist because the left hand limit is not equal to the right hand limit

Q.15 The value of $\lim_{x\to 0}\frac{\sqrt{\frac{1}{2}(1-\cos 2x)}}{x}$

(a) 1 (b) -1
(c) 0 (d) none of these

Q.16 $\lim_{x\to 0}\left\{\frac{1}{1-n^2}+\frac{2}{1-n^2}+\ldots+\frac{n}{1-n^2}\right\}$ is equal to

(a) 0 (b) $-\frac{1}{2}$

(c) $\frac{1}{2}$ (d) none of these

Q.17 If $f(a)=2, f'(a)=1, g(a)=-1, g'(a)=2$ then the value of $\lim_{x\to a}\frac{g(x)f(a)-g(a)f(x)}{x-a}$ is

(a) -5 (b) $\frac{1}{5}$
(c) 5 (d) none of these

Q.18 If $G(x)=-\sqrt{25-x^2}$ then $\lim_{x\to 1}\left(\frac{G(x)-G(1)}{x-1}\right)=$

(a) $\frac{1}{\sqrt{24}}$ (b) $\frac{1}{5}$

(c) $-\sqrt{24}$ (d) none of these

Q.19 There exists a function f(x) satisfying f(0)=-1, $f'(0)=-1, f(x)>0$ for all x and

(a) $f''(x)>0$ for all x

(b) $-1<f''(x)<0$ for all x

(c) $-2\le f''(x)\le -1$ for all x

(d) $f''(x)<-2$ for all x

Section-D
(Board pattern)

Q.1 For what value of *p* does the limit $\lim_{x\to 2} f(x)$ exist where *f* is defined by the rule *f*(*x*) = $\begin{cases}3px+1, \text{if } x<2\\ 1-px^2, \text{if } x>2\end{cases}$

Q.2 Suppose *f*(*x*) = $\begin{cases}p+qx, & x<1\\ 5 & x=1\\ q-px, & x>1\end{cases}$ and if $\lim_{x\to 1} f(x)=f(1)$

What are possible values of p and q?

Q.3 Evaluate $\lim_{x\to 0}\frac{\sin(3+x)-\sin(3-x)}{x}$

Q.4 Evaluate $\lim_{x\to 0}\frac{1-\cos 2x}{1-\cos 4x}$

Q.5 Evaluate $\lim_{x\to 0}\frac{\tan x^{o}}{x}$

Q.6 Evaluate $\lim_{x\to 0}\frac{\sin 3x-4\tan 2x+3\sin x}{x}$

Q.7 $\lim_{x\to 3}\frac{3x^2-5x}{2x+4}$

Q.8 $\lim_{x\to 0}\frac{(x+2)^6-2^6}{x}$

Q.9 $\lim_{x\to 0}\frac{\sin 5x}{7x}$

Q.10 $\lim_{x\to 0}\frac{\sin 4x}{\sin 3x}$

Q.11 $\lim_{x\to 0}\frac{3x+x\cos x}{4\sin x}$

Q.12 $\lim_{x\to 0}\frac{1-\cos 2x}{x}$

Q.13 $\lim_{x\to \pi}\frac{\sin x}{\pi-x}$

Q.14 $\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{\sin x}$

Q.15 $\lim_{x\to -1}\frac{x^3+8x^2+19x+12}{x^3-2x^2-13x-10}$

ANSWER KEY

CHAPTER-01 (CIRCLE)

SECTION-A (Straight objective type questions)

1. (a)	2. (a)	3. (d)	4. (c)	5. (c)	6. (d)	7. (b)
8. (a)	9. (a)	10. (a)	11. (d)	12. (c)	13. (b)	14. (b)
15. (a)	16. (d)	17. (b)	18. (b)	19. (a)	20. (a)	21. (b)
22. (c)	23. (c)	24. (b)	25. (b)	26. (a)	27. (d)	28. (a)
29. (c)	30. (c)	31. (a)	32. (a)	33. (c)	34. (a)	35. (a)
36. (a)	37. (c)	38. (b)	39. (c)	40. (a)	41. (b)	42. (d)
43. (a)	44. (a)	45. (c)	46. (a)	47. (a)	48. (c)	49. (a)

SECTION-B (Assertion-Reason type questions)

1. (a)	2. (a)	3. (d)	4. (a)	5. (b)

SECTION-C (Previous Year questions)

1. (c)	2. (a)	3. (b)	4. (c)	5. (a)	6. (c)	7. (b)
8. (c)	9. (d)	10. (a)	11. (c)	12. (b)	13. (d)	14. (a)
15. (a)	16. (b)	17. (c)	18. (a)	19. (c)	20. (d)	21. (a)
22. (d)	23. (d)	24. (d)	25. (b)	26. (b)	27. (d)	28. (a)
29. (b)	30. (c)					

SECTION-D (School/Board level type questions)

1. $(x-2)^2+(y-2)^2=4$ 2. $(5+10\pi, 5)$ 3. $(x-6)^2+y^2=5^2$ 4. $x^2+y^2-6x-8y+20=0$

5. $(x-2)^2+y^2=5^2$ 6. Centre $(3,-2)$, radius 4 7. $(x-4)^2+(y-3)^2=25$

8. $\left(x-\frac{a}{2}\right)^2+\left(y-\frac{a}{2}\right)^2=\frac{a^2}{2}$ 9. $(x-1)^2+(y-2)^2=25$ 10. Inside

CHAPTER-02 (PARABOLA)

SECTION-A (Straight objective type questions)

1. (c)	2. (c)	3. (c)	4. (a)	5. (a)	6. (b)	7. (c)
8. (d)	9. (b)	10. (b)	11. (b)	12. (c)	13. (c)	14. (d)
15. (b)	16. (d)	17. (d)	18. (d)	19. (a)	20. (b)	21. (b)
22. (b)	23. (c)	24. (c)	25. (a)	26. (c)	27. (a)	28. (c)
29. (a)	30. (c)	31. (c)	32. (a)	33. (a)	34. (b)	35. (a)
36. (a)	37. (b)	38. (b)				

SECTION-B (Assertion-Reason type questions)

1. (c)	2. (a)	3. (d)	4. (c)

SECTION-C (Previous Year Question)

1. (a)	2. (d)	3. (c)	4. (a)	5. (c)	6. (d)	7. (c)
8. (c)	9. (c)	10. (a)	11. (d)	12. (c)	13. (a)	14. (d)
15. (c)	16. (b)	17. (b)	18. (c, d)			

SECTION-D (School/Board level type questions)

1. $|a| \le \frac{1}{\sqrt{2}}$ 2.Vertex : (1, 1), focus : $\left(\frac{5}{4}, 1\right)$, directrix : $x = \frac{3}{4}$, axis : $y = 1$, latus recturm : 1

3. 9/2 4. $y = 2x^2 + 3x + 4$ 5. 16 6. (4, –4)

CHAPTER-03 (ELLIPSE)

SECTION-A (Straight objective type questions)

1. (b)	2. (c)	3. (b)	4. (b)	5. (a)	6. (c)	7. (a)
8. (d)	9. (b)	10. (b)	11. (b)	12. (d)	13. (c)	14. (b)
15. (b)	16. (b)	17. (a)	18. (b)	19. (b)	20. (a)	21. (a)
22. (c)	23. (d)	24. (a)	25. (a)	26. (c)	27. (a)	28. (c)
29. (c)	30. (c)	31. (b)	32. (c)	33. (a)	34. (a)	35. (c)
36. (c)	37. (c)	38. (c)	39. (c)	40. (a)	41. (a)	42. (d)
43. (b)	44. (c)	45. (b)	46. (a)			

SECTION-B (Assertion-Reason type questions)

1. (a)	2. (d)	3. (b)	4. (a)	5. (d)	6. (b)

SECTION-C (Previous Year Question)

1. (a)	2. (c)	3. (c)	4. (b)	5. (d)	6. (c)	7. (c)
8. (d)	9. (a)	10. (b)	11. (a)	12. (*)		

CHAPTER-04 (HYPERBOLA)

SECTION-A (Straight objective type questions)

1. (d)	2. (a)	3. (a)	4. (d)	5. (a)	6. (c)	7. (b)
8. (b)	9. (c)	10. (b)	11. (b)	12. (b)	13. (d)	14. (d)
15. (d)	16. (d)	17. (c)	18. (b)	19. (c)	20. (b)	21. (a)
22. (b)	23. (c)	24. (b)	25. (b)	26. (b)	27. (d)	28. (b)
29. (b)	30. (b)	31. (b)				

SECTION-B (Assertion-Reason type questions)

1. (d)	2. (a)	3. (a)	4. (c)	5. (a)	6. (a)	7. (a)
8. (a)	9. (a)	10. (b)				

SECTION-C (Previous Year questions)

1. (c)	2. (b)	3. (a)	4. (b)	5. (b)	6. (b)	7. (b)
8. (b)	9. (d)	10. (d)	11. (a)			

SECTION-D (School/Board level type questions)

1. Foci : $(\pm\sqrt{13}, 0)$, vertices : $(0, \pm 3)$, $e = \frac{\sqrt{13}}{3}$, latus rectum = $\frac{8}{3}$

2. $\frac{x^2}{25} - \frac{y^2}{9} = 1$ 3. $\frac{y^2}{16} - \frac{x^2}{9} = 1$ 4.$e = \sqrt{13}$ 4. $y = \frac{1}{\sqrt{3}}(x + 3)$ 5. $k \in (4, 7)$

6. $b = 2$ 7. $\frac{x^2}{5} - \frac{y^2}{4} = 1$ 8. $|a| \le \frac{1}{\sqrt{2}}$ 9. 5/12

10. $(2x - 1)^2 + (2y - 7)^2 = 2(x - y + 1)^2$

CHAPTER-5 (ANALYTICAL GEOMETRY 3D)

SECTION-A (Straight objective type questions)

1. (a)	2. (d)	3. (a)	4. (a)	5. (a)	6. (c)	7. (b)
8. (a)	9. (d)	10. (d)	11. (c)	12. (d)	13. (a)	14. (c)
15. (d)	16. (b)	17. (d)	18. (a)	19. (c)	20. (a)	21. (b)
22. (c)	23. (b)	24. (a)	25. (b)	26. (a)	27. (a)	28. (a)
29. (c)	30. (c)	31. (a)	32. (c)	33. (c)	34. (b)	35. (a)
36. (b)	37. (c)					

SECTION-B (Assertion-Reason type questions)

1. (1) 2. (4) 3. (3) 4. (2)

SECTION-C (Previous Year questions)

1. (b)	2. (a)	3. (d)	4. (a,c)	5. (b)	6. (b)	7. (b)
8. (d)	9. (b,c,d)	10. (c)	11. (b,d)	12. (a,b)	13. (c)	14. (d)
15. (a)	16. (d)	17. (c)	18. (a)	19. (a)	20. (a)	21. (c)
22. (c)	23. (a)	24. (d)	25. (a)	26. (a)	27. (b,c)	28. (a,d)
29. (b,d)	30. (b,c)					

31. $62x + 29y + 19z - 105 = 0$ 32. $\frac{9}{2}$ cu units 34. $\alpha = 6, \beta = 5, y = -2; Q(6, 5, -2)$

35. $|d| = 6$

SECTION-D (School/Board level type questions)

1. $\cos^{-1}\left(\frac{1}{6}\right)$ **2.** $\frac{\pi}{3}$ **3.** $\lambda = 8$ 4. $C \equiv \left(\frac{29}{12}, \frac{9}{4}, \frac{25}{0}\right)$ **5.** (–2, –9, 8) **6.** –1 7. $\left(\pm\frac{1}{\sqrt{2}}, 0, \pm\frac{1}{\sqrt{2}}\right)$ **8.** 5 **9.** 3 **10.** $\sqrt{26}$

CHAPTER-06 (LIMITS AND DERIVATIVES)

SECTION-A (Straight objective type questions)

1. (c)	2. (c)	3. (d)	4. (b)	5. (a)	6. (b)	7. (d)
8. (b)	9. (c)	10. (d)	11. (d)	12. (b)	13. (c)	14. (b)
15. (d)	16. (c)	17. (a)	18. (a)	19. (a)	20. (b)	21. (a)
22. (a)	23. (b)	24. (b)	25. (a)	26. (c)	27. (c)	28. (c)
29. (d)	30. (c)	31. (c)	32. (b)	33. (c)	34. (d)	35. (a)
36. (b)	37. (d)	38. (d)	39. (c)	40. (c)		

SECTION-B (Assertion-Reason type questions)

1. (c)	2. (a)	3.(a)	4. (a)	5. (d)	6. (d)	7. (b) 8. (b)

SECTION-C (Previous Year Question)

1. (b)	2. (b)	3. (d)	4. (b)	5. (c)	6. (c)	7. (d)
8. (d)	9. (c)	10. (c)	11. (b)	12. (c)	13. (c)	14. (d)
15. (d)	16. (b)	17. (c)	18. (a)	19. (a)	20. (b)	21. (a,c)
22. (0)						

SECTION-D (School/Board level type questions)

1. $p = 0$	2. $p = 0, q = 5$ 3. 2 cos 3		4. $\frac{1}{4}$	5. $\frac{\pi}{180}$	6. –2	7. $\frac{6}{5}$
8. 192	9. $\frac{5}{7}$	10. $\frac{4}{3}$	11. 1	12. 0	13. 1	
14. 1	15. –1					